Innovate Together, Thrive Sustainably, Advance the World
攜手創新，永續繁榮，領航世界

URBP7008 Strategic and Community Planning Studio

To position San Tin Technopole as a global innovation powerhouse and frontier-creator that fosters talent development, advanced innovation, sustainable prosperity, and regional integration while bridging the GBA with the world.

2026
BASELINE REVIEW & DRAFT SCP

San Tin Technopole and Surrounding Area

San Tin/ Ngau Tam Mei/ Lok Ma Chau/
The loop/ Mai Po

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List of Abbreviations

AAB	Antiquities Advisory Board
AECOM	AECOM Asia Company Ltd.
AMO	Antiquities and Monuments Office
ASA	Archaeologically Sensitive Area
CEDD	Civil Engineering and Development Department
CUHK	The Chinese University of Hong Kong
GBA	Greater Bay Area
GDP	Gross Domestic Product
HFSTC	Hangzhou Future Science Tech City
HKEX	Hong Kong Exchanges and Clearing
HKIAC	Hong Kong International Arbitration Centre
HKSTP	Hong Kong Science and Technology Park
HKU	The University of Hong Kong
HKUST	The Hong Kong University of Science and Technology
HSITP	Hong Kong-Shenzhen Innovation and Technology Park
HUL	Historic Urban Landscape
I&T	Innovation and Technology
ICH	Intangible Cultural Heritage
ICH Database	Hong Kong Intangible Cultural Heritage Database
IIT	Individual Income Tax
IP	Intellectual Property
IPO	Initial Public Offering
IV	Indigenous Village
MEC	Microelectronics Centre
MP	Mai Po
NbS	Nature-based Solutions
NIV	Non-indigenous Village
NM	Northern Metropolis
NTM	Ngau Tam Mei
OECD	Organisation for Economic Co-operation and Development
OZP	Outline Zoning Plan
PBG	Permitted Burial Ground
PCT	The Patent Cooperation Treaty
PD	Planning Department
R&D	Research and Development
SAI	Site of Archaeological Interest
SCP	Strategic Concept Plan
SLR	Sea Level Rise
STT	San Tin Technopole
TBP	Town Planning Board
TCH	Tangible Cultural Heritage
TOD	Transit-oriented Development
UNESCO	United Nations Educational, Scientific and Cultural Organisation
VC	Venture Capital
WIPO	World Intellectual Property Organisation

Contents

Executive Summary

San Tin Technopole (STT) is a strategic and important pillar of the NM. This report takes San Tin Technopole as the core research subject, based on the national 14th and 15th Five-Year Plan, the Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area (GBA), and Hong Kong's Northern Metropolis (NM) Development Strategy. Using the RIG-Triple Loop Synergy Model (RIG-TLS Model), the system conducts a full-dimension baseline assessment, draws on international best practices, carries out a Sustainable Impact Assessment and SWOT analysis, and ultimately proposes a preliminary Strategic Concept Plan (SCP). This report provides a scientific basis and spatial solutions for transforming San Tin Technopole into a global innovation and technology hub.

The study established three major strategic focuses: I&T development and regional synergy, talent-supportive and urban-rural inclusive communities, and environmental sustainability and carbon neutrality. The baseline review has identified several strengths and weaknesses that San Tin possesses. While the site has a favourable Shenzhen-Hong Kong cross-border location and rich rural culture and wetland ecological resources, it also features missing linkages for industrialising research achievements and high housing and living costs. San Tin is also facing different opportunities and threats. On the one hand, macro policy trends and the adoption of smart living technologies provide momentum for optimising the talent ecosystem. But on the other hand, the site is facing serious threats from climate change, especially the risk.

Based on the results of the baseline review, the report proposes a concept of a holistic strategy: in the industry, the deployment of artificial intelligence, life and health, pre-manufacturing, and financial legal services in four major production areas, to complement the cycling and entrepreneurship of the deep harbour, to enhance the movement of transborder elements; in the community and space, the establishment of a three-tier housing system with government funding, market directors, and enterprise/higher education, to create a 15-minute life-cycle business, public services and rural communities, to protect local villages and cultural heritage, to promote the integration of talent and village space; and, at the environmental level, to protect and enhance the ecological value, to build up climate resilience and to adopt concept of circular economy, to establish energy, water, material and environment, and to promote climate change.

The plan is based on the vision of "Innovate Together, Thrive Sustainably, Advance the World" and aims to bring together the world's young talents, spearheading technological development, supporting Shenzhen-Hong Kong collaboration, building a carbon-neutral, world-class innovative hub, unleashing new productive forces and ensuring urban-rural integration, and providing core support for the establishment of an international innovative centre in Hong Kong and for the development of northern metropolitan areas.

1. Introduction

This baseline review report aims to conduct a comprehensive and systematic review and evaluation of the STT project in line with national and regional macro development strategies. At the national level, the 14th Five-Year Plan has clearly supported Hong Kong's development into an international innovation and technology hub and established a "two corridors and two nodes" framework, positioning the Hetao Shenzhen-Hong Kong Science and Technology Innovation Cooperation Zone (the Hetao Co-operation Zone) as a key platform for collaboration at the national level (The State Council of the People's Republic of China, 2021). With the introduction of the 15th Five-Year Plan (2026–2030), the strategic focus has shifted to developing "new quality productive forces" and achieving a high level of technological self-reliance (The State Council of the People's Republic of China, 2026). STT has been positioned as one of the key bases for these emerging productive forces, aiming to connect Mainland China's innovation chain with global markets through the 'dual circulation' framework. At the regional level, according to the Outline Development Plan for the Guangdong-Hong Kong-Macao GBA, STT is located at the core of the "Hong Kong-Shenzhen Close Interaction Circle", which is not only the pillar of the NM development strategy to build an international innovation and technology new city, but also a pilot project to deepen the integration of Shenzhen and Hong Kong in the fields of industry, society and ecology (HKSAR, 2021; The State Council of the People's Republic of China, 2019).

This baseline study adheres to the theoretical framework proposed in the Inception Stage, using the RIG-TLS model as the core tool for diagnosing the current situation. This model advocates the interlocking cycle of "research", "industry" and "governance" driven by the synergy between talent support, urban-rural integration and environmental sustainability. Based on this model, this phase will conduct an in-depth baseline analysis of the six key aspects, including environment, talents, urban-rural integration, research, industry, and governance, together with general contexts such as topography, land use, and demographics. On this basis, the report includes a SWOT analysis and community input to identify planning challenges and potential opportunities, highlighting specific strategic implications that directly inform the formulation of the SCP.

Based on the analysis and vision-oriented nature of the RIG-TLS model, three strategic focuses have been identified, namely I&T development and regional synergy, talent support and urban and rural inclusive communities, and environmental sustainability and carbon neutrality. These focus areas do not exist in isolation but are intertwined through the RIG-TLS model, which collectively guide the formulation of the SCP. The SCP is planned to align with our visionary objectives, which aim to position San Tin and its surrounding areas as a global innovation powerhouse and frontier-creator that fosters talent development, advanced innovation, sustainable prosperity, and regional integration while bridging the GBA with the world. This vision is reflected in our slogan, "Innovate Together, Thrive Sustainably, Advance the World" (攜手創新，永續繁榮，領航世界), which marks the proactive leadership and originator role of STT, emphasising its capacity to generate new knowledge, technologies, and global standards that provide a driving force for global progress.

The scope of this baseline review report covers several strategically important key areas in the NM, including STT, Hong Kong-Shenzhen Innovation and Technology Park (HSITP in the Loop) (The Loop), Ngau Tam Mei (NTM), and San Po Shue (SPS) Wetland Conservation Park. The report will first introduce the analytical framework and key concepts of the three strategic focuses. Then, it gives a review of best practices in the country and overseas, providing key lessons to help the development of the SCP. Subsequently, it conducts a comprehensive analysis of the baseline situation, followed by the proposal of the sustainability impact assessment (SIA) variables and the SWOT analysis. Finally, it details the SCP structured around the three strategic focuses.

2. Analytic Framework and Strategic Focuses

2.1. Analytic Framework

The analytic framework of this baseline review report is rooted in the RIG-TLS Model proposed in the inception stage (**Fig. 2.1**). The model places research, industry, and governance at its core, emphasising the interaction cycle of the three to drive regional innovation, and radiates to dimensions such as talent support, urban-rural integration, and environmental sustainability. This model not only provides a theoretical basis for the development of STT but also directly guides the structural organisation of the baseline review, ensuring that the evaluation process covers all key variables of the I&T ecosystem.

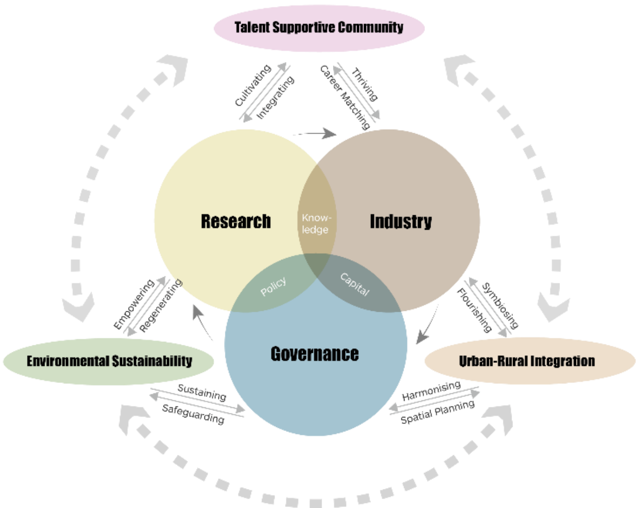


Fig.2.1. The RIG-TLS Model

To implement the above conceptual model, this study constructs a rigorous and comprehensive analytic framework, as shown in **Fig. 2.2**. The framework divides the research scope into six key aspects, supplemented by a general baseline that covers physical and built environment and demographics as the underlying support. For each aspect, specific assessment frameworks with key variables, as shown in **Fig. 2.2,** are adopted to ensure the depth and robustness of the analysis.

As shown at the bottom of the analytic framework in **Fig. 2.2**, the baseline findings collected through the above detailed areas will ultimately lead directly to the three strategic focuses: I&T development and regional synergy, talent support and inclusive communities, and environmental sustainability and carbon neutrality. This progressive structure, from “conceptual model” to “technical scope” to “strategic focus”, ensures that baseline review effectively identifies gaps in the status quo and provides scientific justification support for development of the SCP.

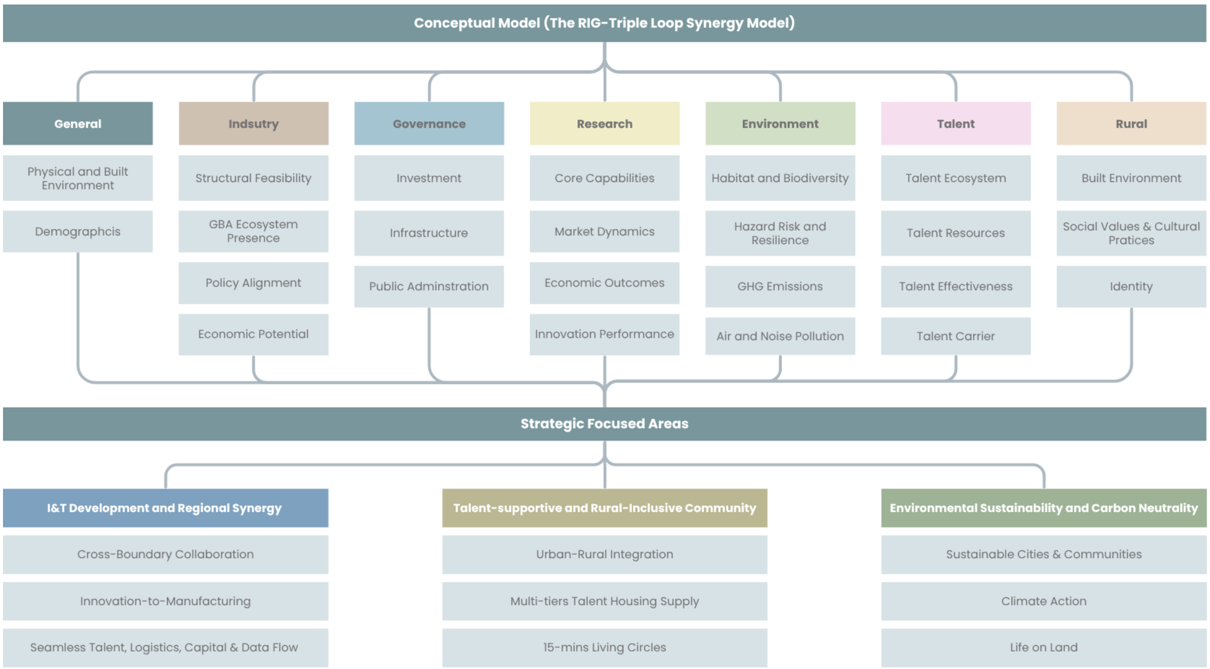


Fig. 2.2. Analytic Framework of Baseline Review

2.2. Core Concepts of Strategic Focuses

<u>I&T Development and Regional Synergy</u>

As shown in **Fig. 2.3**, the core logic of this dimension lies in building a regional innovation ecosystem that can effectively convert knowledge into economic value. Rather than single-point scientific research, this concept emphasises a complete path from basic R&D, application transformation to high-end manufacturing. Also, it leverages the complementarity of Hong Kong and Shenzhen in terms of institutions and resources to achieve a seamless flow of talent, capital, materials, and data between the two places. This concept also stresses the role of government. Through government policies that facilitate the inflow of venture capital and talent, it optimises resource circulation and cross-border collaboration, and promotes the formation of R&D and high-end manufacturing clusters.

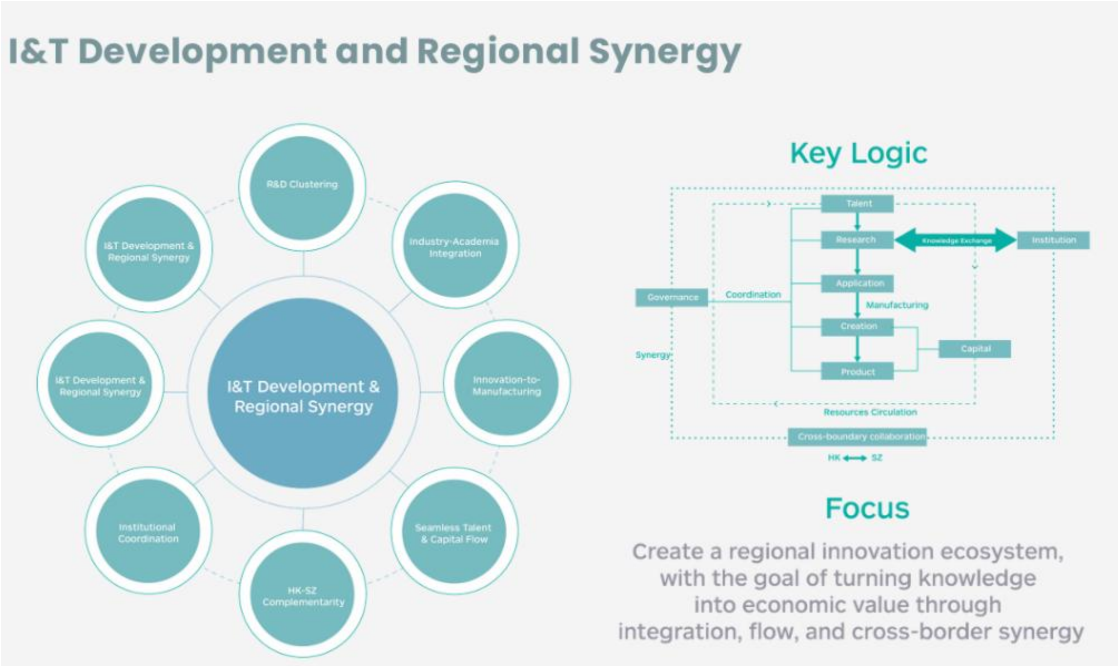


Fig. 2.3. I&T Development and Regional Synergy

<u>Talent-Supportive and Urban-Rural Inclusive Communities</u>

As shown in **Fig. 2.4**, this focus centres on the idea that the community is a place where talent and rural values coexist, performing the function of a “two-way matching engine for talents and rural areas”. This concept highlights that a sustainable community is built on three cornerstones of ecological space, industrial economy, and social governance. It also encompasses a talent supportive ecosystem that can safeguard quality of life and enable career development through policy support to meet talent needs. This concept also emphasises the importance of rural-inclusive governance, highlighting that by allowing equitable participation and local-cultural co-creation, the trust and relationship between the industry and the local community can be built and strengthened, and the adverse social impacts, such as social segregation, can be minimised.

Fig. 2.4. Talent Support and Urban-Rural Inclusive Communities

<u>Environmental Sustainability and Carbon Neutrality</u>

As shown in **Fig. 2.5**, this focus moves away from the traditional linear consumption model and shifts towards a circular economy logic of "Supply-Recover-Restore". It emphasises closed-loop resource circulation, utilising recycling and reuse of resources to maximise resource efficiency. It also stresses designing out waste and pollution from the sources of industry, agriculture, and the living sectors and improving energy efficiency to reduce carbon footprint. This focus also incorporates the concepts of blue-green network and nature-based solutions, advocating the integration of wetland conservation and urban design and the adoption of nature-based solutions to enhance sustainability and climate resilience.

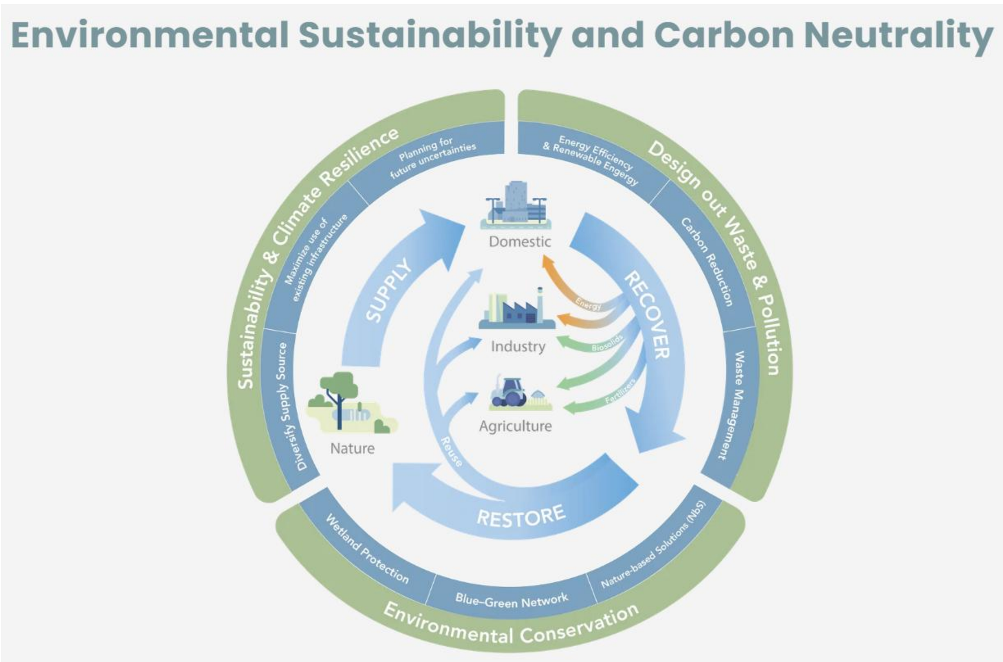


Fig. 2.5. Environmental Sustainability and Carbon Neutrality

3. Review of Best Practice

3.1. Hangzhou Future Science Tech City

Success Factors	Current Performance	Reason Behind It
Policy Proactivity & Continuity	• The local government exhibits high administrative efficiency and provides highly customised, proactive services to enterprises. • Business registration, project approvals, and policy applications are heavily streamlined, creating a highly favourable "pro-business" environment.	• This is largely driven by Zhejiang Province's famous "At Most One Visit" (最多跑一次) administrative reform, which utilises digital governance to minimise bureaucratic red tape. Also, New Enterprise Registration, a 24-hour response mechanism, is providing the fastest service. • "Yushengxin" Digital Platform enables smart policy matching for start-ups and 'apply now, receive now' fund disbursement, reducing payout time by 3+ months.
Industrial Chain	• Upstream-downstream industrial chain system. Within the same park, companies that supply materials, technology or services are physically clustered together. • Low Altitude Economy (China Fly Valley) is commonly used by 100 firms for the supply chain.	• The scale of Urban Commercial Nodes (UCN) has a significant positive contribution to the agglomeration and comprehensive development level of the park (Yue Wu, 2022). • Government-Alibaba construction of 'Moyu Space' provides computing power, data and models from chip designers (Upstream) to algorithm developers (Midstream) and to application makers (Downstream).
Financial & Business Support	• 'Investment loan guarantee subsidy insurance' five-in-one mechanism. Start-ups and mature tech companies have exceptional access to early-stage capital, angel investments, venture capital (VC), and government-backed industry guidance funds.	• The government deliberately engineered this by creating specific financial hubs, such as the "Angel Village" (天使村) within Dream Town, to cluster financial institutions. '3+N' Industry fund cluster, Runmiao Fund solely for start-ups, Science and Technology Insurance covers 50% loss for start-ups.
Talent and R&D	• The park has developed a new generation of information technology innovation clusters focusing on industries like future networks, big data, cloud computing and the internet of things. In 2019, Hangzhou's digital development index ranked first in China.	• Job-housing relationship (Employee balance index, commuting distance and proportion of happy commuters) is a major contributing factor in R&D, confirmed by Yue Wu (2020). • The initial rent cut/housing subsidy attracted many start-ups to gather in Dream Town at the beginning.

Market Access	• Enterprises in HFSTC enjoy unparalleled access to both domestic and international digital markets, particularly in sectors like e-commerce, cloud computing, AI, and fintech. • Enterprises plug directly into the ecosystem of Alibaba, Bytedance and vivo.	• Anchor Firm spillover effect is now multipolar. • The primary reason is the proximity to Alibaba Group's headquarters. It creates a massive spillover effect, allowing start-ups to plug directly into Alibaba's vast ecosystem as suppliers.
Community Living	• Overall job-housing relationship in the corridor is relatively close, and more than three-quarters of the commuters who both lived and worked in the corridor achieved a 'happy commute'.	• Approximately 75.21% of residents live within 5 km of the workplace. • In HFSTC, five 'Feature Towns', including 'Dream Town', play a strategic role as they were intended to concentrate and offer agglomeration benefits to numerous small enterprises.

3.2. Luixiandong Headquarters Base, Shenzhen

Success Factors	Current Performance	Reason Behind It
Policy Proactivity & Continuity	• The local government actively secures premium space for tech firms through innovative land policies like "Joint Upstairs" (聯合上樓) and provides policy-backed R&D spaces with rent discounts of 30% to 70% off market rates.	• Highly mature 'upstream-downstream' vertical industrial cluster. The area forms the Liuxiandong Intelligent Manufacturing Corridor, where HQ and R&D are concentrated along a single axis.
Industrial Chain	• Highly mature 'upstream-downstream' vertical industrial cluster. The area forms the Liuxiandong Intelligent Manufacturing Corridor, where HQ and R&D are concentrated along a single axis.	• It is a core node in Nanshan's 10-kilometre "Headquarters R&D + High-end Manufacturing" industrial belt. About 70% of the industrial agglomeration can be achieved around Liuxiandong Boulevard. • Companies keep their headquarters/R&D in Liuxiandong while utilising nearby manufacturing parks (like Baiwangxin), creating a highly efficient "research-production integrated" closed loop.
Financial & Business Support	• Start-ups and tech enterprises enjoy robust access to capital, incubation funds, and supply chain financial services, supported by both government subsidies and private venture capital.	• Supported by the Xili Lake concept validation platform, which helps projects move from lab to prototype. Venues like the Xili Lake Talent Service Centre offer space, funding, and application guidelines for start-ups.
Talent and R&D	• The base boasts an exceptional concentration of high-tech talent and world-class R&D capabilities, with a seamless pipeline from	• This is due to its immediate geographic proximity to the Xili Lake International Science and Education City and Shenzhen University Town.

	academic research to commercial application.	
Market Access	• Enterprises in Liuxiandong have unparalleled access to global hardware, telecommunications, and AI markets, easily integrating into international supply chains.	• The presence of massive global anchor companies (like Transsion, which dominates the African mobile market, and DJI, which dominates global civilian drones) creates a massive global spillover effect. Additionally, international partnerships, such as the Intel GBA Technology Innovation Centre located in the base, plug local firms directly into global tech ecosystems.
Community Living	• The area features highly integrated, mixed-use urban environments (like Vanke Cloud City) that blend office, residential, and talent apartments, commercial plazas, and green parks into a single ecosystem.	• To attract and retain top-tier global talent, urban planners abandoned traditional, rigid industrial park designs. Instead, they utilised "Cloud City" concepts with three-dimensional green spaces, sunken plazas, and direct MTR subway integration to ensure a vibrant work-life balance and a high quality of life.

3.3. Hi Tech Campus Eindhoven (HTCE), The Netherlands

Success Factors	Current Performance	Reason Behind It
Policy Proactivity & Continuity	• It benefits from national and regional policies specifically designed to de-risk deep tech and hardware R&D.	• The Triple Helix Model: regional governance is shared equally between the local government, knowledge institutes (such as TU/e), and industry leaders. This ensures policy continuity regardless of political shifts. • WBSO (R&D Tax Credit): the Dutch government provides a highly proactive R&D tax credit (WBSO) that significantly lowers the wage costs for R&D employees, directly subsidising innovation at the campus.
Industrial Chain	• Start-ups, component suppliers, and global OEMs are physically co-located.	• HTCE provides shared access to world-class cleanrooms, reliability testing labs, and prototyping facilities that start-ups could never afford alone. • The presence of anchor giants such as ASML, NXP, and Philips creates a massive, localised supply chain.
Financial & Business Support	• The campus provides robust support for early-stage and scaling deep-tech companies.	• Institutions such as the Brabant Development Agency (BOM) and the Deep Tech Fund (backed by the Dutch government) provide crucial

		early-stage capital, loans, and equity. • The campus hosts HighTechXL, a premier deep-tech venture builder that helps entrepreneurs build companies around advanced patents sourced from CERN, the European Space Agency (ESA), and Philips.
Talent and R&D	• Known as the "smartest square kilometre in Europe," HTCE has an exceptionally high density of R&D talent.	• The campus is deeply intertwined with Eindhoven University of Technology (TU/e), Fontys University of Applied Sciences, and research institutes such as TNO, creating a direct pipeline of highly specialised engineering talent.
Market Access	• Enterprises at HTCE have immediate access to the European Single Market and are plugged directly into the global supply chains of the world's leading hardware and semiconductor companies.	• HTCE start-ups plug into the ecosystems of ASML (the world's sole supplier of EUV lithography machines) and Philips. • Located in the Brainport region, it is geographically positioned at the crossroads of Europe's industrial heartland (close to Germany's Ruhr area and Belgium), providing seamless access to European industrial clients.
Community Living	• The campus excels in providing a high quality of life, excellent work-life balance, and a highly engineered sense of community, which results in very high talent retention rates.	• By centralising all dining facilities into one 400-metre building (The Strip), employees from different companies mix daily, creating opportunities for "serendipitous encounters." • Eindhoven boasts world-class cycling infrastructure. A vast majority of campus employees live within a 15–30 minute bike ride, achieving a highly sustainable and stress-free "happy commute."

3.4. Hong Kong Science and Technology Park Ltd

Success Factors	Current Performance	Reason Behind It
Policy Proactivity & Continuity	• The government's new industrialisation agenda is actively supporting advanced manufacturing.	• Lack of a dedicated policy sandbox for cutting-edge tech trials. • Innofacturing Tomorrow: rebranded its three industrial estates (Tai Po, Yuen Long, and Tseung Kwan O) as "InnoParks" in 2021, attracting high-value-added, high-technology content, and advanced processes.

Industrial Chain	• Significant progress but incomplete.	• Outward leakage persists — R&D stays, but mass production still goes to Shenzhen. • High "Outward Leakage" of technology. Core R&D stays, but prototyping and manufacturing move to Shenzhen. • Absence of downstream manufacturing support and "midstream" pilot production facilities.
Financial & Business Support	• Strengthening with targeted intervention.	• Sufficient government grants sustain early R&D, but risk-averse private capital (favouring property/finance) creates a scaling vacuum. • Lack of specialised venturing support to bridge the transition from laboratory prototypes to market-ready products.
Talent and R&D	• Strong with measurable results. With 16,000 R&D professionals and 2,600 companies. • Limited informal knowledge spillovers despite proximity to CUHK.	• The biotech sector grew rapidly from 5 clinical companies to 60. The Shenzhen branch in Futian fosters cross-border collaboration.
Market Access	• Strategic and improving. • Innopole's proximity to Shenzhen provides Hong Kong R&D with mass production in Shenzhen. • China booster programme facilitates market entry via China Resource Group's industrial network.	• Still developing the anchor firm spillover effect (unlike Hangzhou). • Need more government-related opportunity scenarios as first customers.
Community Living	• Talent accommodation (i.e. InnoCell) provision is limited and remains priced above the reach of many researchers, while distant off-site housing forces long commutes. • Lack of "Third Spaces" (cafés, social hubs, late-night venues) that encourage accidental innovation and a humanistic atmosphere.	• Low footfall leads to after-hours facility closures; the "Work-Live-Play" concept fails to materialise as a 24/7 reality. • Despite Hong Kong's infrastructure, the Park's relative isolation from urban cores makes international schools and family amenities difficult to access, hindering senior talent retention.

3.5. Key Performance Drivers for the Cases

<u>Hangzhou Future Science Tech City</u>

The key dominant reason for its success is "Market Access." It is built around Alibaba's HQ, creating a powerful gravitational pull for start-ups. There is frictionless access to Alibaba Cloud, e-commerce platforms, and Ant Group's fintech network. Start-ups skip the market build-up phase by plugging directly into a world-class digital ecosystem.

Luixiandong Headquarters Base

The key dominant reason for its success is its "Industrial Chain" (Research-Production Closed Loop). It seamlessly integrates R&D with manufacturing to create a research-production loop. It anchors Nanshan's 10 km industrial belt, which links hardware giants such as DJI and Transsion. Physical adjacency to factories enables a 'morning design, afternoon prototype' workflow. It leverages Shenzhen's status to minimise supply chain friction and maximise iteration speed.

Hi Tech Campus Eindhoven (HTCE)

The key dominant reason for its success is "Talent-Supportive Strategies and Community Living." The Netherlands offers the 30% rule, a massive tax advantage whereby highly skilled migrants receive 30% of their gross salary tax-free. To remove the language barrier, English was made the medium of communication in the city of Eindhoven. The forced networking technique using The Strip is also remarkable: there is no other restaurant than The Strip, so talents come to the same place to eat and gain opportunities for knowledge co-creation.

Hong Kong Science and Technology Parks

The key dominant bottleneck is the lack of Industrial Chain Completeness. It has strong funding for early-stage research, but a total vacuum in mid-stream infrastructure. The lack of specialised space for pilot production prevents start-ups from scaling locally. Firms externalise production to access a mature supply chain and space lacking locally. The economic harvest of local research is lost to outside regions due to the missing manufacturing link.

3.6. Strategic Implication for San Tin

- Establishment of a very fast industry chain through a research-production closed loop, which will help start-ups to test their prototypes within a few hours. The government should implement a specialised regulatory sandbox that allows for the seamless, pre-approved flow of critical R&D materials between Shenzhen and San Tin.

- Hong Kong can target GBA market integration through simplified cross-border business policies, as it lacks a massive consumer market. This would give international companies the chance to access the Chinese market, and mainland companies the opportunity to reach global markets through Hong Kong.

- Encouraging a tech giant (such as Huawei, BYD, DeepSeek) to open its regional headquarters in San Tin by providing attractive subsidies or incentives would create an "Alibaba effect" for start-ups.

- Creation of a 'Feature Town' in San Tin, following the concept of Dream Town in Hangzhou, to avoid the isolation problem experienced by HKSTP.

- Creating opportunities for 'Forced Networking' by allowing only shared dining spaces across the entire work area, thereby facilitating accidental innovation.

- Hong Kong's biggest deterrent for global talent is exorbitant housing costs. San Tin must create a good community living environment, maintaining a sound job-housing relationship while keeping subsidised housing costs to attract talent.

- Extension of government grants for start-ups to create a risk-compensation pool, matching funds for venture capital, and risk-sharing insurance, similar to the Hangzhou start-ups.

- San Tin should implement a specialised "NM Tech Pass" — a frictionless, fast-track visa and border-crossing mechanism that allows researchers to commute daily between Shenzhen and San Tin.

4. Baseline Review

4.1.Strategic Planning Context

<u>National Level: Strategic Positioning and Emerging Industries</u>

Under the macro framework of national development, the *14th Five-Year Plan (2021-2025)* clearly supports Hong Kong's development as an international innovation and technology hub and establishes a "two corridors and two nodes" framework to position the Hetao Co-operation Zone at the national level (The State Council of the People's Republic of China, 2021). The *15th Five-Year Plan (2026-2030)* further shifted the focus to developing "new quality productive forces" and achieving high-level technological self-reliance, proposing 10 strategic emerging industries (The State Council of the People's Republic of China, 2026). In this context, STT is positioned as a key base for the development of new quality productivity, aiming to connect the Mainland innovation chain with the global market through the "dual circulation" framework. In addition, the implementation of *Ecological and Environmental Code of the People's Republic of China* (《中華人民共和國生態環境法典》) provides a legal bottom line for the green transformation of the I&T industry, requiring large-scale development projects to meet strict standards for pollution control, ecological protection, and carbon neutrality, to ensure that the development of new productive forces goes hand in hand with the construction of ecological civilisation (Ministry of Ecology and Environment of the People's Republic of China, 2026). This lays the foundation for the net-zero emissions and ecological integration goals of the STT.

<u>Regional Level: GBA Integration and Cross-boundary Coordination</u>

The Outline Development Plan for the Guangdong-Hong Kong-Macao GBA positions the GBA as an international innovation and technology hub with global influence, emphasising the construction of a complete innovation value chain. The plan proposes the concepts of "pole-driven" and "axle-supported", in which the "Hong Kong-Shenzhen Pole" leverages Hong Kong's advantages in R&D and international finance, combined with Shenzhen's advanced manufacturing capabilities. STT is strategically located at this pole and directly benefits from the talent flow and knowledge exchange of the Guangzhou-Shenzhen-Hong Kong I&T Corridor (The State Council of the People's Republic of China, 2019). At a cross-border scale, the "Twin Cities Three Circles" spatial framework places STT at the core of the "Hong Kong-Shenzhen Close Interaction Circle" (**Fig. 4.1.1**), supporting the synergistic development of research, education and fintech through infrastructure connectivity (HKSAR, 2021). At the same time, the *Guangdong Provincial 14th Five-Year Plan for Ecological Civilisation Development* and the *Action Plan for Building a Beautiful China Pilot Zone in the Guangdong-Hong Kong-Macao GBA* advocate the construction of a "world-class beautiful bay area" to strengthen the connectivity between wetland protection and regional green networks through cross-border ecological corridor (Ministry of Ecology and Environment of the People's Republic of China, et al., 2025; People's Government of Guangdong Province, 2021).

Fig. 4.1.1. "Twin Cities Three Circles" spatial framework
Source: HKSAR (2021)

<u>City Level - Hong Kong: Spatial Strategy & STT Development</u>

According to Hong Kong 2030+, the Hong Kong Government is committed to building Hong Kong into a globally competitive "Asia World City" and providing land reserves through "capacity creation" (Planning Department, HKSAR, 2021). Under the framework of the NM Development Strategy, STT is positioned as a "New International Innovation and Technology City" at the heart of the "Hong Kong-Shenzhen Close Interaction Circle". The specific development strategy emphasises its role as a base for the transformation and scale of R&D achievements in the Loop region, focusing on industrial clusters such as life and health technology, artificial intelligence, and robotics. To balance development and conservation, the project includes a 338-hectare SPS Wetland Conservation Park to ensure that the I&T capacity is expanded while upholding the commitment of achieving "no net loss" of ecological functions (HKSAR, 2021).

<u>City Level - Shenzhen: Master Planning & Strategic Interfacing</u>

Corresponding to the NM (NM) of Hong Kong, the *Territorial Spatial Master Planning of Shenzhen (2021–2035)* deepens pragmatic cooperation with Hong Kong, optimises the spatial pattern of the "Shenzhen-Hong Kong Twin Cities", and focuses on the development of the urban core area (Futian and Luohu) directly connected to the NM of Hong Kong (Shenzhen Municipal Bureau of Planning and Natural Resources, 2021). The *Futian District Territorial Spatial Plan (2021–2035)* further positions Futian as the "core of the Super Bay Area" and designates the Hetao Co-operation Zone (Shenzhen side) as the "International Advanced Innovation and Technology Rules Pilot Zone" (Futian Administration of Shenzhen Municipal

Bureau of Planning and Natural Resources, 2023). By promoting the convergence of cross-border rules and the construction of the "Shenzhen-Hong Kong Port Economic Belt" (**Fig. 4.1.2**), the Shenzhen side will form a synergistic system with STT with complementary advantages and differentiated development and jointly build an internationally competitive innovation and technology ecosystem.

Fig. 4.1.2. Shenzhen-Hong Kong Port Economic Belt
Source: LegCo (2022)

4.2. General Baseline

Physical Baseline: Topography Analysis

As shown in **Fig. 4.2.1**, the terrain of the study area is extremely flat, with ground elevations generally ranging from +0mPD to +7mPD, with a median of only 3mPD (**Fig. 4.2.2**). The overall landform is slightly sloping from northeast to southwest (towards Shenzhen Bay), lacking significant terrain undulations. This typical low-lying plain characteristic is conducive to the layout of large-scale I&T facilities, but it also suggests that the area is highly sensitive to flooding disasters caused by sea level rise and extreme weather, and subsequent planning needs to focus on its climate resilience.

Fig. 4.2.1. Typography of project area

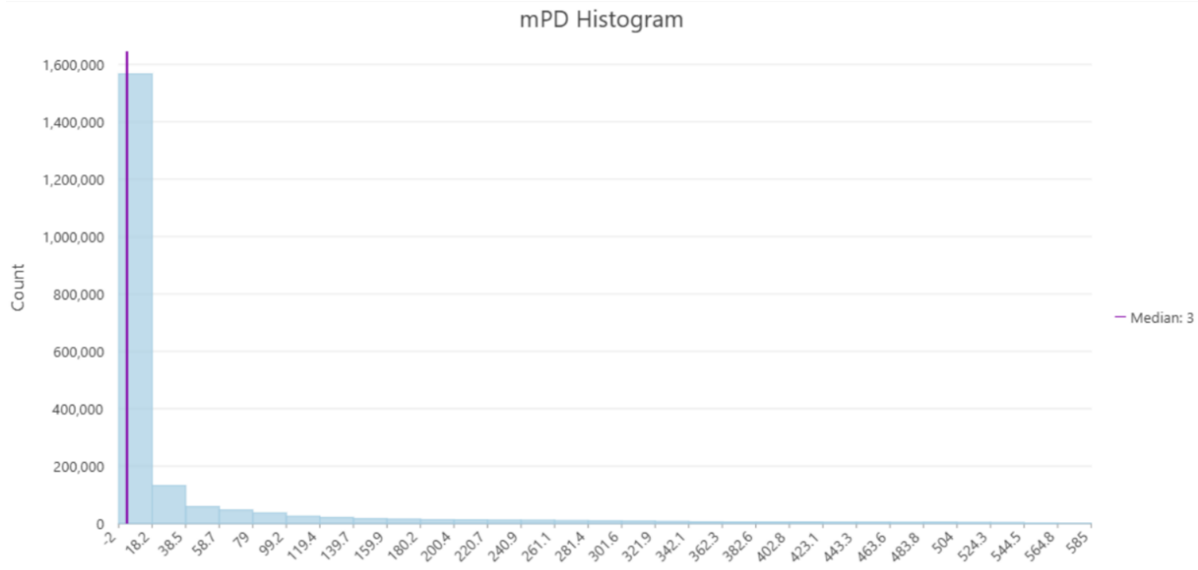


Fig. 4.2.2. Typography of project area

<u>Physical Baseline: Planned Land Use & Functional Transformation</u>

The project area is undergoing a dramatic functional restructuring from a low-intensity rural landscape to a high-intensity I&T hub. The existing land use, as shown in **Fig. 4.2.3**, mainly consists of fishponds, wetlands and agricultural land with high ecological value, of which 43.3% are non-industrial land, reflecting the current rural and conservation nature. However, according to the latest OZPs, as shown in **Fig. 4.2.4**, the future land allocation presents a clear industrial orientation and spatial hierarchy. Other Specified Uses (OU, mainly I&T uses) are expected to be the largest category (estimated to account for about 45-50%), concentrated in the central region and the northern end of the adjacent Loop; residential development (R(A) to R(D)) is distributed in a transit-oriented (TOD) model around the proposed railway station (estimated to account for about 15-20%); while the western and southern edges retain a significant proportion of ecological conservation areas (CAs) and green belts (GB). This large-scale functional transition means that some of the existing purely agricultural/ecological land and a large number of scattered brownfield sites will face comprehensive land consolidation and resumption, and the planning must adopt fine boundary design to mitigate the interference of high-intensity development on the surrounding ecological buffer zone. In addition, the high concentration of I&T functions and residential facilities indicates that the region will transform from a border buffer zone to a resource transformation base connecting global markets, and its success will depend on whether strategic infrastructure can support the huge demand for factor flows as scheduled.

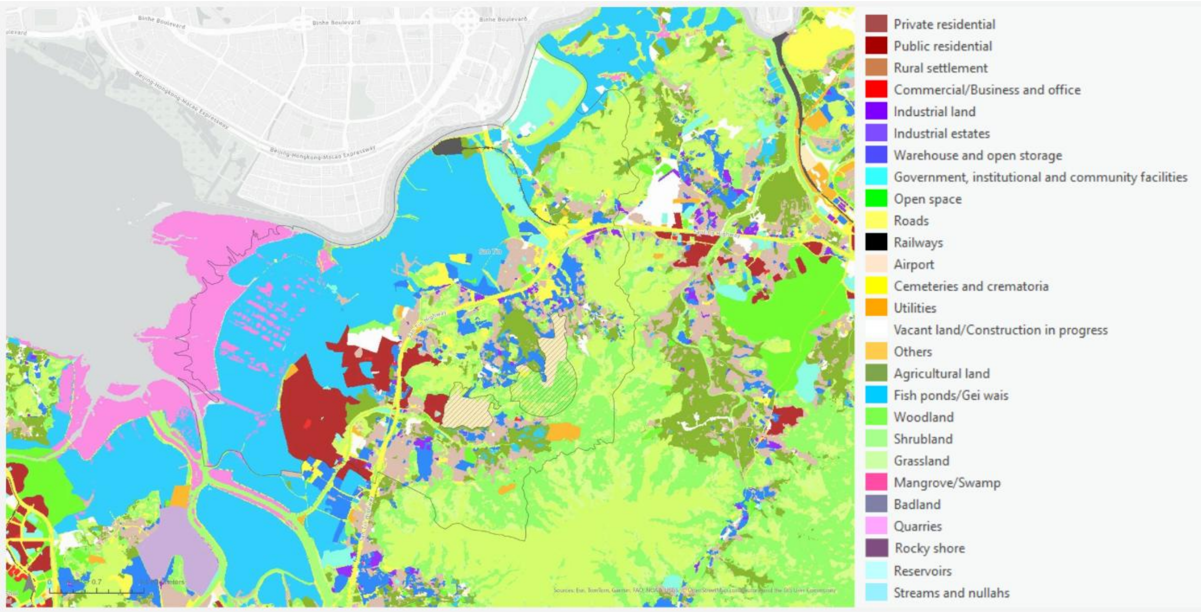


Fig. 4.2.3. Existing land use

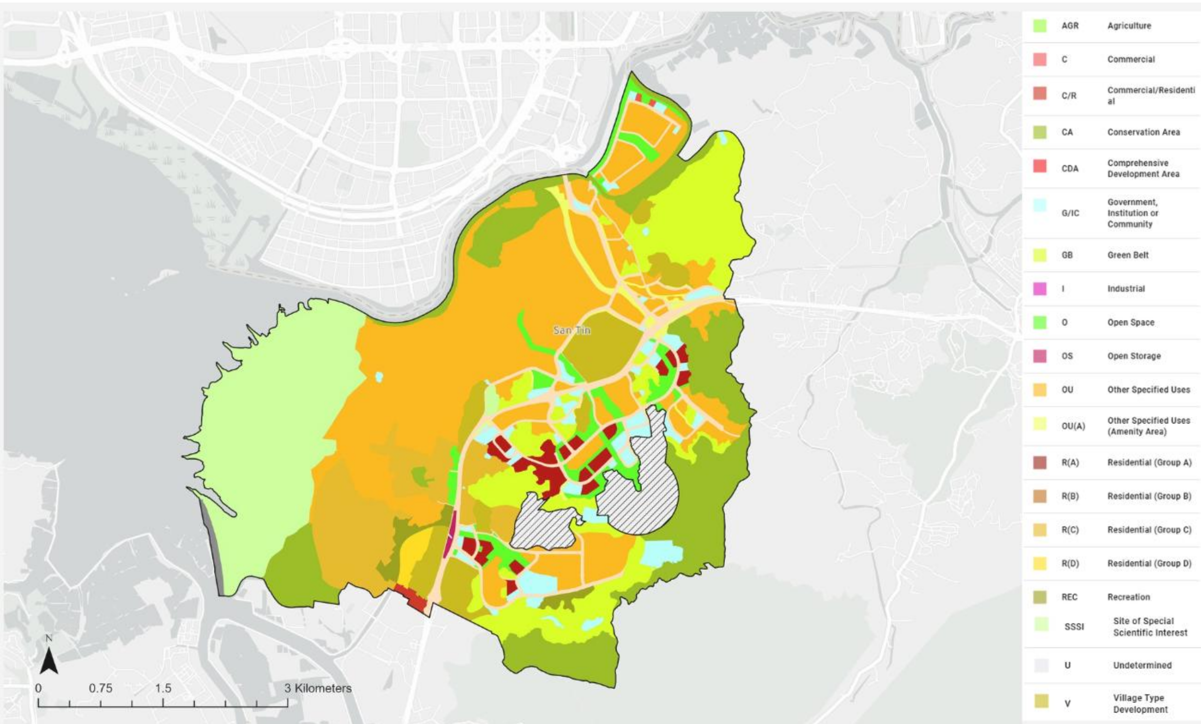


Fig. 4.2.4. Planned land use

Physical Baseline: Building Condition

As reflected in **Fig. 4.2.5**, the buildings in the area exhibit a typical low-density rural form, lacking high-intensity or large-scale urban development. The buildings are mainly composed of well-maintained low-rise village settlements, with a spatial distribution concentrated within the existing village area. According to the statistics of floor distribution, as shown in **Fig. 4.2.6**, the vast majority of buildings are below 4 floors, with 3-story buildings as the core body (more than 70 buildings), followed by 2-story buildings. This low and dispersed architectural feature reflects the stable physical condition of the status quo, but it also indicates that the transformation into a high-density I&T hub will require extensive spatial reshaping based on the existing rural fabric.

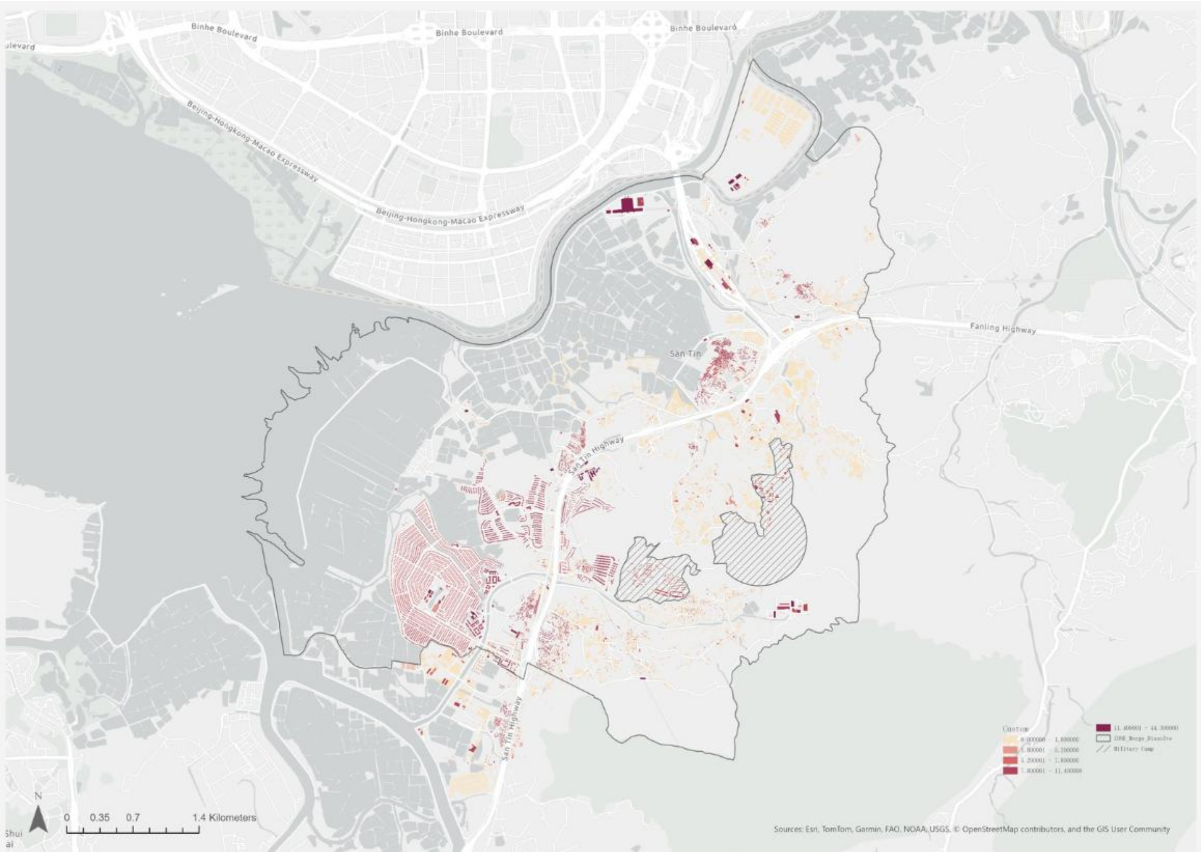


Fig. 4.2.5. Building conditions in the project area

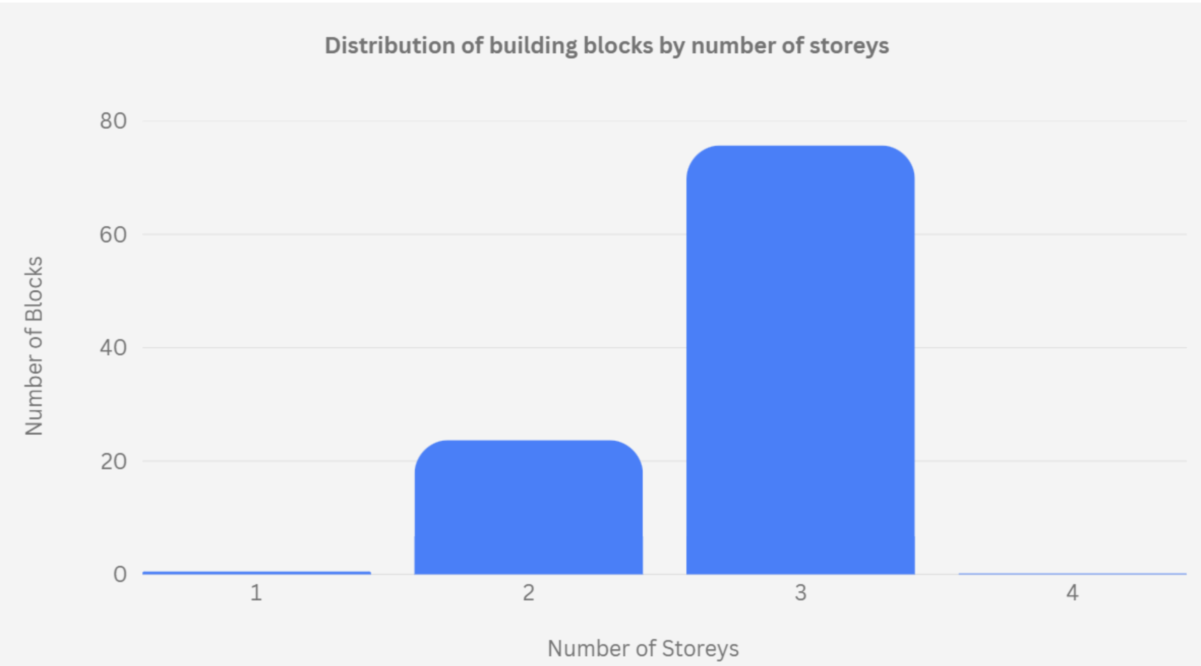


Fig. 4.2.6. Distribution of building blocks by number of storeys in project area

Physical Baseline: Urban Form

The project area shows a representative urban form of rural and special-purpose coexistence, with low density and scattered distribution as a whole, as reflected in **Fig. 4.2.7**. The existing spatial structure can be summarised into four typical models. First, garden-style villas with a regular layout, such as Palm Springs. Secondly, natural villages with organic growth characteristics, showing spontaneous spatial textures. In addition, there are small three-storey houses (small houses) with high coverage, as well as special urban projects established in response to special needs, such as the Central Government Aided Emergency Hospital, which are mostly in the form of regular strip structures. Although this diverse and low-intensity morphological distribution retains the landscape characteristics of the urban-rural transition zone, it also reflects the current situation of land use fragmentation and lack of strong core guidance.

Fig. 4.2.7. Urban/rural form in the project area

Socio-economic Baseline: Demographic Trends & Projections

The development potential of STT is examined in the context of the macro context of its Yuen Long district. According to **Fig. 4.2.8**, the current situation in Yuen Long District shows mature but stressed social characteristics: its median age is 48 years old, of which 57.3% are adults, and the elderly population (over 65 years old) has accounted for 28.1%. In addition, the district has a high degree of ethnic diversity (88.4% of Chinese ethnicity, the rest are mainly Indonesian and Filipino) and a diverse distribution of educational attainments, with about 25.7% of the population having degree qualifications. In terms of population size evolution, as shown in **Fig. 4.2.9**, Yuen Long District's population has maintained steady growth between 2021 and 2026, and is expected to see a significant jump after the implementation of the planning plan in 2029, reflecting strong regional development momentum.

A comparison of 2041 projections for the STT reveals two distinct future paths. As for the without-development scenario, as shown in **Fig. 4.2.10**, if the existing land use is maintained, San Tin is expected to have only about 6,160 residents by 2041. Its population pyramid shows significant contractive characteristics, with the proportion of the population over 65 years old accounting for 40%, and the dependency ratio (0.84) reflects a severe trend of negative population growth and community ageing. As for the planned scenario, as shown in **Fig. 4.2.11**, if the population is fully absorbed according to the government's plan, the population will surge to 165,942 in 2041, with a growth rate of +2,594%. The structure has transformed into an expansionary system with a predominantly 15–64-year-old workforce (70%), aiming to meet the talent needs of the global I&T industry.

During the first phase of the move-in period from 2029 to 2033, STT is expected to welcome the first batch of I&T enterprises and population growth of 147 to 159 thousand. This transformation is expected to create approximately 217 thousand jobs and provide between 50,000 and 54,000 residential units, including 6,400 talent apartments. This leapfrog development from low-density villages to high-intensity I&T hubs largely depends on the precise matching of talent accommodation facilities with I&T job skills, and whether the TOD model can effectively support the surging demand for transportation and public services.

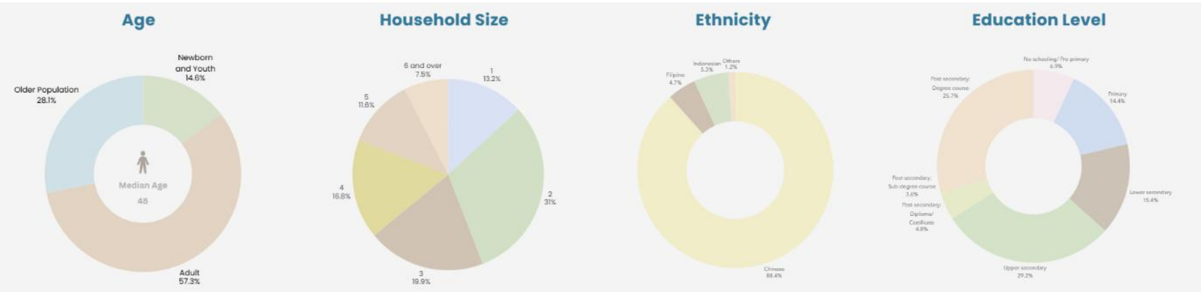


Fig. 4.2.8. Existing demographic characteristics of Yuen Long District

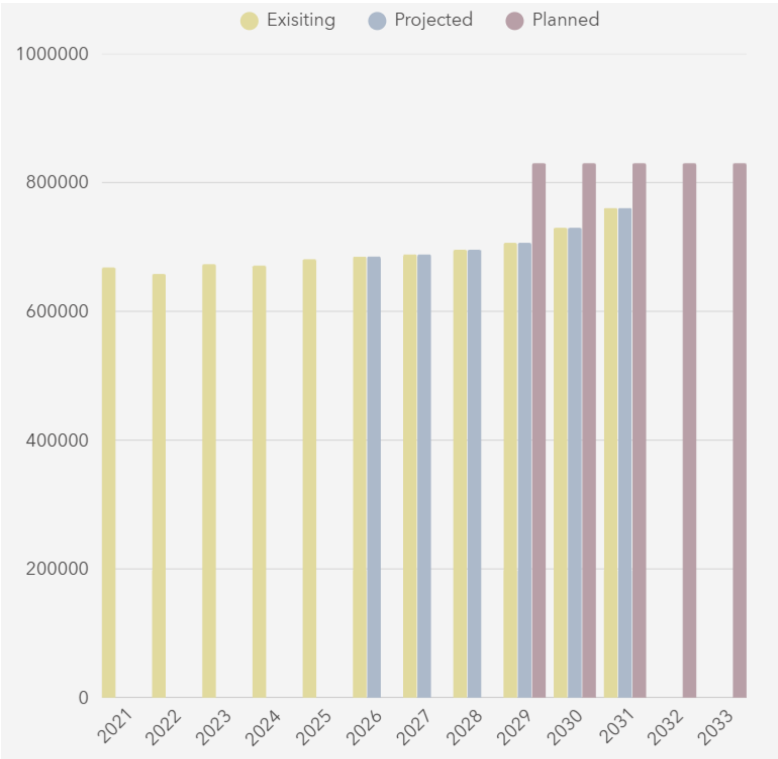


Fig. 4.2.9. Project population trend of Yuen Long District

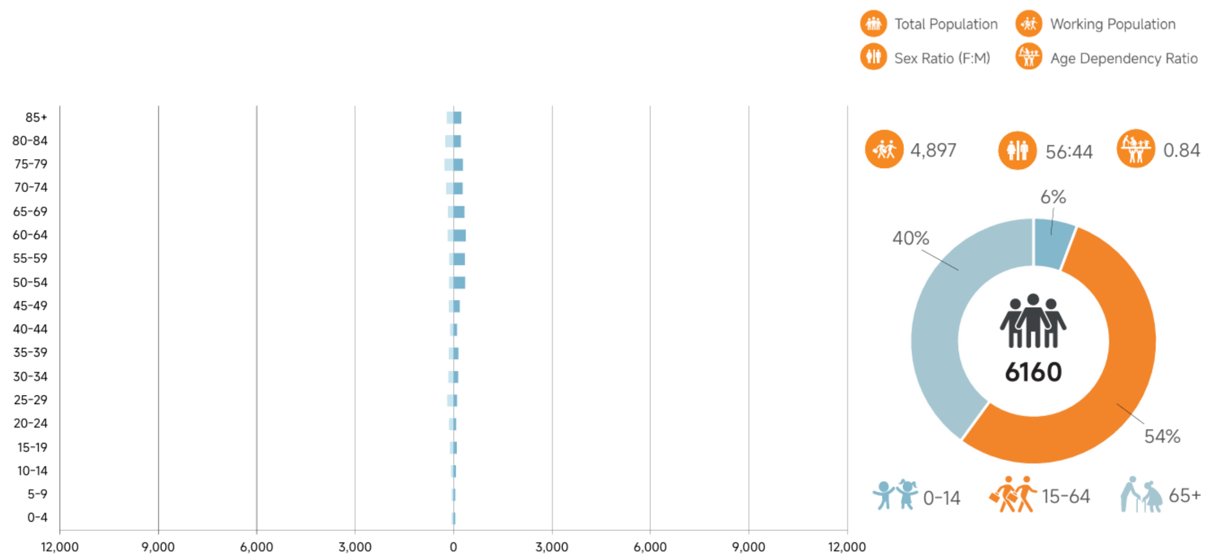


Fig. 4.2.10. Without-development scenario of STT in 2041

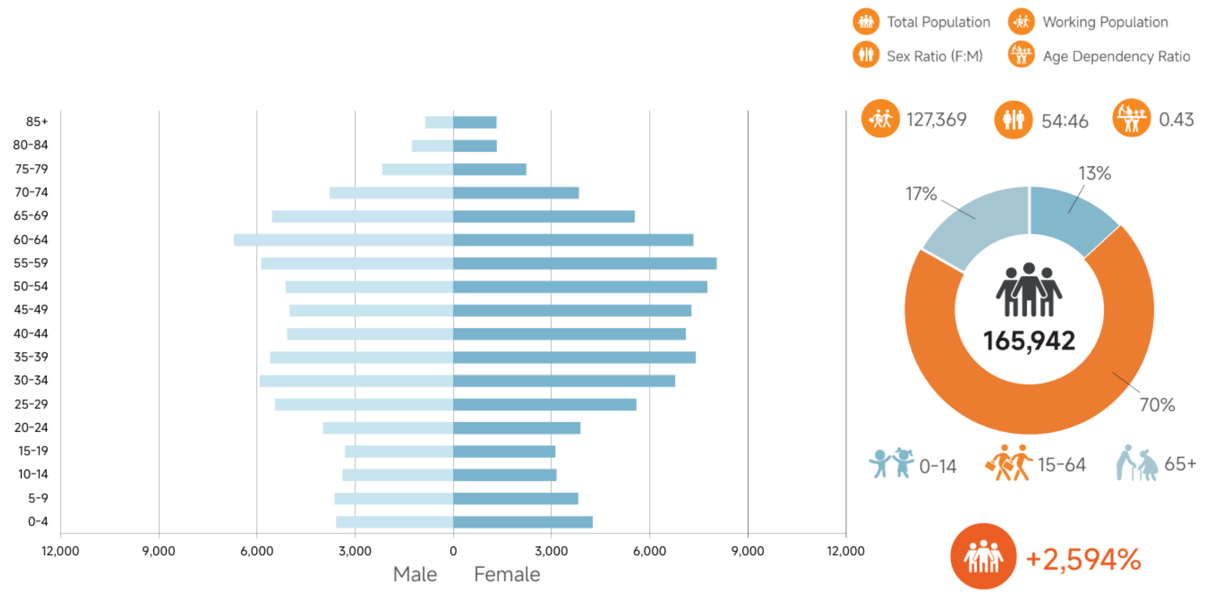


Fig. 4.2.11. Planned scenario of STT in 2041

4.3. Environment

4.3.1. Assessment Framework

For the environmental aspect, this part evaluates the existing environmental conditions of San Tin and its surroundings using a framework structured from three key Sustainable Development Goals **(Fig. 4.3.1)**, which are SDG11: Sustainable Cities and Communities; SDG 13: Climate Action; and SDG 15: Life on Land (United Nations, 2015). Each SDG captures a structurally distinct category to carry a comprehensive analysis of the environmental stress present on the site. Together, the three pillars provide the empirical evidence base that grounds the Ecological Symbiosis Loop of the RIG-Triple Loop Synergy Model, translating the conceptual framework into measurable, place-based conditions.

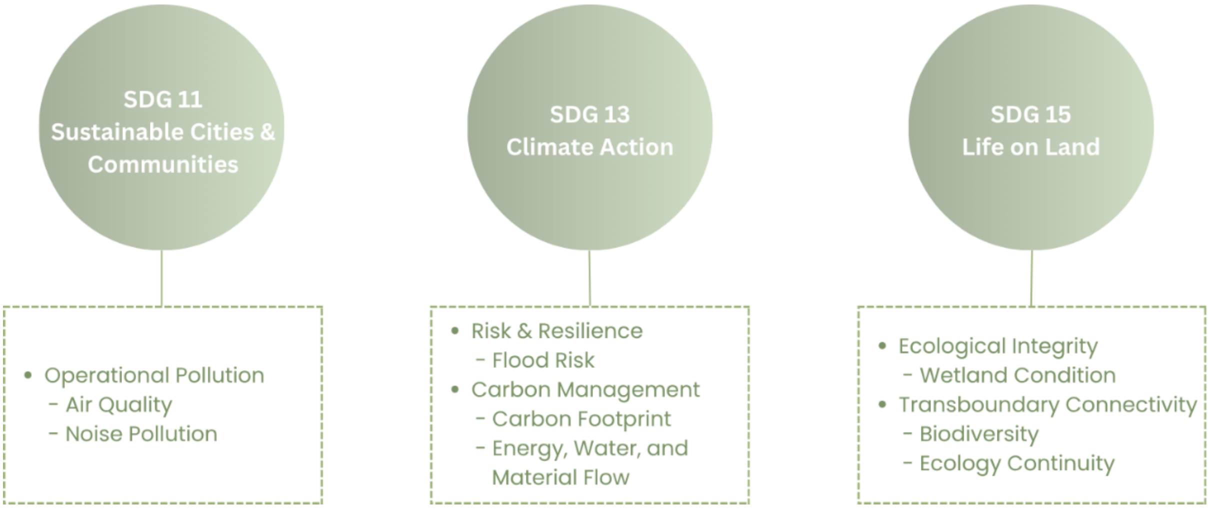


Fig. 4.3.1. Environmental Assessment Framework
Source: United Nations (2015)

4.3.2. SDG 11: Sustainable Cities and Communities: Operational Pollution

<u>Air Pollution</u>
According to the data **(Fig. 4.3.2)**, AQI readings across the region are predominantly in the 60–70 range for key pollutants, including PM2.5, PM10, SO_2, NO_2, CO, and O_3, placing the site in the "Moderate" category (EPD, 2026; Meteorological Bureau of Shenzhen Municipality, 2026), which is acceptable under the Air Quality Index. However, similar readings recorded simultaneously in Shenzhen imply that air quality is significantly shaped by transboundary regional airflow, so implementing local mitigation alone is insufficient. The government proposal acknowledges the issue by committing to no net increase in pollution load to Deep Bay and integrating blue-green infrastructure such as vegetation buffers and breezeways to improve air circulation (EPD, 2024). However, no quantified emission reduction targets and transport emission strategy are established to address the planned high volume of logistics and cross-border vehicle flows. The current proposal solely relies on passive design rather than systemic decarbonised mobility, which reflects a weak alignment with the air quality management dimension of SDG 11 (United Nations, 2015).

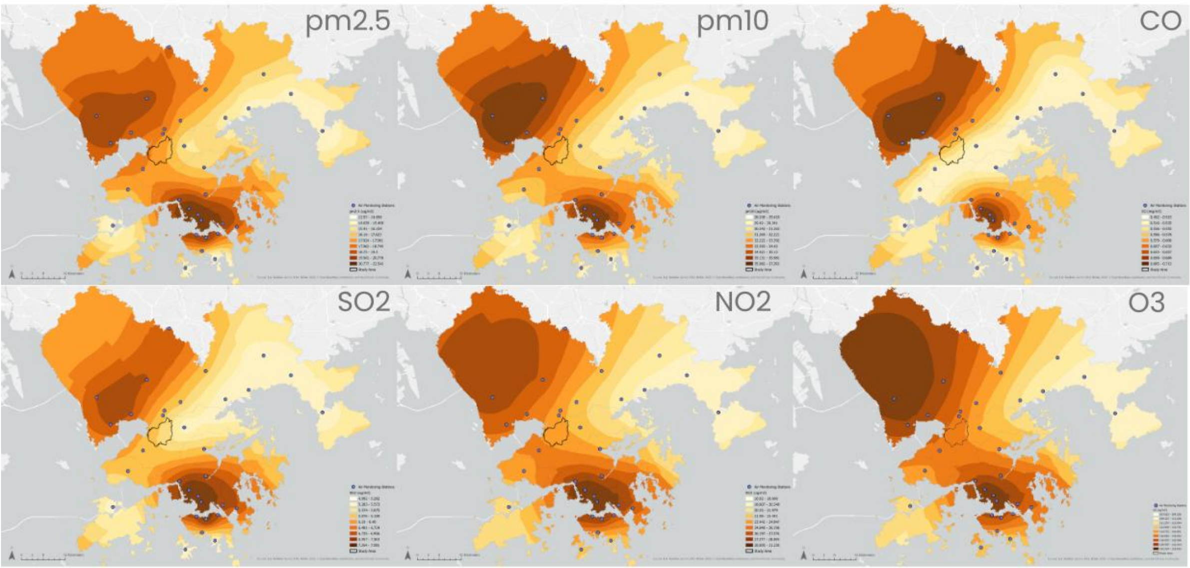


Fig. 4.3.2 Pollution Level of Hong Kong and Shenzhen
Source: EPD (2026); Meteorological Bureau of Shenzhen Municipality (2026)

<u>Noise Pollution</u>

Two main highways, San Tin Highway and Fanling Highway, are located near the study area, which create a dominant noise corridor reaching up to 91 dB through the centre of the site, and it reveals that the noise are bled into residential and conservation areas and it creates a significant impact across planned I&T, residential, and ecological zones **(Fig. 4.3.3)**. While the EIA only identifies the disturbance caused during construction rather than the operation phase, even there is expected to be high traffic volume within the area that drives bird populations away from corridors they currently use. In this circumstance, Nature-based Solutions such as vegetated buffers alongside revitalised drainage channels offer opportunities to improve noise and habitat quality. This addresses and enhances the current policy gap in the urban liveability dimension of SDG 11.

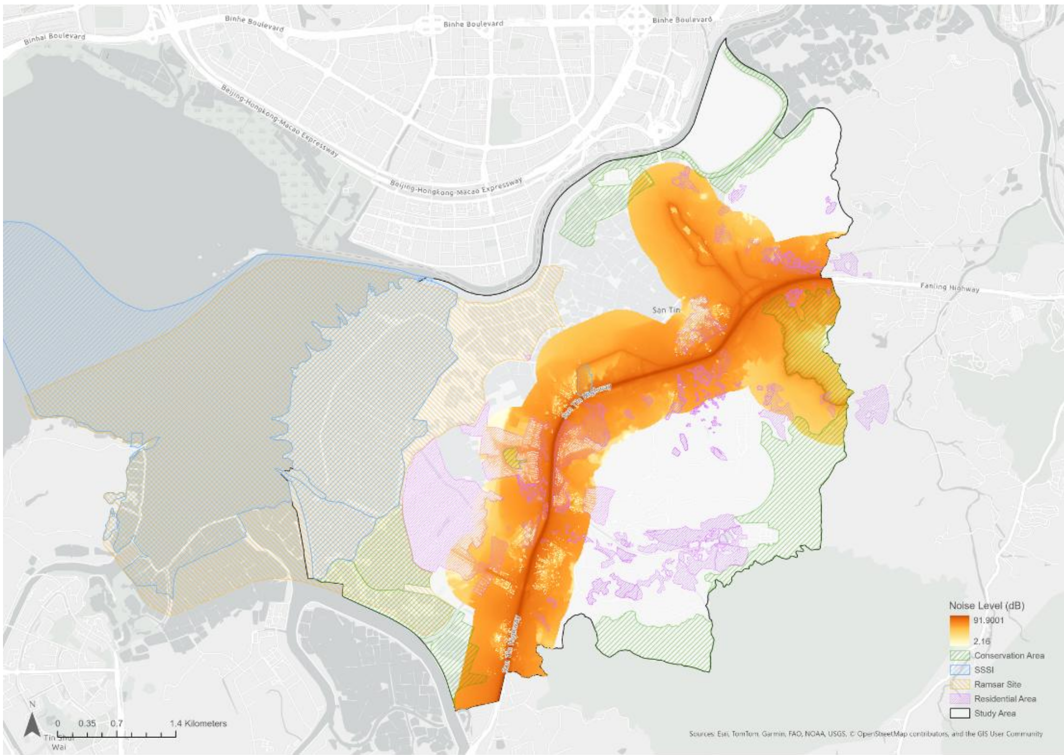


Fig. 4.3.3. Noise Level in the Study Area
Source: Highways Department (2026)

4.3.3. SDG 13: Climate Action: Hazard and Risk Resilience, and Carbon Management

Hazard and Risk Resilience: Existing Coastal Baseline

The Site is located in a low-lying area in the Northwest New Territories with an average mPD of negative 2, and it has immediate proximity to Shenzhen Bay. Therefore, the site is highly susceptible to coastal dynamics, especially tidal fluctuations and storm surges. This baseline review uses the nearby Tsim Bei Tsui Meteorological Station (Station 1366) as the core reference point for sea level observation, as its geographical location in Shenzhen Bay best reflects the actual risk of coastal flooding along the Northwest New Territories coast. According to the records of the Hong Kong Observatory (HKO) (2025c), the current baseline mean sea level (Mean Sea Level) at the site is around 1.4 mPD. In terms of tidal levels, the highest astronomical tide level recorded in recent years was 3.3 metres in November 2025 **(Fig. 4.3.4)**, which is adopted as a benchmark for assessing flooding risk related to tidal regime. In terms of extreme weather benchmarks, the storm surge caused by Super Typhoon Hato in 2017 recorded a historic high sea level of 4.56 mPD at Tsim Bei Tsui **(Table 4.3.1)**, which has become a key benchmark for assessing the current extreme flooding risk.

十一月 NOVEMBER 2025

尖鼻咀 TSIM BEI TSUI

漲潮及退潮的時間及高度

TIMES AND HEIGHTS OF HIGH AND LOW TIDES

星期一 Mon 時間 TIME 米 M	星期二 Tue 時間 TIME 米 M	星期三 Wed 時間 TIME 米 M	星期四 Thu 時間 TIME 米 M	星期五 Fri 時間 TIME 米 M	星期六 Sat 時間 TIME 米 M	星期日 Sun 時間 TIME 米 M
					1 0050 1.6 0602 2.3 1345 0.7 2027 2.1	**2** 0201 1.3 0714 2.4 1433 0.7 2034 2.3
3 0258 1.0 0812 2.4 1515 0.8 2044 2.5	**4** 0351 0.7 0909 2.4 1551 0.9 2103 2.7	**5** 0443 0.4 1009 2.3 ○ 1622 1.0 2130 3.0	**6** 0536 0.2 1116 2.2 1648 1.1 2200 3.2	**7** 0632 0.2 1223 2.1 1709 1.2 2232 3.3	**8** 0729 0.2 1323 1.9 1730 1.3 2306 3.2	**9** 0826 0.2 1421 1.8 1756 1.4 2343 3.1
10 0921 0.2 1524 1.8 1829 1.5	**11** 0026 2.8 1014 0.3 1643 1.8 1912 1.7	**12** 0122 2.5 1108 0.4 ☾	**13** 0254 2.3 1203 0.5 1930 2.0 2356 1.7	**14** 0550 2.1 1256 0.6 2001 2.1	**15** 0123 1.4 0702 2.1 1344 0.7 2018 2.2	**16** 0223 1.1 0758 2.0 1424 0.8 2023 2.3
17 0313 0.9 0853 1.9 1458 0.9 2029 2.5	**18** 0359 0.7 0950 1.9 1525 1.0 2044 2.6	**19** 0443 0.6 1043 1.9 1547 1.1 2104 2.8	**20** 0528 0.5 1130 1.8 ● 1609 1.2 2129 2.9	**21** 0612 0.4 1213 1.8 1632 1.2 2155 3.0	**22** 0657 0.4 1255 1.8 1656 1.3 2224 3.0	**23** 0741 0.4 1336 1.8 1722 1.3 2256 3.0
24 0822 0.5 1421 1.7 1749 1.4 2331 2.9	**25** 0901 0.5 1513 1.7 1819 1.5	**26** 0012 2.8 0939 0.5 1620 1.7 1857 1.6	**27** 0100 2.6 1018 0.5 1742 1.7 2012 1.7	**28** 0202 2.4 1100 0.6 ☽ 1840 1.8 2238 1.6	**29** 0321 2.2 1147 0.7 1909 2.0	**30** 0010 1.4 0459 2.1 1239 0.8 1920 2.1

Fig. 4.3.4. Times and Heights of High and Low Tides recorded at Tsim Bei Tsui in Nov 2025
Source: Hong Kong Observatory (2025b)

Table 4.3.1. Top 5 Storm surge events recorded at Tsim Bei Tsui during tropical cyclones

Rank	Sea Level (Metres Above Chart Datum)	Tropical Cyclone	Datetime
1	4.56	Hato	2017-08-23 13:42
2	4.18	Mangkhut	2018-09-16 17:14
3	3.98	Becky	1993-09-17 11:37
4	3.77	Ragasa	2025-09-24 12:12
5	3.73	Gordon	1989-07-18 08:16

Source: Adapted from Hong Kong Observatory (2025a)

Hazard and Risk Resilience: Future Coastal Baseline Projections for 2100

Considering the life cycle of STT as a strategic development project, this benchmark assessment extends the time frame to 2100 to measure the potential impact of long-term climate change on site stability. **Fig. 4.3.5** shows NASA's sea level rise (SLR) prediction model for Tsim Bei Tsui, with data sourced from IPCC Sixth Assessment Report (AR6). It reveals that the future baseline sea level of Shenzhen Bay will increase significantly, with the scale of increase depending on global emission trajectories. With reference to **Fig. 4.3.5** and **Table 4.3.2**, under the low-emission scenario (SSP1-2.6), the sea level of Shenzhen Bay is expected to increase by 0.35m, while it rises by 0.48m under the intermediate-emission scenario (SSP2-4.5) (i.e., conventional planning baseline). As for the very-high-emission scenario, the sea level is projected to increase by 0.7m by the end of 2100. This projection model suggests that future sea level rise will cause structural changes to the site's existing low-lying landform.

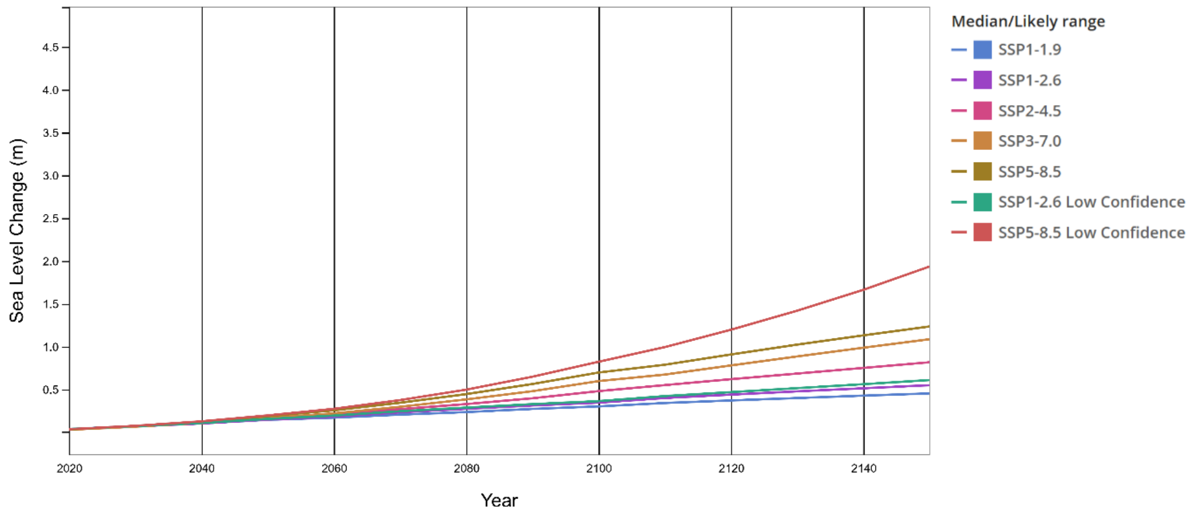


Fig. 4.3.5. Projected SLR at Tsim Bei Tsui tide gauge station under different SSP Scenarios. Source: NASA IPCC AR6 Sea Level Projection Tool (2022).

Table 4.3.2. Sea level projections at Tsim Bei Tsui tide gauge station through 2150.

	Very Low SSP1-1.9	Low SSP1-2.6	Intermediate SSP2-4.5	High SSP3-7.0	Very High SSP5-8.5	SSP1-2.6 Low Confidence	SSP5-8.5 Low Confidence
Total (2030)	0.07	0.07	0.07	0.07	0.07	0.07	0.08
Total (2050)	0.15	0.15	0.16	0.17	0.19	0.16	0.20
Total (2090)	0.27	0.31	0.40	0.48	0.56	0.33	0.65
Total (2100)	0.30	0.35	0.48	0.60	0.70	0.36	0.82
Total (2150)	0.45	0.55	0.82	1.09	1.24	0.61	1.94

Source: Adapted from NASA IPCC AR6 Sea Level Projection Tool (2022)

Hazard and Risk Resilience: Cumulative Flooding Risk Assessment

By overlaying the baseline of future sea level rise with daily tidal and extreme weather events, this baseline review identifies three levels of cumulative risk. First, under the 2100 intermediate-emission scenario, areas below 1.9 mPD would be at risk of permanent inundation, while areas below 2.1 mPD would be inundated under the high-emission scenario (**Fig. 4.3.6 and 4.3.6**). This would directly affect the site's developable land boundaries. Moreover, the daily tidal risk will expand significantly. With the current maximum tide level of 3.3 meters as a benchmark, the daily high tide level could reach about 3.8 to 4.0 mPD in 2100 under the intermediate- and high-emission scenarios (**Fig. 4.3.6** and **4.3.7**). This poses a constant threat of tidal surge to adjacent low-lying areas. Finally, under extreme storm surge scenarios, if compounded by a historical Typhoon Hato- or Mangkhut-level event with record-breaking sea level, the extreme sea level could reach at least 5 mPD under intermediate- or high-emission scenarios (**Fig. 4.3.6 and 4.3.7**). Such extreme water levels will lead to large-scale backflow and inundation caused by severe overtopping waves, posing serious threats to the nearby neighbourhood and I&T zones.

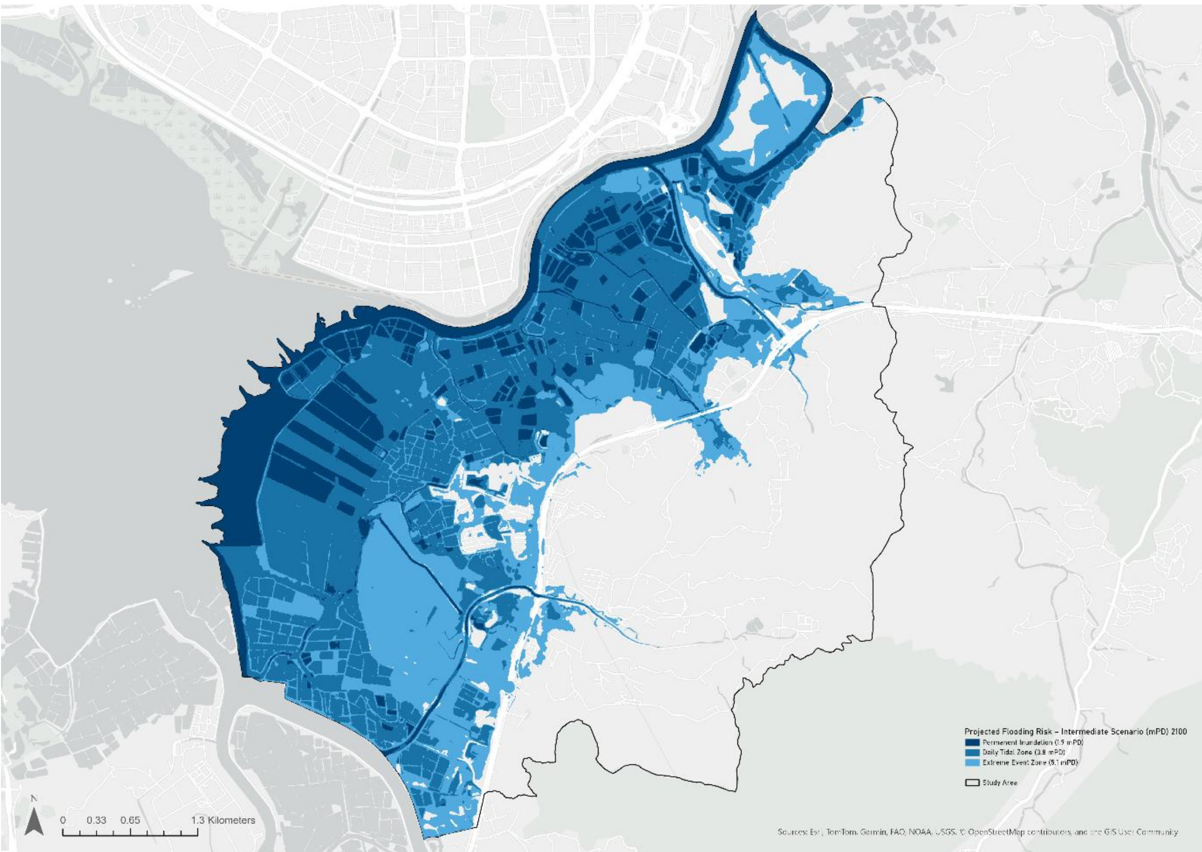


Fig. 4.3.6. Projected Coastal Flood Risk (2100): Intermediate Scenario (SSP2-4.5)

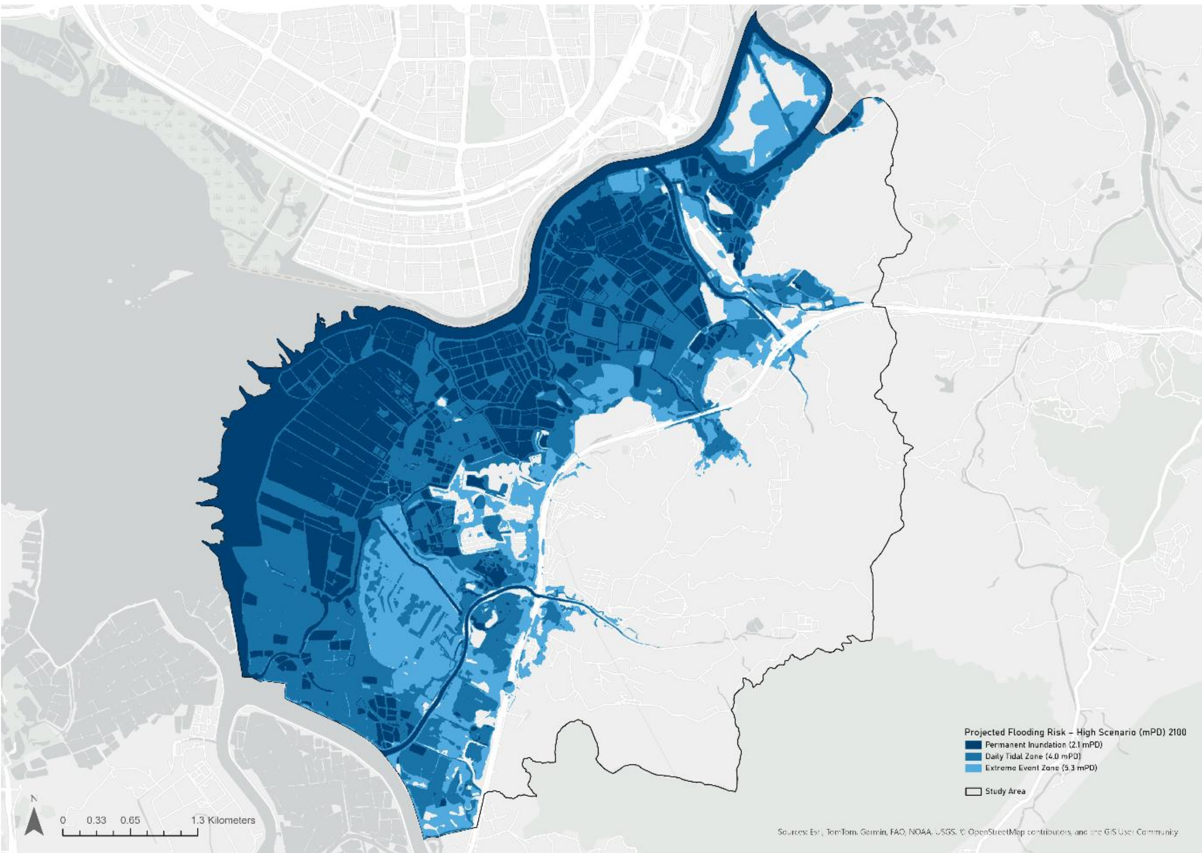


Fig. 4.3.7. Projected Coastal Flood Risk (2100): High Scenario (SSP5-8.5)

Carbon Management: Carbon Footprint

Using a GPC-based assessment, the carbon footprint of the STT reveals a fundamental mismatch between Hong Kong's climate ambitions and project-level planning. The existing areas of San Tin and NTM are currently dominated by Scope 1 emissions from diesel-based brownfield logistics and transport (See **Fig. 4.3.8** and **Table 4.3.3**), yet the proposed Technopole would shift emissions toward Scope 2 due to electricity-intensive innovation and technology industries (ITIB, 2024), and this will significantly increase overall emissions (See **Table 4.3.5**). Critically, while the EIA focuses on construction-related environmental controls such as dust and runoff (EPD, 2024), it fails to establish a carbon baseline and Scope 1–3 inventory, so that the emissions are not quantified for assessment and with climate targets. Although Hong Kong's Climate Action Plan 2050 sets a clear carbon neutrality goal (Environment and Ecology Bureau, 2021), its principles are not translated into the current proposal, which lacks a defined carbon budget, reduction pathway, or sector-specific targets (ITIB, 2024). Although the current plan emphasises "green industries" as an economic strategy and assumes low-carbon outcomes, the high energy demand in the operation is not accounted for. Therefore, threats of high carbon footprint are identified and imply the need for implementing the circular economy to achieve carbon neutrality.

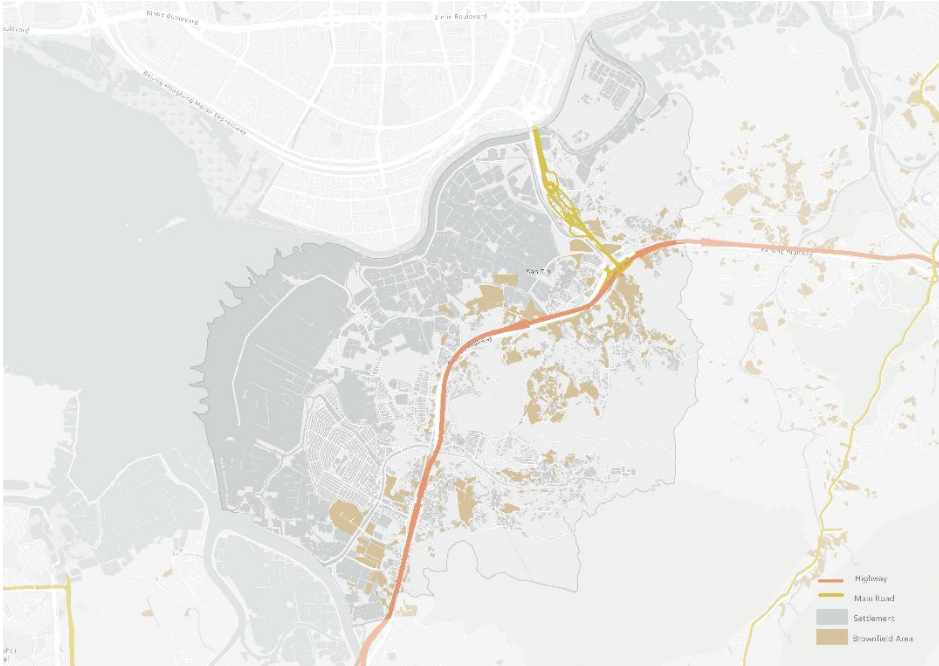


Fig. 4.3.8. Source of Carbon Footprint in the Study Area
Source: Highways Department (2026)

Table 4.3.3. GPC Table for GPC Table 4.3 – San Tin / NTM/ HKSTP (Existing Baseline)

Sector / Sub-Sector	Scope 1 CO_2	Scope 1 CH_4	Scope 1 N_2O	Scope 1 F-Gases	Scope 2 CO_2	Scope 3 CO_2E	$CO_2(B)$	Scope (TCO_2E)
Stationary Energy								
Residential Buildings	35,000	3,000	2,000	0	30,000	10,000	2,000	**80,000**
Commercial & Institutional (HKSTP)	150,000	20,000	10,000	0	200,000	30,000	5,000	**410,000**
Manufacturing & Construction (Brownfield)	130,000	30,000	10,000	0	20,000	30,000	10,000	**220,000**
Agriculture, Forestry & Fishing	10,000	25,000	5,000	0	5,000	5,000	0	**50,000**
Transportation								
On-Road	240,000	40,000	20,000	0	20,000	80,000	0	**400,000**
Railways	5,000	2,000	1,000	0	5,000	5,000	0	**18,000**
Waste								
Solid Waste Disposal	30,000	70,000	10,000	0	0	40,000	0	**150,000**
Wastewater	10,000	10,000	5,000	0	0	5,000	0	**30,000**

Source: EMSD (2025)

Table 4.3.4. Summary by Scope (Existing)

Scope	Emissions (TCO_2E/Year)
Scope 1	~570,000
Scope 2	~275,000
Scope 3	~195,000
Total	**~1,040,000**

Table 4.3.5. GPC Table for Future STT (Operational Scenario)

Sector / Sub-Sector	Scope 1 CO_2	Scope 1 CH_4	Scope 1 N_2O	Scope 1 F-Gases	Scope 2 CO_2	Scope 3 CO_2E	CO_2(B)	Scope (TCO_2E)
Stationary Energy								
Residential Buildings	80,000	10,000	10,000	0	250,000	50,000	5,000	**400,000**
Commercial & I&T (Technopole)	700,000	150,000	50,000	200,000	900,000	200,000	10,000	**2,200,000**
Manufacturing (Light Industry)	100,000	30,000	20,000	30,000	100,000	100,000	5,000	**380,000**
Transportation								
On-Road	300,000	100,000	50,000	0	50,000	200,000	0	**700,000**
Railways	10,000	5,000	2,000	0	20,000	10,000	0	**47,000**
Waste								
Solid Waste	50,000	100,000	30,000	0	0	100,000	0	**280,000**
Wastewater	20,000	40,000	10,000	0	0	20,000	0	**90,000**
AFOLU (Land-Use Change Impact)	10,000	70,000	20,000	0	5,000	80,000	0	**185,000**

Source: ITIB (2024)

Table 4.3.6. Summary by Scope (Future)

SCOPE	EMISSIONS (TCO_2E/YEAR)
Scope 1	~740,000
Scope 2	~1,305,000
Scope 3	~730,000
Total	**~2,775,000**

Carbon Management: Existing Energy Flow

The existing energy structure remains fundamentally fossil fuel–dependent **(Fig. 4.3.9)**, with approximately 72% of electricity generated from fossil fuels imported from Southeast Asia, alongside nuclear imports from the Daya Bay Nuclear Plant (EMSD, 2025) **(Fig. 4.3.10)**. This creates a high carbon footprint, which electricity-intensive industries in STT, such as AI computing centres and data infrastructure, will intensify the carbon emission (ITIB, 2024). Although renewable energy applications (e.g., photovoltaics) are mentioned (CEDD, 2024), there is no specification of generation sources, and uncertainty is raised about whether these "renewable" systems would still rely on fossil fuels. This reflects a linear energy flow, where energy is supplied and consumed without recovery mechanisms. In the absence of a defined renewable energy and circular energy pathway, the current proposal reveals a gap in addressing decarbonisation at a systemic level. So, a circular economy approach to energy is needed to treat the waste heat and excess energy through integrated infrastructure to achieve the goal of carbon neutrality.

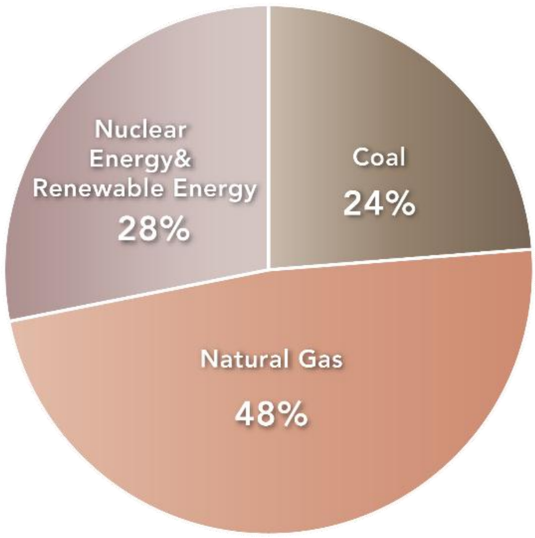


Fig. 4.3.9. Existing Energy Composition in Hong Kong
Source: EMSD (2025)

Fig. 4.3.10. Existing Energy Flow
Source: EMSD (2025)

<u>Carbon Management: Existing Water Flow</u>

Current water flow reveals a linear water metabolism characterised by a heavy reliance on import of freshwater, with 70–80% of freshwater being imported from the Dongjiang and supplemented by local reservoirs and emerging desalination in Tseung Kwan O (See **Fig. 4.3.11**). Generally, Hong Kong follows a "treat-and-discharge" model rather than a closed-loop recovery system (WSD, 2026). Then, there is an opportunity for a shift in water management, which involves the wastewater from domestic and production towards a circular water loop within San Tin in order to maximise the usage of the water resources.

Fig. 4.3.11. Existing Water Flow
Source: WSD (2026)

<u>Carbon Management: Existing Material Flow</u>

Hong Kong's current material circularity is lagging, with the local municipal solid waste recovery rate only for 22% within Hong Kong in 2022 (Legco, 2024), while the rest are exported to other countries or directed to landfill (World Integrated Trade Solution, 2023) (**Fig. 4.3.12**). Current practice shifts the environmental burden rather than resolving it and fails to align with the circular economy goal mentioned in both the 15th Five-Year Plan and Hong Kong's Waste Blueprint 2035 (EPD, 2021; CPG, 2026). Even though there is a presence of local recycling facilities such as EcoPark and the WEEE Park, in the current government proposal, no explicit circular material strategy is identified, and the plan does not propose any connection or collaboration with the existing facilities. Therefore, it is needed to establish on-site material recovery systems, including connections to designing for material reuse to achieve a closed material loop.

Fig. 4.3.12. Existing Material Flow
Source: World Integrated Trade Solution (2023)

4.3.4. SDG 15: Life on Land: Ecological Integrity and Transboundary Connectivity

<u>Ecological Integrity: Biodiversity</u>

According to **Fig. 4.3.13**, the study area showcased a high ecological value with a rich ecological profile with high biodiversity, over 400 bird species, including the globally threatened Black-faced Spoonbill, Eurasian Otters, etc., are recorded (Global Biodiversity Information Facility, 2026; WWF, 2025). Yet, a significant concern is raised during the interview that the EIA's species selection scope is insufficiently broad to prove the effectiveness of compensation measures. For instance, abandoned ponds are regarded as having low ecological value, but in fact serve as important duck habitat (Conservancy Association, 2026). The current government's response is the wetland compensation through ecologically enhanced fishponds and enhanced freshwater wetland habitats. It is highlighted that the Habitat Creation and Management Plan lack long-term financial commitment and resources to ensure long-term conservation management (EPD, 2021; Conservancy Association, 2026). As a result, the current strategy for biodiversity remains reactive as compensation is the main measure rather than avoidance and enhancement.

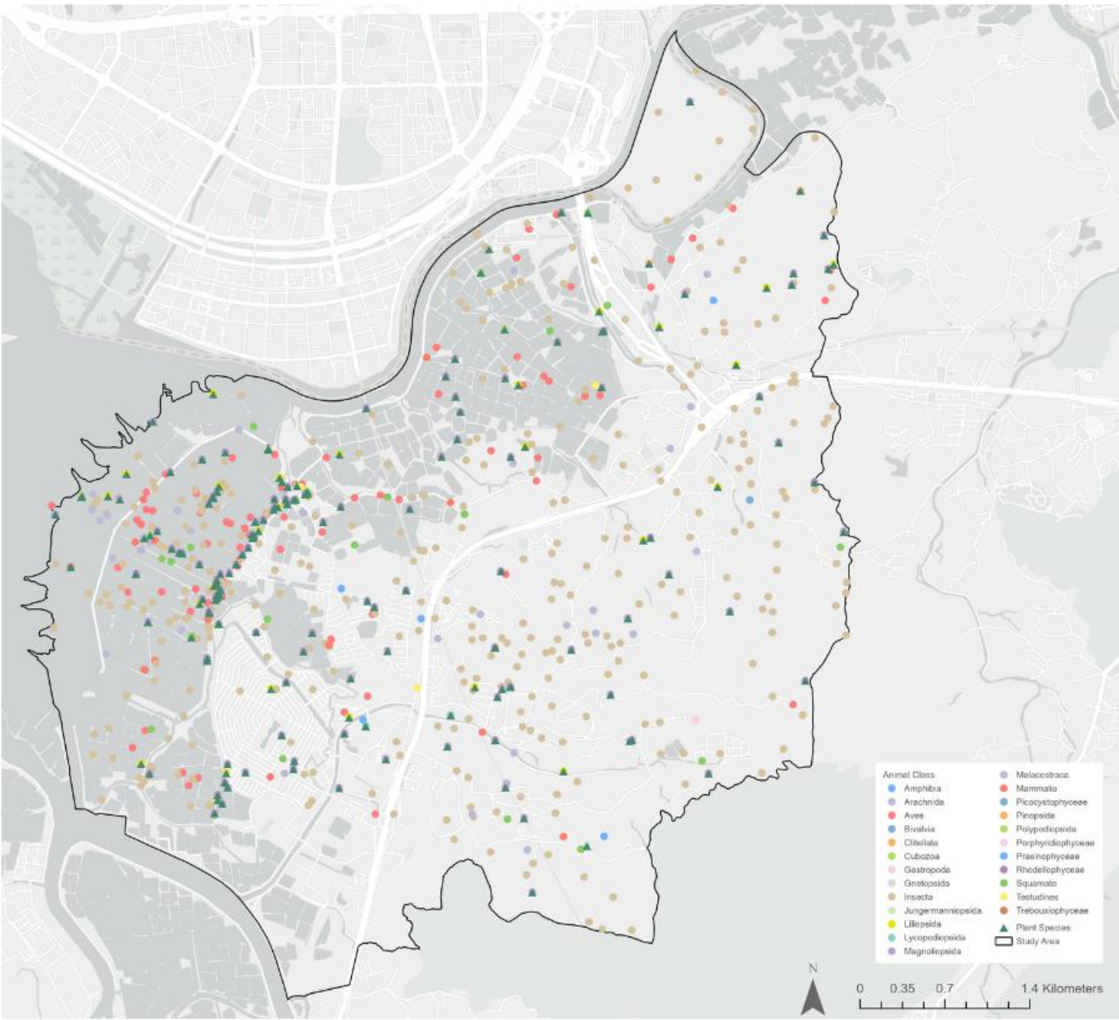


Fig. 4.3.13. Biodiversity of the Study Area
Source: Global Biodiversity Information Facility (2026); WWF (2025)

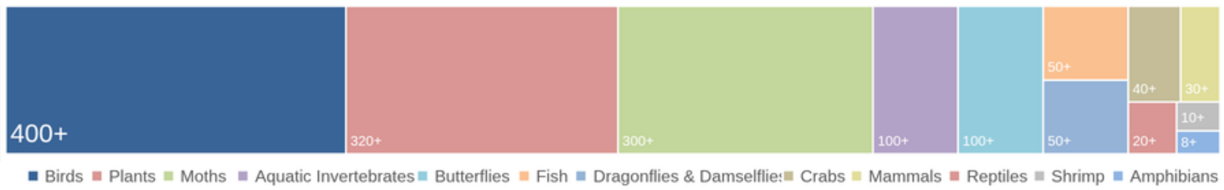


Fig. 4.3.14. Abundance of Species within the Study Area

Transboundary Connectivity: Ecology Continuity

The fragmentation of the wider ecological corridor is a key concern, and San Tin is the pivotal middle segment of a complete transboundary ecological system **(Fig. 4.3.15)**. EIA identifies habitat fragmentation as a moderate to high impact in the northern portion, where high-rise structures would potentially block the west–east avifauna flight path, so that mitigation measures include ecologically enhanced fishponds with improved connectivity, maintenance of the flight corridor under the Revised RODP, and Green Belts with wildlife corridors are proposed as mitigation measures (EPD, 2024). However, the massive development poses a threat to vital wildlife corridors despite these measures. The Conservancy Association questioned the technical feasibility of proposed animal passages for Eurasian Otters, as otter movement is confirmed by physical evidence, including spraints found within the San Tin area. Also, unresolved concerns about passage width and lighting conditions are raised, and the lack of any confirmed similar cases makes the technical feasibility of the proposal questionable (Conservancy Association, 2026).

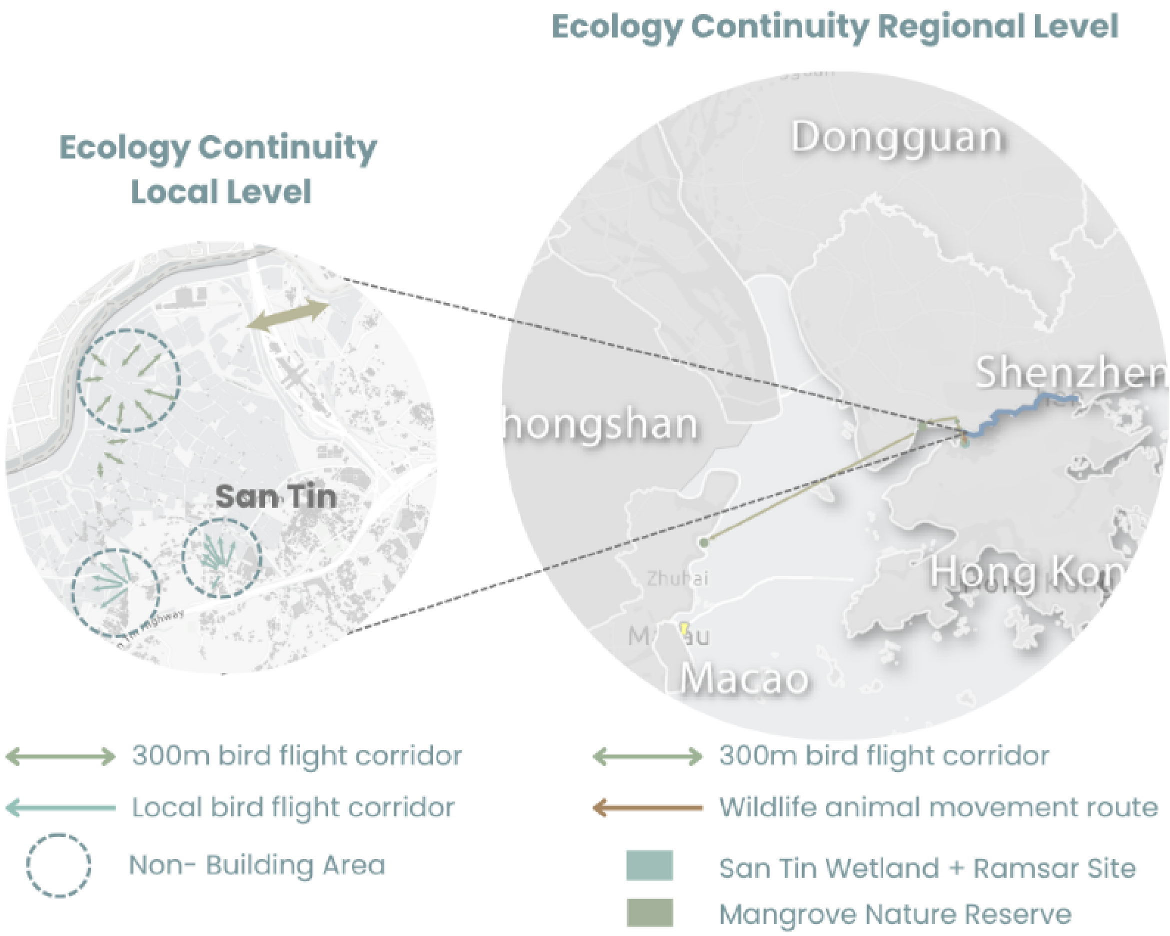


Fig. 4.3.15. Ecology Continuity on the Local Level and the Regional Level
Source: CEDD (n.d.)

Transboundary Connectivity: Wetland Condition

For wetland conditions, EIA confirms construction phase impacts on sites with high ecological value in the northern part **(Fig. 4.3.16)**, including the Conservation Area, SSSI zone, and Ramsar Site. To mitigate, mandatory wetland compensation is required (EPD, 2024). However, the Conservancy Association challenged the definition of the no-net-loss standard, which the current no-net-loss standard focuses exclusively on ecological function rather than area, yet international standards from Taiwan, the United States, and South Korea demonstrate that both function and area are essential to maximising ecological outcomes. Also, the government is criticised for its design conflict in the current proposal, which involves the playground with lighting above the egretry site, directly threatening their breeding behaviour. (Conservancy Association, 2026). Therefore, the current plan for compensation for loss of habitat still has room for improvement. Further studies are needed not only to compensate, but also to plan to enhance the habitat and leverage the overall biodiversity by improving the environmental quality and ecological functions of the habitat.

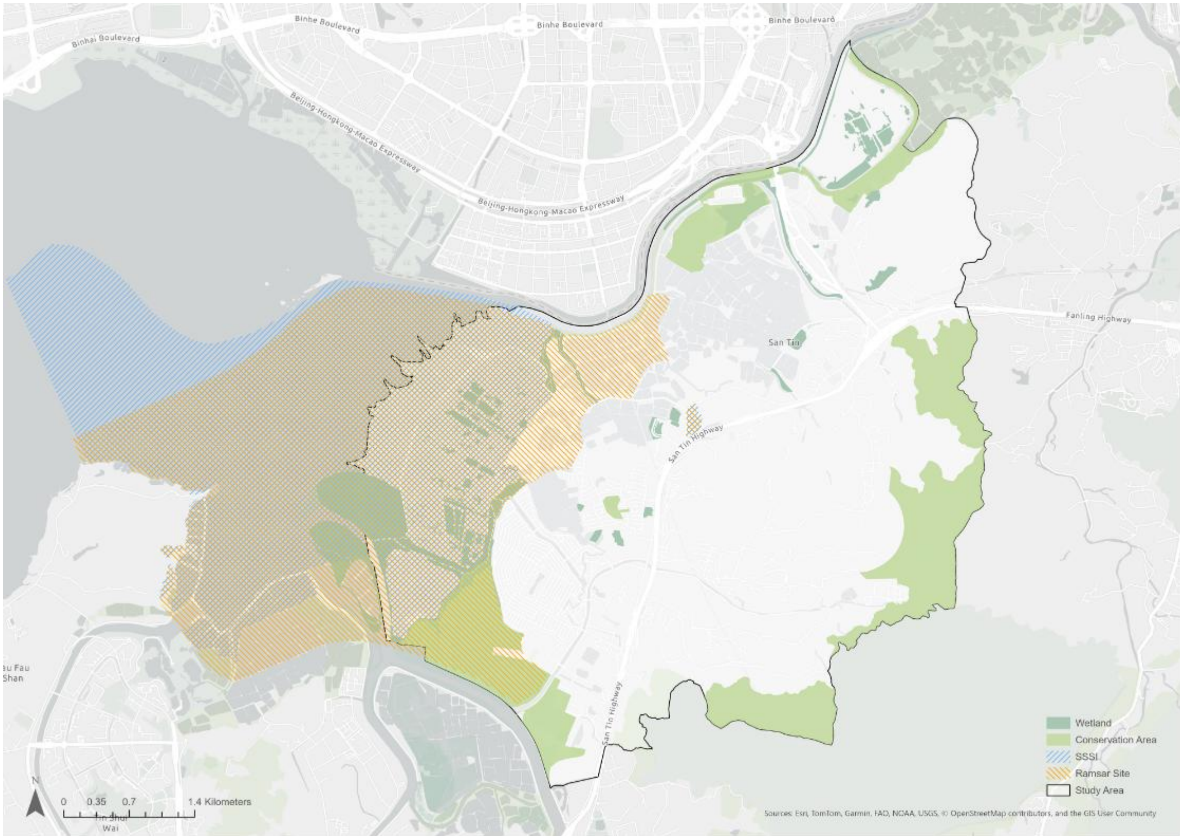


Fig. 4.3.16. Wetland Condition
Source: EPD (2024)

4.4. Urban-Rural Integration

4.4.1. Assessment Framework

To fully establish an urban-rural integration and protect the existing site's integrity and authenticity in the future development of the study area, it is essential to first understand the existing assets, cultural landscape, and community image and identity. In this study, the HUL Approach formulated by UNESCO (2011) is adopted to analyse the site. In this approach, urban areas are studied through dissecting the complex layers and features within the HUL. These features are grouped into seven layers in this study **(Fig. 4.4.1)**. Additionally, the 'Place and Placelessness' framework (Relph, 1978) is used to support the analysis of identity. However, in this section, the focus is on the built environment, social values and cultural practices, and identity, as natural resources and topography, land use, economic processes, and diversity have been discussed in section 4.2. and section 4.3. Subsequently, an analysis of the current proposed land use and OZP, and the spatial implications based on the baseline findings, will be discussed.

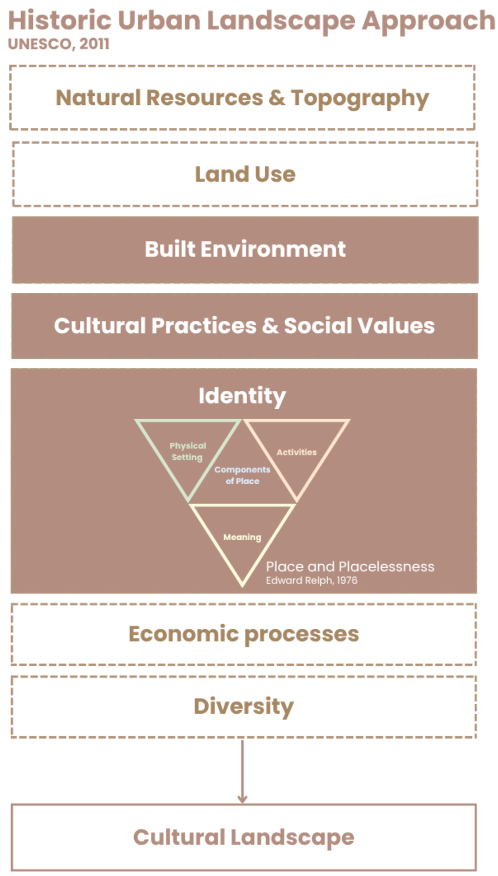


Fig. 4.4.1. Assessment Framework of Urban-Rural Integration

4.4.2. History of the Study Area

San Tin first appeared on a 1866 map **(Fig. 4.4.2a)**; however, its history dates to the 15th century during the Ming Dynasty. The founding ancestor of the San Tin Man-clan, Man Sai-gor, who is the descendant of Man Tin-shui, the brother of a renowned patriot in the Southern

Song Dynasty, Man Tin-cheung, settled in San Tin in the Ming Dynasty (AAB, n.d.). As the years went by, the Man clan expanded and formed ‘three wai and six tsuen’ (三圍六村) (AAB, n.d.). In **Fig. 4.4.2b.**, the map shows that there were paddy fields near the Man villages, and this reflects their agricultural nature. Another village that has existed for half a century in San Tin is Lok Ma Chau Tsuen, located near the border, formed by the Cheung clan from the Zhangjiawei, Getang of Shenzhen in the early Qing Dynasty, and they mostly relied on fish farming to support themselves (AAB, n.d.; CEDD, PD and AECOM Asia Company Ltd., 2024). Mai Po Tsuen is also a village that has been in San Tin for 200 to 400 years, established by a mixture of clans (CEDD, PD and AECOM Asia Company Ltd., 2024).

Fast forward to the 1950s to 1970s, when immigrants from China fled to Hong Kong due to political instability during the post-war period and formed several NIVs **(Fig. 4.4.2c)**, and they either rented land from the Man clan for agriculture or engaged in fish farming (CEDD, PD and AECOM Asia Company Ltd., 2024; Liber Research Community, 2024).

Fig. 4.4.2a. 1866 Map zoomed into the Study Area
Source: Simeone (1866)

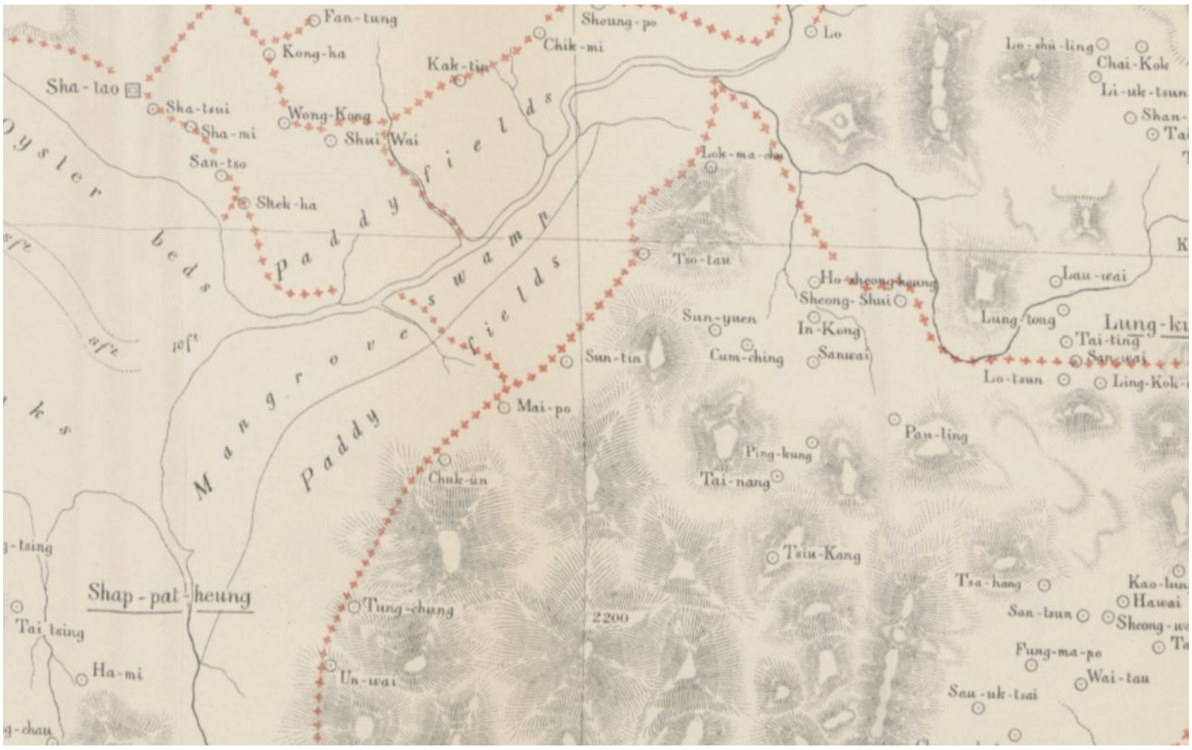


Fig. 4.4.2b. 1897 Map zoomed into the Study Area.
Source: Hurley (1897)

Fig. 4.4.2c. 1985 Map of the Study Area. All the villages are shown on the Map.

4.4.3. Built Environment

<u>Indigenous and Non-Indigenous Villages</u>

26 villages are identified in the study area, of which 15 of them are IV, and the rest are NIV (**Fig. 4.4.3**). The IVs are under legal protection. Hence, they are protected from new development (Shum, 2024). However, NIVs are not yet some of them contain rich cultural value. Therefore, research has to be conducted before making the decision to demolish them (**Table 4.4.1**).

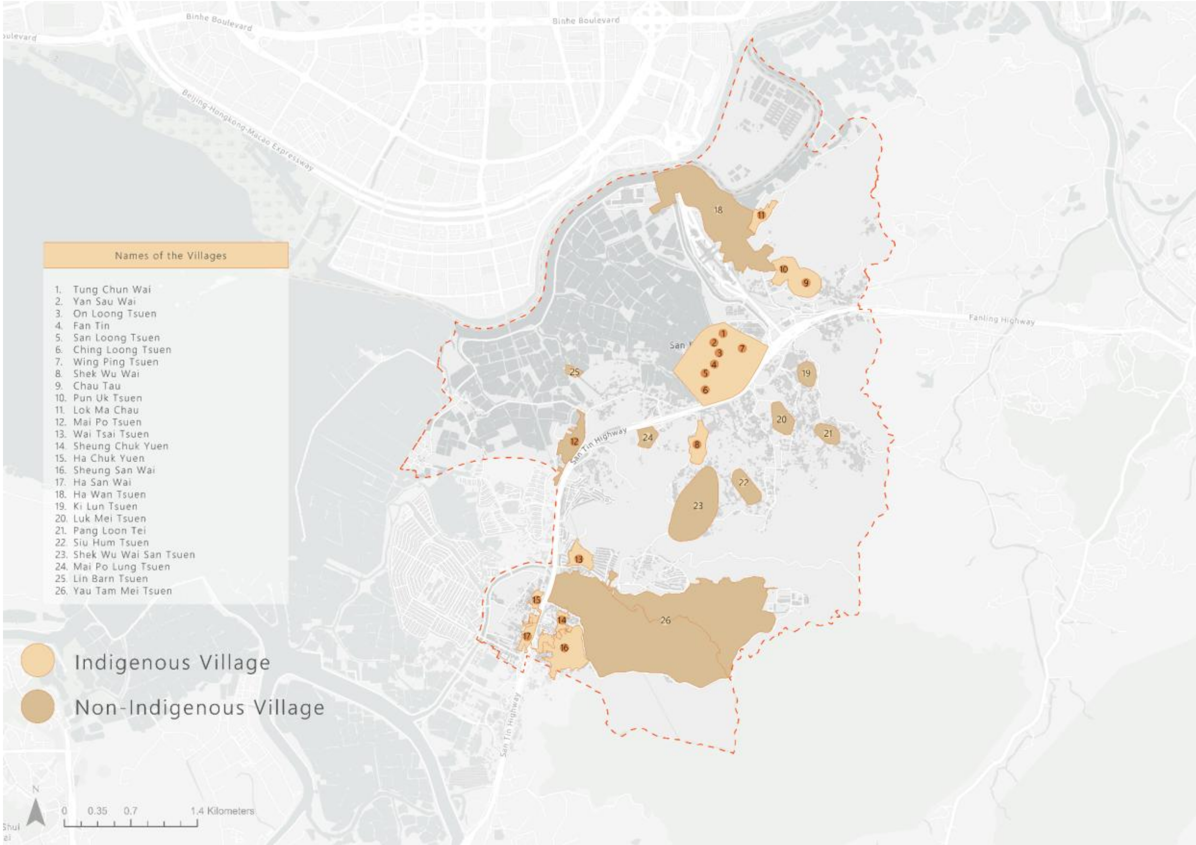


Fig. 4.4.3. Locations of the Villages in the Study Area.
Source: Rural Representative Election (2026)

Table 4.4.1. History of the Non-indigenous Villages in the Study Area

Village Name	History
Ha Wan Tsuen	The exact origin of this village is not known; it is believed to have been established in the 1950s. The villagers relied on fishing and fish farming to support themselves and used traditional fish and shrimp pond operation technique, “gei wai”.
Ki Lun Tsuen	Formed in the 1940s with around 80 families whose livelihoods depended on agriculture and poultry farming.
Luk Mei Tsuen	The origin of this village is not known. However, it is believed that they depended on agriculture, fish farming and fishing, as there are few streams near the village.

Lin Barn Tsuen	The origin of this village is not known. However, the villagers likely relied on fishing, fish farming, and poultry farming due to the presence of fishponds and their proximity to water.
Mai Po Lung Tsuen	Established by immigrants from Huizhou Hoi Luk Fung in China during the 1960s due to political instability, the villagers relied on agriculture to support themselves.
Pang Loon Lei	The origin of this village is not known. However, it is believed that they depended on agriculture, fish farming and fishing, as there are few streams near the village.
Siu Hum Tsuen	Established by immigrants from China during the 1960s due to political instability, the villagers rented agricultural land from the Man clan for farming.
Shek Wu Wai San Tsuen	The origin of this village is not known. However, they are an agricultural village where farming activities persist today.
Sheung San Wai & Ha San Wai	Formed in 1909 by villagers from the Wong clan and Chiu clan, and later the Lees, Wongs and Chans also joined this village. In the 1980s, due to the development of the San Tin Highway, the village was split into two, forming Sheung San Wai and Ha San Wai.
Yau Tam Mei Tsuen	Hakka people from China rented and purchased land from the indigenous villages in San Tin and formed this agricultural village in the early 20th century.

Source: AECOM (2025); CEDD, PD and AECOM (2024); Liber Research Community (2024)

Tangible Cultural Heritage (TCH)

The study area contains two declared monuments, which are under statutory protection and 17 graded historic buildings **(Fig. 4.4.4a)** (AAB, 2026; AMO, 2026). However, many non-listed and non-graded tangible structures are of great importance to the villagers, or places where cultural festivals are hosted, and that reflect the history of the Study Area **(Fig. 4.4.4b.)**.

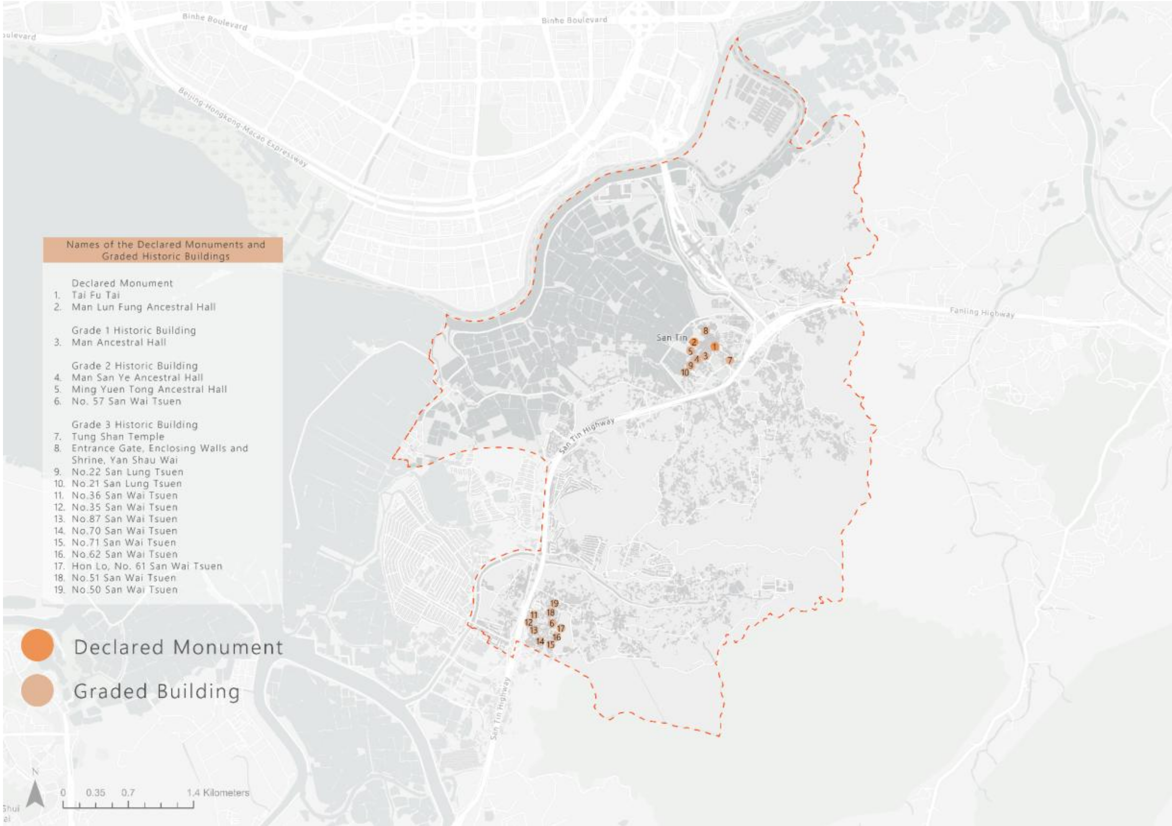


Fig. 4.4.4a. Locations of the Declared Monuments and Graded Historic Buildings
Source: AAB (2026); AMO (2026)

Fig. 4.4.4b. Non-listed Tangible Cultural Heritage
Source: AECOM (2025); CEDD, PD and AECOM (2024); In-media HK (2025)

Archaeology

In 1980 and 1997, Song Dynasty coins and Han Dynasty artefacts were discovered in MP and NTM (AECOM, 2025; CEDD, PD and AECOM, 2024). Therefore, these two sites became SAIs designated by the AMO (2024). In addition, recent studies have identified ten ASAs **(Fig. 4.4.5)**.

There are a few mitigation measures to prevent damage to these sites (AECOM, 2025; CEDD, PD and AECOM, 2024). For archaeological sites where direct impacts from new developments are expected, an archaeological excavation and a survey-and-excavation should be conducted to retrieve data before construction. Sites with restricted access due to private ownership should perform a further archaeological survey before site formation works. Finally, sites that have encountered modern disturbance should have an archaeologist present to monitor the site formation works.

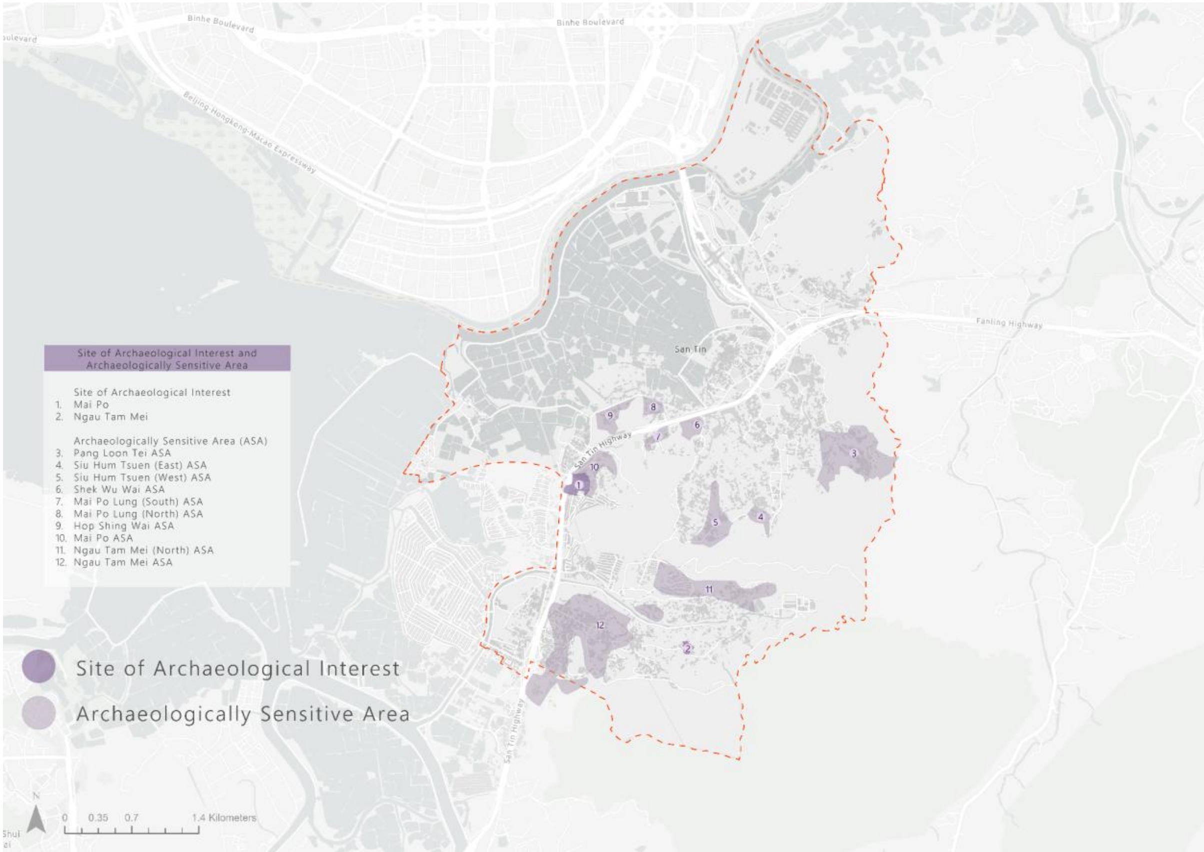


Fig. 4.4.5. Locations of Site of Archaeological Interest and Archaeologically Sensitive Areas. Source: AECOM (2025); AMO (2024); CEDD, PD and AECOM (2024)

Burial Sites

Hillside burial is a custom of the IVs; therefore, the government has put forward the “hillside burial policy” and assigned PBGs on government land for indigenous villagers and locally based fishermen (Hillside Burial, 2026). In our study area, there are a few PBGs **(Fig. 4.4.6.)** and unknown illegal hillside burial sites and graves which should be carefully handled with respect.

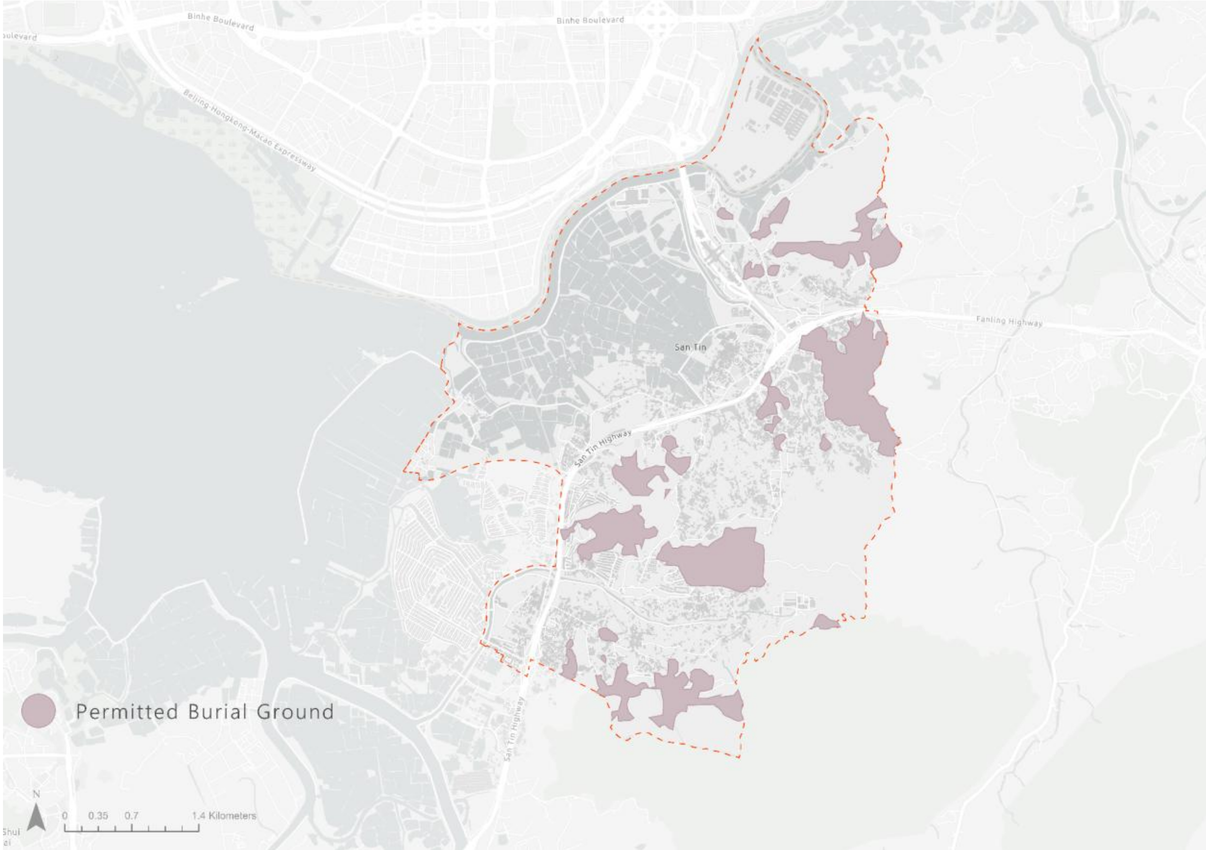


Fig. 4.4.6. Locations of Permitted Burial Grounds.
Source: Home Affairs Department (2024)

<u>15-Minute Neighbourhood</u>

To better understand the facilities and lives of the current residents in San Tin, a 15-minute neighbourhood analysis has been conducted. The 15-minute neighbourhood refers to the accessibility of six core services all reachable within 15 minutes through low-carbon mobility (Moreno, 2024). Hence, this analysis has separated the six core functions into six categories: 1) Living, 2) Healthcare, 3) Education, 4) Public transit, 5) Community support, and 6) Sports and Recreation. **Fig. 4.4.7** reveals the final scores of the 15-minute neighbourhood clusters regarding the provision of the services in the six categories across 24 hours. The final score of our study area falls in the lowest range of scores, implying that the current accessibility and provision of services are insufficient. In an interview (Man, 2026), the interviewee also expressed that they hope the new development can provide more community facilities and services to make their lives better. Hence, the new development must enhance the quality of life of the residents by improving the accessibility and provision of services.

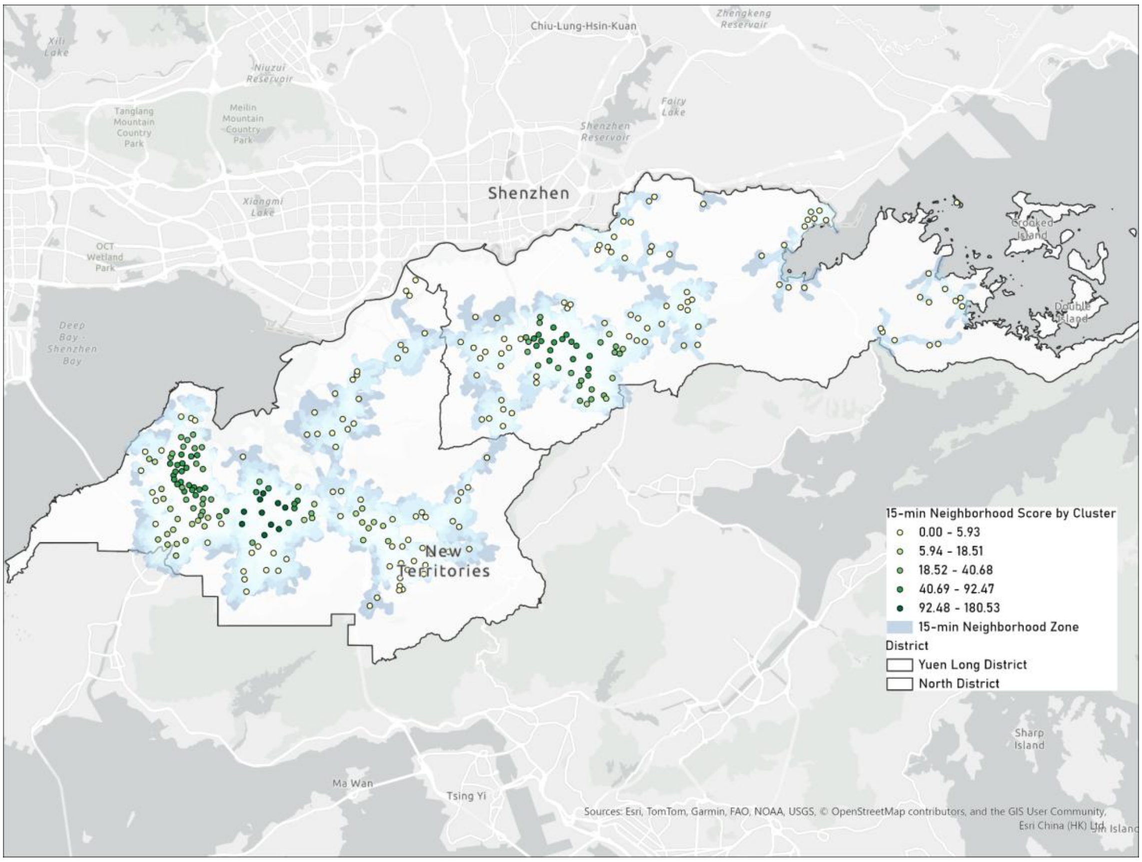


Fig. 4.4.7. The 15-minute Neighbourhood Scores by clusters

4.4.4. Cultural Practices & Social Values

Intangible Cultural Heritage

The ICH in the study area is rich, and they bring life and soul to the above-mentioned TCH **(Fig. 4.4.8)**. Two Man-clan-specific ICH have been listed in the Hong Kong ICH list, which are the Oral Legends of San Tin Man Lineage and Autumn Ancestral Worship of San Tin Man Lineage (ICH Database, 2024a; ICH Database, 2024c). Other general ICH items found in the study area are, for example, basin meals, Earth God Festival and Yu Lan Festival **(Table 4.4.2)**. However, some ICH, like the Birthday of the Thirty-Six Immortal Masters and Luoshan Boxing, are not currently recognised (In-media HK, 2025).

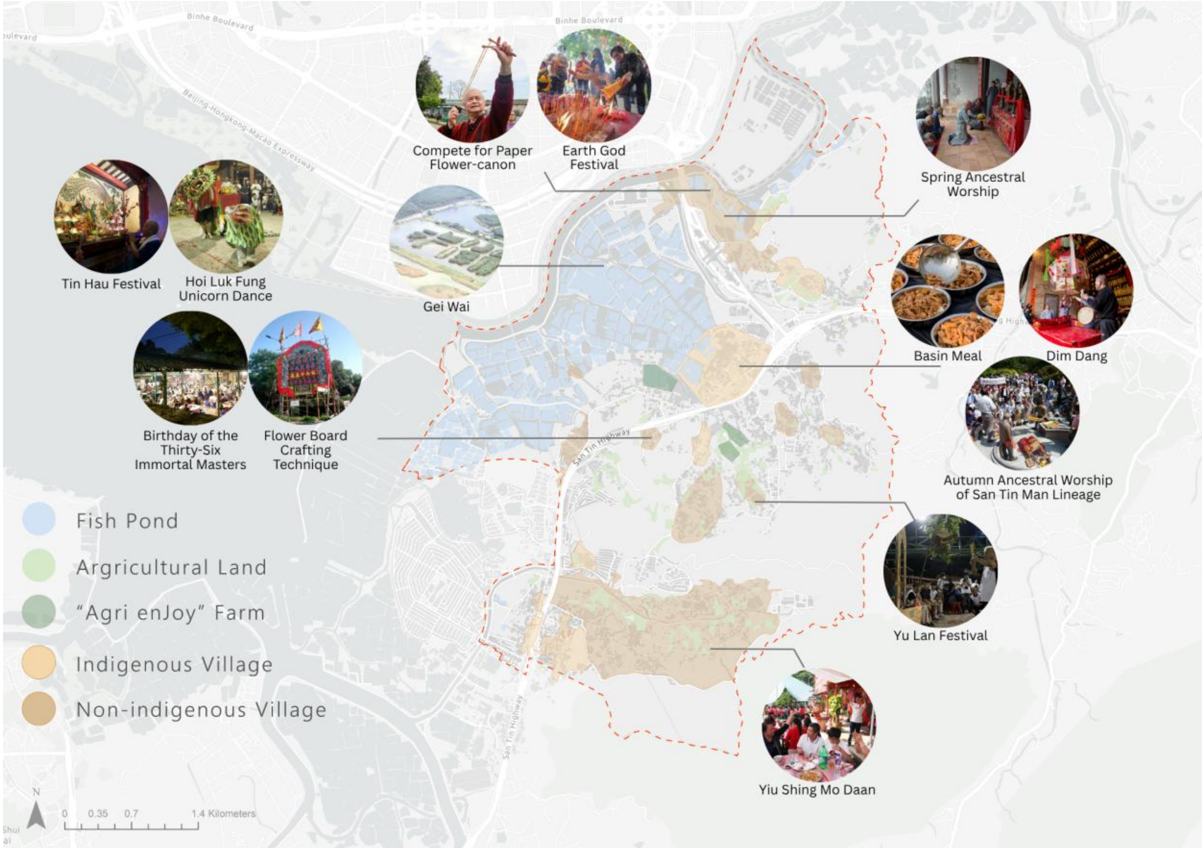


Fig. 4.4.8. Locations of Intangible Cultural Heritage.
Source: AECOM (2025); CEDD, PD and AECOM (2024); In-media HK (2025)

Table 4.4.2. The List of Intangible Cultural Heritage in the Study Area

Intangible Cultural Heritage	Domain	Inventory Code	Description
Oral Legends Of San Tin Man Lineage	Oral Traditions and Expressions	1.10.9	The oral legends of the founding ancestors and the feng shui of the past 500 years have been passed on from generation to generation.
Autumn Ancestral Worship Of San Tin Man Lineage	Social Practices, Rituals and Festive Events	3.9.2.8	On the ninth lunar month, the San Tin Man Clan has an ancestral worship ceremony which includes rituals like a silver band parade, presentation of offerings and sharing of roast pork.
Lantern Lighting Ritual (Dim Dang)	Social Practices, Rituals and Festive Events	3.5	This is performed during the first lunar month to inform the ancestors that a newborn son has become a member of the lineage. There are three parts of this process, which include lantern opening, lantern celebration and lantern completion. Then, a banquet is held after the completion of this ritual.
Earth God Festival	Social Practices, Rituals and Festive Events	3.8	It celebrates the Earth God’s birthday.

Spring Ancestral Worship	Social Practices, Rituals and Festive Events	3.9.1	On the fourth lunar month, the ancestral worship ceremony at ancestral halls or graves is held.
Yu Lan Festival	Social Practices, Rituals and Festive Events	3.30	Different lineage has a different ways to celebrate Yu Lan Festival. However, the purpose of this festival is to pay homage to ancestors, pacify wandering ghosts with clothing and food, and express gratitude to deities.
Tin Hau Festival	Social Practices, Rituals and Festive Events	3.18	On the 23rd day of the third lunar month, it is the birthday of Tin Hau, and it is celebrated at Tin Hau Temples to express gratitude to Tin Hau.
Autumn Ancestral Worship	Social Practices, Rituals and Festive Events	3.9.2	On the ninth lunar month, an ancestral worship ceremony at ancestral halls or graves is held.
Basin Meal	Social Practices, Rituals and Festive Events	3.59	The basin feast is held during celebrations and festivals, and the clansmen sit around the round table to enjoy a meal together.
Hoi Luk Fung / Hoklo Unicorn Dance	Performing Arts	2.4.3	In some Hoi Luk Fung/ Hoklo communities, the unicorn dance with unique movements and music rhythms is performed during festivals and birthdays of deities.
Lion Dance	Performing Arts	2.1	During festivals, birthdays of deities, weddings, celebrations or funerals, lion dance is performed to invoke blessings or express mourning.
Flower Board Crafting Technique	Traditional Craftsmanship	5.44	The making of flower boards involves processes like writing messages, making paper flowers, crafting the structure and assembling different parts. Flower boards are seen on the celebration of deities' birthdays, new building inaugurations and shop openings.
Gei Wai (Inter-Tidal Shrimp Ponds) Operation Technique	Traditional Craftsmanship	5.92	Gei Wai captures fish and shrimp through a system that has a water gate. During high tide, fish and shrimp swim into the gei wai, and the water gate closes so that they breed in it.
Birthday Of The Thirty-Six Immortal Masters (三十六仙師寶誕)	/	/	During the eighth lunar month, the birthday of thirty-six immortal masters is celebrated in the Hoi Luk Fung culture.

Yiu Shing Mo Daan (姚聖母誕)	/	/	The celebration of the birthday of Yiu Shing Mo
Baizi Opera Performance	/	/	A type of traditional Chinese opera that is popular among the Hoi Luk Fung communities.
Luoshan Boxing	/	/	A traditional Chinese martial arts style originated in the Minnan area.
Compete For Paper Flower-Canon (搶花炮)	/	/	The act of competing for the Paper Flower-canon is usually seen during a festival.

Social Values

These ICH items reflect the social values and the culture of the villagers. First, there are a number of festivals related to ancestor worship. This is because there is a strong emphasis on lineage, continuity, and remembrance of one's roots in Chinese culture (Man, 2026). Additionally, rituals and festivals bring people from the same village together. In some villages, festivals act as a binding force to bring people from different villages together. In an interview conducted with the Conservancy Association (2026), the interviewee shared his experience of participating in the Earth God Festival held in the NIV, Ha Wan Tsuen, and was fascinated by their rituals, the whole experience and the close-knit community. This reflects how these festivals act as catalysts to bring people together, thereby helping residents to form a sense of belonging and attachment to the place. The belief in feng shui, the legends of which have been passed down from generation to generation (ICH Database, 2024c), also reflects their social value regarding living in harmony with natural forces and the environment.

4.4.5. Identity

The identities of the site and people are established based on the analysis of the physical setting, activities and meaning (Relph, 1978). First, the study area is a culturally rich site that captures the evolution of Hong Kong across diverse periods, from the Ming Dynasty to the Qing Dynasty to the post-war period to the present, through TCH and ICH. Second, although the agricultural and fish farm identities of the site have been diminishing in recent decades, these activities still exist, and they resonate with the roots of this site. Finally, many of the descendants of different villages identify themselves as members of the clan that they belong to, especially the Man clan, illustrating their sense of belonging and attachment to their lineage.

4.4.6. Cultural Landscape

Combining all the layers of this historic urban landscape, it becomes apparent that there are strong natural-and-cultural, and geographic-and-history relationships, which formed a base for the economic and social development. The geographical location, being situated along the Shenzhen River, became an ideal spot for immigrants from China to settle. In addition, its natural resources, with fishponds and paddy fields, not only act as a beautiful backdrop but have become a productive landscape that has shaped the livelihoods and economy of the villages for the past 500 years. These natural resources and the tranquil environment allowed villages to flourish and pass on their traditional cultural and ecological knowledge of the site, as well as their rituals and traditions. These complex layers have evolved and formed a distinct character that stands out within the cosmopolitan landscape of Hong Kong.

4.4.7. User Needs Analysis

To better understand the villagers and residents' needs, a user needs analysis was conducted based on Maslow's Hierarchy of Needs (1943) **(Fig. 4.4.9)**. They not only need spaces for social and cultural exchange, and cultural and ancestral continuity, but also financial security, professional growth and living security and safety.

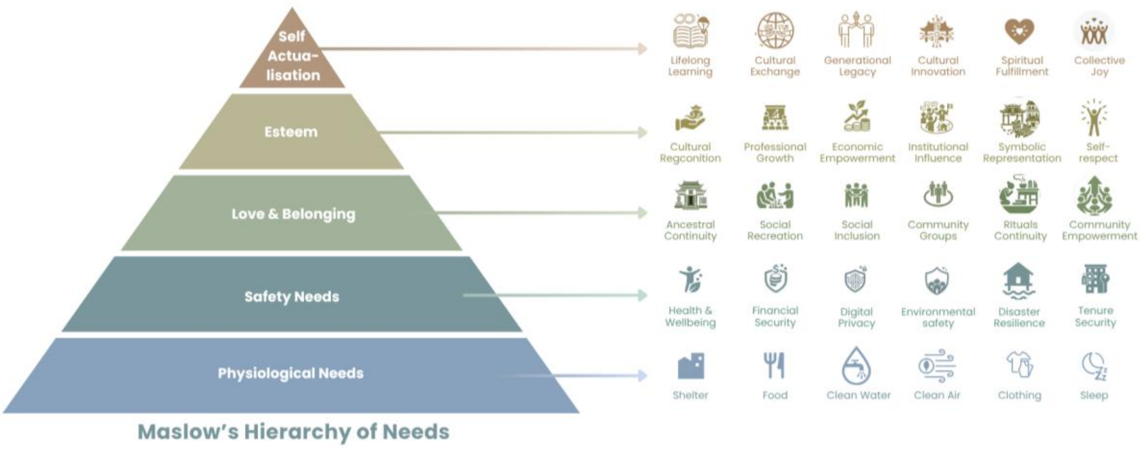


Fig 4.4.9. Villager and Local-resident needs based on Maslow's Hierarchy of Needs.

4.4.8. Existing Plan and OZP

According to the OZP and the proposed outline development plan (CEDD and PD, n.d.; TPB, 2024) **(Fig. 4.4.10a.** and **4.4.10b.)**, only the IVs will be conserved, and their land is zoned as Village Type Development, which aims to control the development of Small Houses in an orderly manner (TPB, 2024). A few commercial and community uses for the villagers are always permitted on the ground floor, while some may be permitted by the TPB after the submission of an application. In an interview (Man, 2026), the interviewee shared that the proposed development was set to place a waste collection station next to a memorial pavilion in the village, which is disrespectful. Therefore, the spatial design must respect the culture and lifestyle of the villages and achieve a harmonious spatial transition from urban to rural areas. In the proposed design, the land use around the villages is mostly Open Space and Green Belt with PBGs, aiming to maintain the natural and tranquil environment of the villages (CEDD and PD, n.d.). In addition, the development has zoned land for Government, Institution or Community (GIC) use, and proposes different recreational activities, which can enhance accessibility to services and enrich the lives of the users.

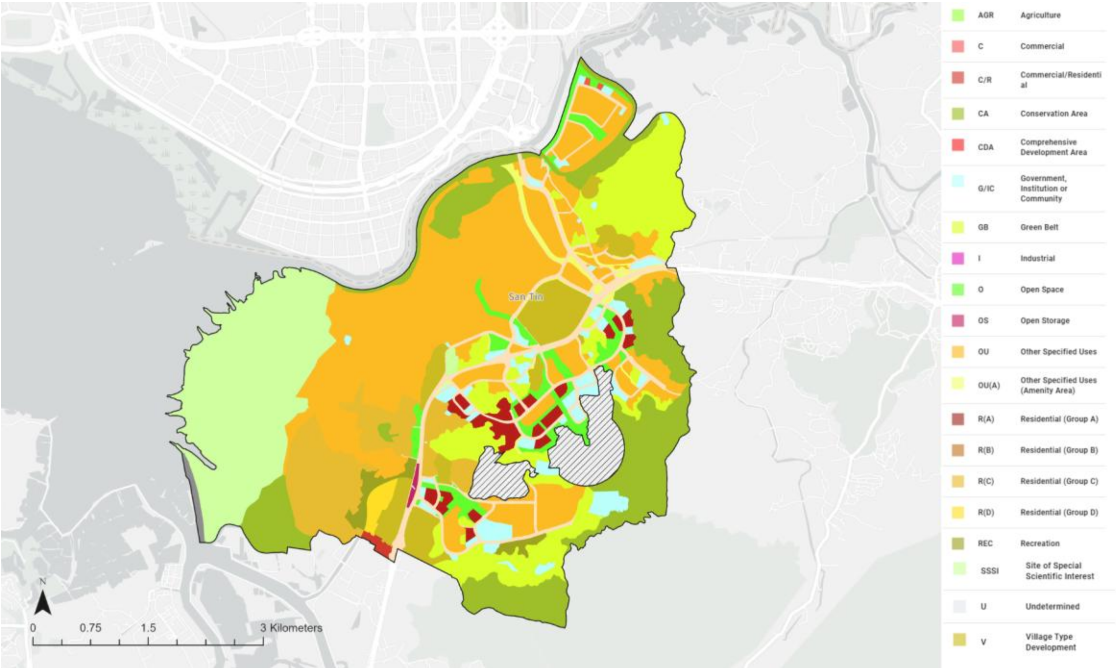


Fig. 4.4.10a. Outline Zoning Plan for the Study Area.
Source: TPB (2024)

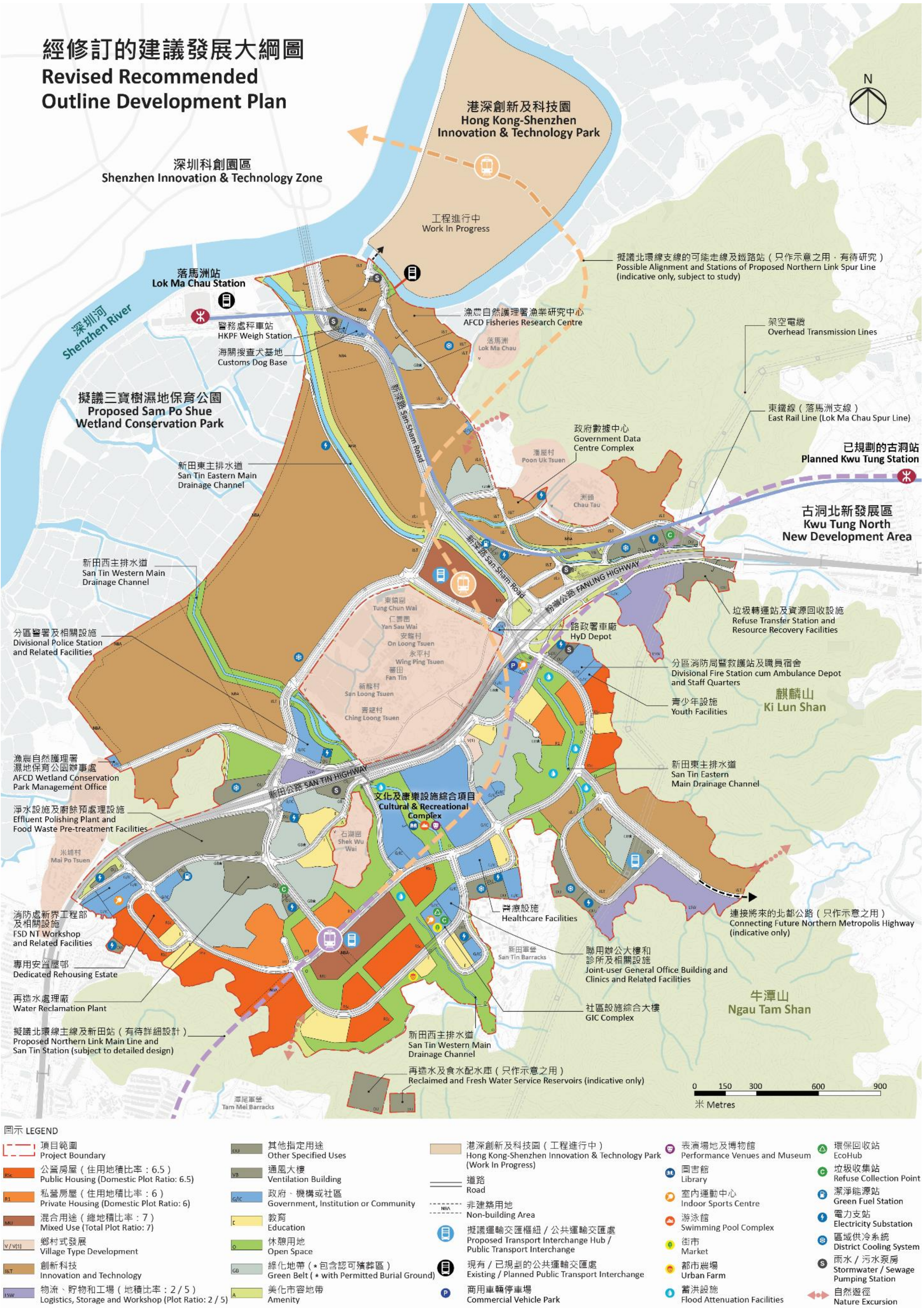


Fig. 4.4.10b. The Existing Recommended Development Plan.
Source: CEDD and PD (n.d.)

4.4.9. Spatial Implications

From the baseline study, areas with great cultural importance are identified in **Fig. 4.4.11.** Not only are IVs protected, but also NIVs that possess strong cultural continuity through festivals and rituals, and act as community-binding catalysts. Additionally, non-graded and non-recognised structures that represent the history of the study area or traditional knowledge are also preserved. However, those places which are no longer in use have to be revitalised through adaptive reuse. To achieve a holistic urban-rural integration, it is crucial to adopt a spatial design that respects the cultural assets, lifestyle and beliefs of the villagers and local residents and does not create a strong contrast between the urban and rural areas. It is equally important to ensure urban-rural integration in social and economic aspects. This can be achieved through policies, schemes and spatial design, which foster the interactions between new residents and talents, and villagers and local residents, and involve villagers and local residents in the new economic ecosystem. Finally, to enhance the quality of life for all users, high-quality services and facilities should be easily accessible within 15 minutes.

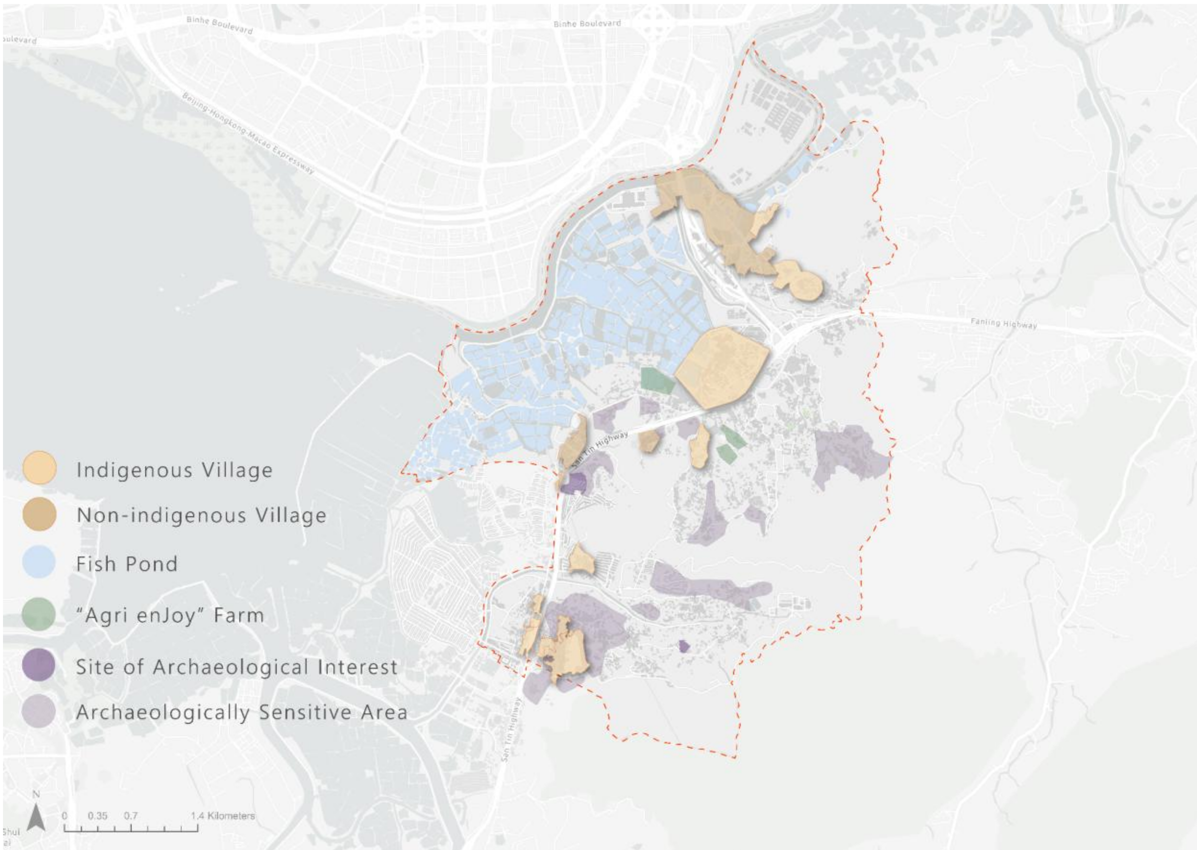


Fig. 4.4.11. Spatial Implications of Urban-Rural Integration after Baseline Study.

4.5. Talent

Talent is the core driving force for the development of STT. STT is a key I&T centre in the North Metropolitan. According to the 2025 Hong Kong I&T Talent Index (**Fig. 4.5.1**) (Hong Kong I&T Talent Index, n.d.), evaluate the baseline situation of Hong Kong talent-related dimensions. The review covers the talent ecosystem, talent effectiveness, talent resources, talent carriers and spatial impact. To determine the advantages, constraints and space needs of San Tin's talent development.

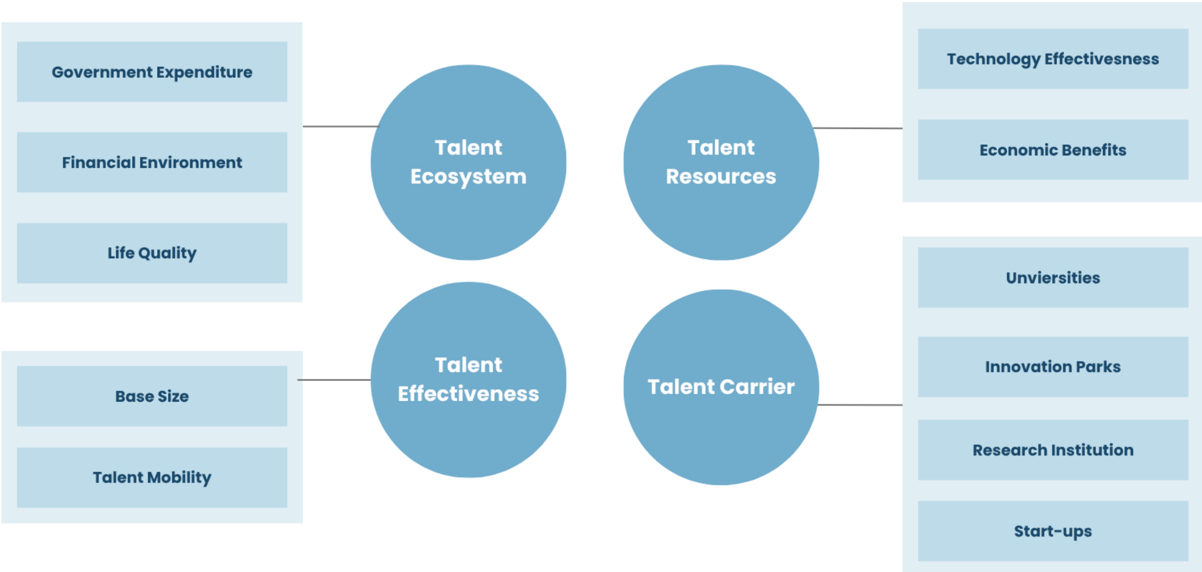


Fig. 4.5.1. I&T Talent Index Assessment

4.5.1. Talent Ecosystem

Hong Kong's I&T talent ecosystem has stable government investment, sound institutional support and a comprehensive urban living environment. However, the structural weaknesses of the cost of living and support facilities are still prominent. In terms of government investment, the total local R&D expenditure in 2023 rose to 3,300.6 billion Hong Kong dollars (**Fig.4.5.2**). The ratio of R&D to GDP reached 1.07% (**Fig. 4.5.4**). The budget of the innovation and technology fund in 2025/26 surged to 12.449 billion Hong Kong dollars. It has increased by 121% compared with 2023/24. Policy support covers multiple channels: the Top Talent Pass Program (TTPS) approved 41,057 people in 2024. It accounts for 30% of the total number of talent admissions; nine occupations will be added to the talent list in 2025. Including senior system architects and patent professionals. Be consistent with the development of high-value I&T. By 2025, the Office of Attracting Strategic Enterprises (OASES) will have introduced 84 key enterprises (**Table 4.5.1**). It is estimated that it will bring about 50 billion Hong Kong dollars in investment and more than 20,000 jobs (**Fig. 4.5.3**).

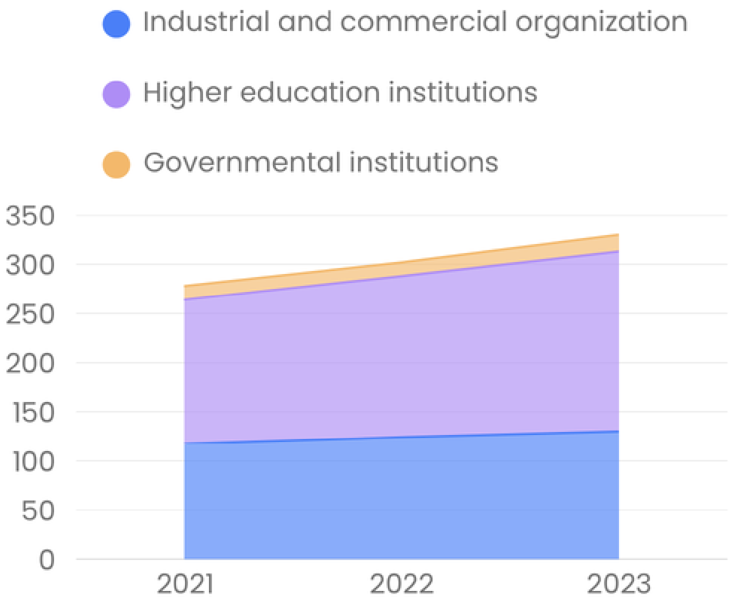


Fig.4.5.2. Total Expenditure of Local R&D in Hong Kong (Unit: Billion HKD)

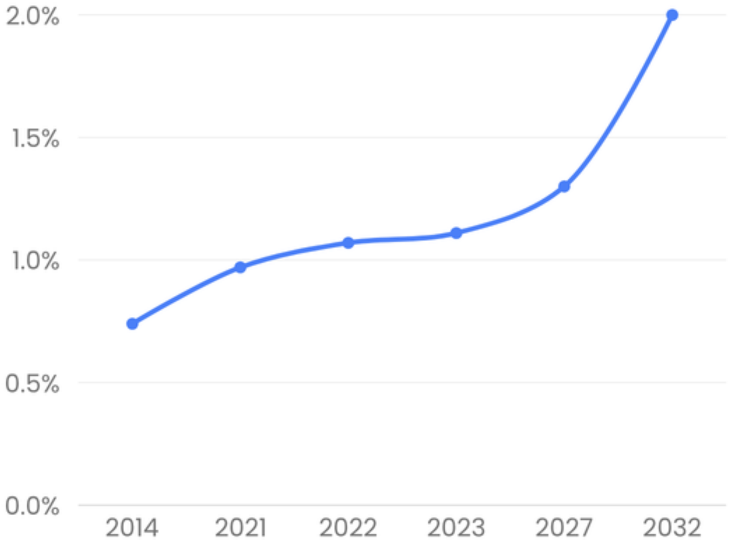


Fig.4.5.3. Ratio of total R&D expenditure to local GDP in Hong Kong

Table 4.5.1. Industrial sectors of OASES' key enterprises

Industry Category	Number Of Enterprises	Enterprise Ratio
Artificial Intelligence And Data Science	28	33%
Life And Health Technology	26	31%
Fintech	15	18%
Fintech	14	17%
Not Announced	1	1%
Total Number	6084	100%

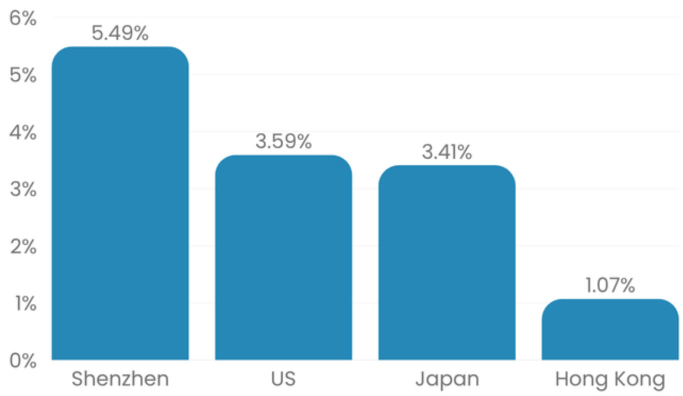


Fig. 4.5.4. The proportion of R&D investment to GDP in designated countries or regions in 2022

In terms of the economy, Hong Kong's capital market has maintained the resilience of I&T development. From 2018 to March 2025, 358 new economic companies will be listed on the Hong Kong Stock Exchange. It raised 1023.3 billion Hong Kong dollars. There are 139 health care and biotechnology enterprises, raising 280.2 billion Hong Kong dollars. Hong Kong Investment Co., Ltd. (HKIC) manages HK$262 billion to provide "patient capital" for the hard technology, biotechnology and new energy sectors. Urban living conditions provide a competitive advantage: the median monthly income of full-time employees in 2024 will reach 21,400 Hong Kong dollars (**Fig. 4.5.5**); there are 22 registered doctors per 10,000 people in the health care system; the per capita education expenditure will reach 10,676.63 Hong Kong dollars.

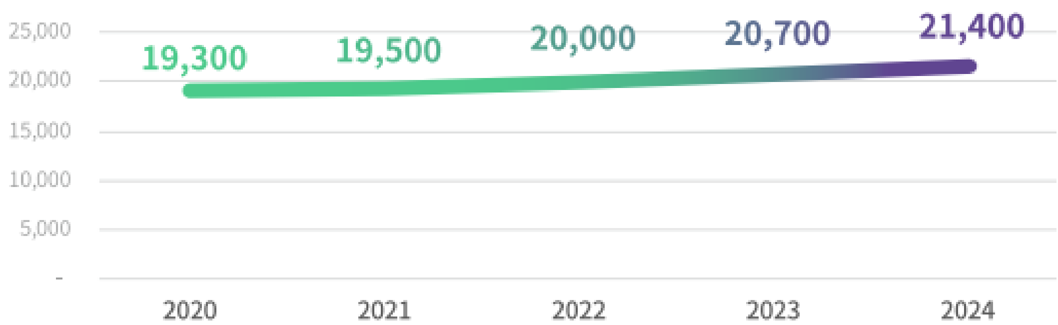


Fig. 4.5.5. Median income of full-time employees in Hong Kong from 2020 to 2024
Source: X Foundation (2025)

However, the high cost of living constitutes the core bottleneck. Hong Kong's house prices are the highest in the world (**Fig.4.5.7**). The ratio of house prices to income will drop to 29.1 in 2025. It still far exceeds most international metropolises. This index tracks the total land area of Hong Kong's I&T parks (**Fig. 4.5.6**). Sufficient land is critical for I&T and new industrialisation, but land shortage has long been a bottleneck. Existing parks are nearly saturated, so Hong Kong is expanding parks to attract more I&T enterprises and talent. Public service support facilities in the surrounding area are unevenly distributed. Cross-border talent support services lack an integrated spatial carrier. It weakens the long-term retention of international and GBA talents.

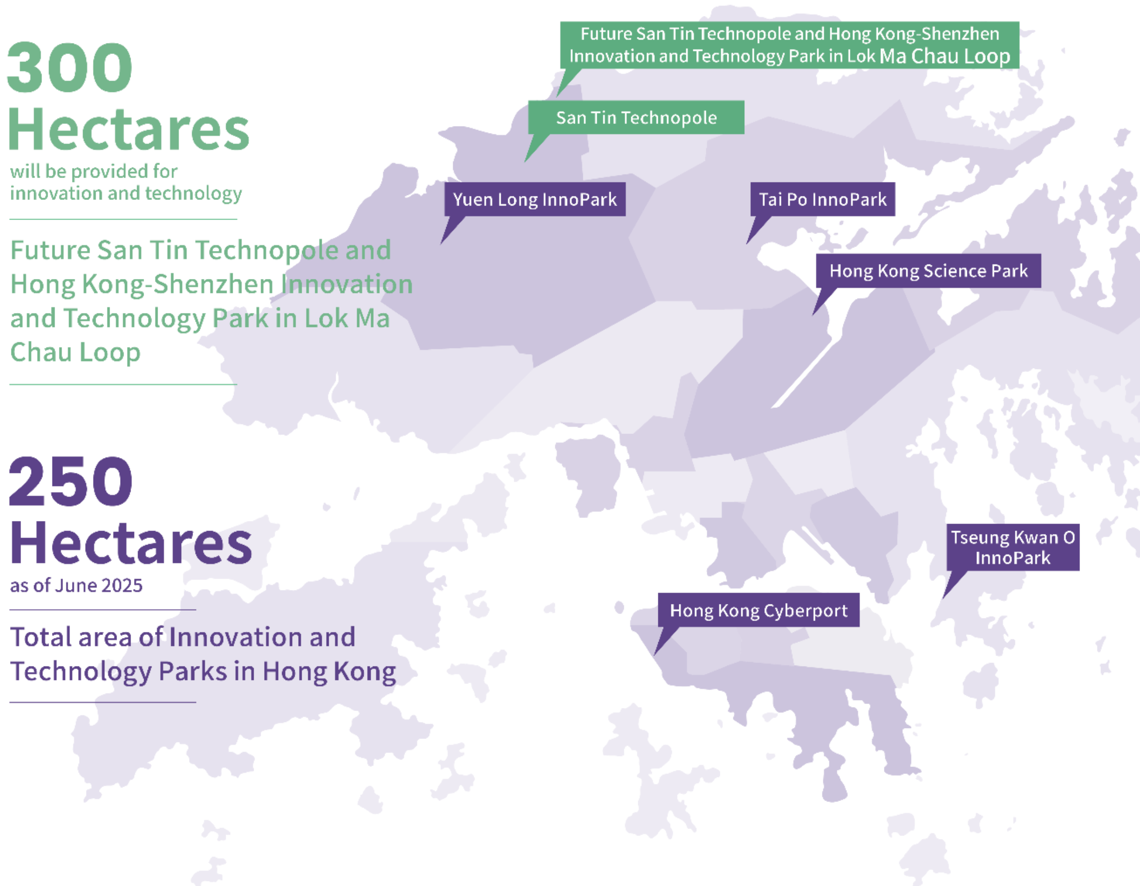


Fig. 4.5.6. Total geographical area of Hong Kong I&T Parks
Source: X Foundation (2025)

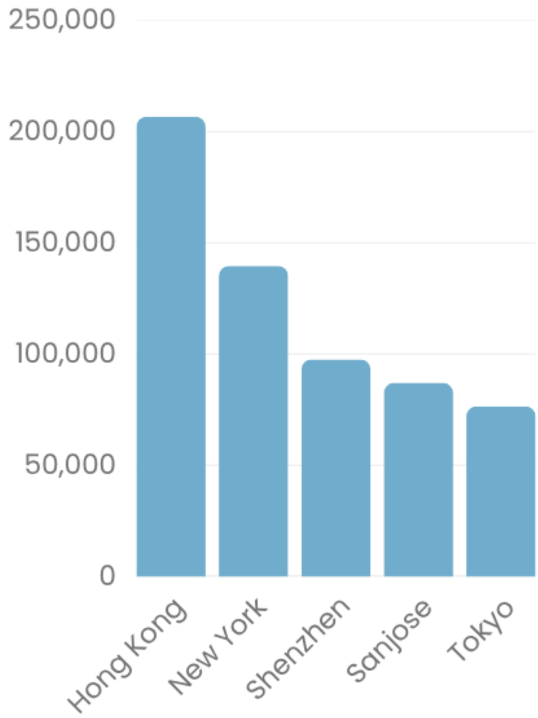


Fig. 4.5.7. Housing price in selected core cities in the four bay areas (Unit: HKD/square meter)

4.5.2. Talent Effectiveness

The effectiveness of I&T talents in Hong Kong shows its strong basic research ability. However, the efficiency of research on commercialisation and economic output conversion needs to be improved. In terms of scientific research output, universities and research institutions maintain high-quality, innovative performance. Five local universities were included in the QS World University Rankings in the top 100 of 2026. Disciplines such as computer science, data science and engineering rank in the top 50 in the world. In 2023/24, university patent applications reached 2,430, an increase of 27% year-on-year (**Fig. 4.5.8**); academic papers in the field of I&T accounted for 74% of the total output (**Fig.4.5.9**). Local scientists have won the National Science and Technology Award, the Future Science Award and the Geneva International Invention Exhibition Award.

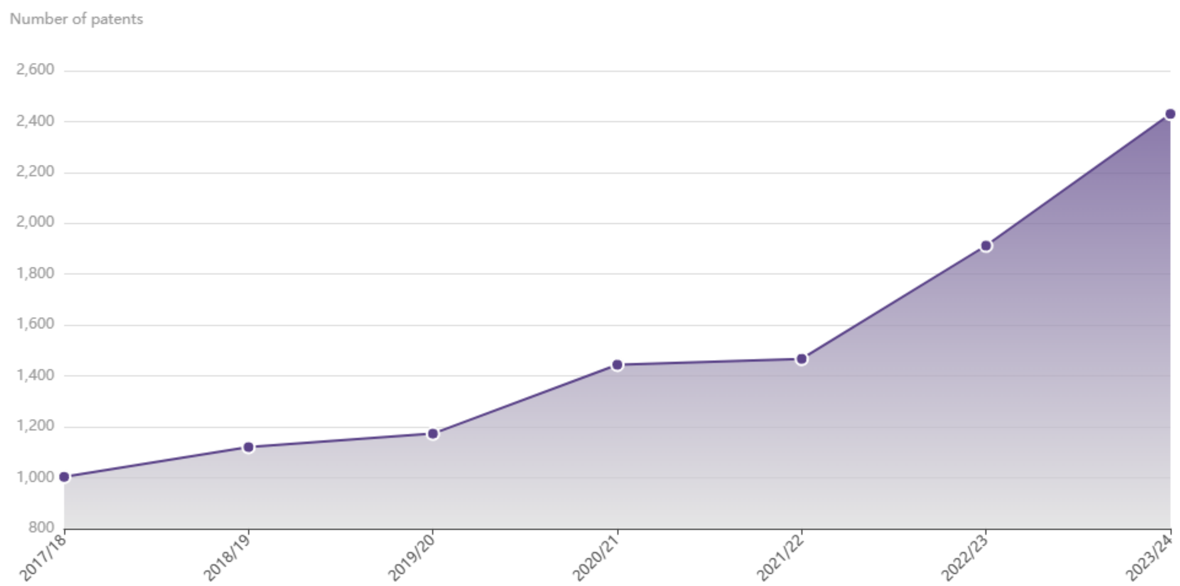


Fig.4.5.8. The change in the number of patent applications by Hong Kong's universities for academic years 2017/18–2023/24
Source: X Foundation (2025)

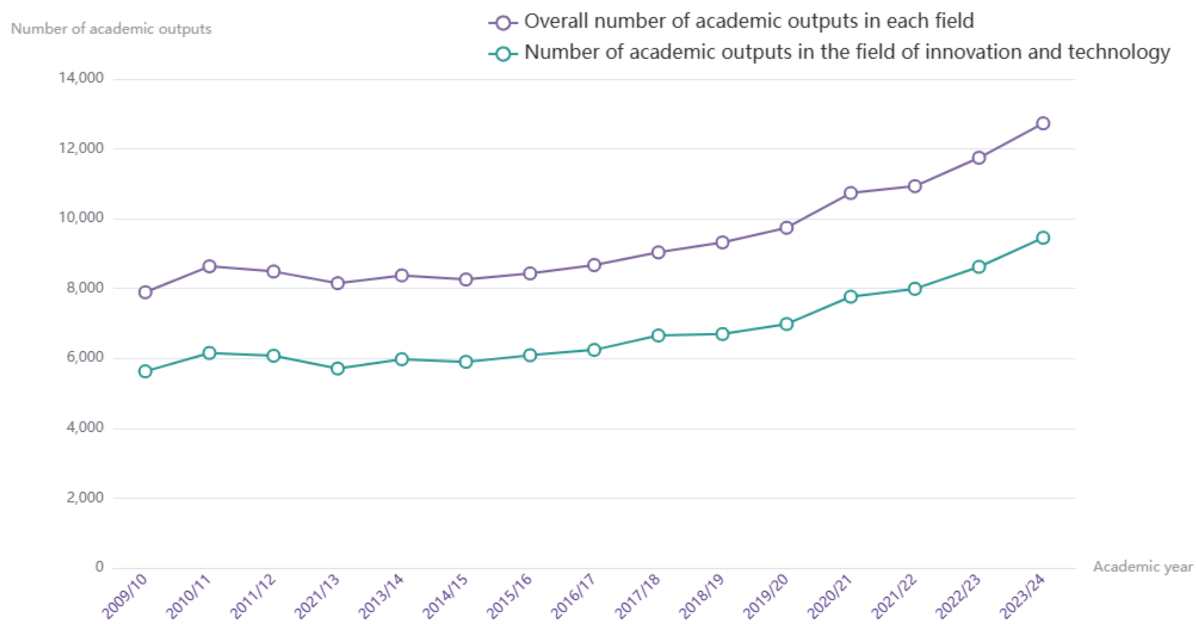


Fig. 4.5.9. Overall Academic Output of Hong Kong Universities from the 2009/10 to 2023/24 Academic Years
Source: X Foundation (2025)

In terms of knowledge transfer and commercialisation, the revenue of university cooperative research projects surged by 74% in 2023/24, reaching 1.912 billion Hong Kong dollars, and intellectual property income increased by 15% to 152 million Hong Kong dollars. With the support of research, the number of university spin-offs of start-ups increased by 8.5% year-on-year. Academic and Industrial Sector One Plus Program (RAISe+). Nevertheless, the gap still exists compared with world-class universities: the total intellectual property income of six local universities is less than one-fifth of that of Oxford University. The industrial chain from the laboratory to the market is incomplete. The commercialisation bottleneck of “1 to 100” limits the transformation of talent efficiency into industrial value.

The economic benefits of talents show a structural imbalance. In 2023, the number of employees in the I&T industry will reach 55,070, accounting for 1.5% of the total number of employees. The average annual growth rate is 5%. However, the average salary of I&T graduates lags behind that of business, medicine and law. It reduces the attractiveness of the I&T department to local young talent. Most talents are concentrated in research and development and research positions. There are not enough positions for pilot production, industrialisation and market application. It restricts the diversification of career paths for talents.

4.5.3. Talent Resources

Hong Kong's I&T talent resources show a trend of quantitative growth and structural upgrading. However, it is under the pressure of population ageing, the decline of the youth labour force and the shortage of structural talents. In terms of basic labour resources, the overall labour force is stable. However, the proportion of the young labour force (15-39 years old) has dropped from 45% in 2018 to 28% in 2024 (**Fig. 4.5.10**). The population aged 65 and above accounts for 23.9%. Enter the super-old society. This limits the long-term supply of I&T talents. The education level of the labour force is constantly improving. In 2024, the proportion of post-secondary education will reach 35.7%, laying a solid foundation for the supply of high-quality talents.

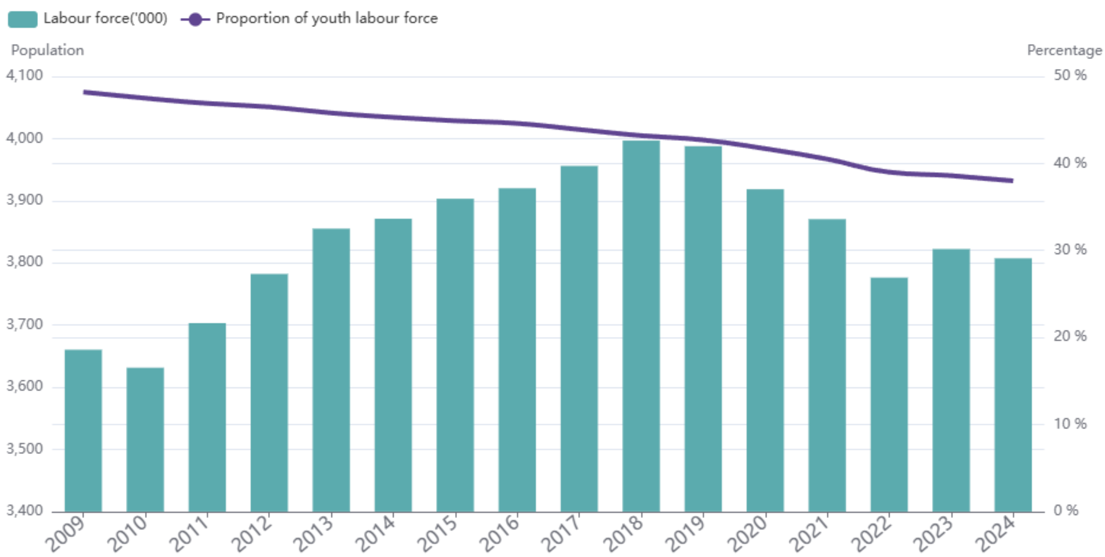


Fig. 4.5.10. Size of the labour force and the proportion of the youth labour force in Hong Kong
Source: X Foundation (2025)

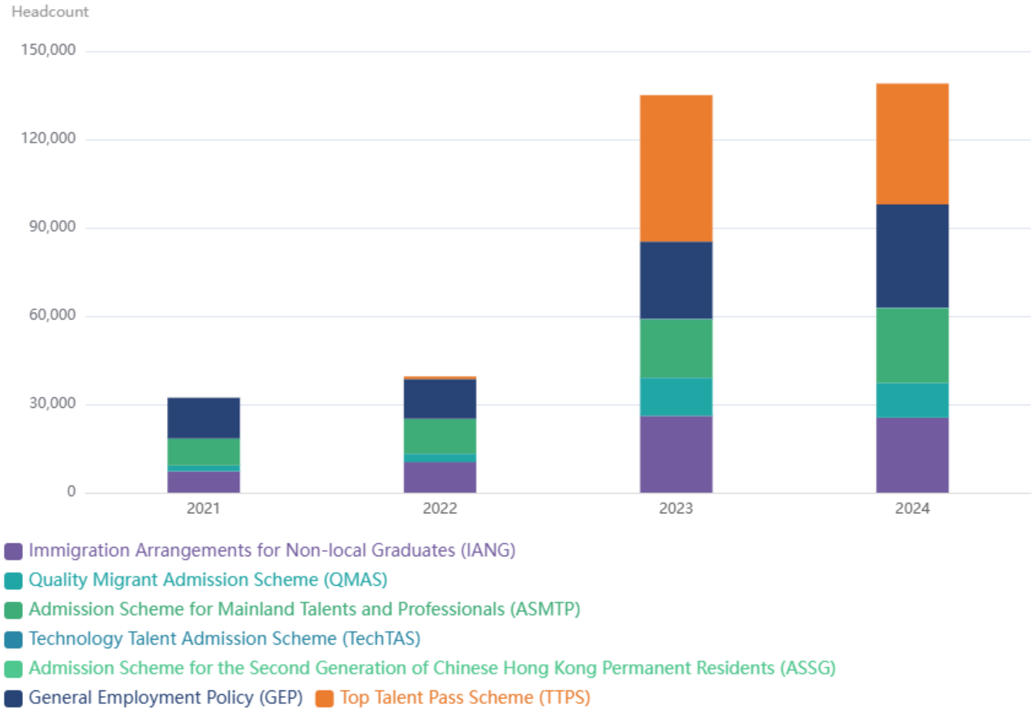


Fig. 4.5.11. Changes in the number of talent imports in Hong Kong
Source: X Foundation (2025)

The supply of talent has improved in both quantity and quality (**Fig. 4.5.11**). The number of R&D personnel increased from 32,355 in 2017 to 43,403 in 2023, of which the higher education sector accounted for 65%. In 2023/24, there were 62,688 students in I&T-related disciplines, an increase of 20% year-on-year. Including 34,814 undergraduates and 27,874 graduate students. Talent inflow remains at a high level: In 2024, 138,980 talents were admitted through various programs, an increase of 3% year-on-year; the number of mainland students coming to Hong Kong in 2024 reached 65,284, and the number of overseas students reached 9,182. It shows recovery and growth.

The shortage of structural talents is becoming more and more prominent. Manpower forecast predicts that by 2028, the demand for I&T talents will soar to 96,400 people, with a shortage of 18,000-23,000 people (**Fig. 4.5.12**). High-demand positions include project managers and data analysts. Artificial intelligence experts, microelectronics engineers and biotechnology experts. The talent source structure is relatively single: most imported talents and international students come from mainland China and the Asia-Pacific region. And the proportion of talents comes from Europe. The slow development of the United States and the Middle East is not conducive to building a diversified global talent centre.

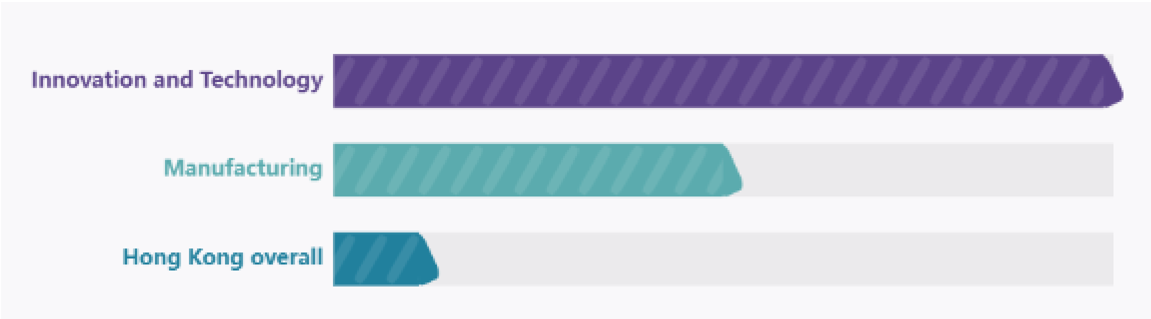


Fig. 4.5.12. Hong Kong projected average annual growth rate of manpower requirement 2023-2028
Source: X Foundation (2025)

4.5.4. Talent Carrier

Hong Kong's talent carrier is diversified and stratified. It covers universities, research platforms, science parks and enterprises. However, the spatial distribution is highly concentrated in the core urban areas. There are not enough carriers in the northern region. Universities are the core carriers for cultivating talents: the five top universities gather strategic scientists. Including 52 academicians of the Chinese Academy of Sciences and 89 overseas academicians. They provide interdisciplinary I&T projects and establish innovative colleges to cultivate interdisciplinary talents. More than 4,900 community public activities are held every year to promote the dissemination of science (**Fig. 4.5.13**).

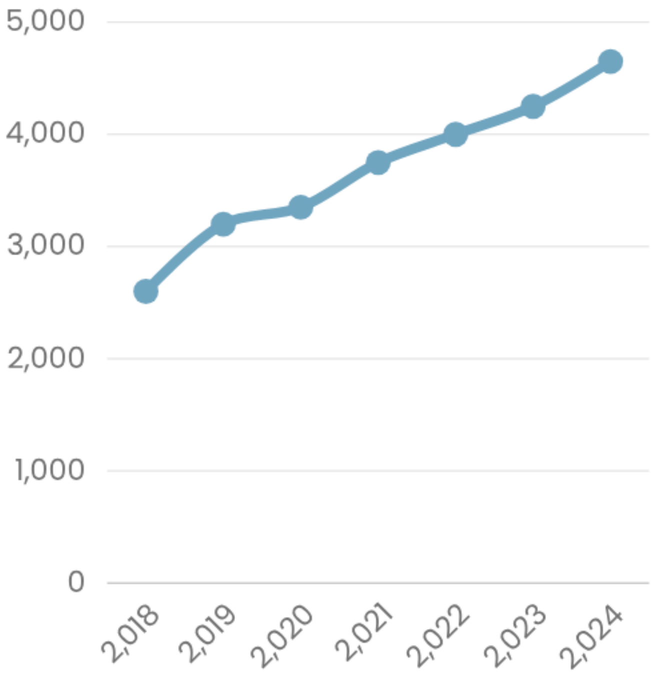


Fig. 4.5.13. Number of start-ups in Hong Kong

The research platform has formed a national-international cooperation network. InnoHK has 30 research laboratories, gathering 2,500 researchers from 12 economies. There are 15 national key laboratories in Hong Kong, 6 branches of the National Engineering Research Centre, and 22 joint laboratories with the Chinese Academy of Sciences, focusing on biomedicine, AI and advanced manufacturing. The Science and Technology Park provides full-cycle support (**Fig. 4.5.14**): Hong Kong Science Park has 22,000 people and 1,700 enterprises; Cyberport has 2,000 start-ups and 7,500 employees. The founders come from 26-27 countries and regions. The cumulative fundraising has exceeded 130 billion Hong Kong dollars.

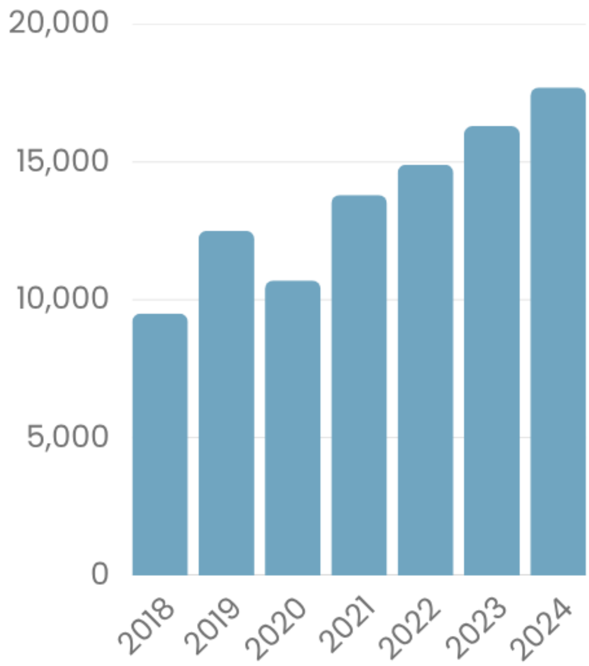


Fig. 4.5.14. Number of employees in Hong Kong start-ups

Enterprise operators promote the gathering of talents. In 2024, the number of local start-ups will reach 4,694. 17,651 people were employed. 28% of them are non-local founders. OASES has introduced key enterprises that focus on artificial intelligence, life and health and advanced manufacturing. It has become a new pole to attract talent. However, high-quality operators are mainly located in Kowloon and Hong Kong Island; there is a lack of large I&T parks, research centres, and enterprise clusters in the northern part of the New Territories, which makes it difficult to form a talent gathering effect.

4.5.5. Spatial Implications

San Tin's baseline space configuration puts forward obvious restrictions on the attraction and retention of talent. Cross-border mobility is only limited by two operational checkpoints. Failed to support GBA's daily talent flow and cross-regional cooperation. Innovative plots are scattered and insufficient. Lack of cluster layout required for comprehensive research and development, and entrepreneurial activities. The bus connection is very weak: a subway line serves the area. It is planned to add only four stations. Commuting to the core of the city and services are still inefficient. The connection in the city centre is insufficient. There is no highway or railway connection to Kowloon and Hong Kong Island. This separates the website from key institutional, commercial and cultural resources. These spatial gaps together hinder the gathering and retention of talents (**Fig. 4.5.15**). The key areas of intervention are defined.

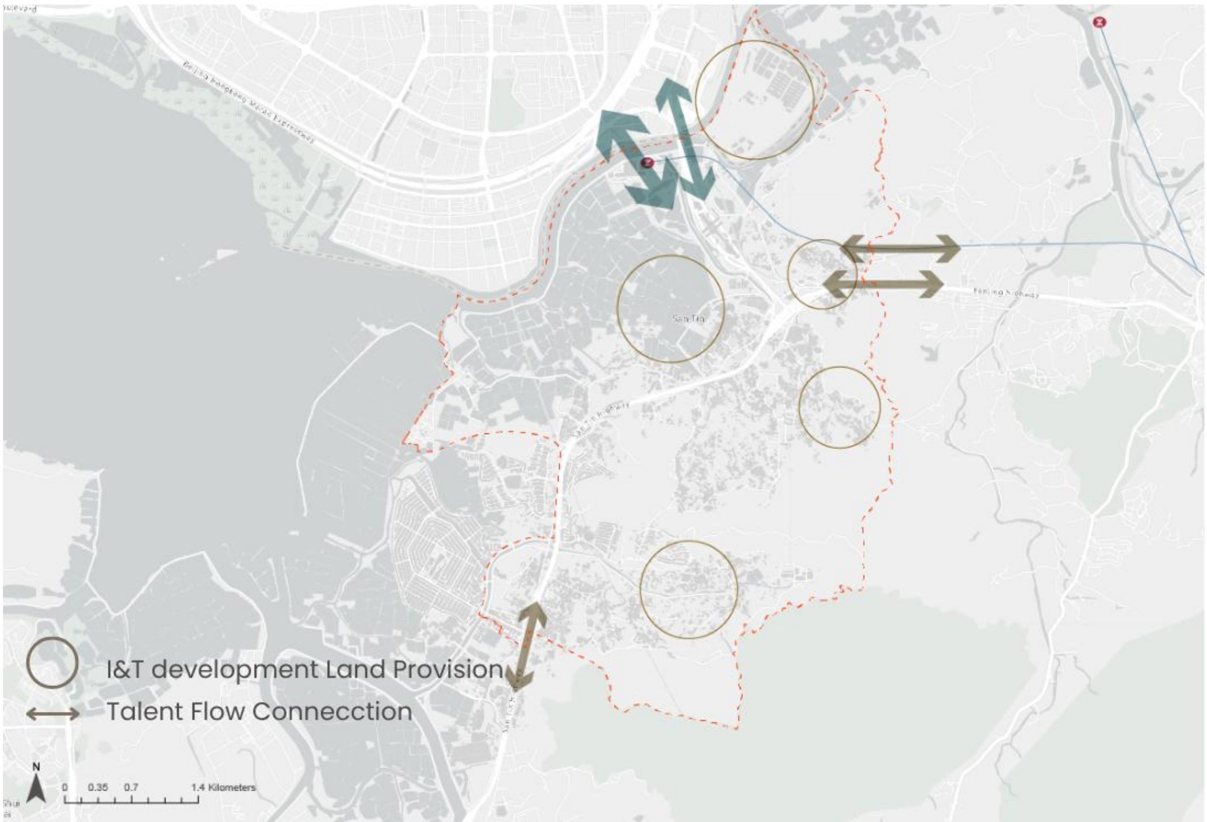


Fig. 4.5.15. Spatial Implications of San Tin in Talent Attraction

4.6. Research

Regarding the scaling up of research into real-world applications to generate economic value under our "1-100 Transformation" strategic planning concept, research currently shows a missing linkage in terms of connecting research output to the market production stage. As STT aims to be a one-stop innovation and production platform that starts from R&D, through pilot testing, and extends to mass production and market launch for commercial sales, research is a key component for knowledge production and technology transfer, which contributes to the final economic output. Based on the review of the existing approach and the spatial plans of our site, this section presents the current condition of Hong Kong's research landscape under a four-factor framework. It also assesses whether the current approved or drafted OZP in STT, NTM and the Loop have enough zoning and policy support to form a connected research-driven innovation system, especially in converting Hong Kong's academic strengths into economic value.

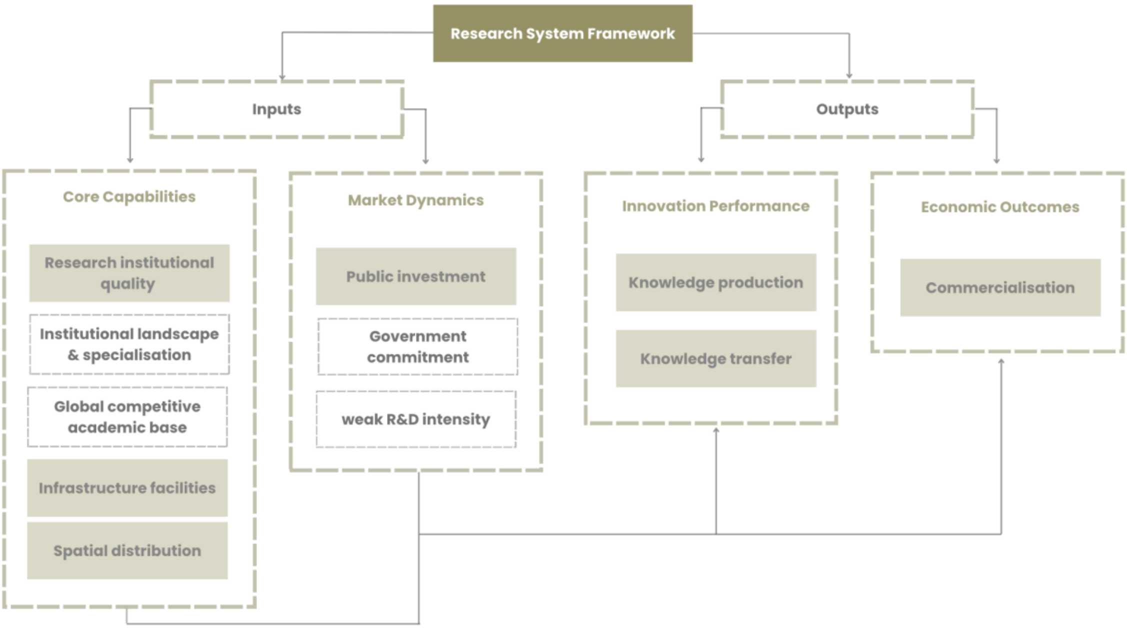


Fig. 4.6.1. Research System Framework
Source: Chatti et al. (2024)

Based on the innovation ecosystem framework proposed by Statistics Canada, a research system framework (**Fig. 4.6.1**) with an input-output structure was developed (Chatti et al., 2024). Under this framework, there are four main indicators categorised as inputs and outputs, which are core capabilities, market dynamics, innovation performance and economic outcomes. This framework focuses on analysing institutional quality and public investment while highlighting the relationship between research production, knowledge transfer and their translation into economic outcomes. Together, these indicators provide an analytical framework for evaluating the competitiveness of the research system across the local, regional and global levels. Understanding its strengths and gaps helps identify opportunities and positions STT more effectively.

4.6.1. Core Capabilities

Research institutional quality

Hong Kong's tertiary education is highly competitive internationally, with a record of strong research output. Not only does this city have several globally recognised universities, but it also has five institutions ranking in the top 100 worldwide (The Standard, 2025). Within these groups, HKU, CUHK and HKUST have consistently stayed within the top 50 over the past decade (The Standard, 2025). The research strength is clear in fields such as biomedical science and engineering. These areas allow Hong Kong institutions to produce high-impact work and attract international collaborations, thus consolidating Hong Kong's role as a knowledge output centre.

At the regional level, Hong Kong controls the major knowledge flows and supports cross-border research activities. As shown in **Fig. 4.6.2**, Hong Kong research institutes play a key intermediary role within the Hong Kong-Macao-GBA research network, with a high betweenness centrality score that is 15 times that of Guangzhou and Shenzhen (Shen et al., 2025).

Indicators	Phase	Indicators explanation	Hong Kong	Macao	Guang zhou	Shen zhen	Fo shan	Hui zhou	Zhu hai	Dong guan	Zhong shan	Zhao qing	Jiang men
Degree centrality	Phase I	Degree	217	10	29	132	0	8	1	43	2	1	11
	Phase II	Degree	308	57	76	183	4	7	25	38	4	2	10
Betweenness centrality	Phase I	Betweenness	31.5	1.5	2	2	0	0	0	2	0	0	0
	Phase II	Betweenness	33	12	0	0	0	0	0	0	0	0	0
Closeness centrality	Phase I	Farness	21	31	27	27	0	29	29	27	29	29	29
	Phase II	Farness	11	14	18	18	20	20	18	18·	13	20	20

Fig. 4.6.2. Node-level characteristics of patent innovation cooperation network in the Guangdong-Hong Kong-Macao GBA
Source: Shen et al. (2025)

Infrastructure facilities

Hong Kong has an advanced and reputable research infrastructure base that is provided by various universities and technology parks. For instance, CUHK has centralised research facilities with specialised equipment, supported by a HK$50 million annual Academic Equipment Grant (CUHK Research Office, 2025). HKSTP's Biomedical Technology Support Centre provides a 22000 square ft. laboratory, while HKU InnoHK clusters have nine research laboratories (HKSTP, n.d.; HKU, 2021). These facilities show that Hong Kong has cutting-edge instruments and plenty of space for laboratories, showing a big effort and commitment of different research parties to advance research development. A solid foundation has been built for research activity in this city, while connecting them to the STT research system will be the challenge.

In the latest approved NTM OZP, the University Town, Third Medical School and Integrated Hospital are confirmed, ensuring the provision of academic and medical research infrastructures (Planning Department, 2025a). This strengthens not only training and research but also brings the life and health technology industry into STT. Besides, the existing Planning and Design Brief for Sites Zoned "OU(I&T)" on the STT OZP stated that STT will provide about 300 ha of I&T land, including 87 ha at HSITP in the Loop with 7 million sq.m. GFA in total, equivalent to 17 HKSTP (Planning Department, 2025b). Supporting infrastructures, like data centres and computing facilities, are explicitly allowed to ensure the smooth I&T hub development. This proposed research landscape shows a strong potential for our site to develop its own research infrastructure base.

Spatial distribution

Although Hong Kong has strong and high-quality research institutions, this research base is not fully embedded in STT yet. A spatial mismatch is identified. As shown in **Fig. 4.6.3**, most existing universities and tertiary institutes are located in the urban core area. The proposed new university town in NTM is far from this existing research cluster, which may result in a disconnect within the collaboration network. However, from a more macro perspective, the new university town is situated in a strategic position between the Hong Kong and Shenzhen research clusters. Hong Kong has well-built institutional quality at the city scale, while the research ecosystem of our site still needs to be strengthened at the site level, although its regional position gives it a clear potential to connect with both sides.

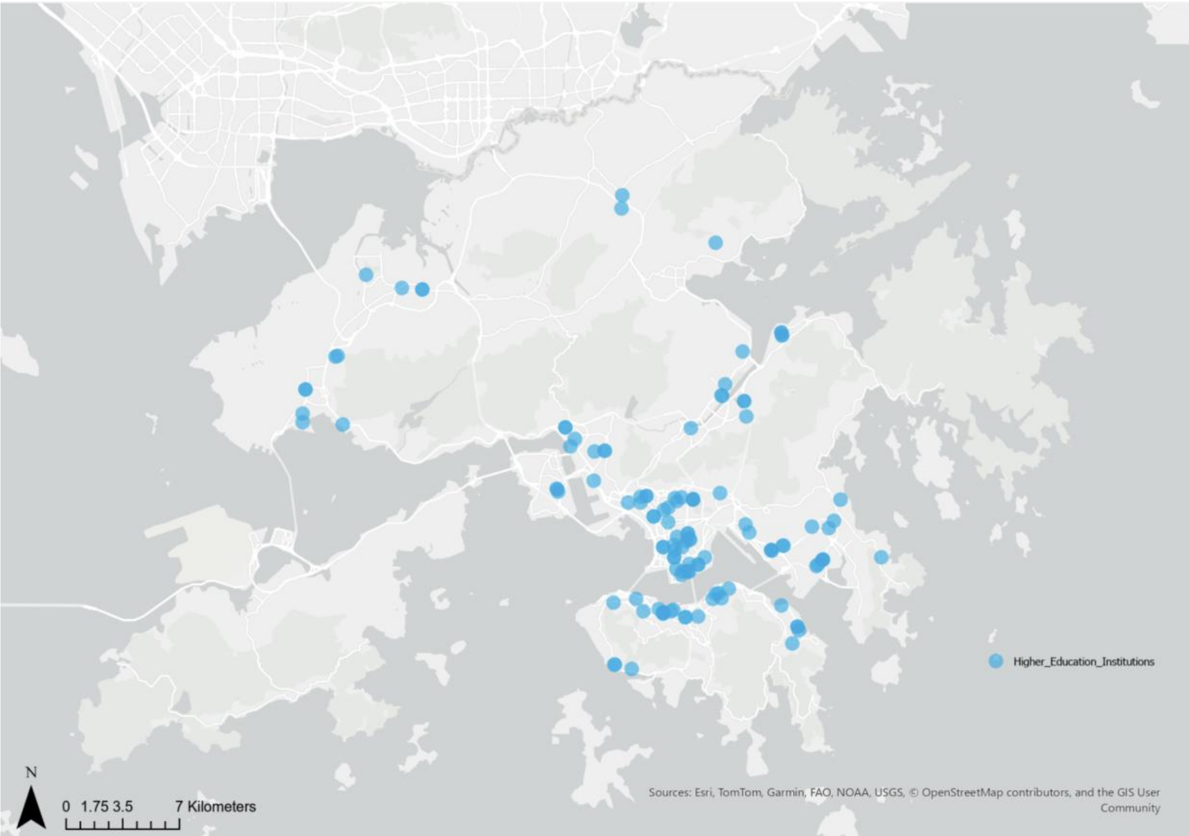


Fig. 4.6.3. Distribution of Research Institute

4.6.2. Market Dynamics

Public investment

The Hong Kong government introduces a range of policies, initiatives and funding measures to support the I&T sector. One example is the Research, Academic and Industry Sectors One-plus Scheme, known as RAISe+. The scheme helps university research teams turn research outputs into real products, thus transferring R&D outcomes closer to market use. It provides funding up to HK$ 100 million for each approved project and encourages start-up development through closer collaboration with government and industry partners (Innovation and Technology Commission, 2023). Investment in the research field also shows a continued growth. The domestic R&D spending reached 35.77 billion in 2024, accounting for approximately 1.13% of GDP, and this marked an 8.4% rise from the year before (Xinhua, 2026). Development plans for the technology-intensive Hetao I&T Zone also set a continuous research direction with continued investment in research, innovation and cross-border technology integration (Xinhua, 2026). These measures show the government's commitment and are putting more resources into the research field, thus showing support for its future growth. However, Hong Kong's R&D intensity is still relatively weak compared to advanced economies such as Germany and Japan, which both have more than 3% of GDP in R&D investment (Bonaglia et al., 2024; **Fig. 4.6.4**). Still, the steady growth in R&D creates opportunities for STT to strategically scale up its research activity and convert public investment into higher economic value.

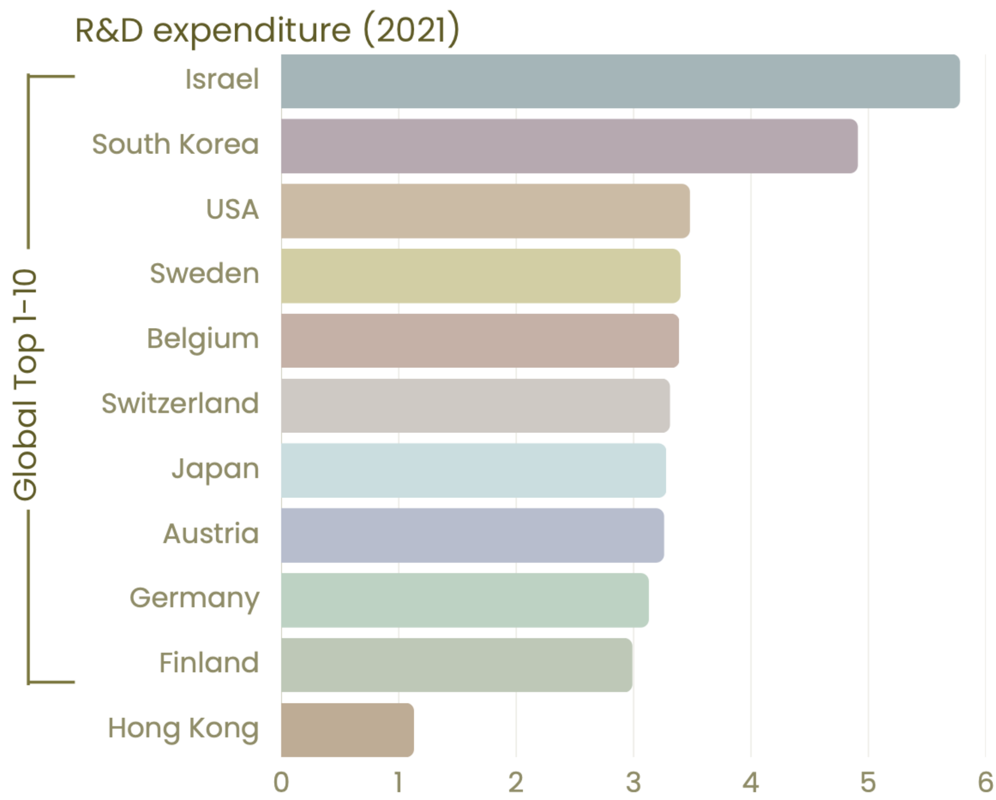


Fig. 4.6.4. R&D expenditure in 2021
Source: The Global Economy (2023); The United Nations (n.d.)

Land-use flexibility and market responsiveness

The current approved and drafted OZP of our site provides a statutory planning foundation for research uses. STT's GIC zone allows 'Research, Design and Development Centre' as Column 1 use, meaning that no additional planning permission is needed (Planning Department, 2025b). From **Fig. 4.6.5**, GIC use is spread across the site of STT and covers a considerable amount of land. In the existing OZP of the Loop, educational institutions are listed as Column 1 uses under Commercial and GIC zones and it has an "Other Specific Uses" zone specifically for "Research and Development, Education, and Cultural and Creative Industries" where educational institutions are also listed in Column 1, making the Loop research friendly with higher education place as a core function (Planning Department, 2018). The current OZP provides a foundation for flexible education-related development within the research ecosystem.

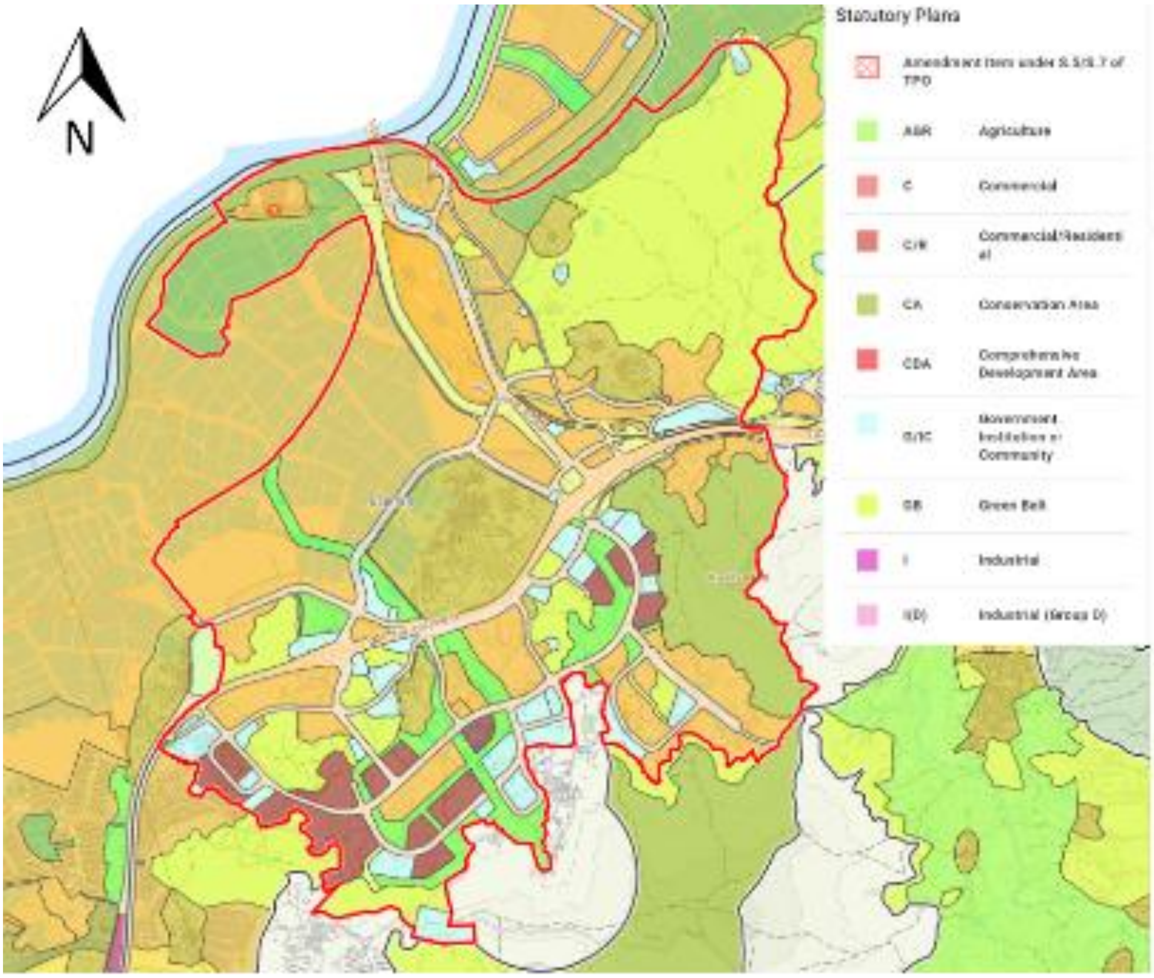


Fig. 4.6.5. Approved OZP No. S/STT/2 of STT

The planned University Town in NTM, which is located one station away from STT, is also expected to complement and support the I&T developments at STT (Planning Department, 2025b). However, its integration with STT might be limited due to spatial separation. Without a comprehensive consideration of spatial integration between research institutions and industry spaces, knowledge exchange may be weakened.

4.6.3. Innovation Performance

Knowledge production

Hong Kong has a strong performance in knowledge production. Scientific publications and patent applications have grown consistently at around 4.7% per year, outperforming the global average (FHKI & HKPC, 2023). Most of this research output comes from educational institutions. The University of Hong Kong, for example, produces about 6700 publications alone each year and has generated more than 1400 patents since 1998 (Nature Index, 2025). Besides, this strength is also visible at the regional scale. GBA contributes about one-sixth of China's total patents, with Hong Kong Universities taking the lead, as shown in **Fig. 4.6.6**. At the global level, Hong Kong ranked 24th as a source of international invention patent applications between 2016 and 2020. This shows Hong Kong has a strong competitiveness in research results and reinforces its role in academic institutions in driving innovation.

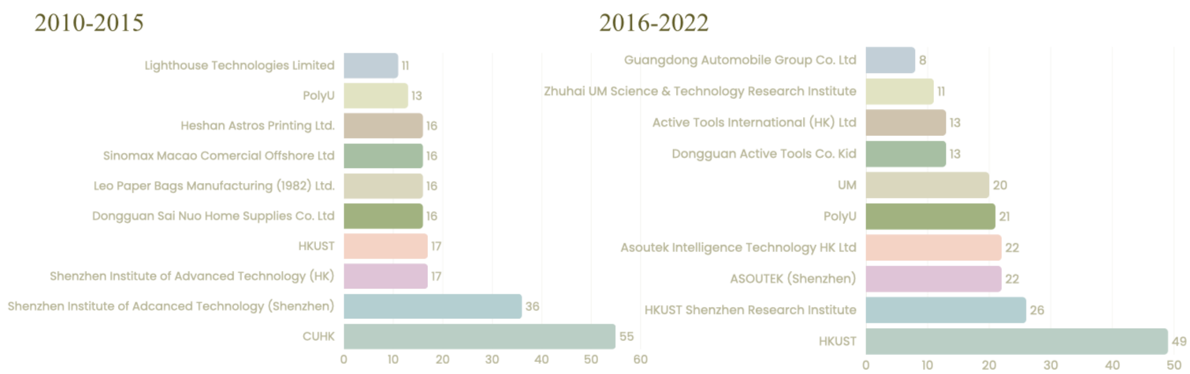


Fig.4.6.6. Core patent applicants of the Guangdong-HK-Macau GBA (2010-2022)
Source: Shen et al. (2025)

Knowledge transfer

Meanwhile, knowledge transfer remains limited in impact. Research activity in Hong Kong universities is strong in volume, but it circulates and stays within the local system. There is little high-value transfer, especially in terms of market use and cross-border spread. For instance, the IP utilisation rate at HKUST is approximately 31%, meaning that more than two-thirds of its research outputs are not applied in practice (HKUST, 2025). This research is academically focused, while not driven by practical application. This kind of lecture-based and publication-driven research is hard to commodify and transfer to marketable products. Under this context and with the support of Shenzhen's strong industrial base to enable production and scaling, STT will be serving as a knowledge transfer platform that supports real application and enhances IP utilisation.

4.6.4. Economic Outcomes

Commercialization

As aforementioned, research activity in Hong Kong universities is very strong in volume; however, most of this circulates remain in the local research system, rather than extending to commercial settings or real market applications. Despite the strong performance of Hong Kong universities in IP and patent production globally, the low IP income generation reveals a commercialisation gap within the research system. As shown in **Fig. 4.6.7**, the University of Hong Kong and the University of Cambridge have a similar number of PCT applications, with HKU only one application behind Cambridge. However, in terms of IP income generation, the University of Cambridge generates around 4 times more IP income than HKU. HKU creates HK$ 0.81 million in IP income per PCT application, CUHK creates HK$3.69 million, and HKUST creates HK$ 1.08 million (LegCo, 2022; Times Higher Education, 2022). By comparison with the top universities, the University of Oxford generates around HK$ 7.85 million while Stanford University generates HK$ 4.7 million per PCT application, showing that globally leading universities tend to achieve strong commercialisation outcomes from their IP too.

Global ranking	University	Number of "Top 2%" scientists	PCT applications	IP income (HK$ million)	Active Spin-offs
1	University of Oxford	904	81	635.8	168
4	Stanford University	1233	194	926.2	-
5	University of Cambridge	599	37	125.7	182
30	University of Hong Kong	272	37	29.0	34
35	The University of Tokyo	458	150	368.7	-
46	Nanyang Technological University, Singapore	105	54	29.0	12
49	Chinese University of Hong Kong	204	17	62.8	10
66	The Hong Kong University of Science and Technology	156	11	11.9	128
351-400	Shenzhen	42	142	-	-

Fig.4.6.7. Research commercialisation performance of selected universities, 2021-2022
Source: LegCo (2022); Times Higher Education (2022)

4.6.5. General Spatial Implications

From the current plans, research-related land exists and is supported by the statutory OZP; however, the roles of different areas remain relatively separated, with STT focusing on applied I&T uses, NTM positioning as a university and talent area and the Loop focusing on the research core. The current research system is strong within each sector, but it may not yet support a well-connected research network that generates higher economic returns. The site-level spatial allocation of education institutes and industrial clusters has to be improved to boost stronger research collaboration and smoother knowledge transfer.

4.7. Industry

4.7.1. Assessment Framework

In order to conduct an all-rounded analysis of the industry of the study area, the initial framework is enhanced through a more regionally grounded and policy-oriented assessment framework. The new framework integrates concepts derived from different streams of literature, namely: Factor-Based Competitiveness Theories (Porter, 1990); Value Chain and Production Networks (Gereffi, 1994; Yeung, 2002), as well as Strategic Industrial Policy and Smart Specialisation Frameworks (Rodrik, 2004; OECD, 2013). Given the above, the industry assessment framework is organised into four major dimensions **(Fig. 4.7.1)**: Factor Conditions, Value Chain Positioning, Policy Alignment, and Market Potential. From feasibility to positioning, and from policy support to economic return, this framework forms a systematic basis for selecting industries in line with Hong Kong's position in the GBA.

Fig. 4.7.1. Industry Assessment Framework

4.7.2. Factor Conditions

Firstly, this aspect looks into whether Hong Kong is capable of supporting an I&T industry system with the necessary cost structure, resources, and abilities, which include land, labour, infrastructure, and institutional power.

<u>Land Costs</u>

Based on the comparison of industrial land cost among the GBA cities, it can be seen that Hong Kong is identified as a city with high industrial land cost, where the average industrial rent is nearly close to HK$ 200 per square meter **(Fig. 4.7.2)**. The cost per square meter of land is much higher in comparison to other nearby cities such as Macau (1.7 times higher at HK$ 117.5/m²) and Shenzhen (3.5 times higher at HK$ 56.5/m²) (CBRE, 2024). In comparison to other major manufacturing cities within the region **(Table 4.7.1)**, the difference becomes more prominent as Hong Kong's industrial land cost is six times higher in comparison to Dongguan (HK$ 32.5/m²) and almost 14 times higher than Zhaoqing (HK$ 14/m²) (CBRE, 2024). As such, there exists a distinct spatial gradient in terms of the cost of industrial land, which emanates from Hong Kong outwardly. This cost factor is a hurdle for large-scale, land-intensive manufacturing operations, which places Hong Kong in a viable position for value-added operations instead.

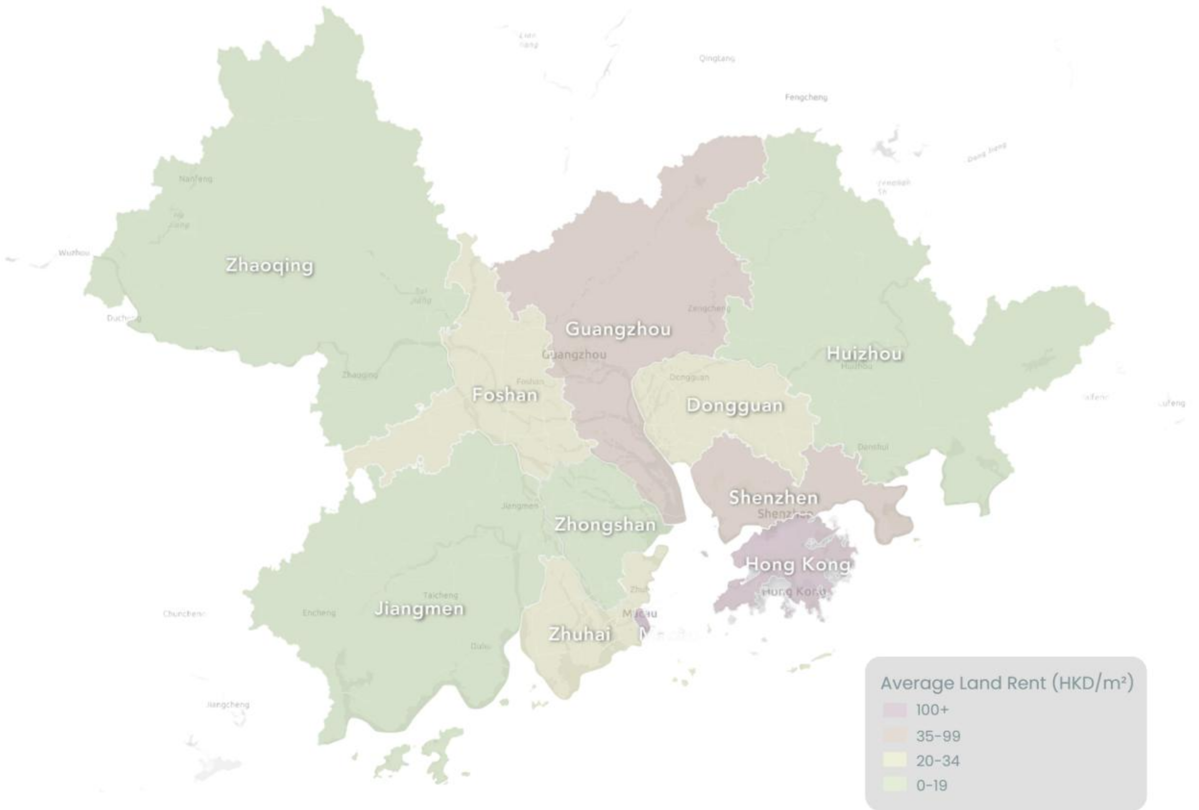


Fig. 4.7.2. Average Land Rent Across the GBA

Table 4.7.1. Average Industrial Rent Across the GBA (HK$/m²)

City	Average Rent (HKD/m²)
Hong Kong	199
Macao	117.5
Shenzhen	56.5
Guangzhou	45
Dongguan	32.5
Foshan	27.5
Zhuhai	24
Huizhou	19
Zhongshan	18.5
Jiangmen	16
Zhaoqing	14

Labour Costs

In addition to land costs, labour costs in Hong Kong are also higher than those of its counterparts within the GBA. In terms of the average monthly salary of employees in Hong Kong, it is HK$18,629, while that of Shenzhen is roughly RMB8,648 (HK$9,912) (Census and Statistics Department, 2024; Shenzhen Municipal Bureau of Statistics, 2024). The difference in costs adds another layer of constraint, as described above regarding land costs. The composition of factor costs encourages industries to move to knowledge-intensive processes.

Innovation and Production Capacity

As per **Fig. 4.7.3** below, Hong Kong's innovation and production capability are spread through a series of specialised nodes. To elaborate, there are three major hubs for I&T, namely, Hong Kong Science and Technology Parks (HKSTP), Cyberport, and the future-to-come Hong Kong-Shenzhen Innovation and Technology Park (HSITP) in Hetao; three InnoParks (Yuen Long, Tai Po, Tseung Kwan O); and several universities undertaking scientific research.

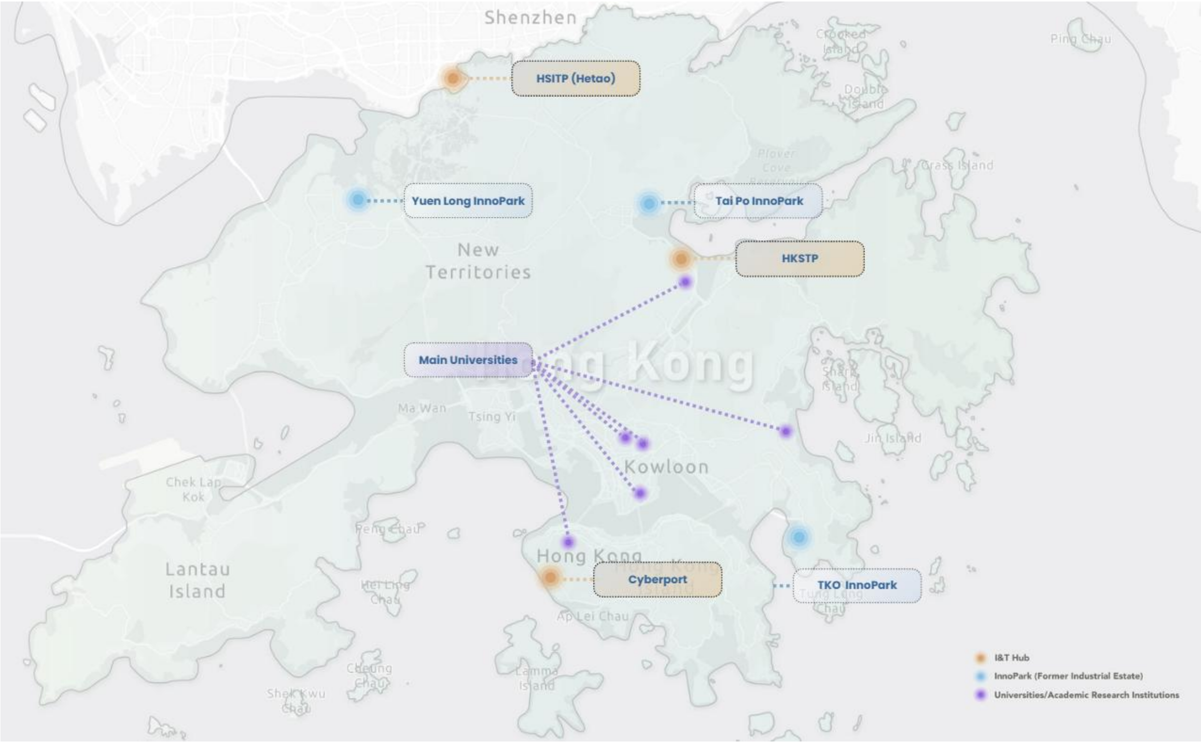


Fig. 4.7.3. Distribution of Hong Kong's I&T Nodes

When analysing the functionality of these I&T clusters, it becomes apparent that there is a concentration of activity towards the upstream of the value chain **(Fig. 4.7.4)**. In this regard, the HKSTP consists mostly of laboratory R&D (basic research plus applied R&D), which accounts for 92% of all activity, while pilot production is limited to just 8% (HKSTP Corporation, 2024). On the other hand, the three InnoParks consist mainly of traditional manufacturing operations that have no link to the overall I&T environment. In this context, the Yuen Long InnoPark is primarily made up of traditional manufacturing, processing, and general industrial uses that account for about 93%, with the remaining 5% relating to pilot production connected to I&T operations, facilitated by the Microelectronics Centre (MEC) pilot facility (HKSTP Corporation, 2024).

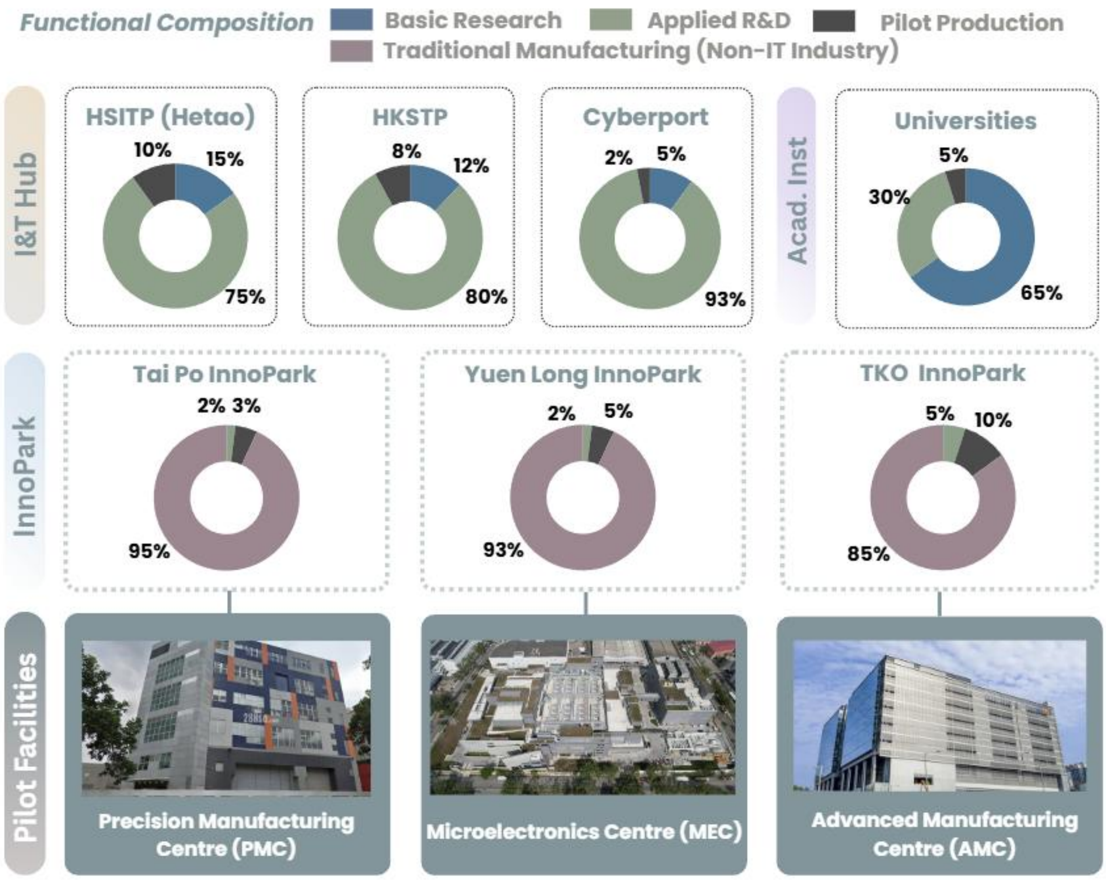


Fig. 4.7.4. Functional Composition & Value Chain Activity of HK's I&T Clusters
There is therefore a void in the middle part of the process chain, between R&D laboratory research and commercial production, in which pilot manufacturing is lacking.

<u>Institutional Strengths</u>

With respect to comparative advantage, Hong Kong's institutional advantages are the "software" that runs the region's "hardware". To explain, Hong Kong is a major centre for capital structuring and venture capital (VC). It acts as the principal initial public offering (IPO) gateway and a VC centre for GBA tech companies. It is a key reason the Shenzhen-Hong Kong-Guangzhou cluster secures the number one spot in the 2025 Global Innovation Index (World Intellectual Property Organisation, 2025). The volume of VC deals is high, being estimated at 135 deals per one million people. Secondly, it offers a robust IP protection and licensing regime within the framework of the common law system, which acts as an international patent filing gateway. Thirdly, it provides strong legal arbitration and contract enforcement services, with the Hong Kong International Arbitration Centre (HKIAC) providing common law contract certainty.

A good example of the synergy between institutional "software" and industrial "hardware" is the Kendall Square model, found in Massachusetts, the United States. The Kendall Square cluster includes over 50 venture capitalist firms in proximity to laboratories, dense law firms specialised in technologies, as well as mixed-use buildings where investors and technologists can co-locate (MIT Innovation Initiative, 2019). Being ready for integration, Hong Kong has over HK$1.5 trillion in investment capital, receives over 30,000 patent submissions per year, and handles over 500 HKIAC disputes each year (Hong Kong Monetary Authority, 2024; Hong Kong International Arbitration Centre, 2024). These turn themselves into a “safe harbour” for businesses based in the GBA area. The combination of such hardware and software creates a “de-risking zone” for global technology firms.

4.7.3. Value Chain Positioning

Secondly, this dimension investigates how the industry sits within the production structure of the region, especially the position of Hong Kong in the upstream, midstream, and downstream stages of production in the GBA.

<u>Regional Innovation Geography</u>

From an examination of the innovation geography in the region, it is found that a complementary labour division exists among GBA cities **(Fig. 4.7.5)**. In this regard, Hong Kong is highly influential in the upstream stage of the value chain through basic and applied R&D activities. Shenzhen possesses remarkable capability in R&D and high-tech manufacturing activities (World Intellectual Property Organisation, 2025). Dongguan stands out in its ability to manufacture electronics machinery and industrial robots, hence creating a robust industrial robot value chain from raw materials to systems (Dongguan Municipal Bureau of Industry and Information Technology, 2023). Foshan excels in producing intelligent equipment for the light industry and metalworking machines with specialisation in industrial robot applications and intelligent manufacturing transformation (Foshan Municipal Government, 2023). Meanwhile, Jiangmen and other western cities are engaged in the downstream process and large-scale manufacturing (Guangdong Provincial Government, 2023). Such a geographical layout forms a fairly complete industry loop within a range of 100 kilometres, which provides a unique advantage for cross-boundary cooperation to create a natural market for the commercialisation of innovative products generated upstream in Hong Kong.

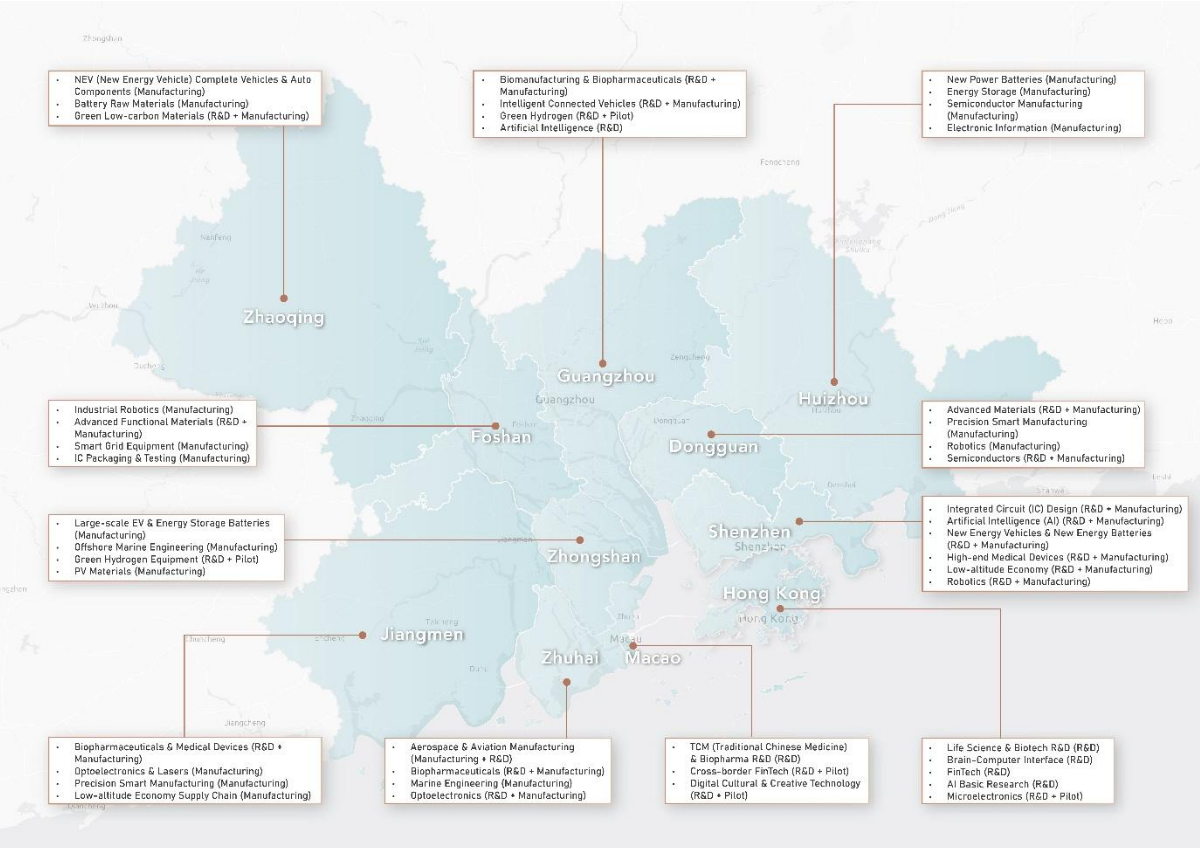


Fig. 4.7.5: Regional Division of Labour: I&T Research and Advanced Manufacturing Specialisations in the GBA

Cross-Boundary Integration

Importantly, a seamless flow of goods, capital, talent, and data is instrumental to a functional integration of the GBA.

Flow of Goods

To start, the GBA possesses an effortless flow of goods. Thanks to the presence of such mega infrastructure as the Hong Kong-Zhuhai-Macao Bridge and Shenzhen-Zhongshan Link, all of which assist in lowering the transport time of commodities **(Fig. 4.7.6)**. Regarding international air transportation, the region includes three international airports: Hong Kong International Airport, Guangzhou Baiyun International Airport, and Shenzhen Bao'an International Airport. They all offer unmatched worldwide coverage for valuable cargo. This integrated system of air transportation is supported by the positioning of I&T zones at the cross-border interface for anchoring the logistics process. That said, challenges remain due to differing customs regimes and the dual certification required for research samples and materials (Constitutional and Mainland Affairs Bureau, 2023).

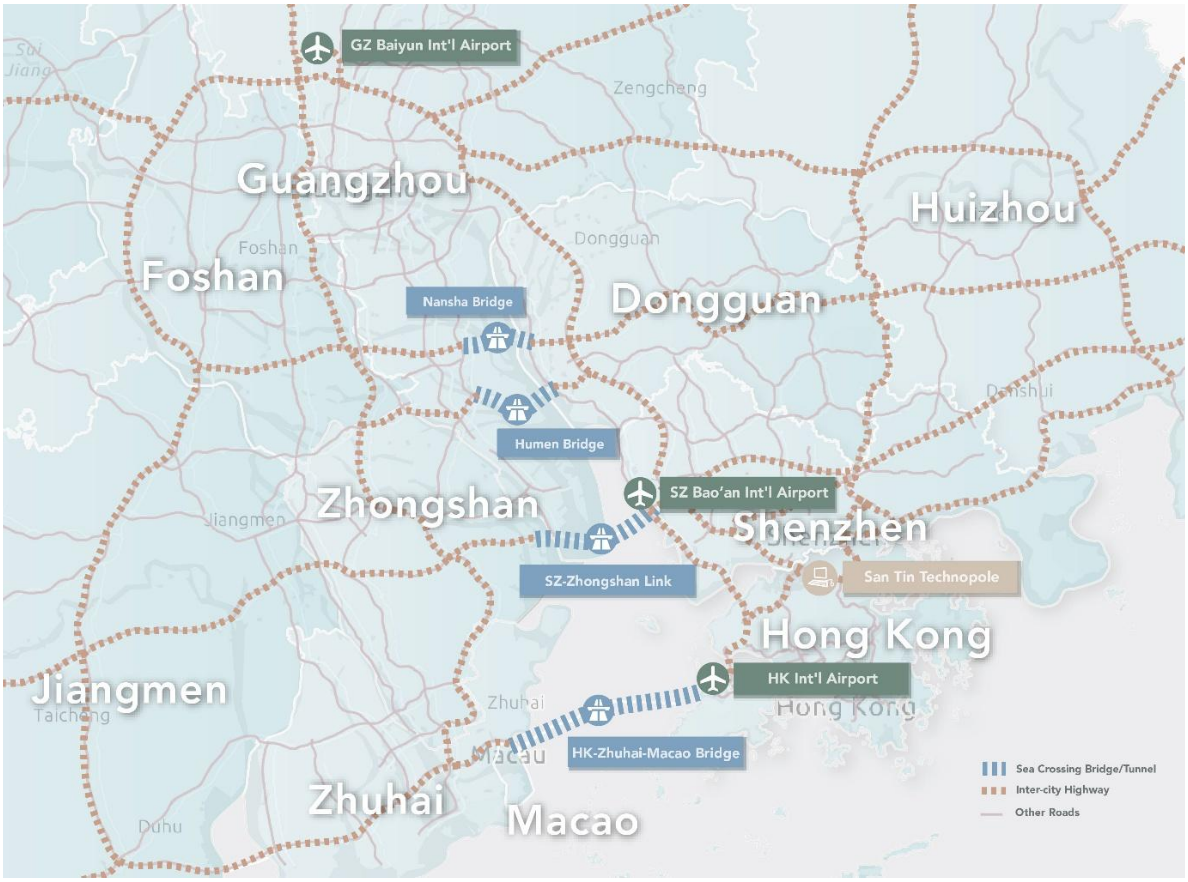


Fig. 4.7.6. Spatial Distribution of GBA's Key Transport Infrastructure and Aviation Clusters

Flow of Capital

Next, there is a moderate to strong flow of capital within the region. On the positive side, there are listing regulations on Hong Kong Exchanges and Clearing (HKEX) (18A/18C) for biotech companies which are yet to make profits and cross-border channels like the Wealth Management Connect and Qualified Foreign Limited Partnership pilots (HKEX, 2023; Hong Kong Monetary Authority, 2024). In the negative aspect, there are strong capital account controls which hinder the process of repatriating renminbi and cross-border funding, regulatory compliance differences, and culturally fragmented venture capital.

Flow of Talent

Besides, there is a moderate flow of talent across the GBA. As a result of aggressive mobility strategies such as the Top Talent Pass Scheme, there is a substantial number of inflows into the area; currently, well over 270,000 talents enter via the mobility policies implemented up until early 2026 (Immigration Department, 2026). In addition to that, the 15% Individual Income Tax (IIT) subsidy provided to high-end technological talents in the Mainland further encourages talent migration (Shenzhen Municipal Bureau of Finance, 2024). Nevertheless, problems of tax residence and cost-of-living constraints at network nodes may complicate smooth talent flow.

Flow of Data

Finally, there is a limited flow of data, which continues to be another drawback. Data transfer across borders is highly limited due to strict Mainland regulations on data localisation, such as the Personal Information Protection Law, that affect industries such as biotech and health care. A special cross-border data channel currently undergoes testing in the Hetao Loop region; yet, there is not one single GBA-wide regulation regarding data governance.

4.7.4. Policy Alignment

Thirdly, this dimension measures the extent to which the industry relates to national strategy, regional priority, and the policy agenda of Hong Kong.

15th Five-Year Plan's 10 Strategic Emerging Industries

The strategic direction of the nation is a significant force in the development of industries. There are 10 Strategic Emerging Industries that are identified by the 15th Five-Year Plan **(Fig. 4.7.7)**. In this regard, the distribution of these sectors is characterised by a clear dominance of the East Bank of the Pearl River **(Fig. 4.7.8)**.

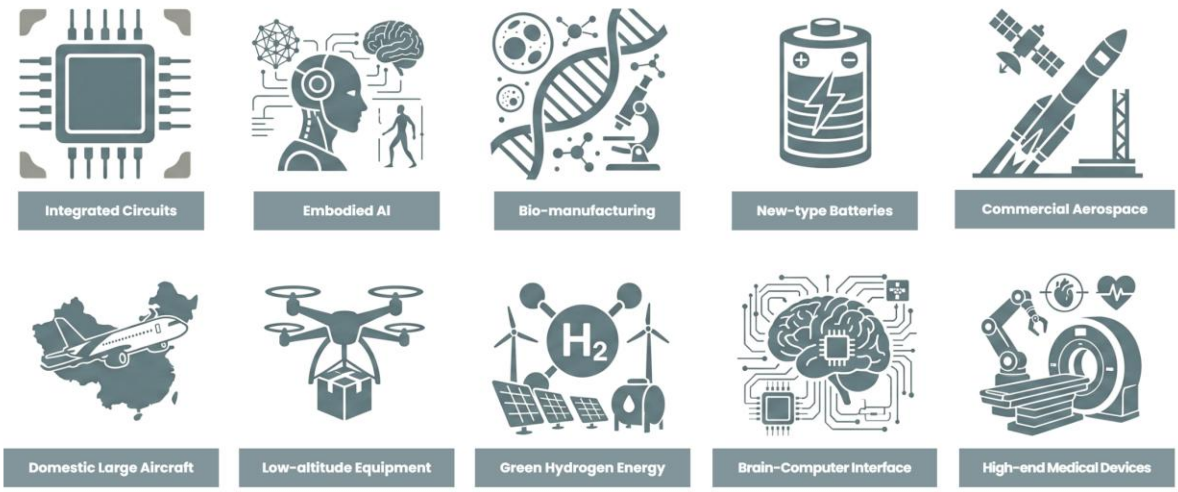


Fig. 4.7.7. 15th Five-Year Plan's 10 Strategic Emerging Industries

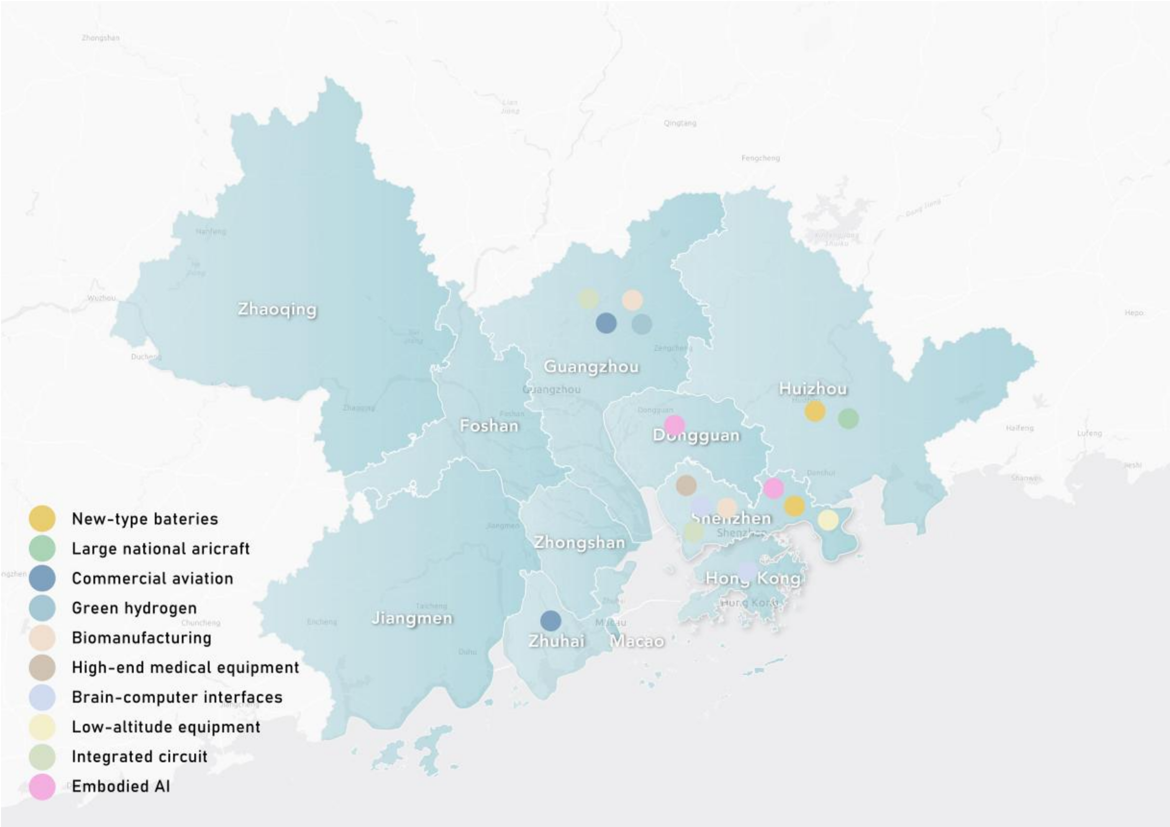


Fig. 4.7.8 Spatial Distribution of Emerging Industries Across GBA

For example, for the East Bank, the industrial sectors of New-type Batteries (Shenzhen/ Huizhou) and Low-altitude Equipment (Shenzhen) already enjoy an advanced score of 4 and are approaching the leading sector position **(Fig. 4.7.9)**. The situation with the West Bank is different (such as Zhuhai), where sectors like Commercial Aviation are still in the developmental niche phase with a score of 3, and Green Hydrogen that is at its earliest stage and earns a score of 2. Future-oriented sectors like Brain-computer Interfaces and Large National Aircraft are also at their earliest R&D phase and earn a score of 1. The detailed breakdown of the scores is in **Table 4.7.2**.

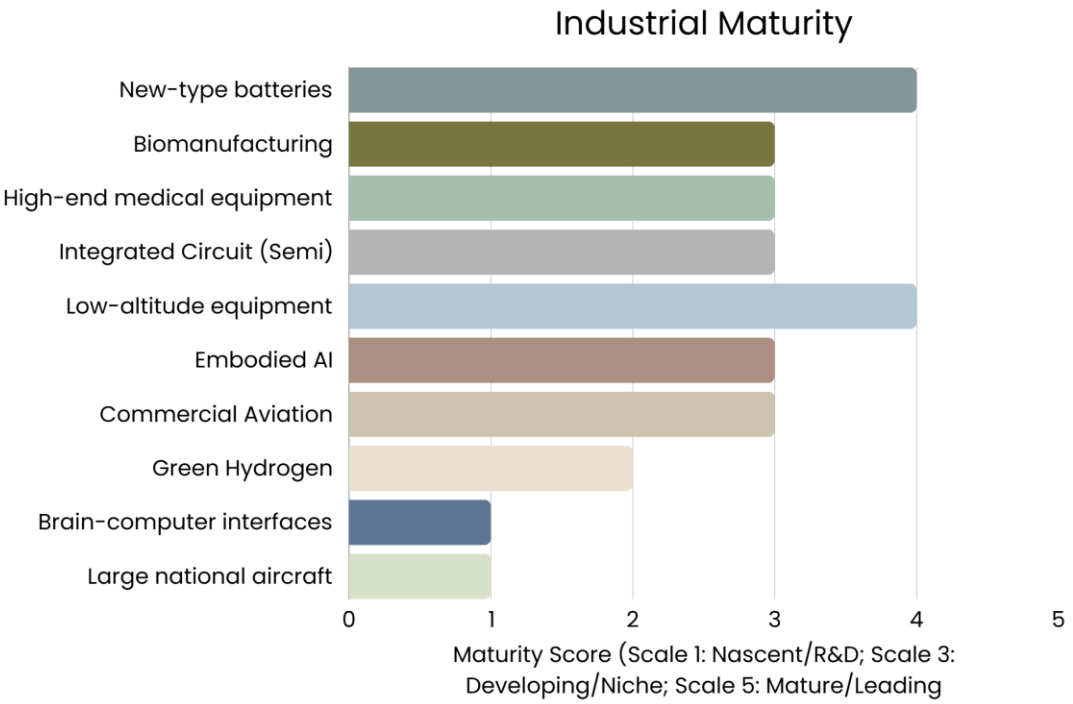


Fig. 4.7.9. Industrial Maturity of Emerging Industries Across GBA

Table 4.7.2. Justification for GBA Industry Maturity Scores

Industry	Score	Reasoning & Source Evidence
New-type batteries	**4**	• The GBA is a major production base for consumer and electric vehicle batteries, concentrated in Shenzhen and Huizhou, anchored by companies like BYD • There is strong R&D focus on next-generation solid-state batteries within the region, although these are still in the pilot and prototype phase globally • Full-scale commercial production of next-generation batteries has not yet been achieved, justifying the score of 4
Biomanufacturing	**3**	• Guangzhou hosts strong research infrastructure, exemplified by the newly inaugurated Max Planck–Chinese Academy of Sciences Center for Synthetic Biochemistry in Shenzhen (April 2026) • The centre is explicitly designed to "drive basic research toward applications," indicating that the regional industry is still at the research-to-application transition stage • Large-scale commercial bioproduction remains limited, justifying a "developing" score of 3
High-end medical equipment	**3**	• Shenzhen-based domestic manufacturers are gaining share in computed tomography and magnetic resonance segments • Government procurement policies are actively accelerating import substitution across China • Despite progress, the high-end segment is still in a catch-up phase with global leadership retained by established multinationals, making a score of 3 appropriate
Integrated Circuit (Semiconductor)	**3**	• The GBA has a strong and growing semiconductor ecosystem spanning Guangzhou and Shenzhen. Huangpu district in Guangzhou alone hosts over 150 enterprises within its 13.2 square-kilometer Bay Area Integrated Circuit Park • CanSemi Technology in Guangzhou is a major manufacturer producing 12-inch wafers for Internet of Things, automotive, and 5G applications, and has filed for an initial public offering. • Another Guangzhou company, ZSemi, has a new 12-inch smart sensor production line with a target of reaching 20,000 wafers per month in 2025 • The region remains focused on mature specialty processes, not advanced nodes, justifying a score of 3

Industry	Score	Reasoning & Source Evidence
Low-altitude equipment	**4**	• Shenzhen is a clear global leader in drones, with DJI dominating the market • Commercial low-altitude logistics routes are already operational. In July 2025, EHang's pilotless logistics aircraft successfully completed an 83-kilometer intercity round-trip cargo flight between Zhuhai and Guangzhou, delivering fresh seafood and medical samples • The score of 4 reflects strong commercial presence; a score of 5 is withheld as electric vertical take-off and landing vehicles have not yet entered fully scaled commercial passenger service
Embodied AI	**3**	• Humanoid robots are being deployed in real-world settings, including border control, airports, patrol, and manufacturing, with activity concentrated in Shenzhen and Dongguan • These deployments remain pilot programmes and niche applications rather than full industrial maturity • The technology is progressing from research and development to niche application, corresponding to a score of 3
Commercial Aviation	**3**	• The C919 aircraft is operational with over 1,000 preliminary orders, which supports the development of aviation-related supply chains in Guangzhou and Zhuhai • The Commercial Aircraft Corporation of China delivered only 15 aircraft in 2025, far below target, indicating production challenges • Dependence on foreign-made engines remains a key vulnerability, meaning full maturity with proven mass-production has not yet been reached
Green Hydrogen	**2**	• China is investing heavily, with over 500 projects launched in 2025, and Guangzhou is developing capabilities in this area • However, actual output is far below capacity as high costs and demand-side challenges persist, keeping the industry at an early scaling stage
Brain-computer interfaces	**1**	• No evidence of commercial deployment exists in the GBA. • Global brain-computer interface technology remains largely experimental, with R&D concentrated in Shenzhen and Hong Kong. • The score of 1 correctly reflects this nascent stage
Large national aircraft	**1**	• The C919 programme is operational but not a mature commercial platform, with low delivery volumes and critical dependence on Western engines • Development activity is positioned in Huizhou • The score of 1 reflects its early operational stage and significant challenges ahead

Source: WIPO (2025); Max Planck–CAS Centre (2026); CanSemi Technology (2025); EHang (2025); COMAC (2025); China Hydrogen Alliance (2025); Guangdong Provincial Department of Industry and Information Technology (2022); National Development and Reform Commission (2025)

In this sense, Hong Kong strategically positions itself by avoiding competition in manufacturing on the mass market level and concentrating instead on developing knowledge-based and highly profitable industries such as Embodied Artificial Intelligence and Biomanufacturing. While these sectors currently hold a score of 3, suggesting they are in the developing phase, they align with Hong Kong's research strengths. By focusing here, Hong Kong drives the "brain" of the value chain for the GBA's more mature production clusters.

<u>Hong Kong's Policy Framework</u>

The connection is also supported by the policies in place in Hong Kong. Most importantly, the I&T Blueprint for Hong Kong provides a detailed blueprint for positioning Hong Kong as an I&T hub around the globe. To highlight, some of the strategic sectors of the I&T Blueprint are life and health technologies, artificial intelligence and data science, and advanced manufacturing and new energy technologies (Innovation, Technology and Industry Bureau, 2022). The blueprint stresses the need to develop midstream value chains and the commercialisation and translation of research results, along with recruiting strategic enterprises in high-value industries. In support of this effort, InnoHK research clusters are formed, which include the Health@InnoHK and AIR@InnoHK clusters with 29 research centres at the end of 2023, along with a third cluster covering advanced manufacturing, materials, energy and sustainable development.

Besides, the scheme called the New Industrialisation Acceleration Scheme, which was announced in September 2024, also allows funding for firms in these strategic industries to establish smart manufacturing factories in Hong Kong. The budget of 2026/27 continues in this vein by committing an amount of HK$ 10 billion each to Hetao and STT, besides providing HK$ 10 billion for the NM University Town.

4.7.5. Market Potential

Fourthly, this dimension examines the capability of the industry to create economic value and growth.

<u>Economic Performance Potential</u>
There are differences between the economic performance potential of the conventional and emerging sectors in Hong Kong. As stated in the interim report of the Federation of Hong Kong Industries, the industry contributes 4.4% to the GDP of Hong Kong by 2023. In detail, “new industries,” which include computer programming, data-related services, industrial internet, research and development, testing and environmental engineering services, make up 1.4%, while traditional manufacturing and publishing and packaging contribute 1.2% and 1.8% to the GDP, respectively (Census and Statistics Department, 2024). New industrialisation increases its contribution to the economy in future years, according to the Hong Kong government.
Regionally, the intelligent equipment industry cluster covering Guangzhou, Shenzhen, Foshan, and Dongguan generated revenues of around 981.7 billion yuan in 2021, which constitutes 10.59% of the country’s total. It includes more than 6,900 companies of designated size, as well as 130 specialised firms within the cluster (Guangdong Provincial Department of Industry and Information Technology, 2022). Traditional industries such as trade, logistics, and local consumer services contribute to job stability. Their future development path can be considered relatively steady. New strategic emerging industries show significant market potential and have a higher chance of earning high-value-added returns by utilising intellectual property and technological innovation.

4.7.6. Assessment of 10 Emerging Industries

To conclude this section, the four assessment dimensions are applied to evaluate the 10 Strategic Emerging Industries against specific criteria **(Table 4.7.3)**.

Table 4.7.3. Assessment Criteria & Details of the 10 Emerging Industries

Aspect	Criteria	What It Checks
A. Factor Conditions	A1. Land Cost	Viability under Hong Kong's high land costs and value-to-footprint ratio
	A2. Labour Cost	Offset potential through productivity, skill level, or automation
	A3. R&D Concentration	Research depth, university strength, and talent pool
	A4. Technology Translation Capacity	Ability to bridge lab-scale innovation to pilot production
	A5. Institutional Leverage	Exploitation of HK's IP regime, financial hub, and legal services
B. Value Chain Positioning	B1. Regional Innovation Geography	Functional fit with HK's upstream R&D role vs SZ, GZ, DG
	B2. Cross-Boundary Synergy	Ability to leverage GBA flows of goods, capital, talent, and data
C. Policy Alignment	C1. National Strategic Priority	Status in the 15th Five-Year Plan or technology sovereignty agendas
	C2. HK Policy Framework Fit	Alignment with HK's I&T Blueprint and local industrial policy
D. Market Potential	D1. Economic Performance Potential	Capacity for technology-driven, high-margin returns with sustained growth

It can be seen from the evaluation matrix **(Table 4.7.4)** that there is a hierarchy within the 10 emerging industries. Integrated Circuits and Embodied AI attain full marks across all criteria, free from any conditional or negative evaluations, and are thus shortlisted for potential development. Meanwhile, Biomanufacturing and High-end Medical Devices are shortlisted based on their performance in meeting eight criteria and satisfying two conditional evaluation criteria. Low-altitude Equipment and Brain-Computer Interface obtain marks for meeting six criteria, and carry four and three conditional evaluations respectively, while one negative evaluation is recorded for the Brain-Computer Interface. In sum, Integrated Circuits, Embodied AI, Biomanufacturing, High-end Medical Devices, Low-altitude Equipment, and Brain-Computer Interface are shortlisted as the priority industries for further strategic development.

Table 4.7.4. Assessment Results of the 10 Emerging Industries

Industry	A1	A2	A3	A4	A5	B1	B2	C1	C2	D1
Integrated Circuits	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Embodied AI	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Bio-manufacturing	~	~	✓	✓	✓	✓	✓	✓	✓	✓
High-end Medical Devices	~	~	✓	✓	✓	✓	✓	✓	✓	✓
Low-altitude Equipment	~	~	~	~	✓	✓	✓	✓	✓	✓
Brain-Computer Interface	~	~	✓	~	✓	✓	~	✓	✓	✓
New-type Batteries	X	X	~	~	X	X	~	✓	~	✓
Green Hydrogen Energy	X	X	~	~	X	X	~	✓	~	✓
Commercial Aerospace	X	X	X	~	~	X	X	✓	~	✓
Domestic Large Aircraft	X	X	X	X	~	X	X	✓	~	~

✓ : Clearly meets the criterion; ~ : Partially meets / Conditional; X : Does not meet

The screening outcomes clearly indicate that industries needing intensive use of factors such as land and labour are subject to inherent feasibility limitations, given the nature of factors in Hong Kong. There are several key similarities among the selected industries: high knowledge intensity; good fit with the upstream capabilities of Hong Kong related to research and development, as well as its institutional advantages; and functional complementarity to GBA partners. AI embodied in products and integrated circuits passes all parameters, making them appropriate sectors for Hong Kong as a high-value command centre. The sectors of bio-manufacturing and high-end medical devices are also very well-fitted to the emphasis on life and health technologies stated in the I&T blueprint.

4.8. Governance

4.8.1. Assessment Framework

In order to evaluate governance, a comprehensive assessment for Government Performance in Technology and Innovation Management is conducted to serve as the indicators (Marinkovic, Rakicevic and Levi Jaksic, 2016). Three stages are examined, namely planning, organising and control, which indicate the input, process and output of the I&T activities respectively. Since STT's development is not completed, only available indicators at the planning and organising stages will be analysed, namely investment, infrastructure and public administration (**Fig. 4.8.1**).

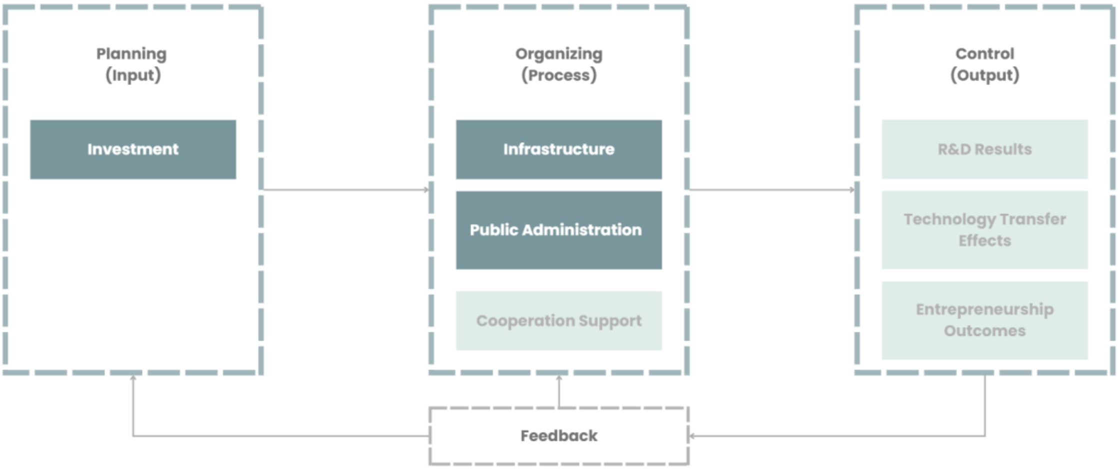


Fig. 4.8.1. The Framework of Assessment for Government Performance in Technology and Innovation Management
Source: Marinkovic, Rakicevic and Levi Jaksic (2016)

4.8.2. Investment

The government is the major investor in the overall I&T ecosystem in Hong Kong. The Innovation and Technology Fund (ITF) established by the government supports six aspects of the I&T industry, including R&D, new technologies through prototype production, smart production, research talents recruitment, start-up incubation, and non-R&D projects. **Table 4.8.1** and **Fig. 4.8.2** showed the list of all funding projects under ITF. The cumulative approved amount for ITF has reached 55,863.4 million in 2026, with one-third going to supporting R&D (ITC, 2026b). Moreover, 31.4% of the funding does not fall within the 6 aspects mentioned (ITC, 2026b). As seen from the diverse funding programmes, allocation of resources on government funding is fragmented and scattered across various aspects. This may not align with the policy in promoting strategic industries such as smart production. Meanwhile, the effort of the government can be seen in consolidating the R&D foundation and promoting knowledge transfer through funding support.

Table 4.8.1. Summary of the funding programme under the Innovation and Technology Fund

Category	Programme
Supporting Research & Development	Innovation and Technology Support Programme (ITSP)
	Mainland-Hong Kong Technology Cooperation Funding Scheme (MHKTCFS)
	Enterprise Support Scheme (ESS)
	Frontier Technology Research Support Scheme (FTRSS)
Facilitating Technology Adoption - Public Sector Trial Scheme (PSTS)	Public Sector Trial Scheme – ITF Projects (PSTS-ITF)
	Public Sector Trial Scheme for Incubatees & Graduate Tenants of Hong Kong Science & Technology Parks Corporation and Hong Kong Cyberport Management Company Limited (PSTS-SPC)
	Public Sector Trial Scheme for Technology Companies Conducting Research and Development (R&D) Activities in Hong Kong (PSTS-TC)
	Public Sector Trial Scheme - Special Call for Projects for the Prevention and Control of Coronavirus Disease 2019 (COVID-19) in Hong Kong (PSTS-COVID-19) [Closed]

Promoting New Industrialisation And Developing New Quality Productive Force	New Industrialisation Acceleration Scheme (NIAS)
	New Industrialisation Support Scheme (NISS) - New Industrialisation Funding Scheme (NIFS)
	New Industrialisation Support Scheme (NISS) - New Industrialisation and Technology Training Programme (NITTP)
	New Industrialisation Support Scheme (NISS)- Pilot Manufacturing and Production Line Upgrade Support Scheme ("Manufacturing+")
Nurturing Technology Talent	Research Talent Hub for ITF projects (RTH-ITF)
	Research Talent Hub for Incubatees and I&T Tenants of the HKSTPC and the Cyberport (RTH-SPC)
	Research Talent Hub for Technology Companies Conducting R&D Activities in Hong Kong (RTH-TC)
	Research Talent Hub for Companies Subsidised under the New Industrialisation Acceleration Scheme (RTH-NIAS)
Supporting Technology Start-Ups	Innovation and Technology Venture Fund (ITVF) [Closed]
	Research, Academic and Industry Sectors One-plus Scheme (RAISe+)
	Pilot Innovation and Technology Accelerator Scheme (PITAS)
Fostering an I&T Culture	General Support Programme (GSP)

Source: ITC (2026a)

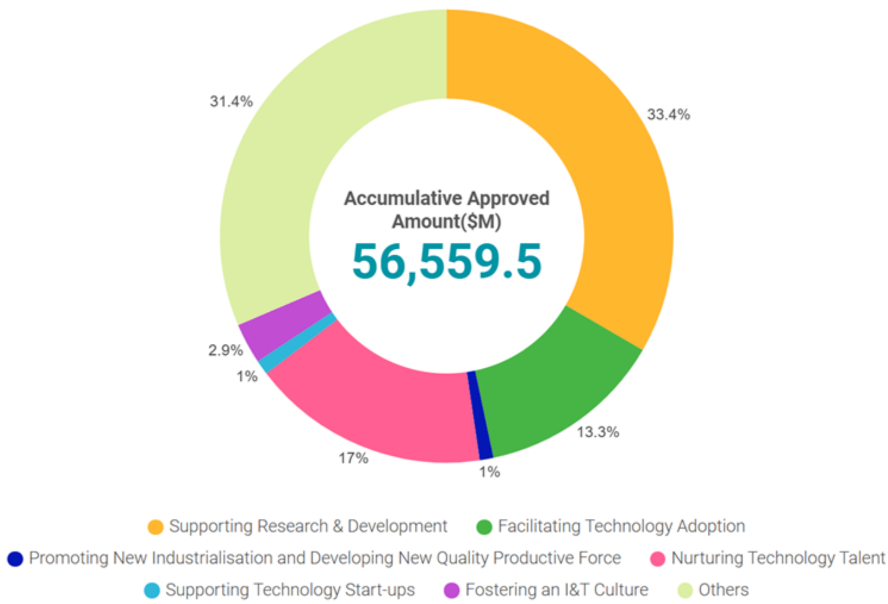


Fig. 4.8.2. The cumulative approved amount of the programmes under the ITF until 2026
Source: ITC (2026)

In addition to ITF, the Innovation and Technology Commission (ITC) also supports other innovation-related investments such as technological entrepreneurship, infrastructural support and quality support (**Table 4.8.2**). The trend of government investment under ITC will increase significantly in 2026, with total expenditure expected to rise from 874.9 million in 2024-25 to 2579.4 million in 2025-26, indicating a growth of 194.8% (FSTB, 2025) (**Fig. 4.8.3**). Such an expected increase in investment is mainly contributed by infrastructure development. 1880.2 million is expected to be invested in infrastructure, including the development of STT, which contributed to 72.9% of the total estimated investment among all 6 aspects (FSTB, 2025).

While the investment trend aligns with the strategic promotion of the I&T industry, the overconcentration of investment in infrastructure may hinder the allocation of resources in other aspects of the I&T ecosystem. Even if there will be sufficient provision of infrastructure in STT, the overall I&T industry ecosystem may not be mature enough for the industry to catch up and fully utilise the provided infrastructure.

Table 4.8.2. Summary of the investment in the ITC from 2023 to 2026

PROGRAMME	2023–24 ($m)	2024–25 ($m)	2025–26 (Estimate) ($m)
(1) Support For Research And Development	103.4	119.6	121.3
(2) Promotion Of Technological Entrepreneurship	38.1	52.3	68.1
(3) Planning For Innovation And Technology Development	109.2	108.9	112.2
(4) Infrastructural Support	57.6	76.2	1880.1
(5) Quality Support	141.2	132.6	180.3
(6) Subvention: Hong Kong Productivity Council, Hong Kong Applied Science And Technology Research Institute Company Limited	371.1	385.3	217.4
Total	**820.6**	**874.9 (+6.6%)**	**2,579.4 (+194.8%)**

Source: FSTB (2025)

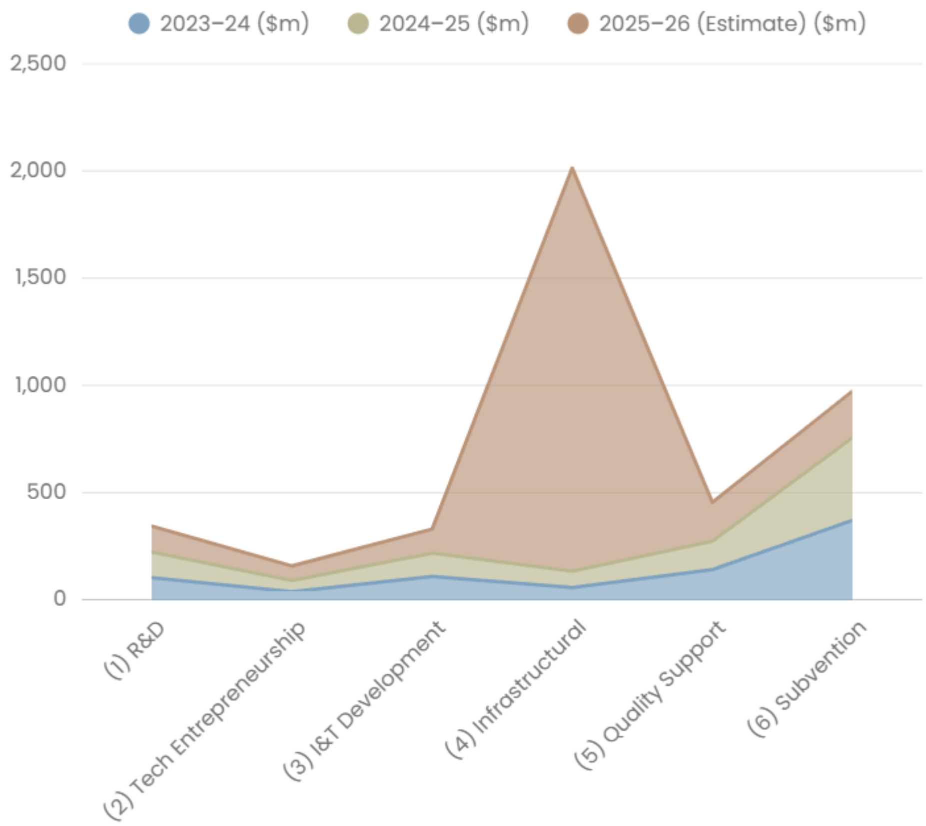


Fig. 4.8.3. Investment trend of the ITC from 2023 to 2026
Source: FSTB (2025)

4.8.3. Infrastructure

Essential infrastructure, including transportation, drainage, electricity, SGR facilities and GIC facilities, will be explored to understand the current and future provision of these infrastructures to support the STT development.

First is the transport infrastructure. There will be a total of three railway lines passing through STT, namely the existing Lok Ma Chau Spur Line (LMC), the proposed Northern Link (NOL) Main Line and NOL Spur Line (**Fig. 4.8.4a** and **b**). The planned NOL is expected to start operation by 2034 (MTR, no date). Among these lines, the LMC Spur Line Control Point and the NOL Spur Line will serve as railway control points that connect to Futian Port and the new Huanggang Port in Shenzhen, respectively (HKSAR, 2025a). Moreover, the High-Speed Rail is intersecting the NOL at around Ngau Tam Mei. This may serve as a potential and faster alternative for connecting STT, Hong Kong urban areas and more cities in mainland China, if an additional High-Speed Rail station is established in STT (**Fig. 4.8.5**). For major roads, San Sham Road allows the linkage to Shenzhen through the LMC Control Point. Other roads, such as San Tin Highway and the planned Metropolis Highway, ensure connectivity within the NM and the New Territories (**Fig. 4.8.6a** and **b** and **Table 4.8.3**). The transport infrastructure demonstrated a good potential for cross-boundary interaction due to the existence of three control points, which are connected by the existing and planned railway and roads. On the contrary, the local connectivity is rather restricted due to the San Tin Highway, leading to the segregation of the I&T sites and the community.

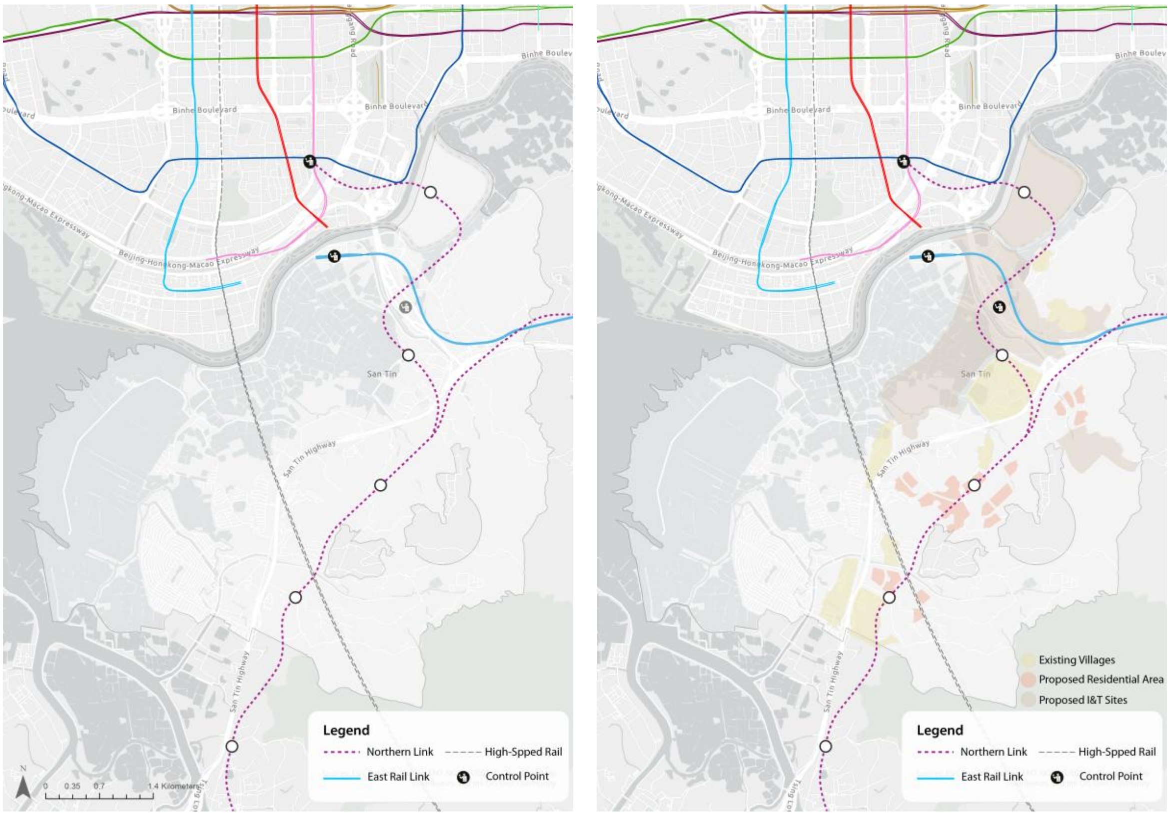


Fig. 4.8.4a and **b**. Existing and planned railway connections in STT without and with the proposed land use

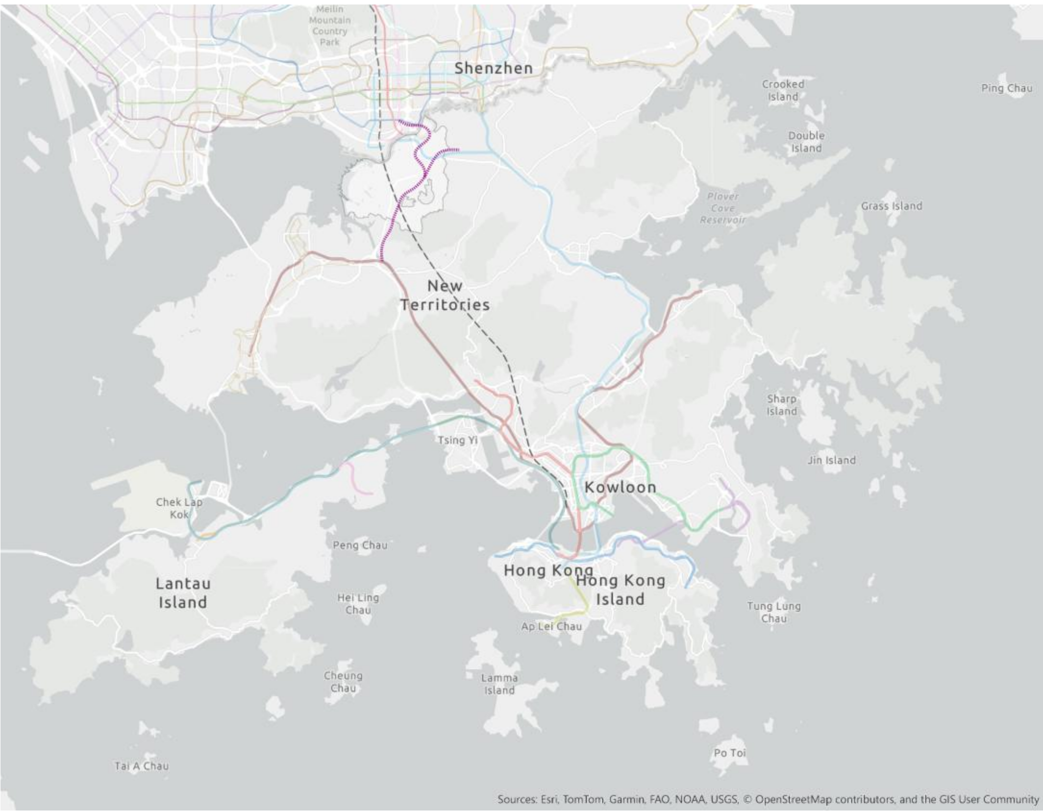


Fig. 4.8.5. High-Speed Rail and other existing and planned railway connections in Hong Kong scale

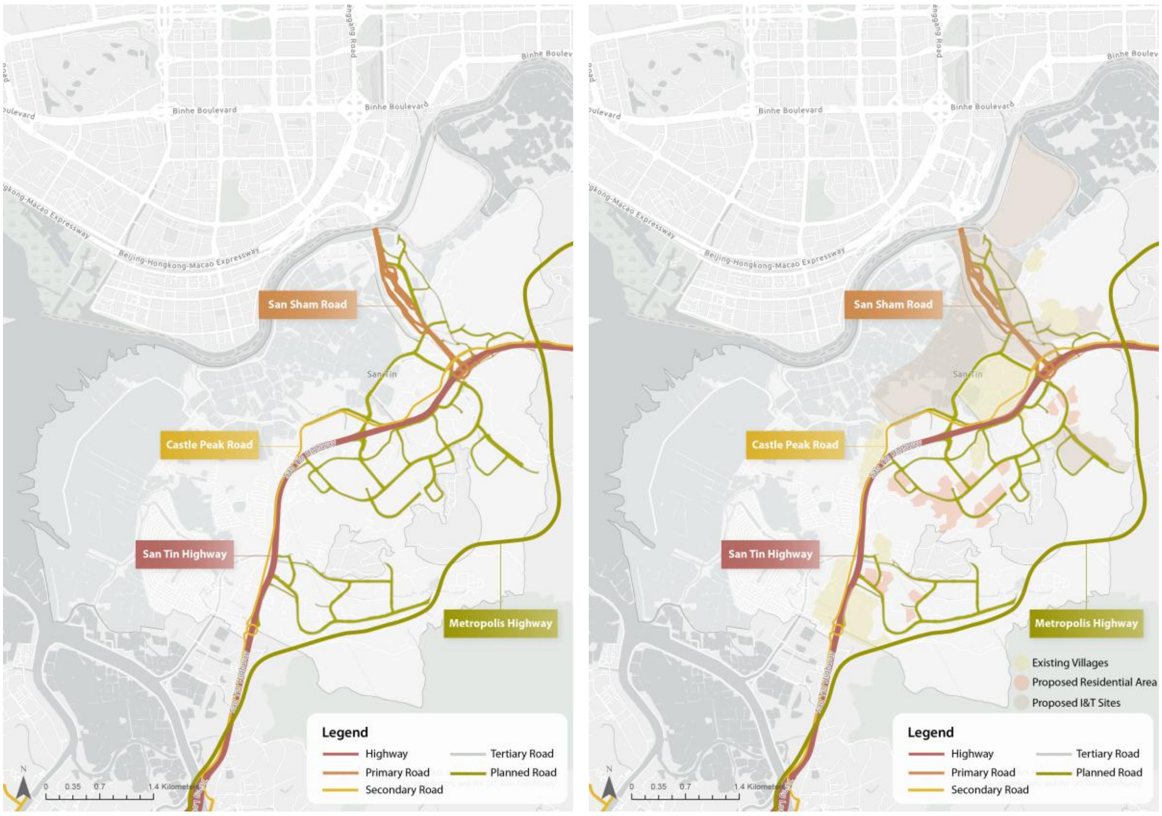


Fig. 4.8.6a and **b**. Existing and planned road connections in STT without and with the proposed land use

Table 4.8.3. Summary of major existing and planned roads in STT

Major Roads	Connection
San Tin Highway (Route 09)	New Territories
San Sham Road	Shenzhen
Castle Peak Road	New Territories & New Kowloon
Planned Metropolis Highway	NM Neighbourhood
Planned road at STT & NTM Neighbourhood	STT And NTM Community

Since there will be an abundant I&T industry with high energy consumption, the electricity infrastructure is also important for STT development. Currently, the existing high-voltage lines and power towers are located on the southern to western side of the STT, which are isolated from the community centres and the planned I&T sites (**Fig. 4.8.7a** and **b**). The village settlements are mainly relying on the local networks branching in spiderweb patterns, rather than showing integration with the urban areas. As for the proposed development plan, there will be a total of 13 electricity substations scattered within the STT to serve the development needs (EPD, 2024). As seen in the plan, the detachment of the I&T sites from the electricity corridor may hinder the high-energy-consuming industry. Moreover, the current focus is more on providing an adequate electricity supply for I&T facilities such as data centres and supercomputers. However, the problem of high electricity consumption and carbon footprint remains unresolved. This indicated the need for greener energy alternatives to be supplied in STT to mitigate the carbon footprint.

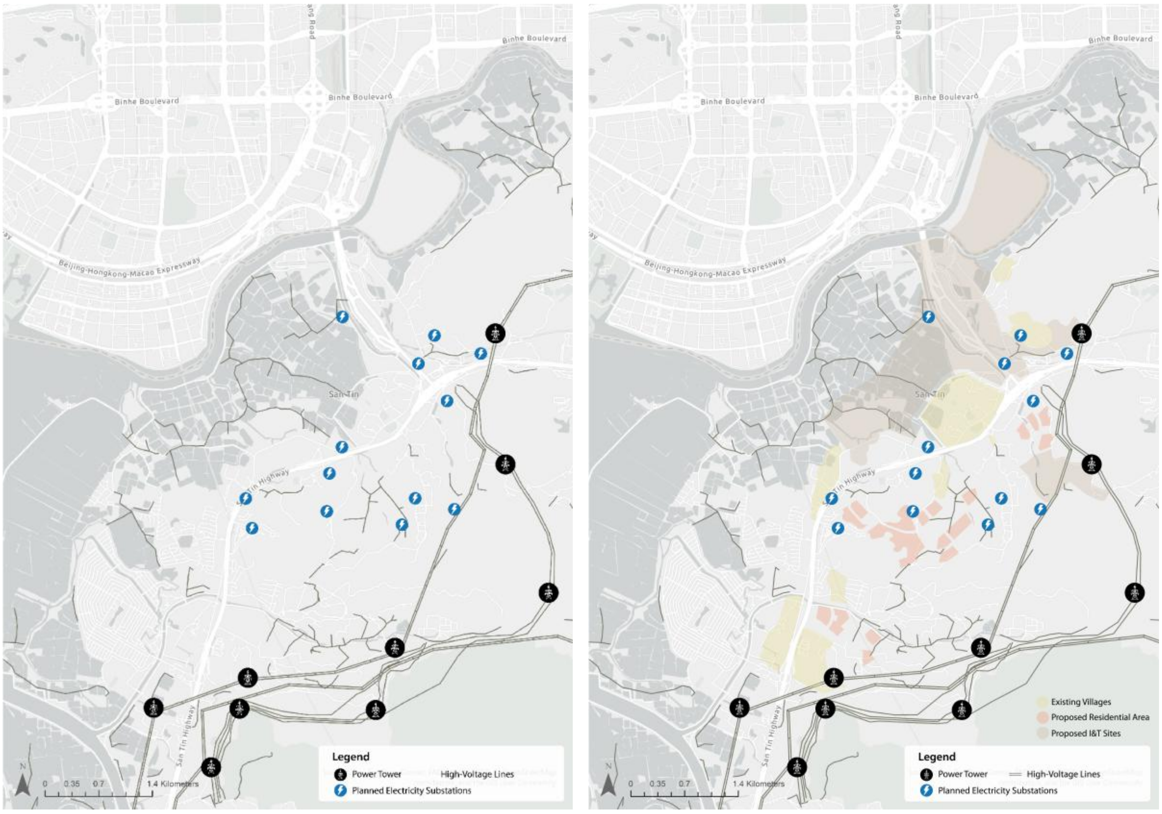


Fig. 4.8.7a&b. Existing and planned electricity infrastructure in STT without and with the proposed land use

Regarding drainage infrastructure, the current I&T site will be located at the fishpond area, which is vulnerable to flooding. The existing drainage system, including the San Tin Storm Water Pumping Station and Polder, is mainly concentrated along the major roads (DSD, 2022). In terms of the proposed plan, the main drainage channel and floodable landscape with flood attenuation facilities are extended further from the San Tin Highway (**Fig. 4.8.8a** and **b**). Nevertheless, the drainage system is not reaching the centre of the fishpond. This raised concerns for flood mitigation, especially under unpredictable extreme weather in the future. This may further hinder the development of the I&T industry due to the potential damage to the expensive I&T equipment.

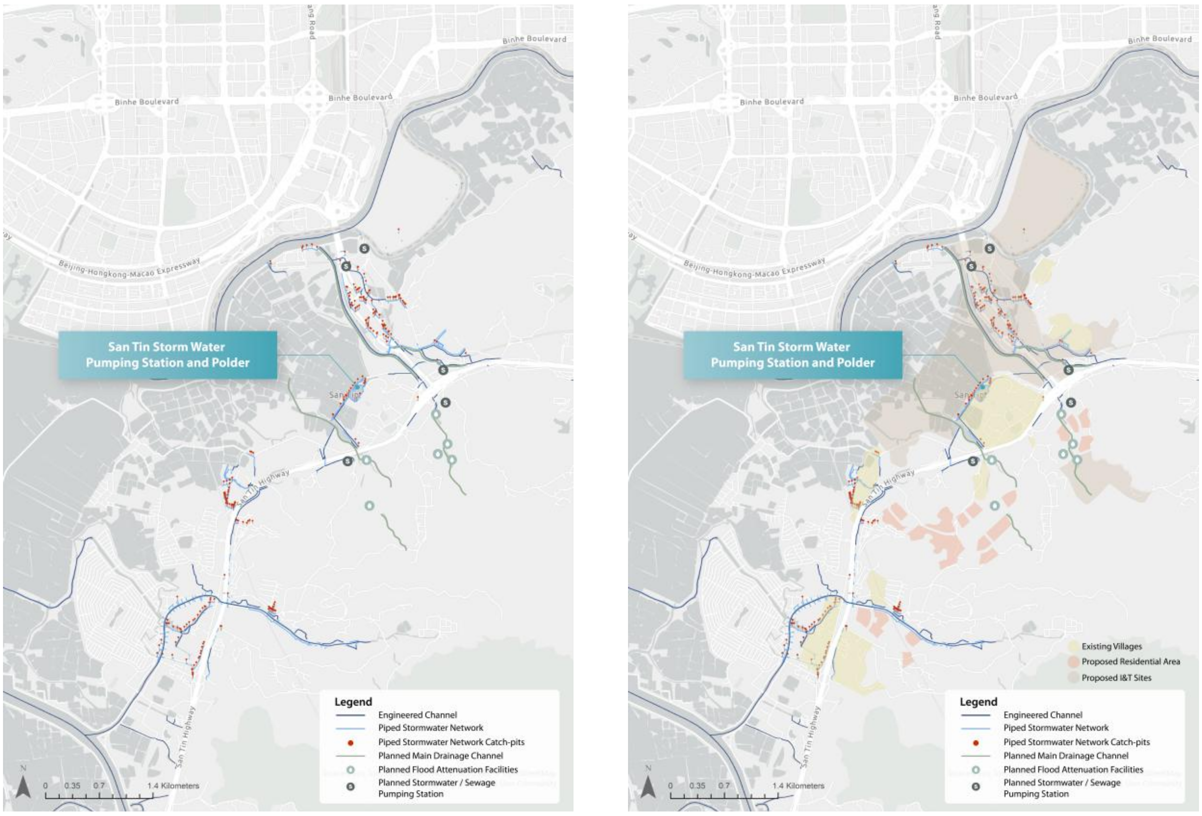
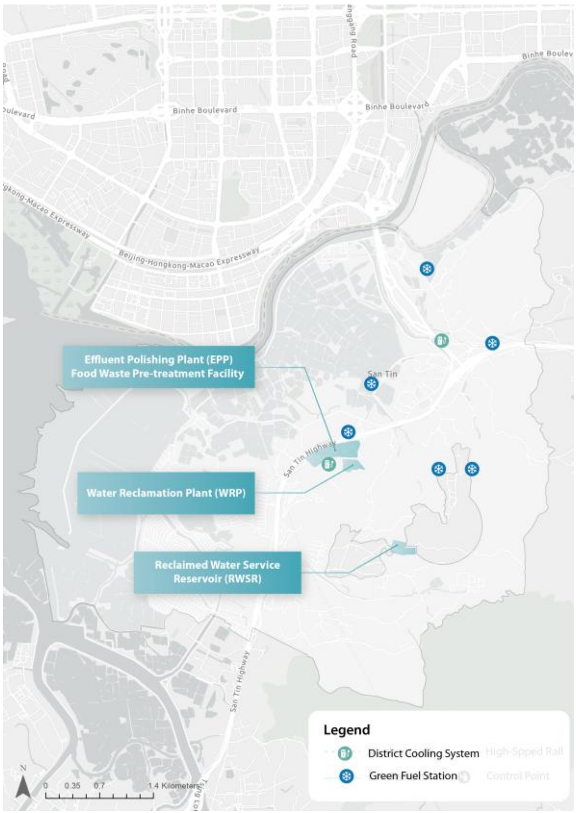


Fig. 4.8.8a and **b**. Existing and planned drainage infrastructure in STT without and with the proposed land use

Beyond the drainage system, the outline development plan of STT emphasised the Smart, Green and Resilient (SGR) infrastructure in encouraging a carbon-neutral community (CEDD and PlanD, no date). The proposed SGR infrastructure includes an Effluent Polishing Plant, Reclaimed Water Service Reservoir, Water Reclamation Plant, Food Waste Pre-treatment Facility, District Cooling System and two Green Fuel Stations scattered within the STT (EPD, 2024) (**Fig. 4.8.9a** and **b**). While these SGR facilities make a step closer to the waste circulation, they mainly focus on wastewater treatment for the industrial sewage, rather than the solid waste or municipal waste generated by the community. The waste treatment facilities are limited at the site level and fail to incorporate waste circulation at a broader scale, making them fail to achieve waste zero or carbon neutral.



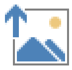


Fig. 4.8.9a and **b**. Planned SGR infrastructure in STT without and with the proposed land use

GIC infrastructure in the sites also demonstrated the insufficiency of supporting the local community. Although there are various GIC facilities existing on the site, they are mainly concentrated within Fairview Garden and the entrances of existing villages (**Fig. 4.8.10a** and **b**). Deficits in recreational, library, and modern civic spaces for talent may hinder the exchange of innovative ideas, making the site less attractive for the talent and their families to stay.

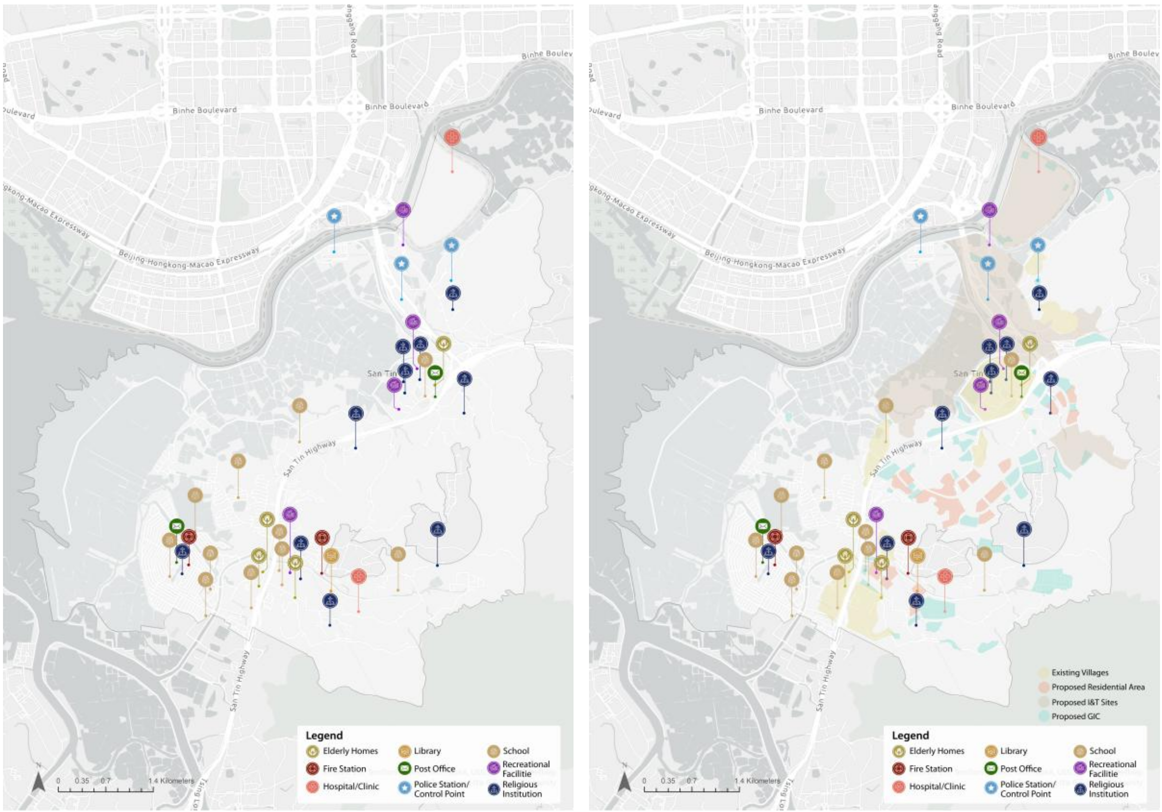


Fig. 4.8.10a and **b**. Existing GIC infrastructure in STT without and with the proposed land use

4.8.4. Public Administration

In order to streamline regulatory and development measures, dedicated legislation is established to accelerate the NM development (Legislative Council, 2026). The measures can be classified into two types: policy and regulatory measures, and development and implementation measures. **Table 4.8.4** showed the detailed measures suggested by the government. With this specialised system, the administration efficiency can be significantly enhanced and aligned with the broader plan in speeding up the NM development.

Table 4.8.4. Summary of the dedicated legislation to accelerate NM development

Policy & Regulatory Measures	Development & Implementation Measures
Empower the Government to devise simplified statutory procedures	Adopt the Fast Track Processing System to reduce construction costs and time
Facilitate cross-boundary flow of people, goods, capital, data and biological specimens.	Integrate construction technologies, materials and equipment from the mainland and overseas.
Speed up the approval of building plans.	Implement a **"phased development" approach** (referencing the mainland's 1.5-level development)
Relax permitted uses in outline zoning plans and adjust development parameters.	Adopt **diverse development models** (in-situ land exchange (pay land premium), large-scale land disposal)

Employ flexible land-grant arrangements (including tenancy (land lease) exceeding 7 years)	**"Pay for what you build" approach** for land premium, and allow payment by phases.
Facilitate compensation payment for land resumption.	/

Source: Legislative Council (2026)

For STT development, different governmental departments and cross-boundary collaboration are fostered. **Fig. 4.8.11** showed the multi-level institutional framework for STT, which showed the hierarchy of different departments. Additionally, the Committee on Development of the NM and three working groups are established to expedite NM's development (DEVB, 2026) (**Table 4.8.5**). However, some government departments may only focus on plan execution rather than constructing the vision together at the planning stage. For instance, EPD mainly focus on EIA approval instead of being involved in the plan formulation, and no representative from EPD is engaged in the Committee on Development of the NM (HKSAR, 2025b). Additionally, regulatory uncertainty due to legal and political differences between HK and the Mainland remains unaddressed and requires further details, such as on information transfer.

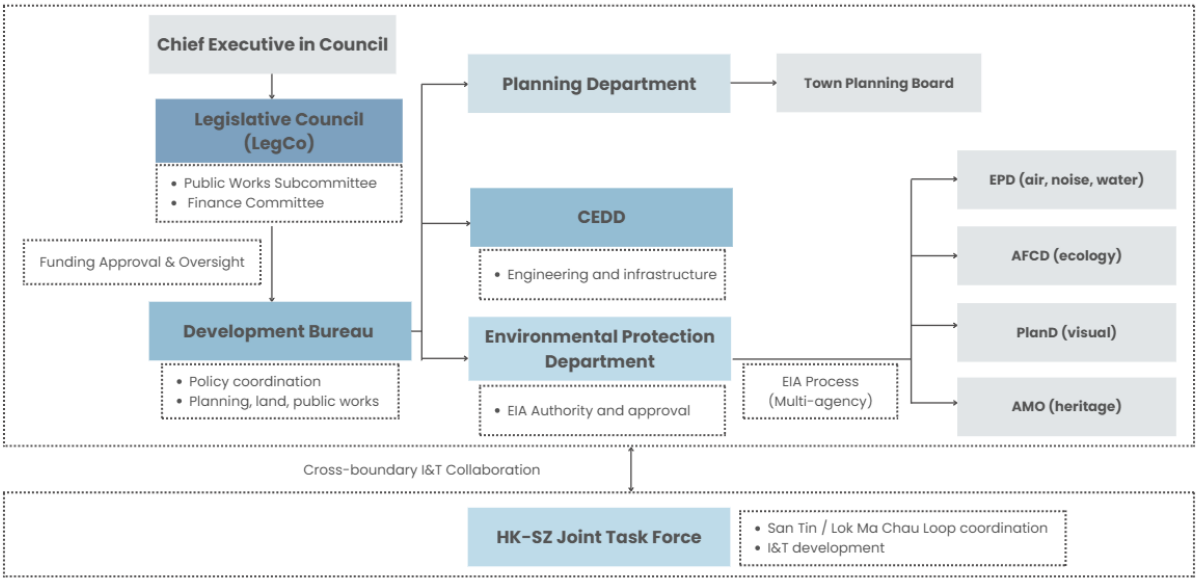


Fig. 4.8.11. Multi-level Institutional Framework for STT

Table 4.8.5. Summary of NM Fast-track Governance Structure

Committee And Working Groups	Leaders	Description
Committee On Development Of The Nm	Chief Executive	To accelerate the development of NM and introduce industries and major projects
Working Group On Devising Development And Operation Models	Financial Secretary	To compose the industry parks' development and operation model in NM

Working Group On Planning And Construction Of The University Town	Chief Secretary for Administration	To investigate the mode of development of the university town
Working Group On Planning And Development	Deputy Financial Secretary	To help with the arrangement in different development aspects such as planning, engineering and environment

Source: DEVB (2026)

4.8.5. Spatial Implication

The baseline review of governance, especially on the hard infrastructure, can be concluded by the spatial implication map (**Fig. 4.8.12**). Firstly, railway-oriented development should be adopted to leverage the cross-boundary integration and the connection to other urban areas. The development of I&T sites and communities along the railway can facilitate local connectivity as they are no longer separated by the major roads. Active transport, such as walking and cycling, can also be encouraged with the extensive railway system serving as the backbone of the STT. The energy-consuming I&T sites should be located along the electricity corridor and further away from the flood-prone fishponds. Moreover, the I&T hub for new energy can be treated as a potential contributor to lowering the carbon footprint in STT development. As for the planned SGR infrastructure, it should go beyond the local level and seek a more proactive approach to net zero and waste circulation. Finally, the existing GIC facilities can be utilised and further enhanced to cater for the needs of the existing and future communities.

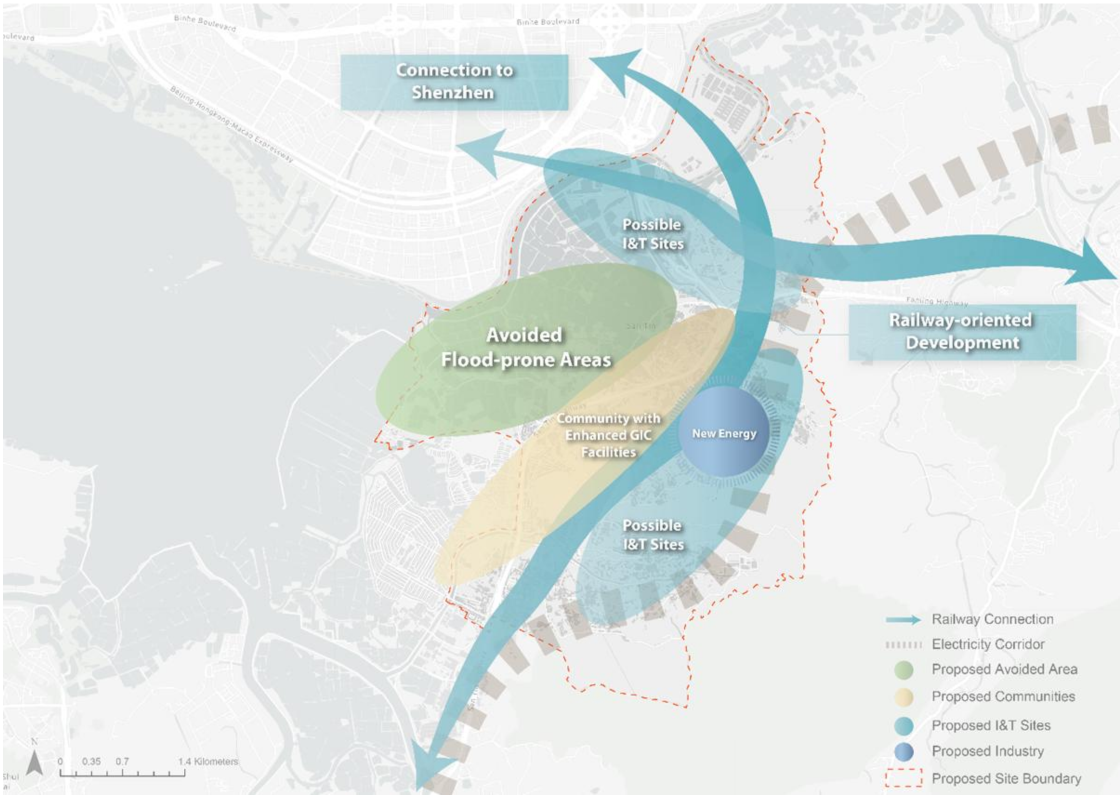


Fig. 4.8.12. Spatial implication of government

5. SIA Variables

Aspect	Sub-Objectives	Sustainability Impact Variables
Economic	Promote Sustainable I&T Growth	Land Area Reserved for Green I&T/Industrial Zones
		GBA Connectivity Infrastructure
	Support Innovation & R&D Clusters	Land Area Reserved for R&D/Incubation Uses
		Number of Anchor Tech Firms
	Integrated Regional & Cross-border Economy	Agglomeration with Shenzhen Tech Clusters
		Cross-border Innovation Corridor Connectivity
	Circular & Low-Carbon Economy	Share of Circular Economy Practices in Industrial Zones
		Waste-to-Energy Infrastructure Capacity
Environmental	Preserve Natural Habitats & Ecological Buffers	Total Land Area of Natural Habitats
		Reserved Ecological Buffer Zones
		Wetland & Biodiversity Protection
	Achieve Net-Zero Carbon & Renewable Energy	Share of Renewable Energy in Mix
		Circularity of Material Flow
		Amount of Carbon Footprint
	Smart Waste & Resource Management	Number of Recycling Facilities
		MSW Recovery Rate
	Climate Resilience & Water Management	Blue-Green Infrastructure Area
		Stormwater Management Facilities
		Flood Risk Mitigation Zones
Social	Enhance Quality of Life & Liveability	Open Space / Greenery per Capita
		15-Minute Amenity Coverage
		Wind & Visual Corridor Preservation
	Provision of Talent & Community Amenities	Facilities/Amenities per Capita
		Multilingual & Family-friendly Services Coverage
	Preserve Local Rural & Cultural Heritage	Land Allocated for Village Heritage & Cultural Activities
		Preservation of Indigenous and Non-indigenous Villages
	Inclusive & Cross-cultural Community	Community Integration Programmes
		Cross-border Talent Support Services

6. SWOT Analysis

Strategic Focus 1 – I&T Development and Regional Synergy

Based on the assessment of the I&T potential of STT, the region has significant strengths in promoting regional synergy. Its core advantages lie in its robust academic and R&D foundation, as well as the potential for strategic cross-border integration. Hong Kong's strong international reputation, robust legal system, and sound infrastructure provide a high-quality environment for attracting global businesses and investment. However, the weaknesses are mainly reflected in the disconnection between "midstream transformation" and commercialisation, as well as problems such as an imbalance in talent structure and fragmentation of resources. The asymmetry in investment allocation and institutional restrictions is still an obstacle to the production of scientific research outcomes. In terms of opportunities, the high alignment of policies and strategies, coupled with the establishment of a professional regulatory system, makes STT poised to become a hub for global expansion and significantly improve the efficiency of knowledge transfer. The potential market growth for commercial products also provides impetus for industrial upgrading. In contrast, STT is facing serious threats, including increasing regional competition and geopolitical risks. Internal demographic pressures, infrastructure strains, and regulatory uncertainties may also affect the achievement of its long-term strategic goals.

Strategic Implications

- Strengthen the integration of industry, academia and research
  Establish closer links between industry and academia and promote technology from the laboratory to the market to overcome the current gap in R&D transformation.
- Strengthening the Link between Innovation and Manufacturing
  Optimise the industrial chain to ensure rapid prototyping and large-scale production of I&T achievements in the region.
- Promote seamless cross-border collaboration
  Reduce barriers to the flow of factors through institutional innovation, leverage the role of "super connector", and strengthen connections with the GBA and the world.
- Cultivate Diverse Emerging Industries
  Support the development of cutting-edge fields such as AI and biomedicine to establish a diverse emerging I&T ecosystem.

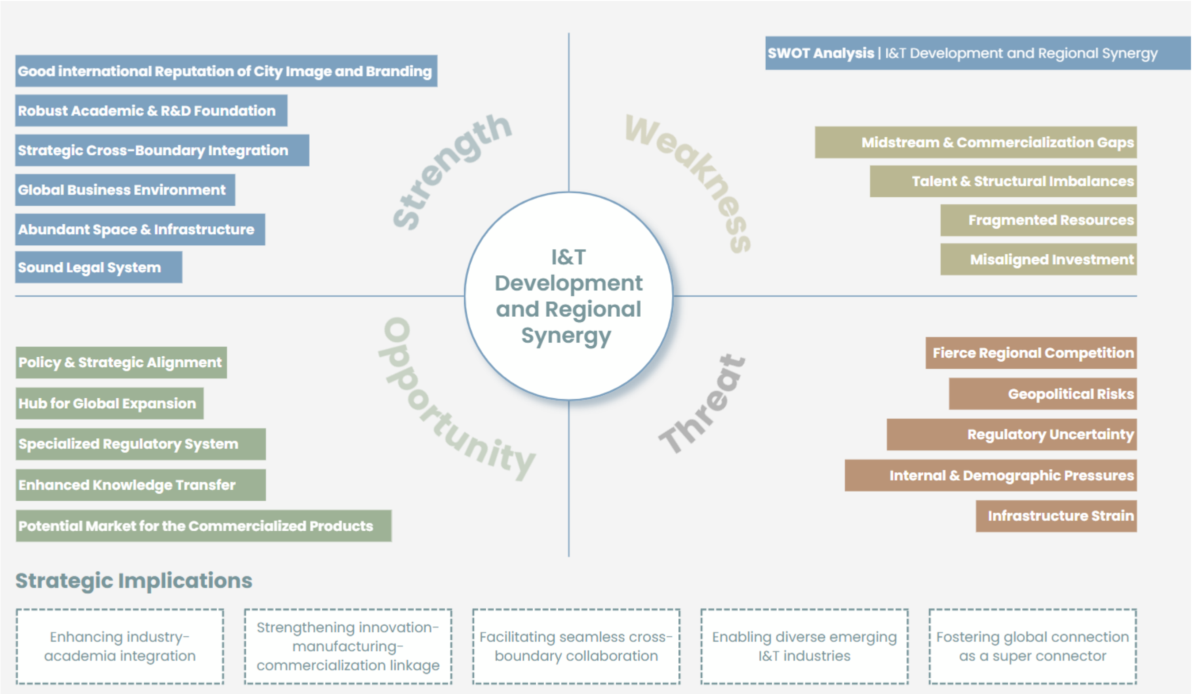


Fig. 6.1. SWOT analysis and strategic implications in terms of I&T development and regional synergy

Strategic Focus 2 – Talent Support and Rural Inclusive Communities

In line to build a talent-supported and inclusive community in rural areas, STT demonstrates unique potential for urban-rural symbiosis. Its strength lies in its rich rural and cultural assets, as well as diverse natural landscapes, which provide a "hard foundation" for creating a high-quality living environment. In addition, the multicultural social structure and favourable talent attraction policies together constitute the "software" that supports the talent ecosystem. However, the current weaknesses cannot be ignored, including the high cost of living, gaps in supporting facilities, and structural employment issues. Meanwhile, the preservation of traditional villages and cultural heritage also faces great challenges. In terms of opportunities, macro policy trends and the application of smart living technology provide momentum for the optimisation of the talent ecosystem. Through cultural integration, STT is poised to shape a more cohesive community identity. Yet, the region is facing multiple threats. Rapid urbanisation pressures could lead to brain drain and social conflict. The negative impacts of geopolitics and potential social isolation might cause socio-economic exclusion and undermine community inclusiveness.

Strategic Implications

- Provide affordable, high-quality housing

  It is essential to address the high cost of living and provide diverse and affordable housing options for talent at all levels and residents.
- Create a Vibrant Living-Work-Learning Ecosystem

  Seamlessly integrate I&T functions with daily life spaces through mixed land use, creating a work-life balance community.
- Promote active interaction among diverse communities

  Promote exchanges between new immigrants and local rural communities and utilise rural and cultural assets to strengthen social resilience.

- Introduce smart technology to enhance inclusivity
  Use smart city technology to improve public services and ensure that different communities can enjoy the convenience brought by technological development.

Fig. 6.2. SWOT analysis and strategic implications in terms of talent support and rural inclusive communities

Strategic Focus 3 - Environmental Sustainability and Carbon Neutrality

In response to the strategic goals of environmental sustainability and carbon neutrality, STT has the potential to transform into a green, sustainable city. Its strengths lie in strategic land use planning, rich ecological profiles, and blue-green infrastructure. Existing regulatory safeguards also provide basic support for environmental protection. However, its development faces significant weaknesses, including irreversible ecological disruption, habitat fragmentation, and trust deficits in environmental protection implementation. In terms of opportunities, STT is poised to make breakthroughs in the path to carbon neutrality through sustainable urban design, cross-border eco-synergy, and the promotion of sustainable industries. In contrast, the region also faces multiple environmental threats. Urban pollution and climate change vulnerability are becoming increasingly prominent. Additionally, resource and capacity constraints, as well as significant energy footprints and supply pressures, pose challenges to achieving long-term environmental goals.

Strategic Implications

- Proactive conservation actions
  Efforts must be made to actively engage in ecological restoration and conservation initiatives to offset the environmental impacts of development.
- Implement a circular economy to achieve carbon neutrality
  Implement circular models such as “supply-recover-restore” in regional planning to reduce resource consumption and waste generation at the source.

- Adopt sustainable design to enhance climate resilience
  Strengthen regional resilience to extreme weather events through resilient infrastructure and sponge city design.
- Leveraging smart technologies to mitigate urban externalities
  Deploy smart sensing and big data analytics to monitor and dynamically optimise urban energy usage and pollution control in real time.
- Strengthen cross-border ecological collaboration
  Jointly establish cross-border ecological corridors with neighbouring regions to ensure the continuity and integrity of biodiversity at the regional scale.

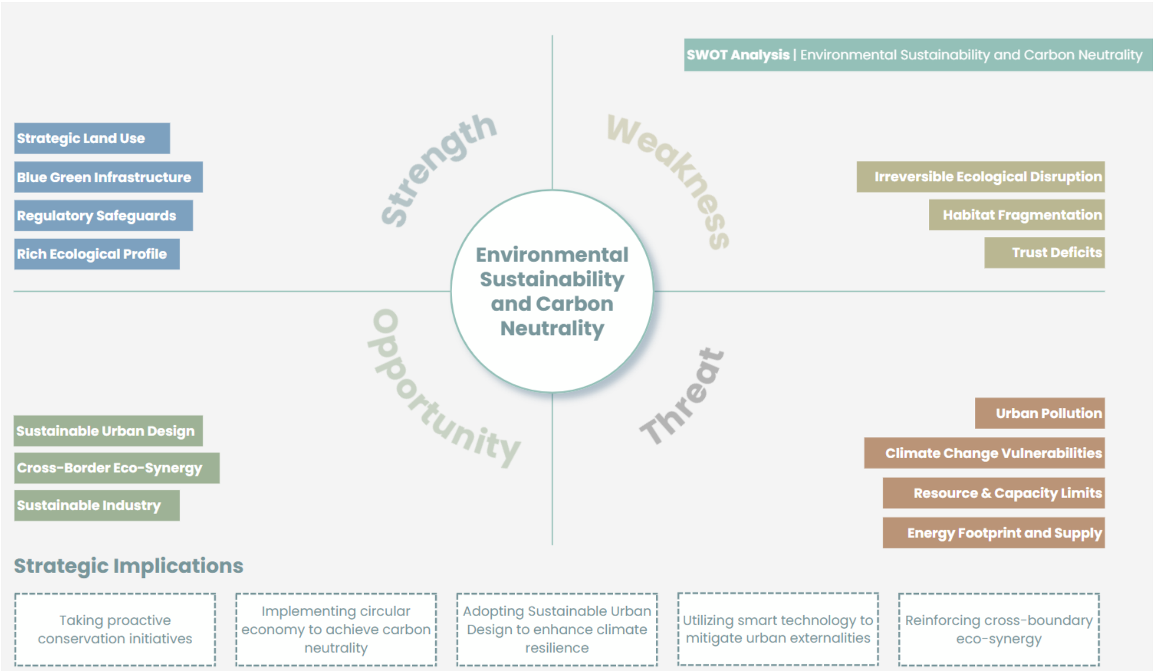


Fig. 6.3. SWOT analysis and strategic implications in terms of environmental sustainability and carbon neutrality

7. Draft SCP

7.1. Vision and Objectives

The vision of the project is **to position San Tin and its surrounding areas as a global innovation powerhouse and frontier-creator that fosters talent development, advanced innovation, sustainable prosperity, and regional integration while bridging the GBA with the world.**

To realise the vision, three categories of objectives have been established, aligned with the strategic focuses identified at the inception stage and reinforced by the strategic implications of the SWOT analysis, as outlined below.

Strategic Focuses	Objectives
I&T Development and Regional Synergy	• To establish STT as a world-class frontier-creator by operationalising agile and flexible planning frameworks designed to anticipate and adapt to rapidly evolving I&T trajectories, setting new global standards for resilient innovation ecosystems within the GBA. • To operationalise a premier innovation powerhouse by architecting a seamless 'Lab-to-Market' engine that originates breakthrough knowledge and scales high-end manufacturing for global impact. • To anchor a pioneering cluster-based for advanced technologies, leveraging specialised cross-boundary synergy in Biotech, New Energy, and Advanced Manufacturing to propel San Tin as a primary generator of global I&T assets.
Talent-Supportive and Rural-Inclusive Community	• To engineer a world-class, innovative 'Live-Work-Learn' environment that attracts global I&T experts and entrepreneurs through affordable, high-quality housing and smart-city infrastructure. • To create a socially sustainable and vibrant society by fostering active interaction between diverse communities and residents through inclusive public spaces and community-building initiatives. • To catalyse harmonious urban-rural integration by implementing spatial planning and policies that respect cultural and historical assets to sustain the site's unique identity and heritage. • To actualise inclusive and resilient communities through community-building initiatives and the revitalisation of decaying villages to ensure social sustainability throughout the development process.
Environmental Sustainability and Carbon Neutrality	• To actualise a territory-wide carbon-neutral and circular economy by integrating site-specific ecological infrastructure into regional resource recovery networks. • To reinforce cross-boundary eco-synergy by operationalising a regenerative loop that designs out waste, restores high-functioning wetland networks, and secures sustainable resource loops to establish a world-class model for rural-urban integration. • To deploy smart technologies to monitor and manage regional environmental health, ensuring resource efficiency and the mitigation of urban externalities through data-driven planning.

7.2. SCP by Strategic Focuses

7.2.1. I&T Development and Regional Synergy

The SCP for the development of I&T in the STT includes the spatial planning of the four specialised I&T Industry-Academia Zones. They include the Artificial Intelligence (AI), Robotics and Data Science Zone, the Finance and Legal Support Services Zone, the Advanced Manufacturing, New Energy, and Materials Zone, and the Life and Health Zone **(Fig. 7.2.1)**. The four zones are arranged on one central corridor, which facilitates high-level industry linkages in the core development zone; simultaneously, two spines extend on either side of the Hong Kong-Shenzhen interface.

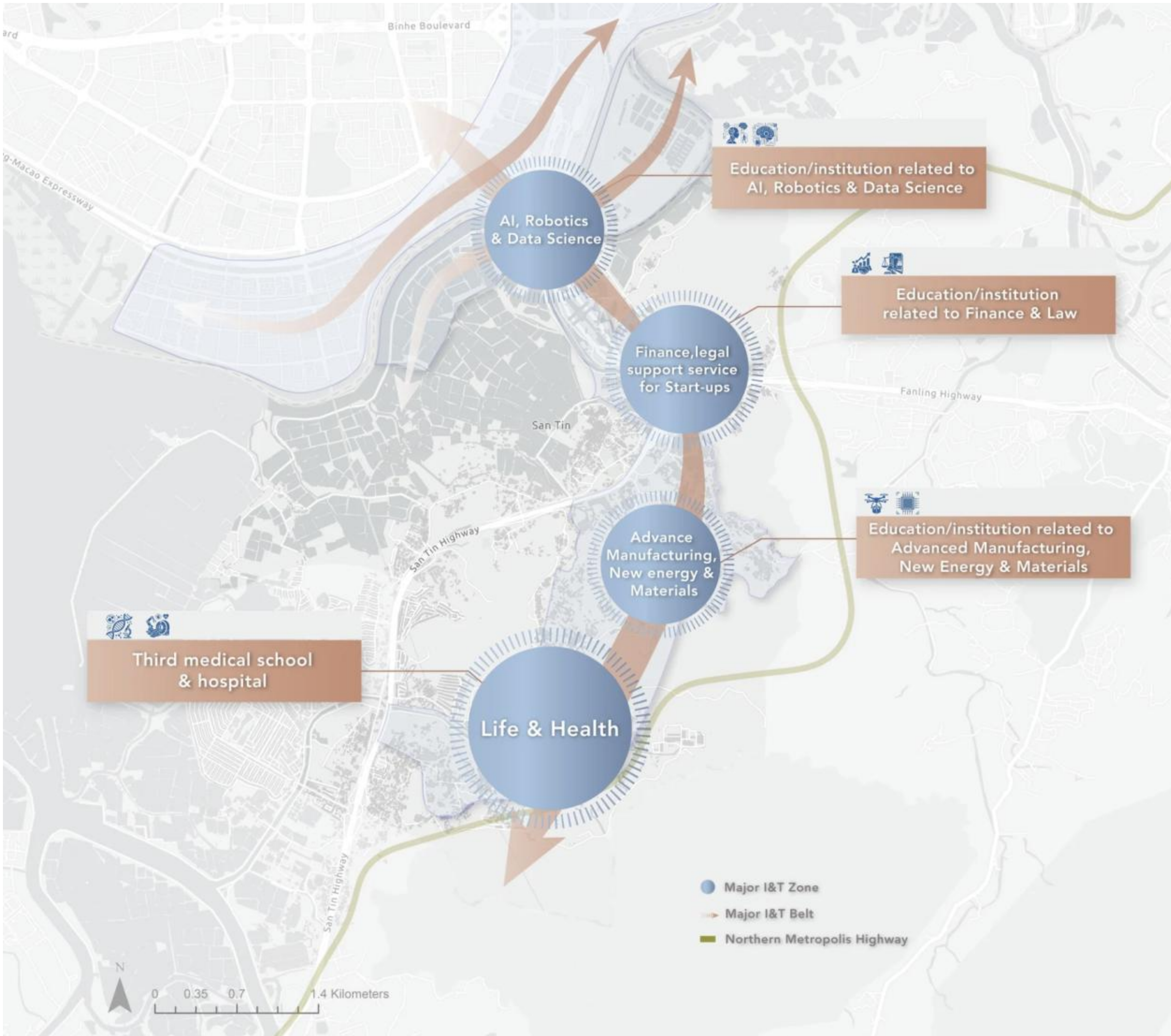


Fig. 7.2.1. SCP of I&T Development and Regional Synergy

<u>Local Scale: I&T Industry-Academia Zones</u>

<u>AI, Robotics, and Data Science Zone</u>
The AI, Robotics and Data Science Zone is situated on the Technology Axis near the Hong Kong Park in the Loop. Such a location helps leverage its closeness to Shenzhen, which ensures faster cooperation with existing hardware and algorithms development infrastructure. Research

labs, prototyping centres and co-working spaces specialising in artificial intelligence and robotics businesses operate in the zone. Examples of the industries that are promoted here include Embodied AI and Brain-Computer Interface, among others, though the list is not exhaustive since technologies develop rapidly and some flexibility is maintained. Finally, an educational/research institute focused on AI, robotics, and data science is founded, providing a skilled workforce and contributing to research development.

<u>Finance and Legal Support Service Zone</u>

The Finance, Legal Support Service Zone is strategically positioned in a prime location of the Technopole that has strong connectivity to new development areas, various established or planned MTR stations (Lok Ma Chau Station, Chau Tau Station), and the Loop, supported by transit and road infrastructure (San Tin Highway, Northern Link, NM Highway). Low-rise buildings are used here since it ensures that the flight paths of migratory birds passing through the site are not hindered. Some of the facilities expected in this zone are those providing patent services, certification, venture capital, and legal advisors who deal with technology acquisitions, among others, which help in developing technology startups and technology transfers. There are plans to establish an educational and research institute dealing with finance and legal studies, which helps in managing intellectual property rights and venture financing.

<u>Advanced Manufacturing, New Energy, and Materials Zone</u>

The Advanced Manufacturing, New Energy and Materials Zone is positioned on the eastern side of the Technopole. The positioning of this zone is based on its proximity to eastern utility routes, which enables the efficient delivery of power loads needed to support high-tech manufacturing processes and materials. Meanwhile, proximity to the San Tin Highway allows for efficient movement of goods and components within the region, making the transportation of products throughout the GBA possible. The zone also features smart factories, pilot manufacturing, and testing facilities for energy and materials. Potential industries to be developed within this zone include, but are not limited to, Low-Altitude Equipment and Integrated Circuits. There is also an establishment of an educational/research organisation specialising in advanced manufacturing processes, new energy sources, and material sciences to encourage cooperation between industries and academia through innovative process implementation and industrial application.

<u>Life and Health Zone</u>

The Life and Health Zone is geared towards Ngau Tam Mei, where a health-tech corridor is established. The underlying reasoning behind such a direction includes the proposed development of a third medical school and hospital at Ngau Tam Mei, which acts as the clinical research base of this particular zone. The proposed facilities include biomedical laboratories, pharmaceutical R&D centres, and health technology prototyping laboratories that are provided with the necessary clinical environment and data by virtue of their close proximity to clinics. Potential industries for this particular zone include, but are not limited to, Bio-manufacturing and High-End Medical Devices.

Comparison with the Government Plan

The planning strategy proposed by the government for the STT involves the development of three hubs from north to south. The three hubs include Hetao Hub, Chau Tau Hub, and Ki Lun Hub (Innovation, Technology and Industry Bureau, 2025). Meanwhile, three industry development corridors are planned: a life and health technology corridor in the north collaborating with the Loop, two corridors south of Chau Tau Hub for AI and robotics, and microelectronics and advanced industries, respectively **(Fig. 7.2.2a** and **b)**.

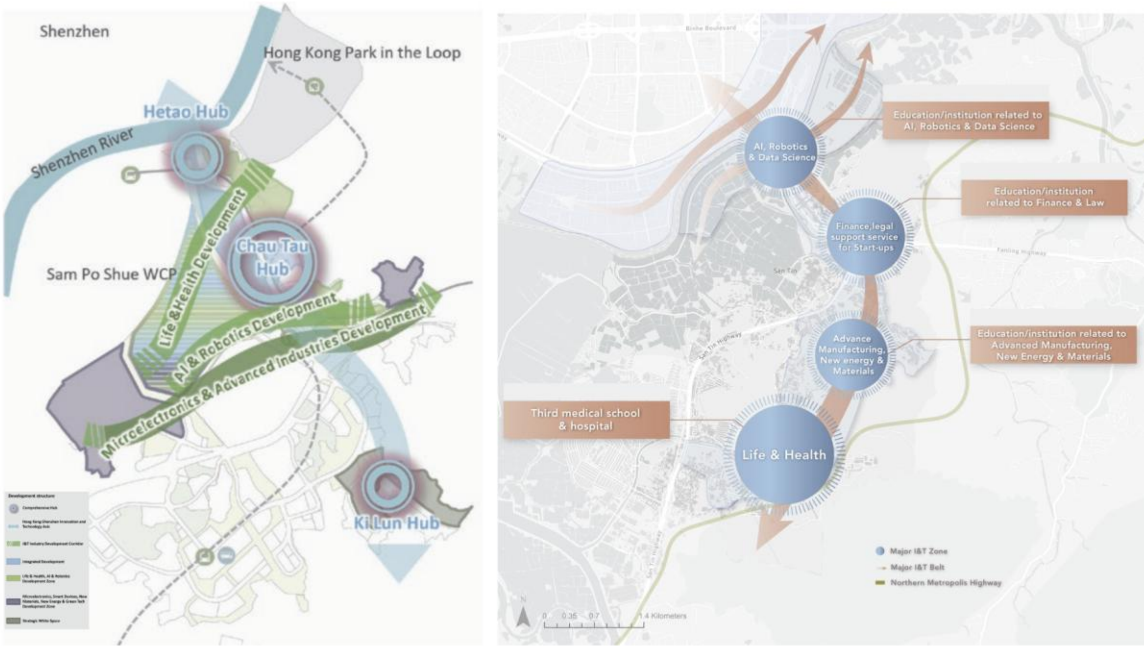


Fig. 7.2.2a and **b.** Comparison of Industrial Layout between the Government Plan and the Proposal

Several similarities and differences can be noted between the conceptual plan and the government plan. First, both plans acknowledge the value of clustering similar industries together via designated corridors and establishing development hubs for coordination. Moreover, both plans list Life and Health Technology, AI and Robotics, and Advanced Manufacturing as key sectors. However, the crucial difference between the two plans can be seen in the addition of an independent Finance, Legal Support Service Zone in the conceptual plan, as opposed to the inclusion of such services in the Chau Tau Hub in the government plan. Another difference is that the conceptual plan stresses the importance of anchoring each zone with its own educational and research institution. In contrast to this, the government plan mentions the possibility of attracting international academic institutions to the Chau Tau Hub; however, no further information on such anchoring is provided in the government plan. In addition, the conceptual plan's spatial configuration of four zones along one central corridor with two spines produces a more structured interface with Shenzhen. This utilises the Hong Kong-Shenzhen boundary to generate cross-boundary I&T collaboration.

Institutional and Soft Infrastructure

In addition to the physical infrastructure, the strategic conceptual framework also comprises institutional and softer forms of infrastructure, which aim to minimise barriers to the translation and commercialisation process. Firstly, flexible and pre-approved compliance procedures allow rapid testing of emerging technologies without having to undergo long regulatory processes. Secondly, secure and dual-compliant digital gateways streamline data exchange and intellectual property protection at the interface between Hong Kong and Shenzhen. Thirdly, a shared facility model eliminates financial barriers associated with initial commercialisation stages by making it possible for spin-off companies to gain access to state-of-the-art prototypes and testing facilities without the need to acquire all the equipment.

Regional Strategic Vision: Cross-border Synergy

Fig. 7.2.3 shows how the proposed industry layout of STT is extended into a cross-border innovation network. Following the selected industries into four strategic zones within STT, this map illustrates the innovation synergy between Hong Kong and Shenzhen with cross-boundary R&D and manufacturing linkages, capital flow and major research clusters. In terms of the spatial location, STT sits at the boundary interface that connects directly to the Loop and is right opposite to Shenzhen's innovation district, such as the Futian and Huanggang districts. The blue and purple arrows indicate the connection between Hong Kong and Shenzhen's technology hubs and research institutions, while the orange arrows show the flow of financial services and seed funding from Hong Kong's international financial hub and Shenzhen's Qianhai innovation-finance centre. Together, these two CBDs ensure a concrete legal and capital support for venture capital and patent protection.

The closest and most strategic collaboration partner would be Hetao SZ-HK Innovation Cooperation Zone. It supports joint R&D and cross-border testing. Besides, Banxuegang Science and Technology Zone provides digital AI platforms and start-up experiences, and various industry clusters complement STT in life sciences and advanced research as well. STT is not meant to be an isolated bubble. It serves as a physical anchor where Hong Kong's research plugs directly into the Shenzhen development corridor. By aligning the specific zones right up against the border as mentioned above, local upstream innovation and Shenzhen's rapid prototyping capabilities are smoothed out.

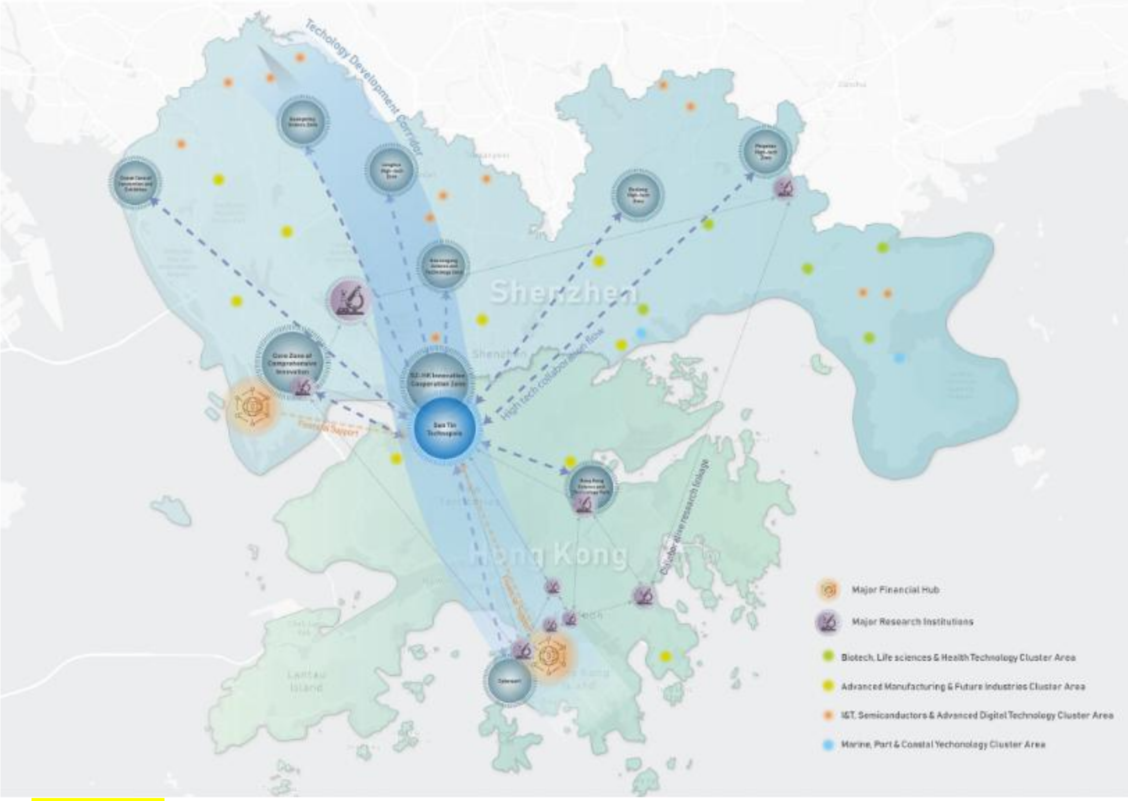


Fig. 7.2.3 HK-SZ Cross-border Innovation Network and Financial Support for STT

<u>Regional Strategic Vision: GBA Synergy</u>

STT not only connect to Shenzhen but also to the wider GBA as a part of the regional innovation-manufacturing network. **Fig. 7.2.4** shows how the proposed I&T zones in STT will be connecting with the well-established manufacturing linkage in GBA. With Hong Kong's high-quality industry and research system, STT specialises in R&D and pilot testing, while the manufacturing process can be complemented by other GBA cities according to their respective strengths. For instance, in health and life sciences, our site focuses on research, clinical testing and pharmaceutical commercialisation, while biomanufacturing and medical device production can be supported by Guangzhou and Jiangmen. With this network, the site's research outputs can be transferred better and commercialised more effectively with the leverage of the mature manufacturing ecosystem in the region.

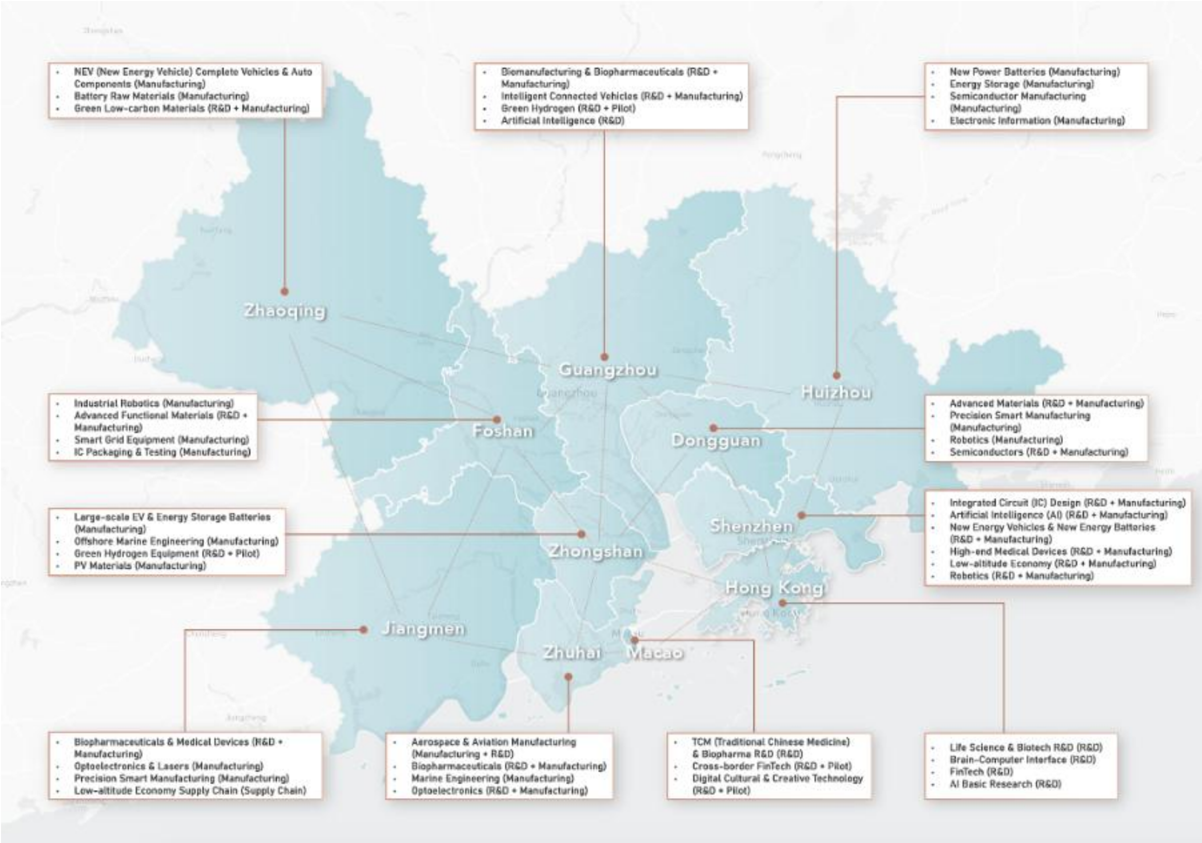


Fig. 7.2.4 Regional Innovation-Manufacturing Network for STT

<u>Global Strategic Vision: Build a transnational innovation ecological network</u>

The vision of STT aims to transform it into a core powerhouse and frontier creator for global innovation. In the context of increasingly fierce global technology competition, STT is no longer just a border belt between Hong Kong and Shenzhen, but a strategic gateway connecting the GBA with the global innovation ecosystem. As shown in **Fig. 7.2.5**, STT is endowed with profound geographical strategic significance, and its core mission is to carry out deep integration between the GBA's advanced manufacturing network and strong supply chain foundation, and global high-end inputs (e.g., capital, knowledge, talent). Under this spatial framework, as demonstrated in **Fig. 7.2.5**, STT has demonstrated strong outward radiation capabilities, establishing stable two-way connections with Europe (e.g., the UK, the Netherlands, Germany, France, etc.), North America (i.e., the United States and Canada), East Asia (e.g., Japan and South Korea), and emerging markets (e.g., Southeast Asia, the Middle East, and South America). This networked layout ensures that STT can optimise the allocation of resources globally. On the one hand, it absorbs the basic research capacity and high-end capital of developed economies. And on the other hand, it exports internationally competitive I&T products to emerging markets, ultimately realising the grand vision of "Innovate Together, Thrive Sustainably, Advance the World".

<u>Global Strategic Vision: Core Element Flow</u>

To achieve its strategic positioning as a "global innovation powerhouse", STT must establish an efficient internal circulation mechanism. According to the key concepts of I&T Development and Regional Synergy, as discussed in Chapter 2, this plan establishes a three-dimensional flow system with knowledge, talent, and capital as the core. The first one is knowledge generation and world-class research collaboration. STT leverages Hong Kong's research foundation of five of the world's top 100 universities and combines the institutional advantages of "One Zone, Two Parks" of the Hetao Co-operation Zone to establish in-depth strategic alliances with world-leading research institutions. Through advanced knowledge exchange mechanisms, STT is committed to promoting original research in cutting-edge fields such as life and health, AI and robotics. This collaboration is not limited to the output of academic papers but also emphasises the closed-loop transformation of "Research-Application-Creation" to ensure that scientific research results have practical market application value. The second one is the seamless talent and capital flow. In terms of talent mobility, STT provides excellent living and R&D space for the world's top scientists, technologists, and entrepreneurs by implementing a high-quality "15-minute neighbourhood living circle" and an international community environment, as illustrated in the previous section. This talent ecosystem is built to attract diverse talents with international backgrounds, promoting knowledge spillover and interdisciplinary innovation. In terms of capital flows, leveraging Hong Kong's status as an international financial centre and offshore RMB centre, STT will attract global venture capital and private equity funds to provide financial support for I&T start-ups covering the entire lifecycle.

Global Strategic Vision: Value Chain Integration

Another major advantage of STT is its ability to integrate the entire value chain of "Innovation-to-Manufacturing". As illustrated in Chapter 2 on the core concepts, the research results will move on to the application and manufacturing stages and become a final product. STT will serve as an industrialisation platform for scaling R&D outcomes from initial prototypes to mass applications, connecting the strong advanced manufacturing foundations of cities in the GBA (e.g., Dongguan and Foshan) to achieve large-scale production and quality control of I&T products.

In terms of product output, STT has demonstrated strong market linkage capabilities. As shown in Fig. x, STT not only targets the high-end consumer and corporate markets in North America and Europe, but also actively expands into countries along the "Belt and Road" and emerging economies. Through Hong Kong's mature international trade network, professional services (e.g., international regulatory certification, intellectual property protection) and superior logistics system, STT's hard-tech products, biopharmaceuticals, and smart solutions can effectively enter the global market and achieve the global empowerment of "new quality productivity"

Fig. 7.2.5. Global I&T Network and Factor Flow Vision for STT

7.2.2. Talent Supportive and Rural Community

Urban-Rural Integration

Urban-Rural Integration is not merely about the harmonious spatial transition between urban and rural areas, but it is also about the social and economic integration (**Fig. 7.2.6**).

In terms of spatial integration, the overall design must respect the cultural assets, lifestyle, feng shui configurations and beliefs of the villagers and local residents. Design strategies and principles, including buffer zones with a 300-foot radius from the edge of the last village-type house, are created. Low-density development with sensible built forms, stepped building height profiles, building setback requirements, breezeways, and view corridors will be adopted. The land use around the villages within and beyond the buffer zones is equally significant. The government's approach is to establish green belts and open space land uses around the villages. Although this can preserve the tranquil and natural landscape of the villages, it may also spatially seclude them. Therefore, other uses and facilities that can encourage new residents and talents to visit should be included. However, these facilities must correspond to the history and culture of the area or possess a similar tranquil nature to the area, for instance, libraries, which are not only tranquil in nature but also resonate with the academic background of the Man Clan.

For social integration, spaces with facilities and services that can be enjoyed by both urban residents and rural residents must be provided. These spaces must also facilitate interaction and allow for shared experiences and should be accessible regularly to these two groups of users. To achieve such goals, 15-minute neighbourhoods are designed across the study area (Moreno, 2024). In these 15-minute neighbourhoods, the six core services are included to cater to the needs of both groups of users, and the concept of them will be further elaborated in a later section.

Finally, for the economic aspect, jobs and high-quality education must be provided for villagers and local residents. According to the 2021 Population Census (Census and Statistics Department, 2021), only 2% of the population in the study area works in the information and communications industry. This reflects that the jobs in the technopole may not align with the skill sets and professions of the villagers and local residents. Therefore, it is crucial to think of ways to integrate them economically. For instance, produce from the local villagers or residents can be sold within the community or to labs for experiments in the life and health industry, and jobs in the real estate industry can also be provided to facilitate the renting and buying of properties within the study area. As residents may prefer to work outside of the study area, a mature and efficient transportation network has to be provided to facilitate their commute. In addition, high-quality education should also be provided within the study area for villagers and local residents to facilitate their professional and educational growth in order to narrow the intellectual gap between the tech talents and local residents.

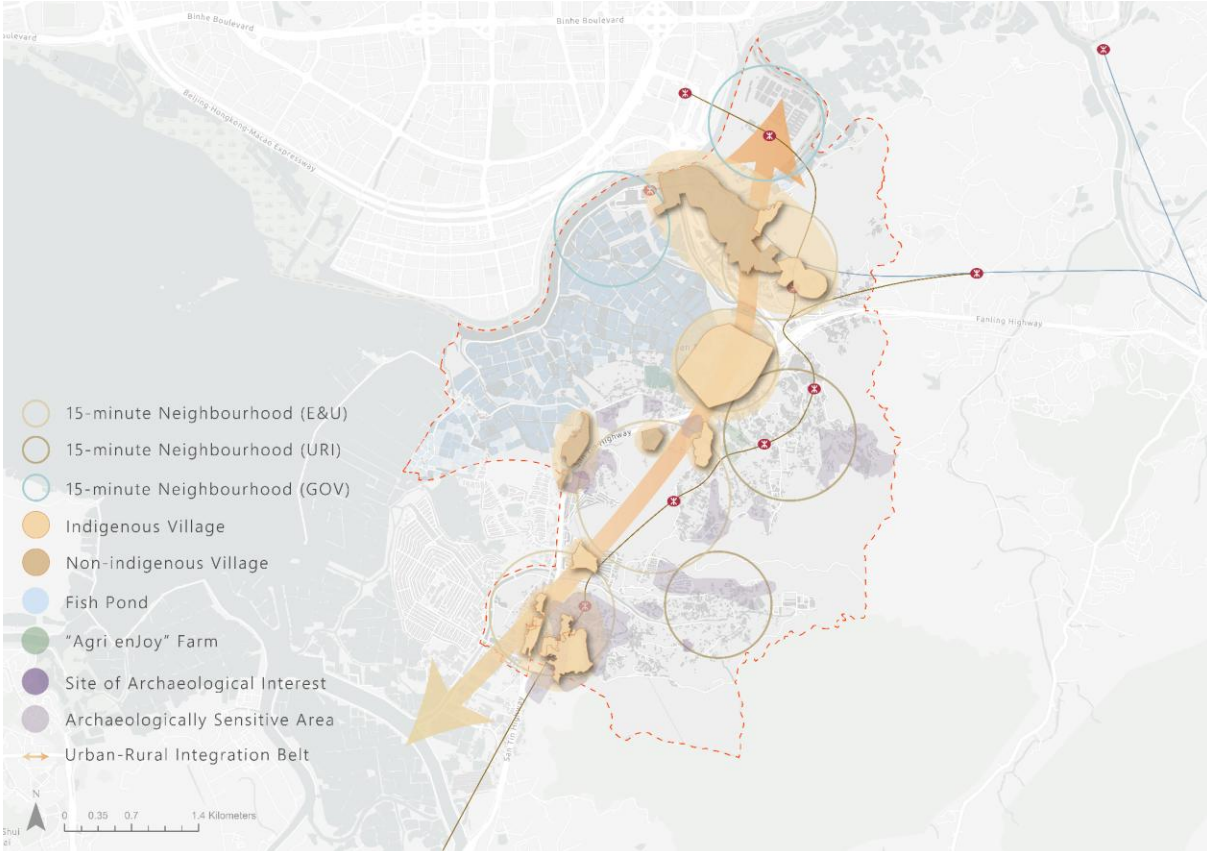


Fig. 7.2.6. SCP for Talent-Supportive and Rural-Inclusive Community

<u>Rural Regeneration</u>

The new development should not only focus on urban-rural integration, but also rural regeneration to enhance the environmental and spatial, social and economic vitality.

In terms of social regeneration, the cultural assets of the villages have to be fully utilised. On ordinary days, the cultural character of the villages may not be prominent. However, during festival days, they create an atmosphere that is rarely seen and experienced in the urban centres of Hong Kong. To ensure the continuity of these festivals and the transmission of cultural knowledge, villagers can actively invite other residents and talents to participate and submerge themselves in the authentic cultural experiences, potentially scaling these festivals to become international cultural festive hotspots like Cheung Chau or Tai Hang.

An adaptive reuse approach for some tangible cultural heritage, the utilisation of fishponds and agricultural land as climate resilience tools and transforming villages into smart and green hubs can be adopted to regenerate the rural areas environmentally and spatially. First, some TCH items no longer serve a functional purpose, but their architecture represents the past of the study area. Therefore, an adaptive reuse approach can be applied to these sites to generate new functions and values for the community. Additionally, fishponds and agricultural lands can serve as components of a sponge city on rainy days, contributing to the climate resilience of the study area, while on dry days, they can serve as productive or recreational spaces. Finally, the villages can incorporate green and new energy solutions developed by the New Energy

Industry in the study area, to evolve and become sustainable and green villages.

Regarding the economic regeneration, ground-level spaces in the villages can be revitalised, and talents and rural residents can collaborate. The villages are often perceived as fenced-up no-go zones for outsiders. To resolve this issue, the ground-floor areas in the villages can be activated with some commercial and creative activities, such as cafes and museums, inviting talents and other residents into the village fabric and enhancing the vibrancy of the area. Furthermore, tech talents and locals can exchange knowledge and tools. For instance, farms can install sensors developed by talents to monitor water and soil quality, while locals can share their traditional ecological knowledge with the talents for their research.

Multi-tier Housing Supply

Drawing on the model of Eindhoven High Tech Campus and the baseline constraints of San Tin, the strategy establishes a three-tier housing system tailored to the needs of I&T talent and existing rural communities.

Government-subsidised talent housing targets high-level R&D and cross-border professionals. Units are located at the northern periphery, buffered from the existing village core to avoid direct early-stage disruption. Rents are set at 30–40% below market rates, with eligibility tied to employment in designated I&T sectors, mirroring Eindhoven's subsidised talent apartments. This tier anchors key talent while minimising conflict with residents.

Market-driven housing combines private developments with small-house conversions, catering to mid-career professionals, families, and existing residents. The model draws on Strijp-S's adaptive reuse of industrial buildings, offering diverse unit types to meet varied needs, including affordable options for young professionals. This tier leverages new and existing land parcels to expand supply, supporting market flexibility and reducing pressure on the subsidised stock.

Enterprise and university accommodation is situated adjacent to workplaces and research hubs, providing short-term and transitional housing for corporate expatriates, interns, and researchers. Following Philips' exclusive staff dormitories and Eindhoven University's shared housing model, large enterprises lease and renovate entire buildings to create low-cost, fully serviced facilities. Partnerships with local universities ensure seamless housing for collaborative researchers, supporting a live-work-learn ecosystem.

The tiered structure balances talent attraction with community stability. By separating initial talent housing from sensitive village areas and embedding market and enterprise units throughout the site, the strategy reduces displacement risks and ensures all talent segments have access to suitable, affordable housing.

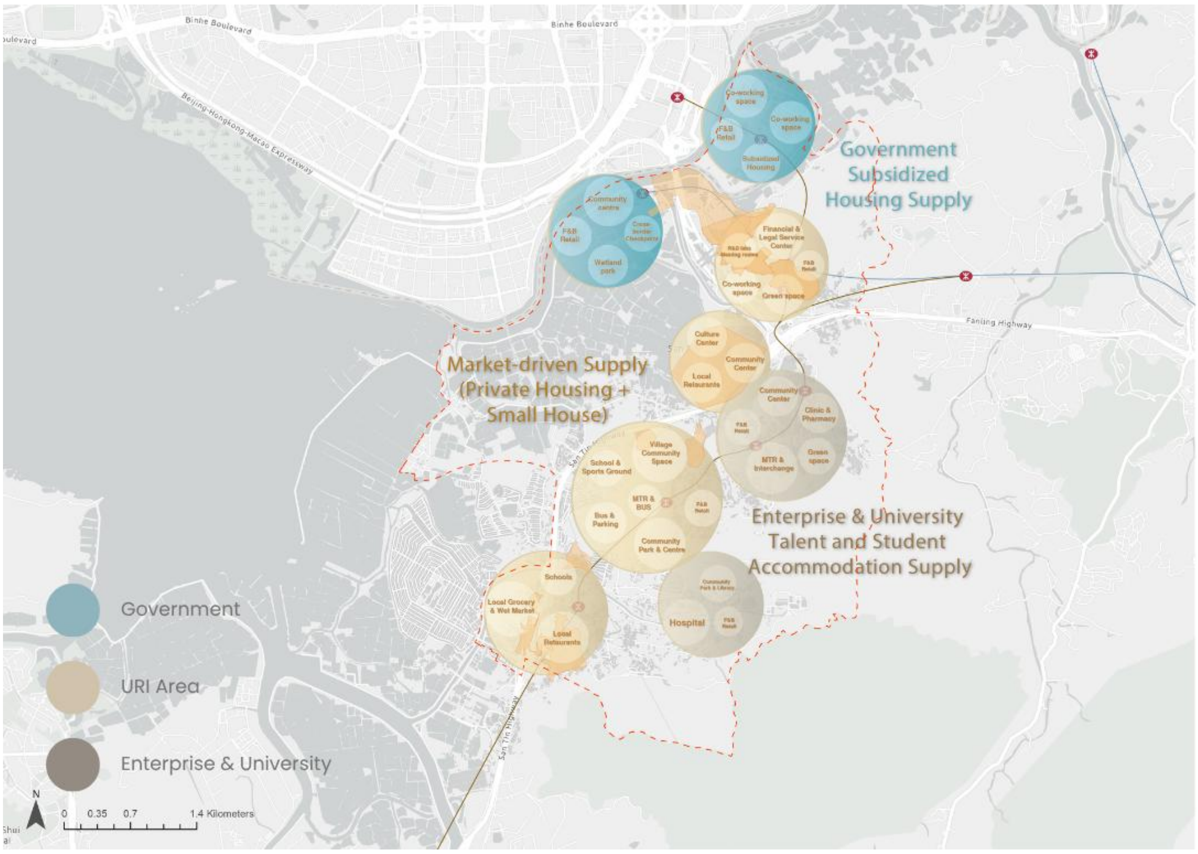


Fig7.2.7. SCP for Multi-tier Housing Supply

<u>15-Minute Living Circles</u>

The 15-minute living circle framework integrates talent housing, workplaces, and rural assets to create inclusive, self-sufficient neighbourhoods. Each circle is planned around three core goals: accessibility, vibrancy, and inclusivity.

Accessibility is prioritised through pedestrian and cycling networks that connect housing, workplaces, healthcare, education, retail, and green spaces within a 15-minute walk or bike ride. This reduces car dependency, cuts commuting times, and lowers carbon footprints. The design leverages the site's existing rural and ecological features to create safe, healthy routes for all users.

Vibrancy is fostered by embedding community and cultural facilities within each circle. Shared workspaces, recreational hubs, and heritage-themed gathering spaces are designed to remain active after work hours and on weekends, catering to students, young professionals, and local residents alike. These spaces draw on Strijp-S's mixed-use model to blend creative studios, commercial facilities, and public spaces.

Inclusivity is central to the design. Facilities are planned for shared use by new talent and existing residents, with multilingual services and programmes that connect the two communities. Circles are linked by regional green corridors and public transport routes, ensuring access to larger employment centres and services. The result is a cohesive ecosystem where talent and local residents can live, work, and thrive together.

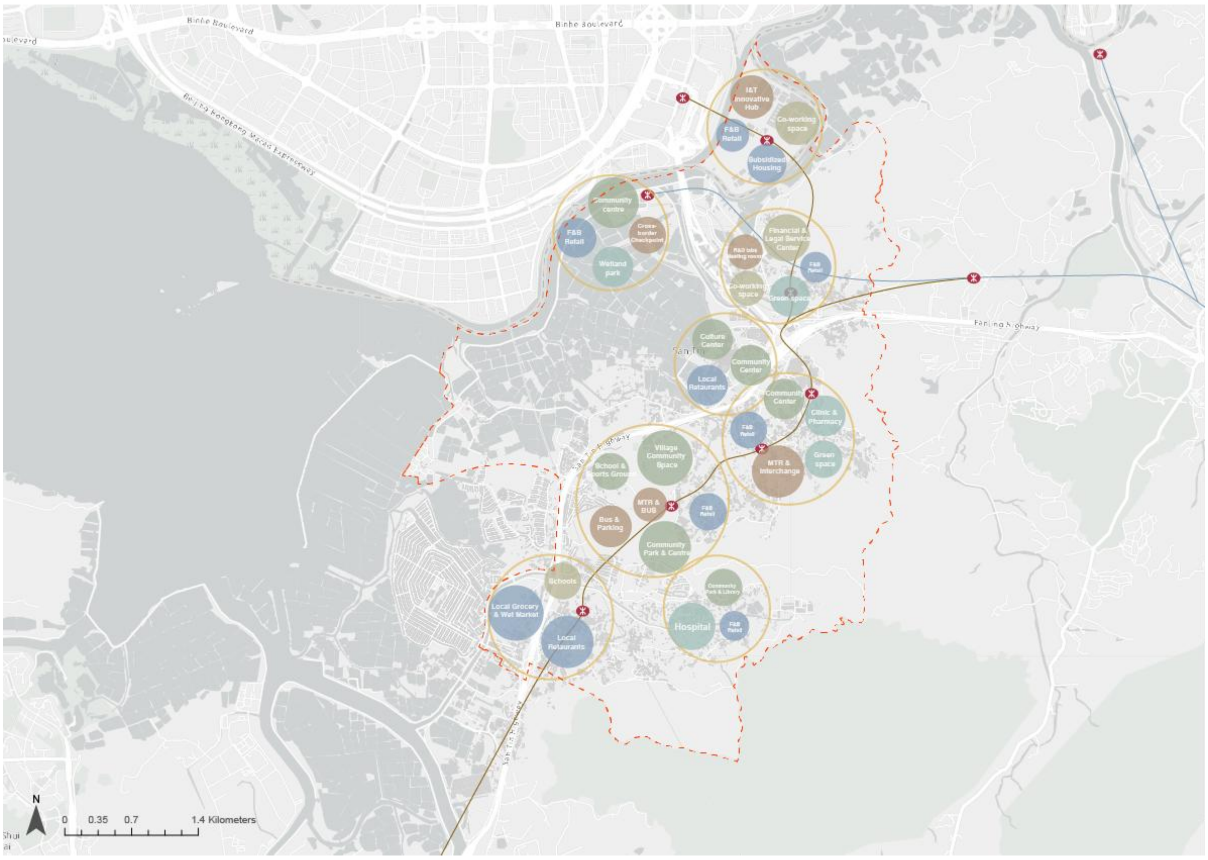


Fig. 7.2.8. SCP for 15-Minute Living Circles

7.2.3. Environmental Sustainability and Carbon Neutrality

<u>Wetland Protection & Avoidance</u>

The avoidance-first approach is introduced to frame the extent of the proposed conservation area in **Fig. 7.2.9** to prioritise degraded brownfield land, keeping the Important Bird Area and ecologically sensitive fishponds outside the area of disturbance. Also, a dual-metric no-net-loss standard will be introduced, which requires both habitat area and ecological function to be maintained concurrently. Drawing on international precedent from Taiwan, the United States, and South Korea, where both area and function are recognised as legal requirements (Conservancy Association, 2026). This standard addresses a critical gap in the government's current approach, which measures ecological function alone and therefore permits habitat area to contract as long as measurable performance is sustained (EPD,2021). The rationale for this strategic direction is grounded in the existing designation map. **Fig. 7.2.10** shows that the site is surrounded by overlapping layers of highly ecologically sensitive areas, including the Ramsar Site, SSSI, and conservation area. These designations form the ecological backbone of the entire ecological corridor of Deep Bay, so it is vital to minimise the disturbance to ecology by providing a habitat with high ecological continuity.

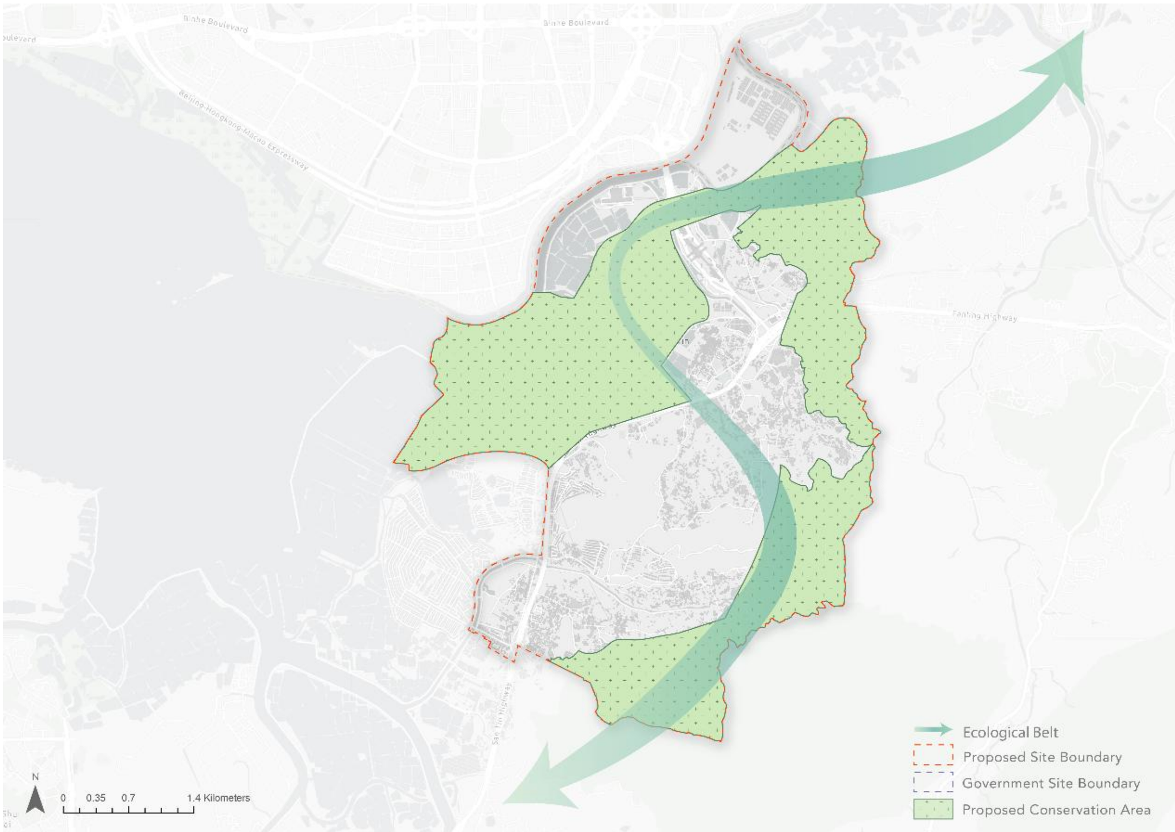


Fig. 7.2.9. Proposed Conservation Area

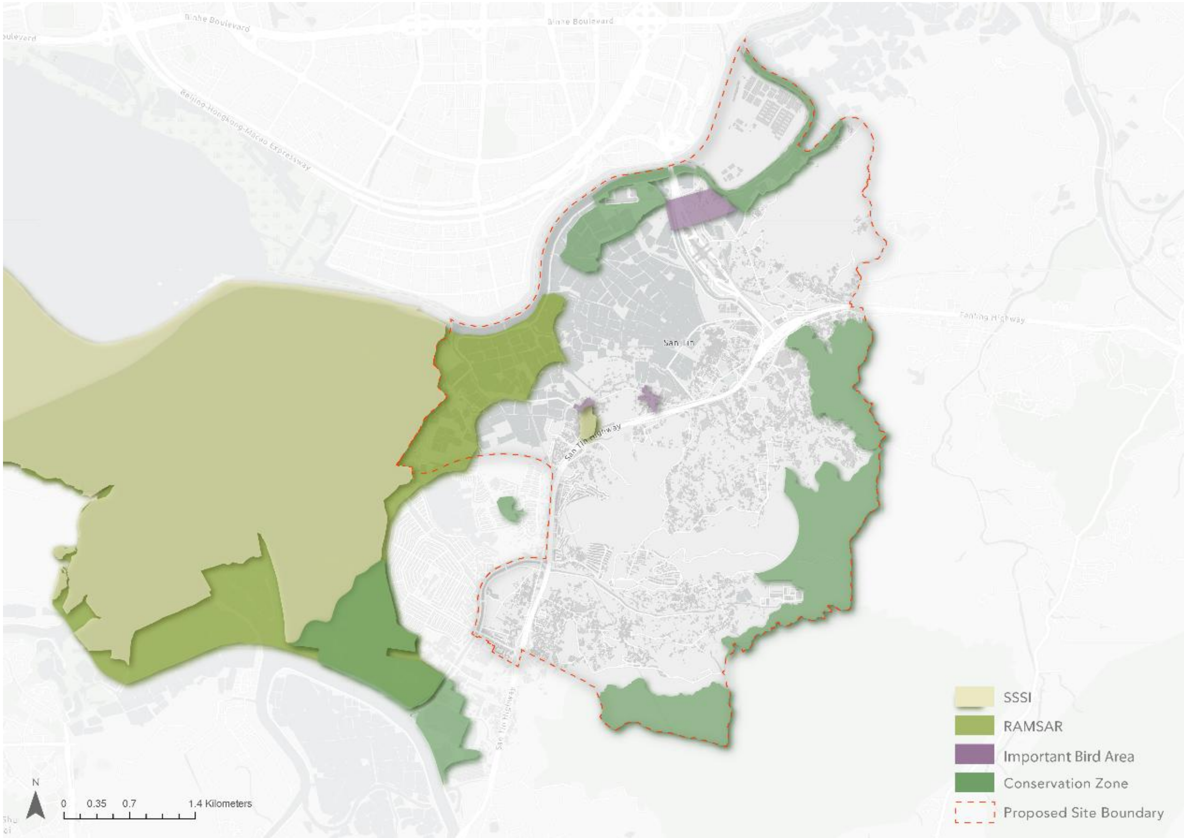


Fig.7.2.10. Existing Ecologically Sensitive Area
Source: EPD (2024)

<u>Blue-Green Network</u>

A continuous Ecological Belt will be introduced to connect every existing conservation area into a single coherent conservation network. Referring to **Fig.7.2.9**, along the western corridor, the conservation area expands beyond the government's SPS WCP boundary, and a continuous wetland spine linking Mai Po Nature Reserve northward to the Shenzhen River is created. Along the northern arc, a formal transboundary designation will be established along the Shenzhen River boundary, creating an enforceable cross-boundary connection to Shenzhen's ecological reserves. In addition, the eastern conservation strip will close the ecological loop along Fanling Highway to link up the eastern conservation area to the wider network.

As San Tin serves as a critical midpoint of a continuous transboundary corridor spanning Hau Hoi Wan, Shenzhen, Mai Po, Nam Sang Wai, Lau Fau Shan, Pak Nai, and Ho Hok Wai. As reflected in **Fig.7.2.9**, the Ecological Belt addresses the current isolated patches to the north and west **(Fig. 7.2.10)** by structurally reorganising the spatial relationship between every existing designation so that species, water, and ecological processes can move continuously across the network.

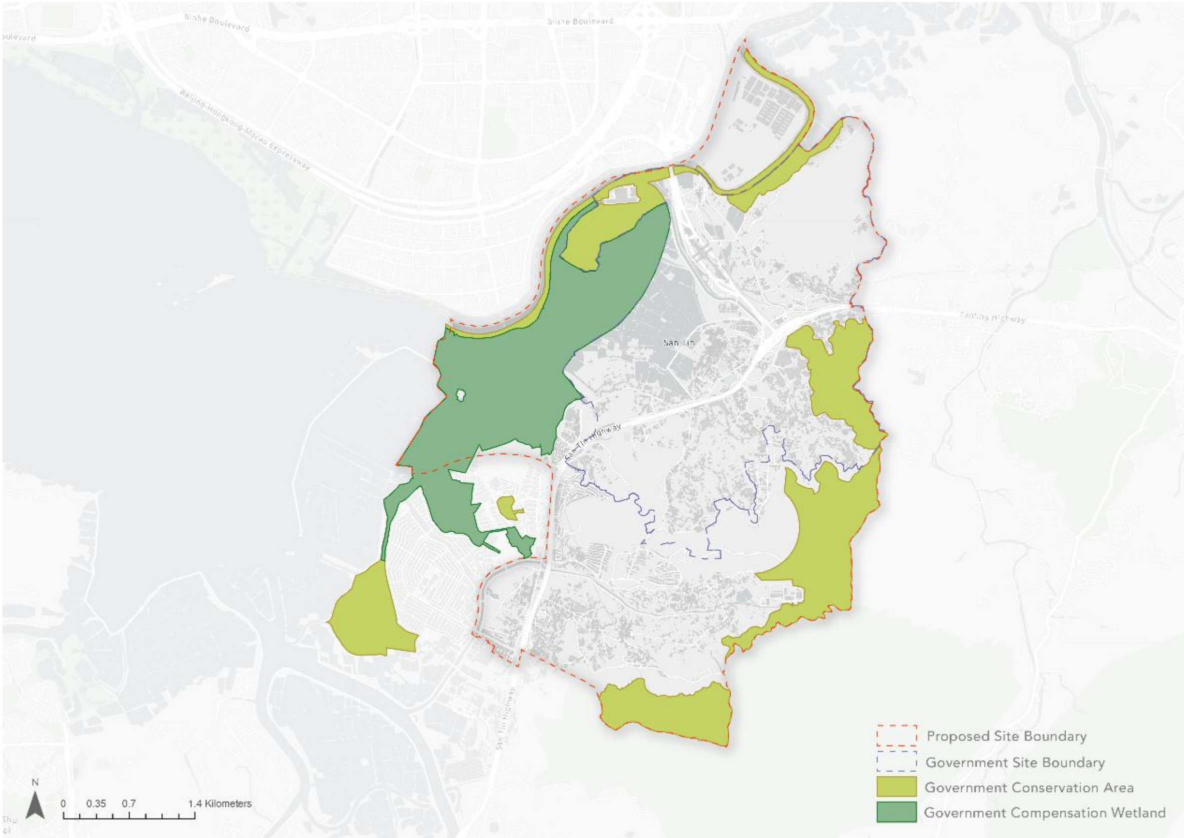


Fig.7.2.11. Government Plan on Ecology (Source: STT, 2025)

<u>Nature-based Solutions (NbS)</u>

Nature-based Solutions will be adopted to create better and even new ecological functions and habitats to enhance the ecological value of the conservation area. At the species level, the targeted species scope will cover a fuller range of species whose presence reflects the overall health of the wetland ecosystem. A wider species coverage provides a more robust and honest basis for assessing whether the network is genuinely functioning and improving over time, rather than simply satisfying the minimum requirement of compensation. At the habitat level, the revitalised drainage channels, managed fishpond sequences, and floodable landscape zones within the conservation belt are planned to generate new ecological functions, including foraging habitat, movement corridors, and seasonal wetland conditions that the landscape does not currently provide. It is expected that the connected network performs at a higher ecological level than the sum of the individual designated areas it brings together and sustains the enhancement over the long term.

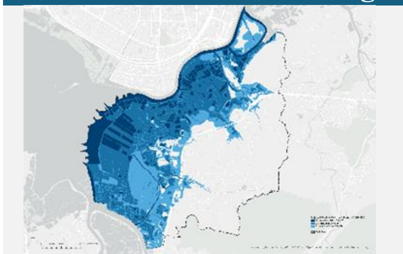
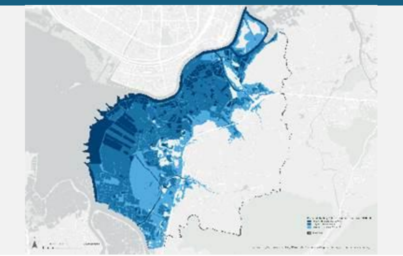
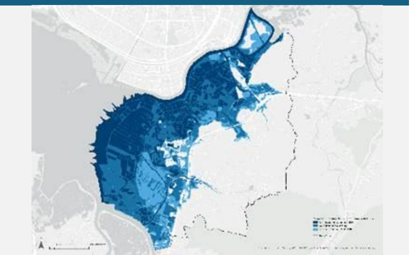
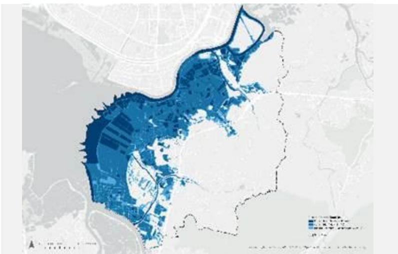
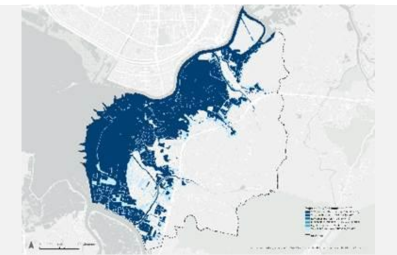
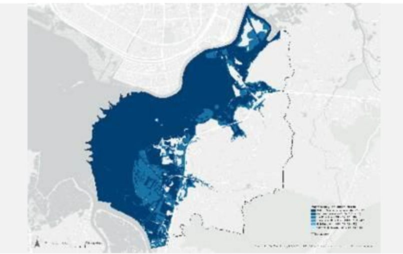
<u>Climate Resilience</u>

In the previous development plan for STT, the government emphasised the SGR initiative to address climate change. However, the current SGR design remains a passive and elementary approach. It focuses on reducing potential losses due to climate impacts, instead of actively enhancing the site's capacity and resilience to cope with climate change. To go beyond mere conservation, this proposal adopts a more proactive "Supply, Recover, and Restore" approach. Based on the Climate Resilient Development Pathways suggested by the IPCC (2023), this climate resilience plan has three main focuses to enhance the climate mitigation and adaptation ability of STT (**Fig.7.3.2.1**).

Flood Mitigation through Scenario-Based Modelling

Future socio-economic shifts will affect the effectiveness of climate mitigation measures under different climate conditions, especially regarding the sea level rise in STT. Having analysed different scenarios (**Table 7.3.1**), this plan adopted the SSP5-8.5 low confidence scenario, which suggests the daily high tide level reaching 4.0 mPD and extreme event reaching 5.1mPD in 2100 (HKO, 2022). To protect the site against extreme climate conditions such as storm surges, existing fishponds should be avoided for development and retained as floodable landscapes. Furthermore, detailed flood mitigation designs and infrastructure at the fishpond, such as sloped pond edges to improve water circulation and underground stormwater storage, should be integrated to enhance flood mitigation capacity under extreme climate events.

Table 7.2.1. Different Scenarios regarding Sea Level Rise in STT

Scenario-Based Modelling		
		
SSP1-2.6	SSP2-4.5	SSP5-8.5
		
Baseline	High Tide	Storm Surge

Nature-based Climate Park for Climate Adaptation

Considering the vulnerability of local ecosystems under climate change, there is an urgent need to adopt NbS to enhance climate adaptation. Besides compensating for the biodiversity loss, NbS also strengthens ecosystem resilience and addresses societal challenges (CEDD and AFCD, 2026). Rather than converting fishponds into I&T sites, the existence of fishponds is needed to safeguard bird flight paths and maintain a continuous ecological corridor. Beyond conservation, active regeneration of the filled or inactive fishponds resolves the problem of low ecological value and further enhances the biodiversity (HKSAR, 2024). Through further integration of wetlands and mangroves, the restored natural vegetation acts as a natural buffer to combat rising sea levels and storm surges. The restored habitats of the fishponds can further be transformed into a climate park that provides both environmental education and recreational space, which enhances the environmental awareness of the community.

Smart Technology and Cross-boundary Flood Monitoring

Apart from climate mitigation and adaptation approaches, an effective monitoring system is important for enhancing future interventions (IPCC, 2023). Given the nature of STT as an I&T hub, it can nurture innovative monitoring systems by adopting smart technology. Moreover, STT shares a similar climatic profile with Shenzhen, located at the opposite side of the Shenzhen River. Shenzhen has already adopted smart technology in monitoring, such as AI-based techniques for heavy precipitation nowcasting and real-time monitoring (WMO, 2025). Therefore, cross-boundary technology and knowledge transfer should be encouraged for regional monitoring, in view of the location proximity and close collaboration between the two places.

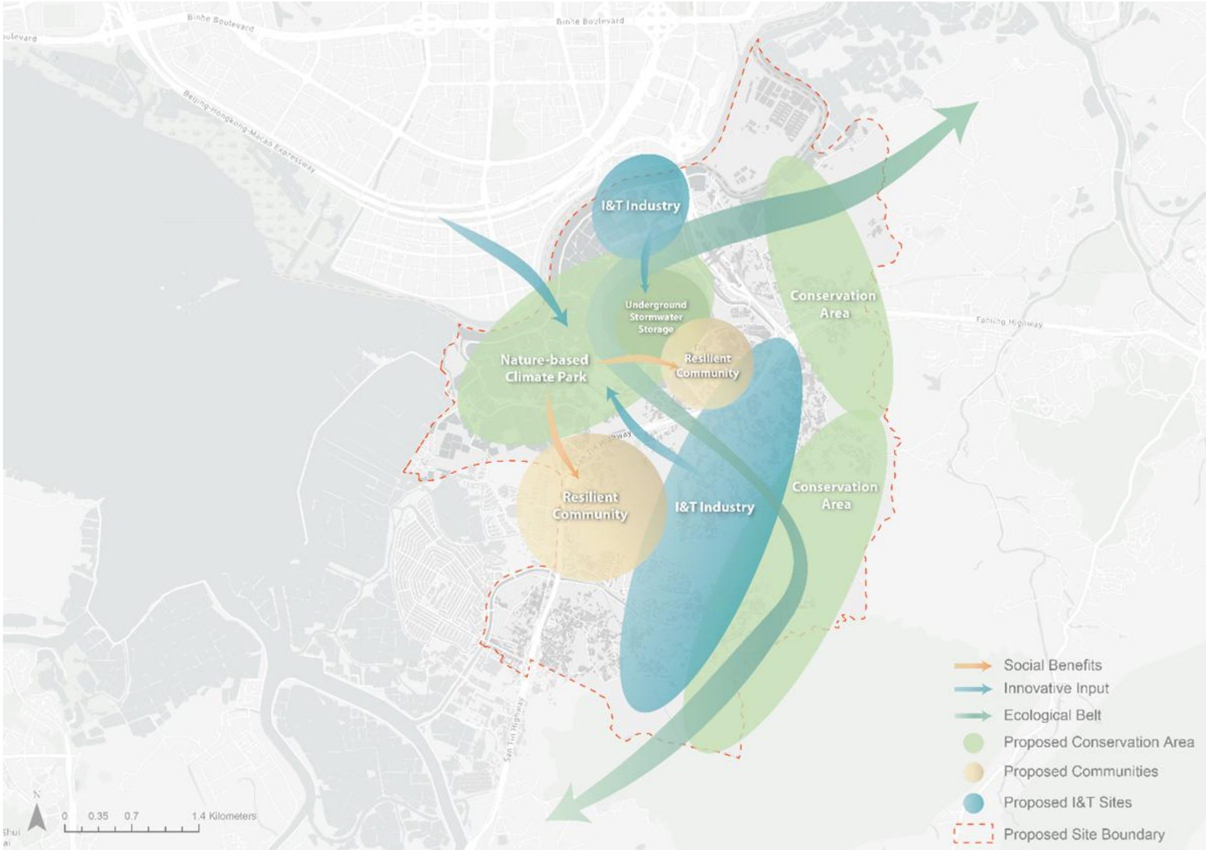


Fig.7.2.12. Climate Resilience Mapping

Carbon Neutrality

To achieve the transition toward carbon neutrality, a Circular Economy Framework is proposed to close the energy, water, and material loops across the Technopole. As illustrated in **Fig. 7.2.13**, resource input, such as renewable energy and raw materials, enters the systems, and the advanced manufacturing maximises the production efficiency and minimises energy consumption at the source. Then, waste from consumption and production is managed for recovery and reuse. Residual waste that cannot be recycled is directed to waste-to-energy facilities to generate energy back into the system. Besides, modern logistics further minimises the carbon intensity of material movement across the supply chain. It is expected that this model could move the Technopole away from the conventional linear model toward a continuously closed loop that supports carbon neutrality.

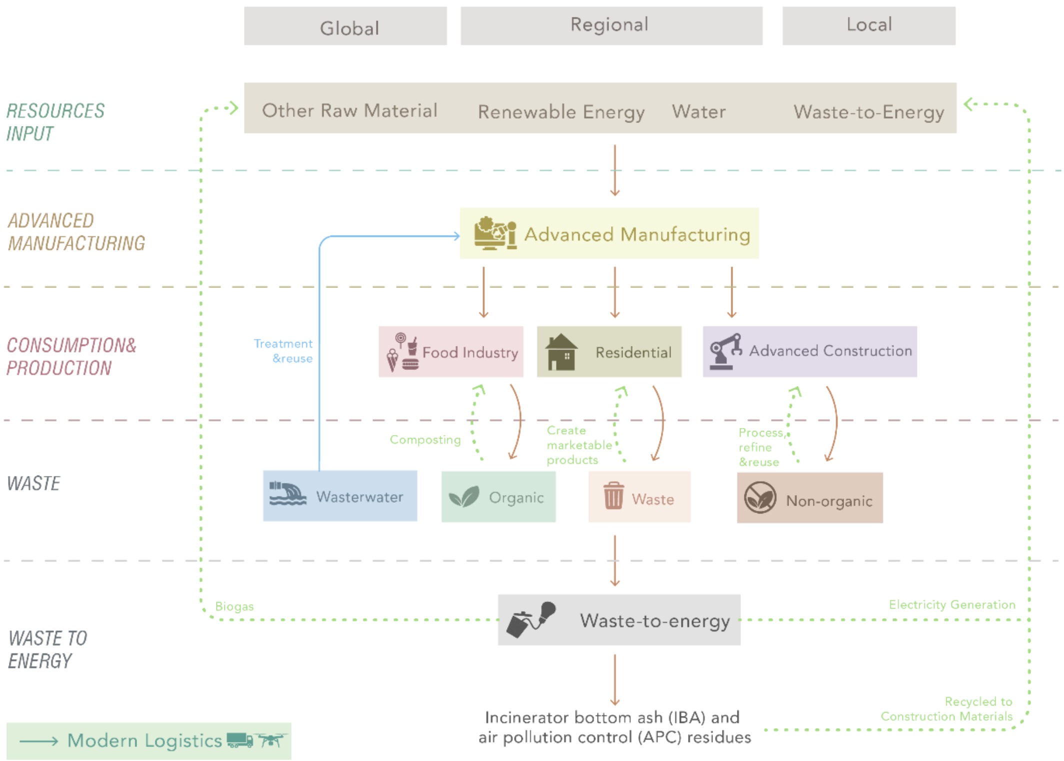


Fig. 7.2.13. Circular Economy Framework

<u>Energy Loop</u>

To supply the framework of the circular economy, it is targeted to use 100% nuclear and renewable energy for the energy consumption in Hong Kong. According to **Fig. 7.2.15**, energy will be sourced from large-scale renewable energy infrastructure in mainland China, such as the Xinjiang and Gobi Desert Solar Complex for solar energy, Fujian Wind Farms for wind energy, and the Daya Bay Nuclear Energy Plant for nuclear energy. Also, for the energy loop **(Fig. 7.2.15)**, renewable energy flows into production to support manufacturing and operational processes. Then waste generated from production is directed to waste-to-energy facilities in Shenzhen and Hong Kong, where organic energy, heat, and recovered materials are fed back into the production and distribution cycle.

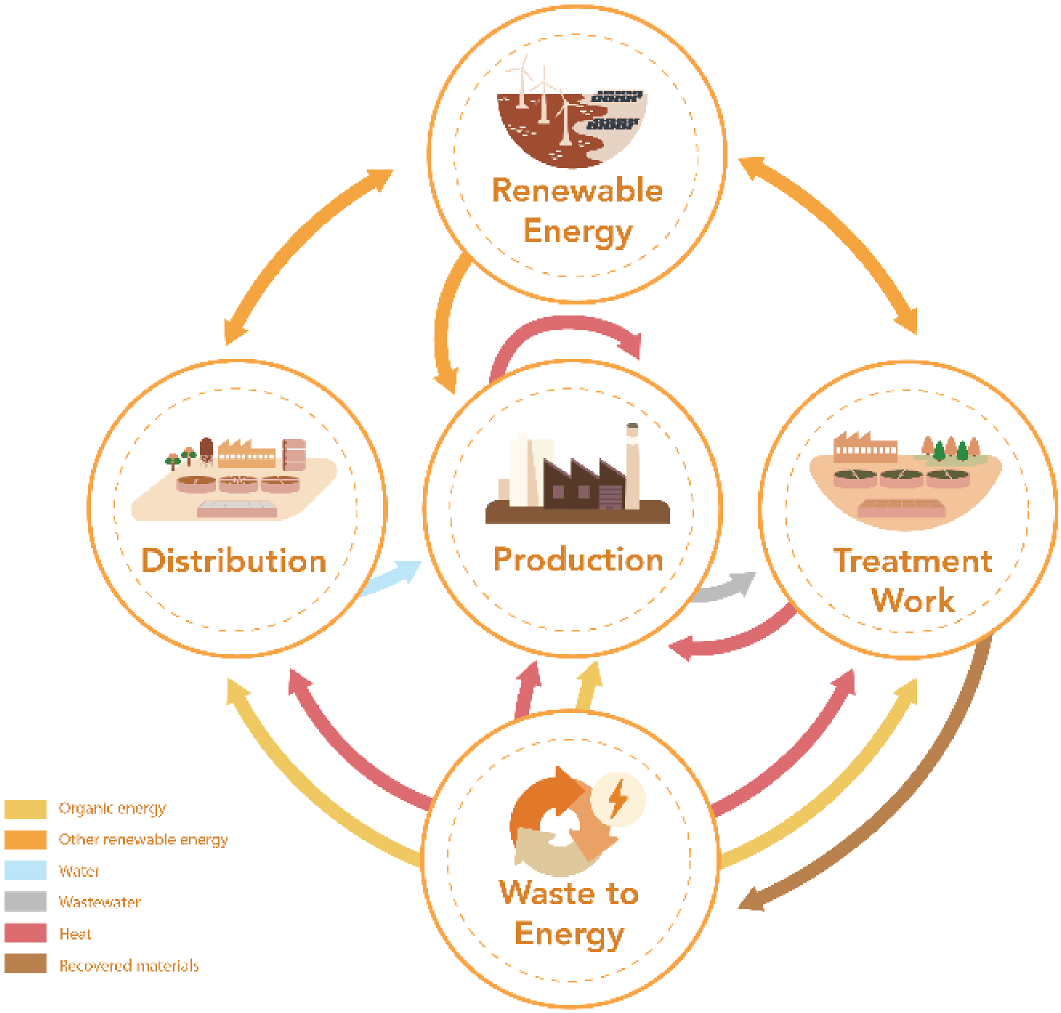


Fig. 7.2.14. Proposed Energy Pathway

Fig. 7.2.15. Proposed Energy Flow

<u>Water Loop</u>

Hong Kong's dependency on the Dongjiang water with linear water metabolism will gradually shift to a closed-loop water system while actively supporting the ecological network. Within the loop, wastewater from domestic, agricultural, and industrial use will be directed to treatment works and processed into reclaimed water for re-consumption (**Fig.7.2.16**). Meanwhile, Nature-based Solutions will be integrated into this process, where constructed wetlands, vegetated drainage channels, and managed fishpond sequences within the Ecological Belt serve as natural filtration systems to purify water through bioengineered systems before it is returned to the environment to generate a freshwater supply and feed back into the habitats to maintain wetland hydrology and biodiversity across the conservation corridor without drawing on potable supply.

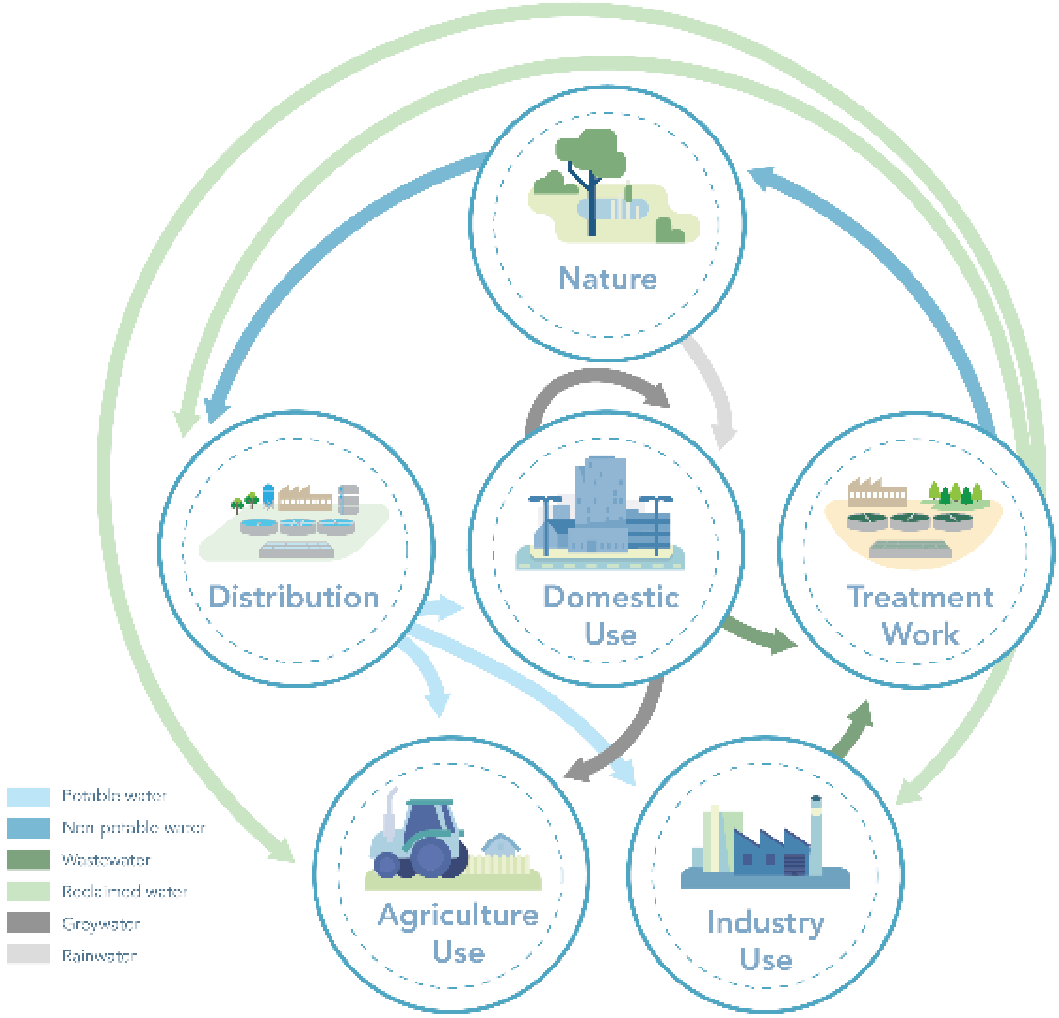


Fig.7.2.16. Proposed Water Pathway

<u>Material Loop</u>

For raw materials, the material pathway shifts from importing from the mainland instead of sourcing worldwide to reduce carbon footprint, which includes key raw materials, rare earth elements, silicon, etc., for targeted industries **(Fig. 7.2.18)**. Then, EcoPark and WEEE·Park will be strengthened as material recovery hubs **(Fig. 7.2.19)**, where recovered metals from electronic waste are reprocessed and reintroduced as secondary raw materials. After the production, in the waste aspect, waste is managed to maximise recovery and minimise leakage. Organic waste will be directed to O·PARK1 and O·PARK2 for biological treatment, producing outputs that re-enter the local cycle. Non-organic industrial and municipal waste will be redirected to Shenzhen's Energy Ecological Parks **(See Fig. 7.2.19)** for waste-to-energy conversion, keeping energy recovery within the GBA rather than exporting waste burden globally to close the material loop within the region.

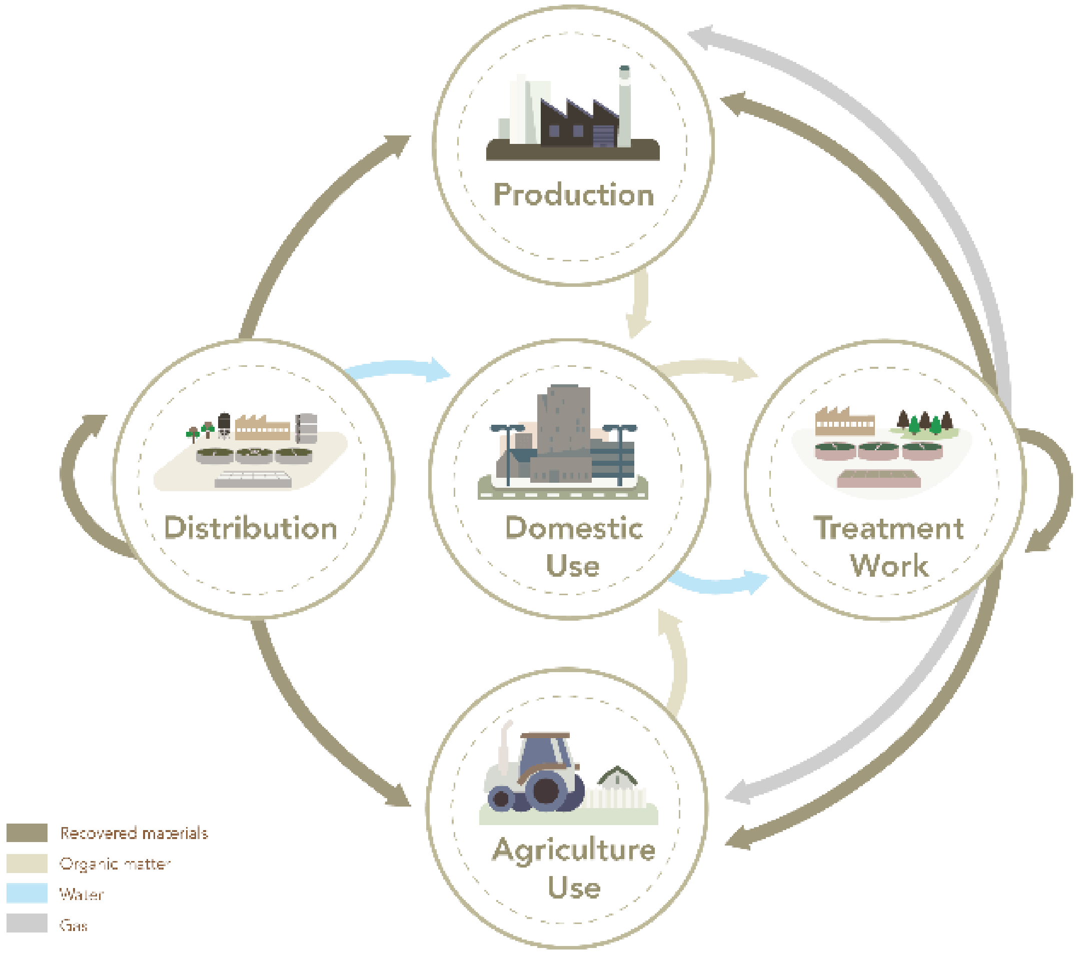


Fig.7.2.17. Proposed Material Pathway

Fig 7.2.18. Proposed Material Flow (Source of Raw Material)

Fig 7.2.19. Proposed Material Flow

7.3. Overall Draft SCP

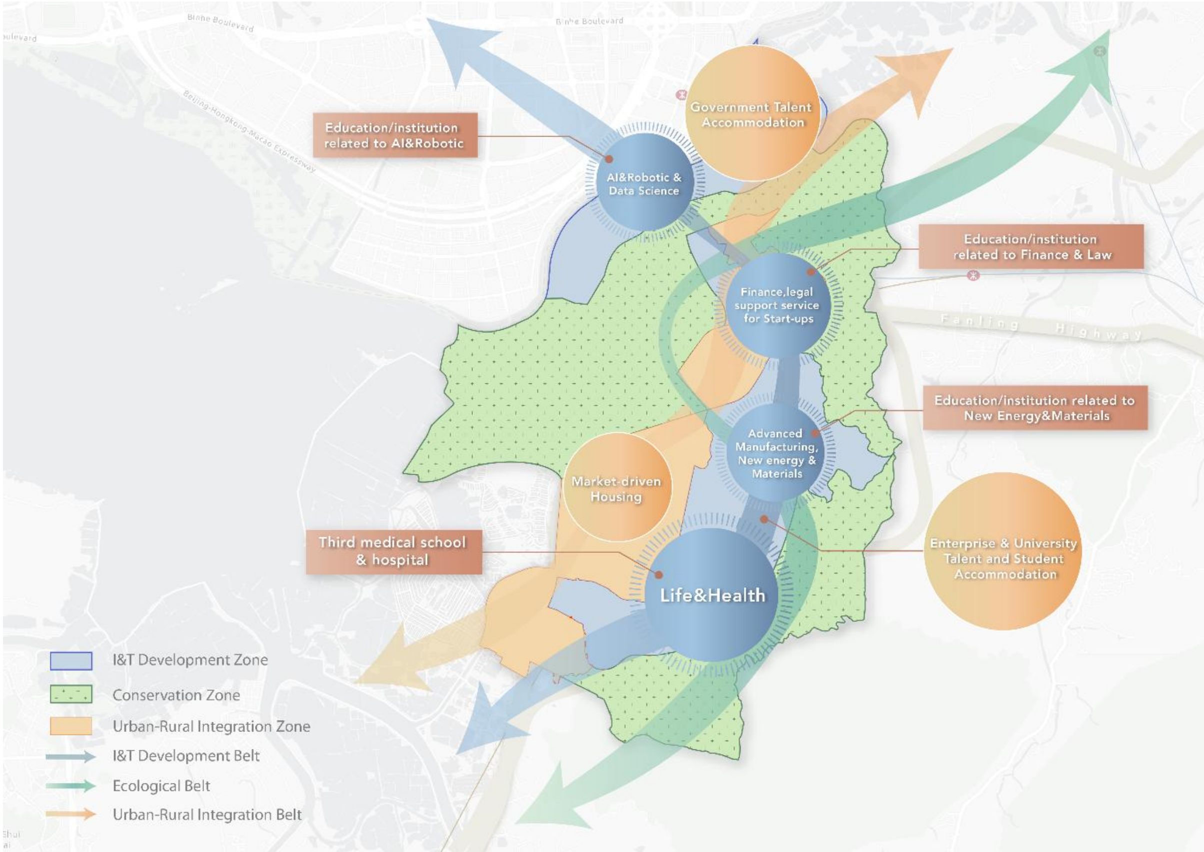


Fig. 7.3.1. Proposed Draft SCP

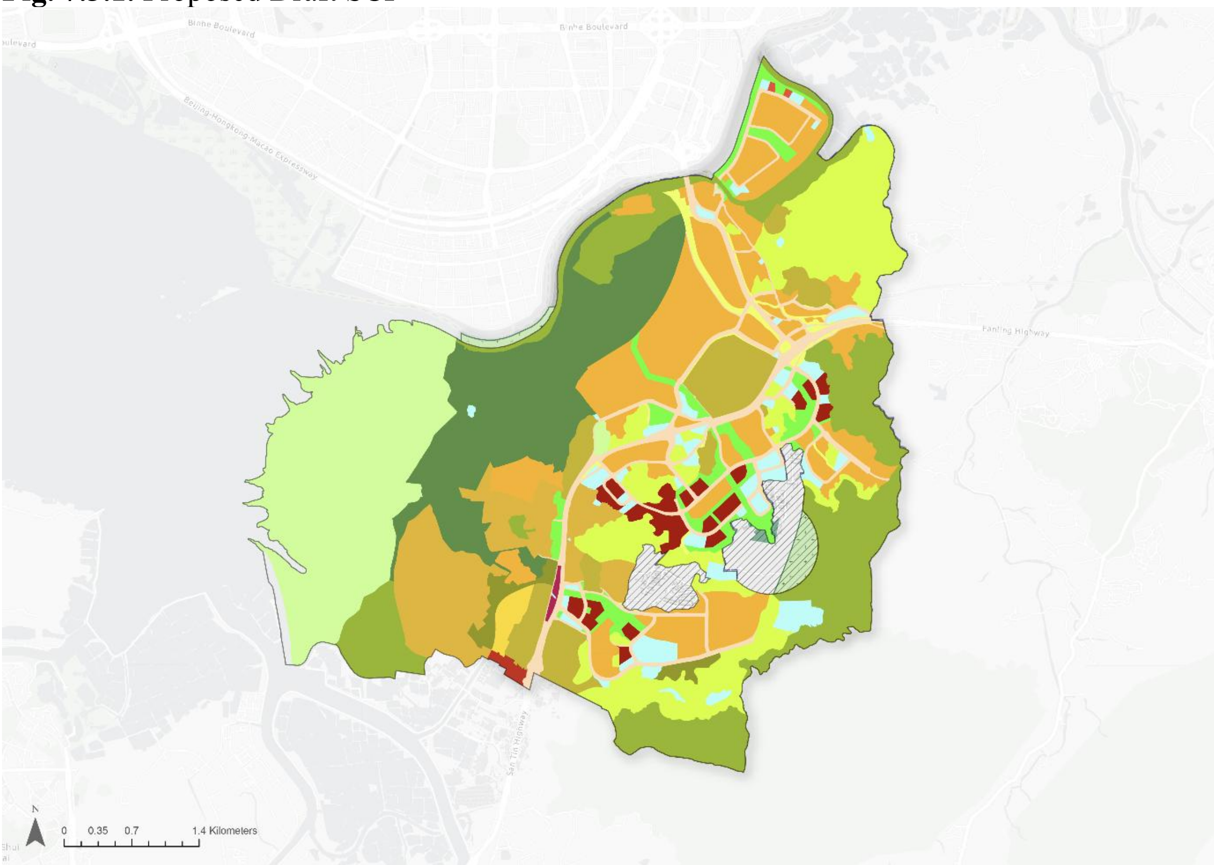


Fig. 7.3.2. Government-Planned OZP

8. Conclusion

This study, which revolves around the strategic positioning of STT as the core of the NM of Hong Kong, completes a comprehensive analysis covering baseline assessment, reference to best practices, and preliminary SCP. The research has shown that STT boasts cross-border locational advantages spanning Shenzhen and Hong Kong, top scientific resources, rural culture and ample development space, all of which are the key factors for Hong Kong to break through in innovative industries, attract international talent, and drive sustainable development.

The baseline assessment clearly revealed the development of the bottleneck: a broken industrial transformation chain of scientific research from laboratory achievement to large-scale commercial application (i.e., “1 to 100”), soaring housing costs for talents, inadequate supporting public services, fragmentation of the ecological corridors, and the systematic and spatial barriers of cross-border movement and integration. The international case comparison shows that the breakthroughs in the new field are: the construction of a holistic “research-testing-production” industry, the implementation of a tiered and categorised talent housing system, the construction of a 15-minute life circle for urban and rural integration, and the implementation of a living balance between conservation and development with a blue-green network.

The strategic planning takes “Innovate Together, Thrive Sustainably, Advance the World” as its vision, and establishes three major strategic focuses: I&T development and regional synergy, talent-supportive and urban-rural inclusive communities, and environmental sustainability and carbon neutrality. Measures have been proposed not only to overcome the baseline shortcomings but also to align with the national strategy and Hong Kong’s positioning as an innovation and technology hub.

The development of STT is not just a land development project, but also an exploration of a model for Shenzhen-Hong Kong collaborative innovation, harmonious coexistence between urban and rural areas, and ecology-first development. The key to its success lies in the collaboration between institutional innovation and spatial design, which is achieved through cross-border integration and a balanced system that connects talent with residents, and development with protection. Looking ahead, with its planned implementation, STT is expected to become the core engine that supports Hong Kong’s innovation transformation, leading the GBA to a new phase of upgrading and achieving coordinated improvements in economic, social, and environmental values.

Appendix

Appendix I

Summary of Interviews Conducted

Name / Organisation	Background / Position	Field
The Conservancy Association (長春社) Rep: Mr. Roy H.M. Ng	Campaign Manager	Environment; Rural
Dr. Ming Tao	Founder of Atom Xquare Limited	IT
Dr Francis Cheung Neoton	Chairman of Doctoral Exchange; Member of the Chief Executive's Policy Unit Expert Group; Advisor of Our Hong Kong Foundation	Governance
Mr Wong, Kam-Sing	Former Secretary for the Environment	Environment
Proptech Smart Space Rep: Mr. Henry Jiang; Dr. Godfrey Ho; Mr. Perry Chan; Mr. Wk Wong	Startup Entrepreneurs	Talent
Prof. Man Hau-Chung	Cheng Yick-chi, Chair Professor in Manufacturing Engineering; Chair Professor of Materials Engineering; Dean of the Faculty of Engineering, and Director of University Research Facility in 3D Printing; Local villager	IT; Rural

Appendix II

Interview Minutes (1)

Interviewee: Mr Roy H.M. Ng (The Conservancy Association)
Interview Place: Jockey Club – The Conservancy Association Urban Forestry Green Hub
Interview Time: April 9, 2026, 11:00 AM

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1. Key Concerns: Santin Development & Ecological Impact

- **Large-scale development:** ~90 hectares, one of the largest in recent years.
- **Significant opposition:** Due to ecological sensitivity.
- **Site Location:** Located within an Important Bird Area (IBA) and a major bird corridor.
- **Regional Ecological Context:** Not limited to Santin; part of a broader ecological network:
    - Hau Hoi Wan – Shenzhen – Mai Po
    - Nam Sang Wai, Lau Fau Shan, Pak Nai, Ho Hok Wai
    - Functions as a continuous ecological corridor.
- **Core Issue:** Santin sits in the middle of this corridor, causing fragmentation.
- **Key Question:** How to ensure ecological connectivity?

2. Eurasian Otter Concerns

- **Evidence from green groups** (camera traps) shows presence:
    - Sam Po Shue (north of site)
- **Contradiction:** EIA report claims absence.
- **Physical evidence:** Otter spraints (poo) found within Santin.
- **Significance:** Species is:
    - High conservation value
    - On the red list
- **Impact Issues:**
    - Direct and indirect impacts not sufficiently addressed.
    - Mitigation measures lack depth and diversity.

3. Habitat Creation & Management Plan (HCMP)

- **~13 pages outlining:**
    - Habitat types (e.g., marsh, ponds)
    - Target species
- **Concerns:** Requires:
    - Careful ecological design
    - Long-term management
- **Key Gap:** Lack of clarity on:
    - Long-term funding
    - Financial sustainability
    - Resource commitment

4. "No Net Loss" Principle

- **Current guideline:** Prioritises ecological function over area.
- **Concerns:** Area still matters for ecological performance.
    - International references (Taiwan, US, South Korea):

    - Both area + function are essential.
- **Key Question:** How is “no net loss” defined and measured?

5. Mitigation Measures & Technical Feasibility
- **Proposed Measures:** Animal passages for otters.
- **Concerns:** Lack of proven cases.
- **Unclear technical details:**
    - Width
    - Lighting conditions
- **Other Issues:** Limited mitigation strategies overall. Need for:
    - More ponds
    - Better habitat accessibility

6. Flood Control vs Ecology
- **Santin East River:** Includes flood prevention infrastructure:
    - Water fences may block otter movement.
- **Conflict:** Climate resilience vs habitat connectivity.
- **Suggestion:** Seasonal or flexible operation:
    - Open only during extreme weather.
- **Issue:** Technical details not yet provided; approach: pass EIA first, design later.

7. Biodiversity Considerations Beyond Birds
- **Fish Ponds:** Even abandoned ponds:
    - Provide habitat for migratory birds (e.g., ducks).
    - Misconception: “low ecological value”.
- **Species Coverage:** Current mitigation targets are too limited.
    - Insufficient to prove effectiveness.
- **Other Species:** Fireflies (e.g., in Mai Po, Hau Hoi Wan):
    - Potential presence in Santin.
- **Lighting Impact:** Need detailed control of:
    - Light intensity
    - Direction
    - Example conflict: Bird habitats located near playgrounds with lighting.

8. Alternative Development Strategy
- **Question necessity:** Of developing Santin.
- **Suggested Alternatives:** Develop:
    - San Tin city centre
    - Pan Wu Tei greenbelt (for innovation/commercial use)
    - NTM
- **Key Idea:** Strategic spatial planning is critical for industry formation.

9. Community & Cultural Impacts
- **Affected Villages:** Ha Wan Tsuen:
    - Non-indigenous village
    - Strong social networks across villages
    - Unique traditions (e.g., 搶花炮)
- **Cultural Practices:** 土地誕, 盂蘭節, local festivals.
    - Social rather than purely religious significance.

- **Recommendations:**
    - **Preserve:** Landmarks (village office, trees, signage).
    - **Include:** Multi-purpose plaza for cultural activities.
    - **Engage:** Social work teams as mediators.

10. Planning & Design Considerations

- **Preserve village identity:** (e.g., archways, symbolic elements).
- **Retain agricultural character:** In NTM.
- **Maintain local institutions:**
    - 信芯園
    - 菜站 (if applicable)

11. Sam Po Shue & Economic Activities

- **Not only ecological importance:** Fishing/agri-business should be preserved.
- **Engagement:** Need direct dialogue with operators.
- **Reference Model:** Long Valley conservation model seen as a positive example.

12. Stakeholder Engagement & Governance

- **EIA included:** 4–5 rounds of consultation.
- **Mechanisms:**
    - Environmental committee with representatives.
    - Regular and informal meetings.
- **Remaining Issues:** Some conflicts unresolved (e.g., village relations).

13. Suggestions for Improvement

- **Reduce development scale.**
- **Strengthen:**
    - Ecological mitigation strategies
    - Species coverage
    - Technical design details
- **Address:** Sam Po Shue compensation.
- **Improve:** Long-term conservation funding mechanisms.

Interview Minutes (2)

Interviewee: Dr Ming Tao (Founder of Atom Xquare Limited)
Interview Place: 16 Science Park West Avenue, HKSTP
Interview Time: April 10, 2026, 10:00 AM

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1. Startup Funding & Support

- **Funding as the Most Important Need:** The journey from ideas to commercialisation is long; the funding gap is the biggest challenge for survival.
- **Government Support:** Subsidies, rental concessions, and supporting resources are essential to bridge the profitless intermediate phase.
- **National Policy Impact:** Projects falling within nationally supported industries attract more VC investment ("following the trend"), though this increases competition.
- **ITF Matching Funds:** Easier to obtain when collaborating with universities, as government risk is reduced by academic leadership.

2. The Science Park Ecosystem

- **Network Value:** The park's strength lies in its ecosystem of 10,000+ people, making it easy to connect with partners, clients, and suppliers.
- **Dual Company Structure:**
    - **Incubatees:** Receive comprehensive subsidies (rent, patents, exhibitions).
    - **Ordinary Tenants:** Pay full rent but leverage the "Science Park" brand and title.
- **Primary Inconvenience:** The lack of an MTR station remains a significant drawback for commuters.

3. Cross-Border Collaboration & Supply Chain

- **Regional Focus:** Collaboration is centred on Guangdong (especially Shenzhen) due to the dense factory network and mature supply chain.
- **R&D Distribution:**
    - **Mainland:** Lower costs make it a primary site for engineering and general R&D.
    - **Hong Kong:** Focused on high-end R&D requiring specific expertise.
- **External Connections:** Overseas exhibition tours organised by the park provide flight/booth subsidies but are of limited help to mature companies with existing contacts. No direct policies facilitate exchange with South Korea or Taiwan.

4. University-Industry Collaboration Models

- **Model 1: Technical Validation:** Leveraging academic endorsement and data testing for marketing credibility.
- **Model 2: Funding Facilitation:** Partnering with professors to secure government matching funds (ITF).
- **Model 3: IP Licensing:** Directly licensing university-owned patents and technology for commercialisation.
- **Flexibility:** Partners are not fixed; selection depends entirely on the relevance of the specific project.

5. Location & Relocation Factors (San Tin Technopole)

- **Tangible Benefits > Geographic Location:** Proximity to Shenzhen alone is insufficient to justify a move.
- **Decision Drivers:** Relocation depends on concrete incentives:

    - Significant rent reductions (affordable rent is the priority).
    - Easier funding application processes and salary subsidies.
    - Exhibition support and higher quotas for engineer funding.
- **Space Requirements:** Massive space is only a priority for manufacturing; R&D-focused startups prioritise policy and software support.
- **Non-Factors:** Recreational facilities are "nice to have" but not a deciding factor for business location.

6. Hong Kong's Strengths & Emerging Challenges

- **Fading Traditional Advantages:** Free port status, simple tax, and free capital flow are losing impact as HK is increasingly labelled as "part of Mainland China."
- **International Decline:** Geopolitical uncertainties have eroded international confidence and created "de-risking" concerns for global clients.
- **The "Involution" Problem:** Extreme competition from Mainland China (lower costs, higher efficiency) means HK startups cannot survive with "me-too" products.
- **Survival Strategy:** Must possess unique, high-end technology that cannot be easily replicated.

7. Integrated Support Needs (Hardware/AI Startups)

- **The Skill Gap:** Many AI/software startups excel at algorithms but lack knowledge of hardware, mechanics, or supply chains.
- **Prototyping Difficulty:** Transitioning from an algorithm to a physical demo prototype is a major hurdle.
- **Proposed Solution:** A "one-stop" prototype development support system—including mechanical design, PCB prototyping, and firmware development—would be a major differentiator for the park.

8. Strategic Recommendations for Improvement

- **Offer Differentiated Value:** Focus on concrete rental concessions and simplified funding processes.
- **Infrastructure:** Improve transportation connectivity (MTR).
- **Hardware Support:** Provide specialised services for mechanical design and prototype development to support software-to-hardware integration.

Interview Minutes (3)

Interviewee: Dr Francis CHEUNG Neoton (Chairman of Doctoral Exchange; Member of the Chief Executive's Policy Unit Expert Group; Advisor of Our Hong Kong Foundation)
Interview Place: The Foreign Correspondents' Club
Interview Time: April 10, 2026, 2:30 PM

1. Development of NM

- **Actual development area:** Much less than expected.
- **Lack of coordination:** Poor alignment with other development areas in NM (e.g., NTN).
- **Strategic Need:** Requires planning at the border level (overall Northern Metropolis) rather than fragmented plots.

2. I&T Industry

- **Economic Sustainability:** The current operation mode of the Loop may not be financially viable long-term.
- **Lack of Anchor Players:** Missing key leaders in the industry chain/ecosystem to create a clustering effect.
- **Leveraging HK's Advantages:** Should utilise specific strengths compared to Shenzhen (SZ):
    - Target: International market vs. SZ targeting the mainland.
    - Framework: Law, tax, language, and education systems.
- **Planning vs. Industry Cycles:** * Future industries have short cycles (e.g., 6 months).
    - Execution of San Tin Technopole (STT) may take 10+ years.
- **Suggestions:**
    - **Flexible land use:** Move away from three restricted corridors; favour mixed-use industry development.
    - **Stakeholder Engagement:** Interview industry professionals to define future demands.
    - **Functional Space:** Provide appealing virtual and face-to-face meeting venues.
    - **Soft Capital:** Prioritise education institutions alongside physical infrastructure.
- **Community Building:** Create a foreign-friendly environment by clarifying "red lines" vs. "green zones" under the National Security Law to reduce confusion.

3. Problems of STT

- **Connectivity Issues:** Lack of immediate metro connection; reliance on shuttles (potential 2km distance to stations).
    - Example: Long waiting times for buses in Science Park as a warning.
- **Demand-Led Planning:** Identify anchor players/large businesses first to understand their demands before finalising infrastructure.
- **Regulatory Limitations:** Constraints of the Statutory Outline Zoning Plan (OZP).
    - **Suggestion:** Implement large-scale mixed-use land parcels.
    - **Reference:** Singapore's approach to buying development rights.

4. Urban-Rural Integration

- **Village Exclusion:** Old villages are excluded from the main plan but surrounded by high-density builds.
- **Topographical Conflict:** Surrounding buildings will be elevated ~7m for flood prevention, isolating the original villages.
- **Poor Interface:** Lack of buffer zones leads to "fake" integration.
- **Suggestion:** Focus industry development along the Shenzhen River and leave fish ponds near villages for preservation and a natural transition.

5. NTM University Town

- **Land Allocation Mode:** The current model (single plots for single schools) is too rigid.
- **Collaboration:** Should allow multiple institutions to co-organise executive programs.
- **Curriculum Breadth:** Include Arts/Philosophy (for management roles) rather than just STEM.
- **Operation:** Consider renting land instead of selling.
- **Spatial Integration:** NTM is located too far from STT; suggest integrating educational institutions directly into the Technopole.

6. Twin Cities Cross-Boundary Issues

- **Movement of Goods/People:** Implement "white lists" or visa-free access for samples and personnel.
- **Systemic Synergy:** Combine the legal/market advantages of HK with the scale of SZ.
- **Land Disposal Challenges:** Large-scale land sale may fail because:
    - Developers are wary of residential demand.
    - Competition from cheaper housing in Shenzhen.
    - Future supply peaks in other HK areas.
    - Delayed metro provision.
- **Solution:** Anchor large industries in STT first; clustering will naturally drive housing demand.

7. Housing Issue

- **Addressing the SZ Preference:** How to retain residents who prefer living in Shenzhen for lower costs.
- **Value Proposition:** Provide a high-end environment that justifies HK’s higher housing prices.
- **Flexibility:** Ensure planning is attractive enough for industries to relocate to STT, creating a local "live-work" ecosystem.

Interview Minutes (4)

Interviewee: Mr Wong, Kam-sing (former Secretary for the Environment)
Interview Place: Tseung Kwan O
Interview Time: April 12, 2026, 3:30 PM

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1. Implementation of Ecological Conservation

- **Long Valley NDA (Long Yuen):** Considered a successful case of ecological conservation. It is the first instance where high ecological value land was proactively protected and represents a milestone in Hong Kong's approach to valuing ecological conservation.
- **Main Barriers to Implementation:**
    - **Conflicting Stakeholder Interests:** Involves landowners, NGOs, and farmers (e.g., disputes over farmland distribution).
    - **Governance Challenge:** Planning may be technically sound, but it lacks social consensus.
    - **Financial Constraints (Primary Challenge):** Conservation land does not generate short-term economic returns. Large land areas result in high costs and slow implementation.
- **Institutional Structure:** Not fragmented; there are only two main departments in this case:
    - **AFCD Role:** Responsible for monitoring.
    - **CEDD Role:** Responsible for construction.
- **On Biodiversity (e.g., Eurasian otter):** Government transparency in data is adequate (database established a few years ago), and the policy framework is generally sound. The main issue lies in implementation effectiveness. Bird species are more sensitive ecological indicators than otters.

2. Blue-Green Network Integration

- **Existing Fish Ponds:** Currently not optimal habitats for birds, and are economically unsustainable for fishermen as birds eat their fish.
- **Proposed Transformation:**
    - Shift towards more sustainable aquaculture practices (learning from the GBA).
    - Introduce sloped pond edges and improved water circulation to enhance both fish survival and bird habitat quality.
- **Key Principle:** Ecological value and intensity are more important than total area.
- **Spatial Strategy:** Build new fish ponds in buffer zones near the Technopole to allow higher development density in other areas (permitting taller buildings). This achieves both land-use efficiency and ecological conservation.
- **Implementation Approach:** Trial-and-error through pilot projects followed by on-site testing and scaling led by AFCD.

3. Net-Zero and Circular Economy Governance

- **Main Barriers:** High financial costs (infrastructure and logistics), operational complexity of waste collection systems, and low public participation (the most significant issue), including NIMBY (Not In My Backyard) resistance.
- **Policy Effectiveness:** Government blueprints and climate strategies have had a positive social influence. While targets are not always achieved, influencing public behaviour is considered more important than strictly meeting numerical targets.

4. Cross-Cutting Insights

- **Implementation Gap:** Policy vision and planning frameworks are in place; the main challenge lies in execution and delivery.
- **Economic Constraints:** Financial viability is a key limiting factor across ecological conservation, blue-green infrastructure, and circular economy systems.
- **Shift Towards Active Planning:** Movement from passive conservation to actively redesigning ecological systems. Integration of land use, ecology, and infrastructure is required.
- **Importance of Social Factors:** Stakeholder conflicts, public participation, and community acceptance are often more critical than technical challenges.

Interview Minutes (5)

Interviewee: Mr Henry Jiang (Head of Product and Co-founder, Billie); Dr Godfrey Ho (CEO and Founder, Bitlliant); Mr Perry Chan (CEO and Co-founder, DefyPay); Mr WK Wong (CEO and Founder, Filix Medtech)
Interview Place: 2/F, Hong Kong Housing Society (HS) Centre, 65-67 Ka Shing Street, Fanling
Interview Time: April 22, 2026, 11:00 AM

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1. The Co-working Space: Smart-Space PropTech

- **Strategic Positioning:** The first PropTech-themed co-working space in both the NM and Hong Kong. It serves as a "terminal station" (終點站) connecting local startups to the GBA.
- **Partnership Model:**
  - **HS Role:** Provides capital, manpower, and land/space.
  - **Cyberport Role:** Leases and manages the park; brings in a network of over 2,000 companies.
- **Facilities & Capacity:**
  - Located on the 7th and 8th floors (freed up from traditional housing use).
  - Offers 107 workstations; occupancy is secondary to its role as a networking hub and showroom.
- **The Program:** A 5-year initiative aiming for 42 startups across 3 cohorts. Focuses on both hardware and "software" (business support).

2. Role of the Housing Society (HS) in Innovation

- **Modernisation:** Moving away from 70 years of traditional IT systems (finance-heavy) toward active Property Technology.
- **The "Pain Point" Solution:** HS provides the most critical resource for PropTech—**Proof of Concept (PoC)**.
  - Allows startups to conduct field visits and pilot testing in actual HS estates.
  - Startups can interview frontline staff to validate if their tech is practical for real-world operations.
  - **Success Cycle:** HS often purchases successful solutions; booths are set up to sell these solutions to major developers (e.g., SHKP, Sino).

3. Startup Ecosystem & Support

- **Cyberport Incubation Program (CIP):** Provides a pipeline from Micro-fund to full incubation funding.
- **Strategic Support:**
  - Partnerships with tech giants (e.g., Microsoft providing Copilot free trials).
  - International networking, including overseas business trips and government connections.
- **Trend in PropTech:** First and second cohorts showed that while construction tech is mature, **Property Management tech** is a growing priority for management-level stakeholders.

4. Key Success Factors & Regional Competitiveness

- **The "Hong Kong Advantage":**
  - International positioning and "same culture/language" efficiency.
  - Easier access to diverse talent and transnational corporations.
  - **AI Integration:** Shift toward using AI for labour-intensive and repetitive tasks (wide-coding trends).
- **Mainland vs. Hong Kong (Qianhai Comparison):**
  - **Qianhai:** Strong on "Talent Housing" (free/cheap rent) and direct government-to-startup feedback.
  - **Hong Kong:** Strong on networking, business lineup (connecting startups with firms like 3HK or China Mobile), and acting as a high-end showroom for clients.

5. Urban Planning & Connectivity (Future Outlook)

- **Accessibility is King:** In the NM, transportation/accessibility is more important than "community" feel.
  - *Example:* Success of shuttle buses in TKO vs. lack of transport in Shekou K11.
- **San Tin Technopole:** Potential to attract a cluster of incubation labs to create a "gravity" effect for the area.
- **Living-Working Balance:** Noted trend of "Live in Shenzhen, Work in HK." Accommodation for students and young professionals is the key to talent retention.
- **Spatial Needs:** Future demand for "Business Lounges" over traditional meeting rooms; spaces that function as showrooms rather than just fixed desks.

Interview Minutes (6)

Interviewee: Mr Man Hau-chung (Cheng Yick-chi Chair Professor in Manufacturing Engineering; Chair Professor of Materials Engineering; Dean of the Faculty of Engineering, and Director of the University Research Facility in 3D Printing)
Interview Place: Zoom
Interview Time: April 16, 2026, 05:30 PM

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1. Chau Tau Village

- **Heritage and Values to be Safeguarded:**
  - Continuity of the Man lineage.
  - The Chinese culture of respecting and remembering ancestors and roots.
  - Protection of burial sites, temples, and agricultural land.
- **Spatial Urban-Rural Interface:**
  - Implementation of low-density development.
  - Urban design that remains harmonious with the existing village context.
- **Intangible Cultural Heritage:**
  - Preservation of traditional practices: Dim dang, Ching Ming Festival, and Chung Yeung Festival.
  - Maintaining the village as a focal point for the diaspora to return for ancestral worship and the Chinese New Year.
- **Existing Infrastructure:**
  - Hygiene and streetlights are currently sufficient.
  - Demand for new GIC (Government, Institution, or Community) facilities within the development to improve the quality of life.
- **Villagers' Views on Development:**
  - General understanding and acceptance of the need for development, influenced by the visible modernisation across the river.
- **Cultural Integration:**
  - Openness to sharing local festivals and cultural heritage with future residents of the San Tin Technopole.
- **Community Engagement:**
  - Essential requirement for planners to communicate directly with villagers to identify and protect structures of local importance.

2. Industry

- **R&D Support:** Necessity for robust research and development support systems tailored to companies within the San Tin Technopole.
- **The R&D Ecosystem:**
  - While funding is crucial, a "middleman" institution is required to handle the transitional stages from research to a finished product.
  - **Reference Model:** Suggests looking to the Fraunhofer Institute in Germany as a benchmark for applied research.
- **Hong Kong's Competitive Edge:**
  - **International Context:** Leveraging HK's global position by actively collaborating with overseas research labs.
  - **Trustworthy Branding:** Capitalising on Hong Kong's image of reliability.
- **Strategic Suggestion:** Development of a robust certification system and industry; San Tin is identified as the ideal location to establish this trust-based industrial hub.

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